



OXFORD DICTIONARY OF

BIOCHEMISTRY AND MOLECULAR BIOLOGY



**Fully
up-dated
revised
edition**

Đây là bản dịch một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:
Huỳnhnguyên Co. LTD, Biotechnology, Trades and Services, 201 Lương Nhữ Học, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com



OXFORD DICTIONARY OF
**BIOCHEMISTRY AND
MOLECULAR BIOLOGY**



Editors

General Editors

Dr A. D. Smith (Managing Editor) *Emeritus Reader in Biochemistry, and Honorary Research Fellow, Department of Molecular Pathology, University College London Medical School*

Professor S. P. Datta *Emeritus Professor of Medical Biochemistry, and Honorary Research Fellow, Department of Biochemistry and Molecular Biology, University College London*

Dr G. Howard Smith *Honorary Research Fellow, Department of Biochemistry and Molecular Biology, University College London*

Professor P. N. Campbell *Emeritus Professor of Biochemistry, and Honorary Research Fellow, Department of Biochemistry and Molecular Biology, University College London*

Professor R. Bentley *Professor Emeritus of Biochemistry, Department of Biological Sciences, University of Pittsburgh*

Dr H. A. McKenzie *Fellow, School of Chemistry, University of New South Wales, Australian Defence Force Academy, and Visitor, Molecular Medicine, John Curtin School of Medical Research, The Australian National University, Canberra*

Subject Editors

Dr D. A. Bender *Senior Lecturer, Department of Biochemistry and Molecular Biology, University College London*

Dr A. J. Carozzi *NH and MRC Research Officer, Centre for Molecular and Cellular Biology, University of Queensland*

Professor T. W. Goodwin FRS *Emeritus Johnson Professor of Biochemistry, University of Liverpool*

Dr J. H. Parish *Senior Lecturer, School of Biochemistry and Molecular Biology, University of Leeds*

Dr S. C. Stanford *Senior Lecturer, Department of Pharmacology, University College London*



OXFORD DICTIONARY OF
BIOCHEMISTRY AND
MOLECULAR BIOLOGY
REVISED EDITION

Managing Editor **Dr A D Smith** University College London

General Editors **Professor S P Datta** University College London

Dr G H Smith University College London

Professor P N Campbell (Chairman) University College London

Dr R Bentley University of Pittsburgh

Dr H A McKenzie Australian Defence Force Ac.

Subject Editors **Dr D A Bender** University College London

Dr A J Harris University of Queensland

Professor T W Goodwin University of Liverpool

Dr H A McKenzie Australian Defence Force Ac.

Dr J H Parish University of Leeds

Dr C Stanford University College London

OXFORD

UNIVERSITY PRESS

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:
Huynguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynguyen@huynguyentld.com.



OXFORD

UNIVERSITY PRESS

Great Clarendon Street, Oxford OX2 6DP

Oxford University Press is a department of the University of Oxford.
It furthers the University's objective of excellence in research, scholarship,
and education by publishing worldwide in

Oxford New York

Athens Auckland Bangkok Bogotá Buenos Aires Calcutta
Cape Town Chennai Dar es Salaam Delhi Florence Hong Kong Istanbul
Karachi Kuala Lumpur Madrid Melbourne Mexico City Mumbai
Nairobi Paris São Paulo Singapore Taipei Tokyo Toronto Warsaw

with associated companies in Berlin Ibadan

Oxford is a registered trade mark of Oxford University Press
in the UK and in certain other countries

Published in the United States
by Oxford University Press Inc., New York

© The General Editors, 1997

The authors have asserted their right under the Copyright, Designs and Patents Act, 1988,
to be identified as authors of this work

First published 1997

Revised edition 2000

All rights reserved. No part of this publication may be reproduced,
stored in a retrieval system, or transmitted, in any form or by any means,
without the prior permission in writing of Oxford University Press,
or as expressly permitted by law, or under terms agreed with the appropriate
reprographics rights organization. Enquiries concerning reproduction
outside the scope of the above should be sent to the Rights Department,
Oxford University Press, at the address above

You must not circulate this book in any other binding or cover
and you must impose this same condition on any acquirer

A catalogue record for this book is available from the British Library

Library of Congress Cataloging in Publication Data

(Data applied for)

ISBN 0 19 850673 2

Typeset by Market House Books Ltd, Aylesbury

Printed in Great Britain by

Butler & Tanner Ltd, Frome



Preface

Nearly twenty years ago one of us (S. P. D., soon joined by G. H. S.), began a distillation of the elements of biochemistry into an alphabetical arrangement. The task was formidable and eventually other editors were recruited, an editorial board was established, and now the work is offered as the *Oxford Dictionary of Biochemistry and Molecular Biology*. It is hoped that the dictionary will serve the needs of the research biochemist or molecular biologist, as well as teachers of the subject and their students. In addition, it should prove of value to practitioners of other fields of study or work seeking the meaning of a biochemical term.

An important function of a dictionary is to provide guidance on current usage in the field within its scope. The original 12-volume *Oxford English Dictionary* was compiled from about five million slips of paper bearing sentences or phrases extracted by some thousands of 'readers' from classical works of literature and those of the best contemporary authors. It was thus firmly based on good usage. In scientific subjects, specialist terminology is often codified in sets of recommendations regarding nomenclature, meaning, abbreviations, symbols, and so on. These have been agreed by international commissions (e.g. those of The International Union of Pure and Applied Chemistry and The International Union of Biochemistry and Molecular Biology) as a means of preserving order and facilitating communication between scientists. We have striven to conform as far as possible to the relevant international recommendations, but in some cases, where usage so frequently diverges from a recommendation that adherence to it would seriously detract from ease of use of the dictionary, we have kept to the principle that the dictionary should reflect usage (see the definitions of *lexicographer*). This does not extend, of course, to cases where usage, however widespread, contradicts sound scientific principles. The internationally agreed recommendation is always also listed. The various compilations of these recommendations that have been drawn upon are listed in Appendix B, together with a number of other sources of information on nomenclature.

Biochemistry is the discipline that embraces the study of the structure and function of life-forms at the molecular level.

Molecular biology is a closely related discipline that originates in the study of DNA and its metabolism, and now embraces all those investigations that exploit the technology that has resulted from this work. Both disciplines aim to explain the behaviour of life-forms in molecular terms, and are so closely interrelated that separation is barely possible. It is inevitable that the content of this dictionary is to a degree arbitrary, but it is hoped that all important aspects of these subjects have received consideration. The compilers have attempted to offer a broad coverage of terms encountered in the literature of biochemistry and molecular biology by including an appreciable number from cognate sciences. Although the compilation is designed primarily to serve readers of contemporary material, the needs of those who turn to older literature have also been borne in mind. Some of the entries thus have a historical flavour, some obsolete terms are included (e.g. *zymase*), and in some cases a historical approach has been used as the best means of presenting an explanation of a term, as for example in the case of the entry for *gene*. The value of a scientific dictionary is enhanced by inclusion of contextual information as well as mere explanations of meaning or terminology. This dictionary will be found to have some of the attributes of an encyclopedia, although the extent to which it veers in this direction has varied with the whim of its compilers. It is our hope that in a single volume the reader has easy access to basic definitions as well as a generous helping of other information.

In the present-day world, we are assailed by floods of 'information'. It has been suggested that the average weekday edition of a newspaper of record (e.g. *The New York Times*) provides more information than Shakespeare and his contemporaries would have acquired in a lifetime. With the availability of much information through the Internet, it may be asked whether a dictionary in paper form is actually necessary. In answer, we note that the Internet can be slow, and is not readily accessible in some parts of the world; the databases may be inadequate, and although usually very up-to-date, the high cost of their maintenance restricts them to specialized knowledge in a limited number of fields. Moreover, books have a



Preface

quality of their own, which is enabling them to maintain their popularity. It appears that the increasing use of the Internet is actually paralleled by the rate of publication of printed dictionaries; in an information-hungry age, there cannot be too many sources of good-quality information.

We are deeply indebted to the Leverhulme Trust for the award of an emeritus fellowship to one of us (A. D. S.), to University College London, which has provided us with friends and expert colleagues, and to Dr O. Theodor Benfey, Dr Mary Ellen Bowden, and Professor Arnold Thackray, The Beckman Center for History of Chemistry, Chemical Heritage Foundation, Philadelphia, and Dr John Edsall, Harvard University for assistance with biographical data.

Particular thanks are due to Dr H. B. F. Dixon for much advice on nomenclature and related matters. Help on questions of nomenclature from Dr G. P. Moss and Dr A. D. McNaught is also acknowledged. We are grateful to Dr D. H. Jenkinson for his help with the recommendations of the International Committee on Nomenclature in Pharmacology. We are also grateful for the valuable advice of Professor K. W. Taylor and Dr J. L. Crammer, on clinical topics, and Professor M. C. W. Evans, on plant biochemistry, and to Dr Margaret McKenzie, for reading the proofs.

During the earlier stages of the project, Mrs S. Gove gave much valuable assistance and Miss A. Straker was most helpful in suggesting terms for inclusion. We also wish to thank all those other friends and colleagues, in addition to those separately listed, who have unstintingly given us help and advice.

We are pleased to acknowledge the collaboration and material support given to us by Oxford University Press. We also acknowledge the very friendly cooperation of Market House Books, especially the patience and good humour of Dr John Daintith through all the complications of the production. The copy editors, Robert Hine and Jane Cavell, made a number of helpful suggestions.

The compilers offer no apology for their failure to include many deserving terms in the dictionary, but would be pleased to have their attention drawn to errors and to receive suggestions for additional entries in any future edition.

Correspondence concerning the dictionary is best addressed to the Managing Editor, *Oxford Dictionary of Biochemistry and Molecular Biology*, c/o Oxford University Press, Great Clarendon Street, Oxford OX2 6DP

January 1997

A. D. Smith
S. P. Datta
G. H. Smith
P. N. Campbell
R. Bentley
H. A. McKenzie

This whole book is but a draught—nay, but the draught of a draught. Oh, Time, Strength, Cash, and Patience.

Herman Melville (1851) *Moby Dick, or The Whale* (ed. T. Tanner, 1988, p. 147, Oxford University Press).

Preface to the revised edition

It must be inevitable with any work of this nature that a number of imperfections and errors occur. So the opportunity provided by the need to reprint this dictionary has been taken to effect some improvements within the limitation imposed by retention of the original pagination. As well as the correction of a variety of minor misprints and other minor defects, over four hundred entries have been either revised or completely rewritten, and fifty or so new entries have been provided, some to remedy deficiencies and others to provide additional terms that have become of topical interest. To help make way for the new ones, about half as many original entries have been

deleted. In addition, Appendices B, C, and D have been updated, and Appendix B has been expanded and provided with all the relevant Internet addresses available at the time of writing.

Valuable comments on the original edition by a number of readers are gratefully acknowledged, and thanks are again due to Dr. H. B. F. Dixon for advice on aspects of nomenclature as well as to Oxford University Press and Market House Books for their much appreciated cooperation.

September 1999

A. D. S.

Note on proprietary status

This dictionary includes some words which are, or are asserted to be, proprietary names or trade marks. Their inclusion does not imply that they have acquired for legal purposes a non-proprietary or general significance, nor is any other judgement implied concerning their legal status. In cases where the editor has some evidence that a word is used as a proprietary name or trade mark this is indicated by the designation *proprietary name*, but no judgement concerning the legal status of such words is made or implied thereby.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.



Guide to the Dictionary

1. Alphabetical order

1.1 Main order of headwords

Alphabetical order is determined on a letter-by-letter basis, not word by word; spaces are disregarded:

acid
acid anhydride
acid–base balance
acid–base catalysis
acid dissociation constant
acid dye
acidemia

1.2 Nonalphabetic characters

Numbers, hyphens, primes, and subscript/superscript text are ignored for the purpose of indexing; an example is the following sequence of entries:

FSH-RH
F1 sphere
F' strain
F-type pentose phosphate pathway
ftz

1.3 Locants and modifiers

In chemical names, any locants and other hyphenated modifiers such as *cis*-, *trans*-, *p*-, and alphabetic Greek characters are not used to determine primary alphabetical order; hence the following entries all appear under the letter A:

N-acetylgalactosamine
p-aminobenzoic acid
γ-aminobutyrate shunt
6-aminohexanoic acid

However, the unhyphenated letters 'c' in 'cDNA' and 'd' in 'dCTP', for example, are treated as integral parts of the word and are used to determine alphabetical order.

1.4 Secondary order involving locants

When such modifiers constitute the only difference between two headwords, they determine the alphabetical order of the entries:

benzodiazepine	encephalitis
o-benzoquinone	3'-end
p-benzoquinone	5'-end
benzoyl	end+

1.5 Format differences in headwords

The order for entries where the headword is identical except for format is

b, b, b-, b-, -b, -b, B, B, B-, B-, -B, -B

1.6 Subscripts and superscripts

Single letters with subscripts or superscripts are treated as single letters for the purposes of indexing, so entries for ***k*_{cat}** and ***K*_m** will both be found in the list of single-letter entries at the beginning of the letter K. The primary order of these single-letter entries is determined by their format (*see* section 1.5); where there is more than one entry with a given format (e.g. italic, lower case ***k***), these are arranged by alphabetical order of their subscripts/superscripts.

1.7 Greek letters

- Where Greek letters form part of a chemical name, they are not used to determine alphabetical order (*see* section 1.3). Otherwise they are written out in full in the headword, e.g. **nu body**, **beta strand**.
- The names of the letters of the Greek alphabet, together with their English transliterations used in etymologies, are listed in Appendix A. The meanings assigned to Greek alphabetic characters used as symbols are also given in Appendix A.
- Greek characters are set in italic type when the character represents a variable or locant and in roman type when it represents a unit or subtype e.g. of a protein or particle.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynguyen@huynguyentld.com.

Guide to the Dictionary

2. Format of entries

2.1 Summary of typefaces

- The following distinguishing typefaces are employed in addition to the text light serif typeface used for definitions:

large bold sans serif	headwords
text bold serif	alternative terms for and variant spellings of headwords; hidden entries; run-ons
text bold sans serif	cross-references
<i>text italic serif</i>	usage notes and field labels; parts of speech; foreign language terms (including scientific and medical Latin); symbols for physical quantities and fundamental physical constants; stereochemical prefixes and alphabetical locants

2.2 Headwords

- For each entry, the headword is in bold, sans serif type.
- Upper-case (capital) initial letters are used only for proper names (or terms derived from them) and for proprietary names. Abbreviations and symbols are printed in upper and/or lower case as appropriate.
- If a term would normally be set in bold type, this is indicated in the entry:
B symbol for 1 Napierian absorbance (see **absorbance**).
2 magnetic flux density (bold italic).
- Where the same basic term is used in different typefaces, such as roman/italic, or upper case/lower case, or as a prefix or suffix, each usage is given as a separate headword. For example, **h**, *h*, **H**, and *H* each have a separate entry.
- The order in which such entries are given is listed in section 1.5.

2.3 Alternative terms and variant spellings

2.3.1 Choice of headword

Where alternative terms for a headword, or variant spellings of it, exist (see section 1.3), the headword selected for the main entry is generally the recommended or preferred term, or the one judged to be the commonest. Exceptions to this generalization are those instances where the name of a Greek alphabetic character is written out for convenience of indexing:

beta globulin or **β globulin**....

2.3.2 General

- Any alternative terms and alternative spellings are listed following the headword in bold, serif type:

retrovirus or **ribodeoxyvirus** or **RNA-DNA virus** any virus belonging to the family Retroviridae....

- Notes regarding the usage of these alternatives may be given in brackets and in italics; for example

DNA glycosylase or (sometimes) **DNA glycosidase** any of a group of enzymes....

bacteremia or (esp. Brit.) **bacteraemia** the presence of live bacteria in the blood

bilirubin or (formerly) **bilirubin IX α** the recommended trivial name for the linear tetrapyrrole....

- These alternative terms and spellings also appear as entries in the alphabetical sequence, with a cross-reference to the main entry where the term is defined, unless the variant would appear close to the main entry. Additional information is given where appropriate:

demoxycocin an alternative name for **deaminococytocin**.

fructose-1,6-diphosphatase a former name for **fructose-bisphosphatase**.

lipide a variant spelling of **lipid**.

molecular exclusion chromatography a less common name for **gel-permeation chromatography**.

oleomargarine an alternative name (esp. US) for **margarine**.

penatin an obsolete name for **glucose oxidase**.

2.3.3 Chemical names

- Synonyms may be given following the headword, in the order: other trivial names (if any); the semi-systematic or semi-trivial name(s); older systematic name in style, if still in widespread use; the systematic name in currently recommended style.
- The headword used to represent a chemical compound that can exist in ionized form(s) is in most cases the name of its physiologically predominant form. So, for example, an entry is headed 'succinate' rather than 'succinic acid'.

2.3.4 Enzyme names

Alternative names may be listed following the headword, which is normally the recommended name; otherwise alternative names include the recommended name (if the headword is the common name), the systematic name, and other names. The EC number is also given.

2.4 Multiple definitions

- Where a term has more than one meaning, the different senses are numbered with bold Arabic numerals.

blockade 1 (in pharmacology) the saturation of a specific type of receptor with an antagonist to its normal agonist. **2** (in immunology) the overloading or saturation of the **reticuloendothelial system** with inert particles, such as carbon particles. **3** to impose any such blockade.

- The order of the numbered entries is generally determined by their biochemical significance.
- The different senses may be further subdivided into def. 1a, def. 1b, etc.

di+ comb. form **1** (in chemical nomenclature) (distinguish from **bis+** (def. 2)) **a** indicating the presence in a molecule of two identical unsubstituted groups, e.g. diethylsulfide, 1,3-dihydroxyacetone. **b** indicating the presence in a molecule of two identical inorganic oxoacid residues in anhydride linkage, e.g. adenosine 5'-diphosphate. **2** or **bis+** (def. 1) denoting two, twofold, twice, doubled.

- Homographs are not distinguished.

2.5 Hidden entries

Hidden entries are terms that are not defined at their normal headword position. Instead they are treated (implicitly or explicitly) as



plicitly) at some other headword. They are set in bold serif type. In the following example, 'bentonite flocculation test' is the hidden entry:

bentonite a colloidal, native hydrated aluminium silicate clay consisting principally of montmorillonite, a complex aluminosilicate, $\text{Al}_2\text{O}_3 \cdot 4\text{SiO}_2 \cdot \text{H}_2\text{O}$, which has marked adsorptive properties. It is used as an inhibitor of nucleases and also in the **bentonite flocculation test**, a passive agglutination test in which antigen-coated bentonite particles are used to detect specific antibody.

2.6 Other information

2.6.1 Plurals

The plural form (or forms) of a headword is (are) given in parenthesis following the headword if its formation is non-standard, e.g. for Latin headwords, or where there is more than one form of the plural.

medulla (*pl. medullas or medullae*) the innermost part of an organ, tissue, or structure; marrow, pith. — **medullary** *adj.*

2.6.2 Affixes and combining forms

- In common with other dictionaries, this Dictionary lists and defines many word elements that are used to compose terms or to modify existing terms. These are either combining forms (which are derived from parent words) or affixes (infixes, prefixes, and suffixes, none of which have parents).
- The usual lexicographical convention is to add a hyphen to suffixes and combining forms when listing them as headwords, although generally the hyphen is omitted in formation of composite terms. However, chemical and biochemical terminology also includes a considerable number of specialized affixes that retain the hyphen in the formation of composite terms (e.g. 'meso-' in 'meso-cystine'). In order to make an explicit distinction between these alternatives, this Dictionary departs from the common convention by adding a hyphen to an affix in a headword only when a linking hyphen is retained in a combination:

meso- *abbr.: ms-; prefix (in chemical nomenclature) designating a substance whose individual molecules contain....*

By contrast, combining forms (e.g., 'meso' in 'mesoderm') together with affixes producing unhyphenated composite terms, are listed with an added plus sign, placed after and/or before the word-element as appropriate:

meso+ or (sometimes before a vowel) **mes+** *comb. form* denoting middle, or intermediate.

+agogue or (US) **+agog** *suffix* denoting an agent that elicits or enhances the secretion of....

2.6.3 Abbreviations and symbols

- Where a term may be abbreviated or indicated with a symbol, this is noted after the headword.

nuclear magnetic resonance *abbr.: NMR or nmr*; the phenomenon that occurs when atomic nuclei....

electric potential or **potential** *symbol: V or f*; the work done in bringing unit electric charge....

- The distinction between an abbreviation and a symbol is a little blurred, since some abbreviations (e.g. lg) also may be

used as symbols. In general, the term 'symbol' is used here for

units and their decimal prefixes (e.g. m, mol; μ , M)

physical quantities and constants (e.g. *a*, *H*; *k*, *R*)

mathematical functions (e.g. exp, ln)

chemical elements (e.g. K, Mg)

groups of letters that can be used in place of a chemical group or compound in an equation or formula (e.g. CoA, Me)

recommended abbreviations for nucleotides, bases, or amino acids.

- The symbols for SI base and derived units and their decimal prefixes are mandatory; all other symbols are recommendations of IUBMB or IUPAC. In conformity with these recommendations, symbols for physical quantities and fundamental physical constants are printed in a sloping (italic) typeface.
- No distinction is made between acronyms, contractions, abbreviations, etc. All are classed as abbreviations. Abbreviations formed from the initial letters of two or more words are printed without periods (full-stops), in line with contemporary practice, but abbreviations that are shortened forms of single words have a terminal period.
- In addition to recommended abbreviations, the Dictionary lists a selection of others commonly encountered in the scientific literature.

2.6.4 Derived terms

Derived terms not meriting separate definition are listed at the end of the entry for the parent term, preceded by a bold em dash and followed by an abbreviation indicating the part of speech.

bactericide or **bacteriocide** any agent (biological, chemical, or physical) that destroys bacteria. —**bactericidal** or **bacteriocidal** *adj.*

2.6.5 Etymology

- Generally, the derivation of words is not explained in entries. The exceptions are for eponymous terms and other entries of particular etymological interest.
- The etymology is given within square brackets at the end of the entry.

ångström or **Ångstrom** *symbol: Å*; a unit of length equal to 10^{-10} metres. ... [After Anders Jonas Ångström (1814–74), Swedish physicist noted for his work on spectroscopy.]

- Greek elements of etymologies are transliterated:

chirality topological handedness; the property of non-identity of an object with its mirror image. ... [From Greek *kheir*, hand.]

2.6.6 Usage

- The field within which the term is used may be specified in italics and in parenthesis before the definition.

malonyl **1** (*in biochemistry*) the univalent acyl group, $\text{HOOC}-\text{CH}_2-\text{CO}-$, derived from malonic acid by loss of one hydroxyl group. **2** (*in chemistry*) the bivalent acyl

Guide to the Dictionary

group, $-\text{CO}-\text{CH}_2-\text{CO}-$, derived from malonic acid by loss of both hydroxyl groups.

- Notes may also be given regarding the use of alternative terms and variant spellings: *see* section 2.3.2.

2.7 Cross-references

2.7.1 Format

- Cross-references are set in bold sans serif type, e.g. **thio-uridine**.
- Where a cross-reference refers to only one sense of a word with multiple definitions, this is indicated as in the following example:

siderophage *an alternative name for* **siderophore** (def. 1).

2.7.2 Types of cross-reference

- There are cross-references from a variant spelling, or a less commonly used term, etc., to the entry where the term is defined. For examples, *see* section 2.3.2.
- Some cross-references are to related entries giving more information. These may be either embedded in the text:

octulose any **ketose** having a chain of eight carbon atoms in the molecule.

or listed at the end of the entry:

vacuum evaporation a technique for *See also* **shadow casting**.

- Cross-references may also be used to draw attention to contrasting terms:

heterochromatin ... *Compare* **euchromatin**.

or to pairs of easily confused terms:

prolidase *another name for* **X-Pro dipeptidase**. *Distinguish from* **prolinase**.

prolinase *the recommended name for* **Pro-X dipeptidase**. *Distinguish from* **prolidase**.

3. Abbreviations

<i>abbr.</i>	abbreviation
<i>adj.</i>	adjective
<i>adv.</i>	adverb
<i>Brit.</i>	British
<i>comb. form</i>	combining form (<i>see</i> section 2.6.2)
<i>3-D</i>	three-dimensional
<i>def.</i>	definition
<i>e.g.</i>	[Latin, <i>exempli gratia</i>] for example
<i>esp.</i>	especially
<i>etc.</i>	etcetera
<i>Fr.</i>	French
<i>i.e.</i>	[Latin, <i>id est</i>] that is
<i>max.</i>	maximum
<i>n.</i>	noun
<i>pl.</i>	plural
<i>sing.</i>	singular
<i>sp. or spp.</i>	species (singular and plural respectively)
<i>US</i>	United States
<i>vb.</i>	verb

Other abbreviations are defined in the text itself.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

4. Other conventions

4.1 Spelling and hyphenation

4.1.1 Spelling

- For chemical and biochemical terms, recommended international usage is followed; thus, for example, 'heme' is used rather than 'haem', 'estrogen' rather than 'oestrogen', 'sulfur' rather than 'sulphur', and 'oxytocin' rather than 'ocytocin'. All variants are listed as headwords, however, with cross-references to the corresponding main entries.
- For common terms, e.g. 'colour', British spelling is used.

4.1.2 Hyphenation

- Hyphens are used attributively:
'T cell' but 'T-cell receptor'
'amino acid' but 'amino-acid residues'
- This also applies to enzyme names; thus for example, there is no hyphen following the 'glucose' in 'glucose 6-phosphate', but where this substrate forms part of an enzyme name, it is hyphenated, e.g. in 'glucose-6-phosphatase' or 'glucose-6-phosphate isomerase'.

4.2 Nomenclature

In most cases, headwords conform with the recommendations of the various nomenclature bodies of IUB, IUBMB, and IUPAC (*see* Appendix B for information regarding their nomenclature publications). Other usages are, however, given (*see* Preface). The phrase 'not recommended' has been used to indicate that certain forms are not the recommendation of one of these nomenclature bodies.

4.2.1 Drug names

The recommended international nonproprietary names are used (*International nonproprietary names (INN) for pharmaceutical substances*. World Health Organization, Geneva, 1992); hence, for example, main entries are found under epinephrine and norepinephrine rather than under adrenaline and noradrenaline.

4.2.2 Proprietary names

A few commonly used proprietary names are included; these may be listed at the end of an entry if considered to be of particular interest, especially to non-scientists:

acetaminophen or **paracetamol** ... *Proprietary names:* Tylenol, Panadol. It inhibits

or may be the main headword:

Sephadex.

4.2.3 Other substances

The main entry is under the name used most widely in the scientific literature. Where this is not the recommended name, a cross-reference is given from the recommended name to the main entry. For example, the name 'follicle-stimulating hormone (FSH)' is widely employed instead of the recommended name 'follicitropin', hence the former name has been used as the main headword. In such cases there is a cross-reference from the recommended name back to the entry where the substance is defined:

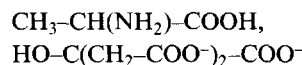
follicitropin *the recommended name for* **follicle-stimulating hormone**.



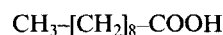
4.3 Representation of chemical structures

4.3.1 Typeset formulae

In conformity with IUPAC nomenclature recommendations for typeset chemical formulae, parentheses (round brackets) indicate a side chain:



and square brackets indicate a condensed chain:



4.3.2 Carbohydrates

- The cyclic forms of monosaccharides are depicted by Haworth representations as are some other compounds; for clarity, the carbon atoms of the heterocyclic ring, and their attached hydrogen atoms, are not shown. See the **Haworth representation** entry for more detail.
- Where an abbreviated terminology is included for oligosaccharide chains, the extended or condensed forms described in the publication entitled *Nomenclature of carbohydrates (recommendations 1996)* (see Appendix B, item 22) are variously used.
- Wherever possible, structure diagrams show absolute configurations.

4.4 Periodic table of the elements

The group numbers used in the text are those of the 18-column format of the table given in the 1990 edition of the IUPAC 'Red Book' (see Appendix B, item 2). The correspondence between this and other versions of the table is described in the **periodic table** entry and shown below the table displayed on the endpapers.

4.5 Amino-acid sequences

- For peptide sequences of up to 15 amino-acid residues, the three-letter code is used; longer sequences are given in the one-letter code.
- Motifs are given in the one-letter code.
- The full sequences of many proteins can be found in protein sequence databases, and database codes are given to facilitate access to these. The database codes relate to a number of different databases. The style of the code gives an indication of the database from which the data origi-

nate, but if the user does not recognize the code, it is necessary to search for it in a composite database that integrates data from all the major databases. Further information is offered in Appendices D and E.

4.6 Genes

- The accepted format of gene names (i.e., whether lower case or upper case or a mixture) varies between different organisms. Where an entry covers genes from various species, the convention for human genes is generally followed in the headword, i.e. all letters are given in upper case, e.g. '*JUN*'.
- However, when an entry refers only to a gene from a specified organism, the accepted convention for that organism is followed.

4.7 Names of organisms

- Those organisms whose Latin names are used frequently are listed in Appendix H.
- Where a binomial Latin name is repeated within an entry, the genus name is abbreviated after the first occurrence of the name; for example, the full form '*Escherichia coli*' is used when first mentioned in any entry, but subsequent references to this organism in the same entry are abbreviated to '*E. coli*'.

5. Appendices

A number of appendices have been included after the main alphabetical text, as follows:

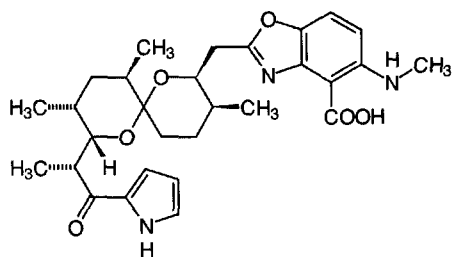
- Appendix A – The Greek alphabet and Greek characters used as symbols (page 697).
- Appendix B – Nomenclature publications (page 700).
- Appendix C – Organizations that are helpful to biochemists and molecular biologists (page 705).
- Appendix D – The Internet (page 709).
- Appendix E – Exploring the language of bioinformatics (page 715).
- Appendix F – Restriction enzymes and methylases (page 725).
- Appendix G – Sequence-rule priorities of some common ligands in molecular entities (page 739).
- Appendix H – Species names (page 740).



This page intentionally left blank

Aa

- a** 1 *abbr.* for adsorbed. 2 symbol for atto+ (SI prefix denoting 10^{-18}). 3 axial. 4 year.
- a'** symbol for pseudoaxial.
- a** symbol for 1 absorption coefficient. 2 acceleration (in vector equations it is printed in bold italic (*a*)). 3 activity (def. 3). 4 van der Waals coefficient. 5 as subscript, denotes affinity.
- a₀** symbol for Bohr radius.
- A** symbol for 1 acid-catalysed (of a reaction mechanism). 2 a residue of the α -amino acid L-alanine (alternative to Ala). 3 a residue of the base adenine in a nucleic-acid sequence. 4 a residue of the ribonucleoside adenosine (alternative to Ado). 5 uronic acid. 6 ampere.
- A** symbol for 1 absorbance. 2 activity (def. 2). 3 affinity. 4 Helmholtz function. 5 mass number/nucleon number.
- A_r** symbol for relative atomic mass.
- A_s** symbol for area.
- [A]_{0.5}** or **[A]_{1/2}** symbol (in enzyme kinetics) for the value of the concentration of a substrate, A, in mol dm^{-3} , at which the velocity of the reaction, v , is half the maximum velocity, V ; i.e. when $v = 0.5V$.
- [A]₅₀** symbol for the molar concentration of an agonist that produces 50% of the maximal possible effect of that agonist. Other percentage values (**[A]₂₀**, **[A]₄₀**, etc.) can be specified. The action of the agonist may be stimulatory or inhibitory. Compare **EC₅₀**.
- 2'-5'A** symbol for any member of a series of oligonucleotides of the general formula $p_aA[2'p5'A]_n$, where p and A are phosphoric and adenosine residues, respectively, and a and n are small integers ($a = 1, 2$, or 3 and n commonly $= 2, 3$, or 4). Potent inhibitors of protein biosynthesis *in vivo* and *in vitro*, they are believed to mediate the action of interferon on virus-induced cells.
- A23187** or **calcimycin** a toxic and weakly antibiotic substance isolated from cultures of *Streptomyces chartreusensis*. It is a lipophilic 523 Da monocarboxylic acid of complex structure, two molecules of which form stable lipid-soluble complexes at pH 7.4 with one atom of certain divalent metal cations, especially Mn^{2+} , Ca^{2+} , and Mg^{2+} ; monovalent cations are bound only weakly. It also forms lipid-soluble complexes with certain amino acids. It is used experimentally as a calcium ionophore.



- Å** symbol for ångström (unit of length equal to 10^{-10} m).
- aa** 1 symbol for an unknown or unspecified aminoacyl group when acting as a substituent on a base or internal sugar in a (poly)nucleotide. 2 *abbr.* for amino acid.
- AA** (formerly) symbol for an unknown or unspecified amino-acid residue. See **Xaa**.
- AAA** a codon in mRNA for L-lysine.
- AAC** a codon in mRNA for L-asparagine.
- Aad** symbol for a residue of the α -amino acid L- α -aminoadipic acid, L-2-aminohexanedioic acid.

- β Aad** symbol for a residue of the β -amino acid L- β -aminoadipic acid, L-3-aminohexanedioic acid.
- AAG** a codon in mRNA for L-lysine.
- A antigen** the antigen defining the A blood group. See also **blood-group substance**, **ABH antigens**.
- AAU** a codon in mRNA for L-asparagine.
- Ab** *abbr.* for antibody.
- abamectin** or **avermectin B₁** a metabolite of *Streptomyces avermitilis* used as an acaricide, insecticide, and a veterinary anthelmintic.
- A-band** an anisotropic band in a sarcomere.
- Abbe refractometer** a refractometer in which the critical angle for total reflection at the interface of a film of liquid between two similar glass prisms is used in determining the refractive index of the liquid. [After Ernst Abbe (1840–1905), German physicist famous for his researches in optics.]
- ABC** *abbr.* for 1 antigen-binding capacity. 2 ATP-binding cassette (see **ABC transporter**).
- ABC transporter** membrane transport proteins having the ABC molecular domain, named after ATP-binding cassette, characteristic of all members of a large superfamily of membrane transport proteins that hydrolyse ATP and transfer a diverse array of small molecules across membranes. See also **CFTR**, **MDR protein**, **sugar transporter**.
- abductin** an insoluble, rubber-like protein from the internal triangular hinge ligament of scallops.
- Abe** symbol for abequeose.
- abequose** symbol: Abe; 3,6-dideoxy-D-xylo-hexose; 3,6-dideoxy-D-galactose; a deoxysugar that occurs, e.g., in O-specific chains of lipopolysaccharides in certain serotypes of *Salmonella*. For the L enantiomer see **colitose**.
- ABH antigens** one of the systems of **blood group** antigens having determinants associated with oligosaccharide structures. It is the basis of the **ABO system**, which was the first human blood group antigen system to be detected, by Landsteiner in 1901, and it remains the most important in blood transfusion. Individuals having neither A nor B antigen express the H antigen, the product of an independent gene belonging to the Hh system. Antigens of the ABH system are oligosaccharide chains, in the erythrocyte carried on band 3 (the anion transporter) and band 4.5 (the glucose transporter), or on ceramide. A highly branched N-glycan, consisting of a trimannosyl-di-N-acetyl-chitobiosyl core with Gal(β 1-4)GlcNAc(β 1-3) repeats, forms the basis of ABH antigens. The H determinant is the precursor; antigen A is formed by addition of N-acetyl-D-galactosamine by fucosylgalactose α -N-acetyl-galactosaminyl-transferase (EC 2.4.1.40); antigen B is formed by addition of D-galactose by fucosylglycoprotein 3- α -galactosyltransferase (EC 2.4.1.37). The terminal sugar residues of importance are: H determinant, Fuc(α 1-2)Gal β -R; A determinant, GalNAc(α 1-3)(Fuc α 1-2)Gal β -R; B determinant, Gal(α 1-3)(Fuc α 1-2)Gal β -R. The enzyme responsible for adding the terminal fucosyl residue of the H determinant is galactoside 2- α -L-fucosyltransferase (EC 2.4.1.69). See also **A-transferase**, **B-transferase**.
- abiogenesis** or **spontaneous generation** the discredited doctrine that living organisms can arise from nonliving materials under current conditions. Compare **biogenesis** (def. 2).
- abiotic** characterized by the absence of life.
- abl** an oncogene from murine Abelson leukemia virus. The viral oncogene (v-*abl*) resulted from recombination of the Maloney murine leukemia virus with the protooncogene (c-*abl*). The protooncogene encodes a 145 kDa protein, which, together with the highly related *arg* gene, forms a unique family within the nonreceptor tyrosine protein kinases; database code **KABL_HUMAN**, 1130 amino acids (122.81 kDa). It was one



of the first protooncogenes identified by the study of retroviruses. The transforming ability of *v-abl* results from increased tyrosine kinase activity of the *gag-abl* p160 oncoprotein. In humans, inappropriate activation of *c-abl* occurs via a reciprocal translocation between chromosomes 9 and 22 in which *c-abl* (9q34) is joined at the **breakpoint cluster region** (bcr) of the *ph1* gene on chromosome 22(q11). In the translocation, a 3' portion of *c-abl* is spliced in-frame to a 5' portion of the *ph1* gene, resulting in an altered chromosome 22, which is referred to as the **Philadelphia chromosome** (*Ph*¹). The protein product of the spliced genes in the *Ph*¹ chromosome is a molecule of 210 kDa, which has increased tyrosine kinase activity compared with the *c-abl* product and is similar in structure to the *v-abl* product in that the N-terminal portion of Abl is removed. The *Ph*¹ chromosome occurs in more than 90% of individuals with chronic myelogenous leukemia and was the first description of a specific genetic translocation associated with a human malignancy. *c-Abl* can potentially regulate cell growth and may participate in growth regulation at multiple cellular locations, interacting with different cell components. It possesses SH2 and SH3 domains (see **SH domains**) and also domains involved in binding to F-actin and DNA. Thus it has been found in both cytoplasmic and nuclear locations and its DNA-binding activity appears to be cell cycle regulated by Cdc-2-mediated phosphorylation; it binds the **retinoblastoma protein** indicating involvement in transcriptional regulation.

ABM abbr. for 2-aminobenzyloxymethyl, a group used for derivatizing cellulose or paper. It is converted by diazotization into **DBM**.

abortive complex or **dead-end complex** or **nonproductive complex** any enzyme-substrate complex in which the substrate is bound to the enzyme in a manner that renders catalysis impossible so that products cannot be formed.

abortive infection infection of a bacterium by phage lacking phage DNA, e.g. in generalized **transduction**.

abortive transconjugate see **transconjugate**.

abortive transduction a type of **transduction** in which the donor DNA is not integrated with the recipient chromosome but persists as a nonreplicating fragment that can function physiologically and can be transmitted to one daughter cell at each cell division.

ABO system one of the systems of human **blood groups**, of great importance in blood transfusion because human plasma contains natural antibodies against A and B blood group antigens of the ABH system (see **ABH antigens**). The antigens on the red blood cells and the plasma antibodies corresponding to the various phenotypes are shown in the table. See also **A-transferase**, **B-transferase**, **O antigen** (def. 2).

Phenotype (blood group)	Antigen on red cells	Antibody in plasma
A	A	anti-B
B	B	anti-A
AB	A and B	neither
O	H	anti-A + anti-B

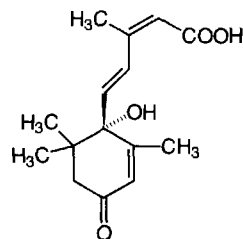
abrin a highly toxic ≈65 kDa glycoprotein obtained from the seeds of jequirity or Indian liquorice (*Abrus precatorius* L.), a tropical Asian vine that also occurs in Florida. It consists of an acidic A chain, ≈30 kDa, and a neutral B chain, ≈35 kDa, held together by disulfide bonds. The A chain is a powerful inhibitor of protein synthesis while the B chain functions as a carrier to bind abrin to the membrane and perhaps to assist penetration of the A chain into the cell. One well-chewed seed can be fatal. The A and B chains are derived from a common precursor: database code ABRA_ABRPR, 528 amino acids (59.24 kDa). Compare **abrine**, **ricin**. See also **ribosome-inactivating protein**.

abrine trivial name for *N*^α-methyl-L-tryptophan, *α*-methyl-tryptophan. Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynguyen@huynguyentld.com.

amino-β-(3-indole)propionic acid; an imino acid obtained from seeds of jequirity (*Abrus precatorius*). Not to be confused with **abrin**.

abscisic acid or (formerly) **abscisin II** or **dormin** abbr.: ABA; 5-(1-hydroxy-2,6,6-trimethyl-4-oxocyclohex-2-en-1-yl)-3-methylpenta-2,4-dienoic acid; a chiral sesquiterpene. The naturally occurring form, the 2*Z*,4*E*,*S* isomer, also designated (*S*)-abscisic acid, is a phytohormone formed by the degradation of carotenoids. It controls abscission in flowers and fruit but probably not in leaves, and is also implicated in geotropism, stomatal closure, bud dormancy, dormancy of seeds requiring stratification (i.e. those that will only germinate after exposure to low temperatures), and possibly tuberization.



abscissa the horizontal or *x* coordinate in a plane rectangular (Cartesian) coordinate system. Compare **ordinate**.

abscission the natural shedding of leaves, fruits, and other parts by a plant.

absolute 1 pure, unmixed; e.g. absolute alcohol. 2 not relative; e.g. absolute configuration. 3 describing a measurement defined in fundamental units of mass, length, and time that does not depend on the characteristics of the measuring apparatus; e.g. absolute temperature.

absolute alcohol the common name for pure ethanol, i.e. ethanol that has been freed of water. It may contain small amounts of benzene that have been added to aid in removing water. Substances may be added to absolute alcohol to render it unfit for human consumption and hence free of excise duty: **industrial spirit** contains 5% v/v methanol, while **methylated spirit** also contains pyridine, petroleum oil, and methyl violet dye, and **surgical spirit** also contains castor oil, diethyl phthalate, and methyl salicylate.

absolute configuration the actual three-dimensional arrangement of the atoms in a chiral molecule.

absolute reaction rate theory a theory that sets out to predict the absolute reaction rate of a chemical reaction from the quantum mechanical description of the potential energy changes during the interaction between chemical species. It is most widely drawn upon in applying thermodynamic reasoning to equilibria between reactants in the ground state and chemical species in the activated state or transition state.

absolute temperature see **thermodynamic temperature**.

absolute zero zero **thermodynamic temperature**, i.e. 0 K or -273.15 °C.

absorb see **absorption**.

absorbance symbol: *A*; a measure of the ability of a substance or a solution to absorb electromagnetic radiation incident upon it. It equals the logarithm of the ratio of the radiant power of the incident radiation, Φ_0 , to the radiant power of the transmitted radiation, Φ . For a solution, absorbance is expressed as the logarithm of the ratio of the radiant power of light transmitted through the reference sample to that of the light transmitted through the solution, the observations being made using identical cells. (Traditionally, radiant intensity was measured instead of radiant power, which is now the accepted form.) Two quantities are defined: (**decadic**) **absorbance** (symbol: A_{10} or *A*), and **napierian absorbance** (symbol: A_e or *B*).

$A_{10} = \lg(\Phi_0/\Phi) = \lg T^{-1} = -\lg(1 - \rho)$, and



$$A_c = \ln(\Phi_0/\Phi) = \ln T^{-1} = -\ln(1 - \alpha),$$

where T is the (internal) **transmittance** and α is the **absorptance**. These definitions suppose that all the light incident upon the sample is either transmitted or absorbed, reflection and scattering being negligible. The more general term **attenuance** should be used when scattering is considerable, as when the quantity $\lg(\Phi_0/\Phi)$ is measured to estimate the cell density of a culture. The older terms **absorbancy**, **extinction**, and **optical density** should no longer be used. *Compare* **absorption coefficient**.

absorbancy or absorbcency (formerly) an alternative term (no longer recommended) for **absorbance**.

absorbate a substance that is absorbed into another substance.

absorbed dose (in radiation physics) a measure of the energy deposition produced by **ionizing radiation** in any (specified) medium as a result of **ion-pair** formation. The CGS unit of absorbed dose is the **rad**; the SI derived unit is the **gray** (symbol: Gy); *compare* **exposure** (def. 3). *See also* **dose equivalent**, **dose rate**.

absorbence a variant spelling of **absorbance**.

absorbent 1 a substance that absorbs another substance. 2 having the capacity to absorb another substance.

absorptance or absorption factor symbol: α ; the ratio of absorbed to incident radiant or luminous flux. A dimensionless physical quantity given by $\alpha = \Phi_{\text{abs}}/\Phi_0$, where Φ_0 and Φ_{abs} are the incident and absorbed radiant powers, respectively.

absorptiometer 1 an apparatus, frequently a photoelectric device, for measuring light **absorption** by solids, liquids, or gases. 2 an apparatus for measuring the amount of gas absorbed by a liquid.

absorption 1 the act or process whereby one substance, such as a gas or liquid, is taken up by or permeates another liquid or solid. *Compare* **adsorption**. 2 the retention by a material of energy removed from electromagnetic radiation passing through the material. 3 the removal of any form of radiation, or the reduction of its energy, on passing through matter. 4 the process whereby a neutron or other particle is captured by an atomic nucleus. 5 a (in cellular physiology) the uptake of fluids by living cells or tissues. b (in animal physiology) the totality of the processes involved in causing water, the products of digestion, and exogenous substances of low molecular mass such as drugs, salts, vitamins, etc. to pass from the lumen of the gastrointestinal tract into the blood and lymph. c (in plant physiology) the uptake of water and dissolved salts through the roots. 6 (in immunology) the process of removing a particular antibody (or antigen) from a mixture by adding the complementary antigen (or antibody) and discarding the antigen-antibody complex so formed. *Compare* **immunosorption**.

absorption band or absorption line a region of darkness or absorption of radiation in the spectrum of heterochromatic radiation that has passed through an absorbing material.

absorption coefficient four different coefficients are defined. The (linear) **decadic absorption coefficient** (symbol: a) is defined by $a = A_{10}/l$; units m^{-1} . The (linear) **napierian absorption coefficient** (symbol: α) is defined by $\alpha = A_e/l$; units m^{-1} . The **molar (decadic) absorption coefficient** (symbol: ϵ) is defined by $\epsilon = a/c = A_{10}/cl$; units $\text{m}^2 \text{mol}^{-1}$. The **molar napierian absorption coefficient** (symbol: κ) is defined by $\kappa = \alpha/c = A_e/l$. A_{10} and A_e are the decadic and napierian absorbances respectively (*see absorbance*), l = path length, and c = amount-of-substance concentration.

absorption cross-section the probability that a photon passing through a molecule will be absorbed by that molecule multiplied by the average cross-sectional area of the molecule. The **net absorption cross-section** (symbol: σ_{net}) is defined by $\sigma_{\text{net}} = \kappa/N_A$, where κ is the molar napierian **absorption coefficient** and N_A is the Avogadro constant.

absorption factor an alternative name for **absorptance**.

absorption index symbol: k ; it is given by $k = \alpha/\lambda n^2$ where α

is the (linear) napierian **absorption coefficient** and $\tilde{\nu}$ the wavenumber in vacuum of the radiation.

absorption line an alternative name for **absorption band**.

absorption spectrometry the process of measuring an absorption spectrum with a **spectrometer**. **Absorption spectrophotometry** is a related process employing a spectrophotometer. *See also* **absorbance**, **absorptivity**.

absorption spectroscopy the spectroscopy of an absorption spectrum.

absorption spectrum a spectrum produced when electromagnetic radiation is absorbed by a sample. The frequencies of the radiation absorbed are those able to excite the atoms or molecules of the sample from their ground states to excited states. The frequency, ν , at which a particular absorption line occurs depends on the energy difference, ΔE , between that of a particular ground state and that of the corresponding excited state. It is given by $\nu = \Delta E/h$, where h is the Planck constant. *Compare* **emission spectrum**.

absorptivity a measure of the ability of a material to absorb electromagnetic radiation. It equals the absorbance of a sample of the material divided by the optical path-length. For very low attenuation, it approximates the absorption coefficient. Use of the term is not recommended.

Abu symbol for a residue of the α -amino acid L-2-aminobutanoic acid L- α -aminobutyric acid.

A₂bu or **Dab** symbol for a residue of the α,γ -diamino acid L- α,γ -diaminobutyric acid, L-2,4-diaminobutanoic acid.

abzyme abbr. for antibody enzyme (an antibody with enzyme activity; also known as catalytic monoclonal antibody).

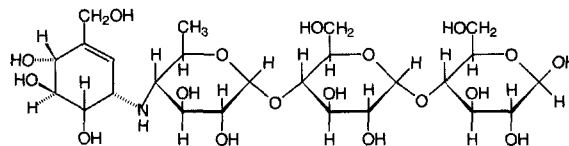
a.c. or **AC** or **ac** abbr. for alternating current.

Ac symbol for 1 actinium. 2 the acetyl group, $\text{CH}_3\text{CO}-$.

ACA a codon in mRNA for L-threonine.

acanthosome an organelle of fibroblasts isolated from the dermis of hairless mice after chronic UV irradiation. It exists as a spinous membranous vesicle.

acarbose a pseudotetrasaccharide, O-4,6-dideoxy-4-[[[1S-(1 $\alpha,4\alpha,5\beta,6\alpha$)]-4,5,6-trihydroxy-3-(hydroxymethyl)-2-cyclohexen-1-yl]amino]- α -D-glucopyranosyl-(1 $\rightarrow$ 4)-O- α -D-glucopyranosyl-(1 $\rightarrow$ 4)-glucose, that inhibits α -glucosidase, thereby reducing gastrointestinal absorption of glucose. It is a putative antidiabetic agent.



ACC a codon in mRNA for L-threonine.

a.c. calorimetry a technique in which the thermal response of a sample to an oscillating heat signal is measured in the form of a temperature wave propagating through the sample. The technique allows the measurement of the heat capacity of the sample on both cooling and heating and the monitoring of its isothermal time-dependence. It is useful in the study of phase transitions in solids and in gel to liquid-crystal systems.

accelerator 1 (in chemistry) **catalyst**, especially one that increases the rate of a polymerization reaction. 2 (in physics) a device or machine used for imparting high kinetic energy to charged subatomic particles, e.g. electrons, protons, or alpha particles, by means of electric and/or magnetic fields.

accelerator globulin an alternative name for factor V. *See* **blood coagulation**.

accelerin an alternative name for factor Va. *See* **blood coagulation**.

acceptor 1 (in chemistry) a chemical entity that in a chemical reaction receives an electron, atom, or group of atoms. *Compare* **donor**, **donor atom**. *See also* **electron acceptor**. 2 (in physiol-

accessory cell

ogy) a receptor that binds a hormone without a biological response being demonstrable. **3** (*in pharmacology*) a receptor that binds a drug but has no identified endogenous ligand.

accessory cell any one of various types of cell that assist in the immune response. The term includes **antigen-presenting cell**, **basophil**, **eosinophil**, **mast cell**, and **platelet**.

accessory chromosome an alternative name for **1** a **B chromosome**. **2** a **sex chromosome**.

accessory DNA surplus DNA present in certain cells or during certain stages of cell development owing, for example, to **gene amplification**.

accessory food factor or **accessory growth factor** a term originally used to describe any unknown substance – subsequently called **vitamin** – found in small amounts in some foods, such as milk, that was necessary for the normal growth of animals fed on diets of purified carbohydrates, fats, proteins, and salts.

accessory pigment any of the pigments, such as the yellow, red, or purple **carotenoids** and the red or blue **phycobiliproteins** in photosynthetic cells. The carotenoids are always present, whereas the phycobiliproteins occur only in algae belonging to the Rhodophyceae, the Cyanophyceae, and the Cryptophyceae. Strictly speaking, chlorophyll b is also an accessory pigment.

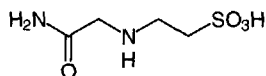
ACE *abbr. for* **1** amplification control element (a DNA sequence in vertebrates that functions as the origin for **amplification**). **2** **angiotensin converting enzyme**.

ACeDB or **acedb** a partly object-oriented database of genetic and sequence information devised for the nematode *Caenorhabditis elegans* but applied to other eukaryotes including *Arabidopsis thaliana*.

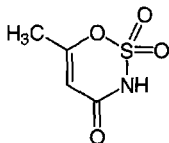
A cell or (*formerly*) **alpha cell** or **α cell** one of the three main histological cell types found in the islets of Langerhans of the pancreas, also found in the gastric oxyntic mucosa. A cells produce, store, and secrete the hormone **glucagon**.

A₁ cell (*formerly*) an alternative name for **D cell**.

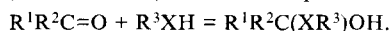
Aces or **ACES** *abbr. for* *N*-(2-acetamido)-2-aminoethane sulfonic acid; 2-[(2-amino-2-oxoethyl)amino]ethane sulfonic acid; a **Good buffer substance**, pK_a (20 °C) = 6.9.



acesulfame 6-methyl-1,2,3-oxathiazin-4(3*H*)-one 2,2'-dioxide; a sweet-tasting material that, as the potassium salt, has been used in foods and cosmetics.



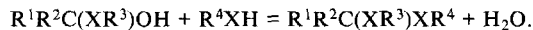
acetal any member of a class of organic compounds having the general formula $R^1HC(OR^3)OR^4$ or $R^1R^2C(OR^3)OR^4$ – in a **thioacetal** the corresponding formulae are $R^1HC(SR^3)SR^4$ or $R^1R^2C(SR^3)SR^4$ – where R^3 and R^4 are alkyl groups (or R^4 is H in a **hemiacetal** or **hemithioacetal**). An acetal molecule is formed by the acid-catalysed combination of the carbonyl group of an aldehyde or ketone molecule with either one or two alcohol (or mercaptan) molecules (which may be the same or different), or with a diol (or dithiol), by a reaction of the following general type, where X is O (or S in a mercaptan or dithiol):



The hemiacetal (or hemithioacetal) so formed may then undergo a further reaction:



or



In carbohydrates such compounds are formed at the carbonyl group of the acyclic form of a saccharide or saccharide derivative. The terms 'ketal' (or 'thioacetal') and 'hemiketal' (or 'thiohemiketal'), may be applied respectively to any acetal of general formula $R^1R^2C(XR^3)XR^4$ or $R^1R^2C(OH)XR^3$, i.e. to those derived from ketones. These terms, at one time abandoned, have recently been reintroduced as the respective names of subclasses of acetals and hemiacetals and as functional class names.

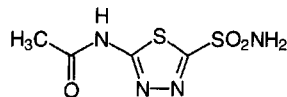
acetaminophen or **paracetamol** 4-acetamidophenol; *N*-acetyl-*p*-aminophenol; *N*-(4-hydroxyphenyl)acetamide; a drug widely used as an analgesic and antipyretic. It inhibits formation of prostaglandins within, but not outside the brain. It is metabolized within the liver mostly to glucuronide and sulfate conjugates. A small amount is oxidized to a highly reactive intermediate, *N*-acetylbenzoquinoneimine, that is normally detoxified by conjugation with glutathione. If it is produced in excess of the capacity of the liver to detoxify it, hepatic necrosis can result. It can be administered with methionine, which increases glutathione in the liver. *N*-Acetylcysteine is administered in cases of poisoning to act as a glutathione substitute. *Proprietary names include:* Panadol, Tylenol.

acetate **1** the traditional name for ethanoate; the anion, CH_3COO^- , derived from acetic acid (ethanoic acid). **2** any salt or ester of acetic acid.

acetate-CoA ligase EC 6.2.1.1; *systematic name:* acetate:CoA ligase (AMP-forming); *other names:* acetyl-CoA synthetase; acyl-activating enzyme; acetate thiokinase; acetyl-activating enzyme. An enzyme that catalyses a reaction between ATP, acetate, and CoA to form AMP, pyrophosphate, and acetyl-CoA. It is an important enzyme in organisms (e.g. *Escherichia coli*, many fungi, protozoans, algae) that utilize acetate as a carbon source. Example from *E. coli*: database code ACUA_ECOLI, 652 amino acids (72.01 kDa). Distinguish from acetate-CoA ligase (ADP-forming), EC 6.2.1.13.

acetate thiokinase *see* **acetate-CoA ligase**.

acetazolamide an inhibitor of carbonic anhydrase that is useful as a diuretic. It acts by preventing bicarbonate reabsorption in the proximal tubules of the kidney.



(+)**aceto+** *comb. form* denoting the acyl group derived from acetic acid.

acetoacetate-CoA ligase EC 6.2.1.16; *other name:* acetoacetyl-CoA synthetase; an enzyme that catalyses the formation of acetoacetyl-CoA from ATP, acetoacetate, and CoA with release of AMP and pyrophosphate. In bacteria that carboxylate acetone to acetoacetate, it activates the latter for further metabolism. It is also present in animals, but utilization of blood acetoacetate after its entry into tissues involves **3-oxoacid CoA-transferase**.

acetoacetyl acetyltransferase *see* **acetyl-CoA C-acetyltransferase**.

acetoacetyl-CoA synthetase *see* **acetoacetate-CoA ligase**.

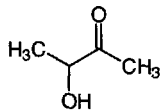
acetoacetyl-CoA thiolase *see* **acetyl-CoA C-acetyltransferase**.

acetogenin any substance built up of two-carbon units that may formally be considered to derive from a polyacetyl chain intermediate; the carbon atoms derived from the carboxyl carbon atoms of acetic acid frequently remain oxidized. It is not a recommended term. *See* **polyketide**.

acetoin 3-hydroxy-2-butanone; a compound formed by action



of acetolactate decarboxylase (EC 4.1.1.5) and, under some conditions, pyruvate decarboxylase (EC 4.1.1.1).



acetone body *see* **ketone body**.

acetone powder any preparation of ruptured cells obtained from a tissue or single-celled organisms that involves dehydration with acetone to form a powder. It is relatively stable, and is used in the preparation of some enzymes.

acetyl the acyl group ethanoyl, $\text{CH}_3\text{CO}-$, derived from acetic acid (= ethanoic acid).

acetylation an **acylation** reaction in which an acetyl group, $\text{CH}_3\text{CO}-$, is introduced into an organic compound. —**acetylated** *adj.*

acetylation coenzyme *the original name for coenzyme A.*

acetylcholine receptor *see* **cholinoceptor**.

acetylcholinesterase *abbr.*: AChE; EC 3.1.1.7; *systematic name*: acetylcholine acetylhydrolase; *other names*: true cholinesterase; cholinesterase I; an esterase enzyme that catalyses the hydrolysis of acetylcholine to choline and acetate; it also acts on a variety of acetic esters and catalyses transacetylations. It is found in or attached to cellular or basement membranes of presynaptic cholinergic neurons and postsynaptic cholinceptive cells. A soluble form occurs in cerebrospinal fluid and within cholinergic neurons. It is inhibited by a number of drugs, e.g. physostigmine, and by several **organophosphates**. Example from human: database code ACES_HUMAN, 614 amino acids (67.72 kDa). The 3-D structure is known for fragments obtained from the electric ray (fish).

acetylcholine transporter protein an integral membrane protein of synaptic vesicles of cholinergic neurons. It transports newly synthesized acetylcholine molecules into the synaptic vesicles in exchange for protons, thereby replenishing vesicular stores of the neurotransmitter. Example from *Caenorhabditis elegans*: database code UN17_CAEEL, 532 amino acids (58.58 kDa).

acetyl-CoA *abbr. for* **acetyl coenzyme A**.

acetyl-CoA C-acetyltransferase EC 2.3.1.9; *other names*: acetoacetyl acetyltransferase; acetoacetyl-CoA thiolase; an enzyme that catalyses the formation of two molecules of acetyl-CoA from CoA and acetoacetyl-CoA. During beta oxidation it catalyses the formation of acetyl-CoA from acetoacetyl-CoA, whereas it acts in the reverse direction to form acetoacetyl-CoA during ketogenesis. It is important in regulating the metabolic pathways for the production of acids, i.e. acetate, butyrate, or solvents, i.e. acetone, butanol, ethanol, during the growth of *Clostridium acetobutylicum*. Reduced activity of the enzyme favours production of acetate and ethanol, while increased activity favours production of butyric acid, butanol, and acetone. Example from human (mitochondrial, precursor): database code THIL_HUMAN, 427 amino acids (45.15 kDa).

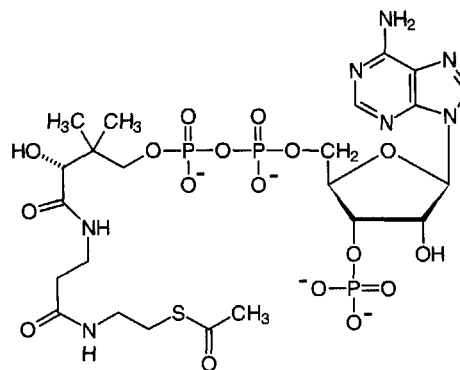
acetyl-CoA C-acyltransferase EC 2.3.1.16; *systematic name*: acyl-CoA:acetyl-CoA C-acyltransferase; *other names*: 3-ketoacyl-CoA thiolase; β -ketothiolase; an enzyme that catalyses the formation of acyl-CoA and acetyl-CoA from CoA and 3-oxoacyl-CoA. This is the concluding reaction of each cycle of the fatty acid oxidation pathway (**beta oxidation**). Different enzymes exist in the mitochondrion and peroxisome, both being included in the **thiolase** family. Examples include rat peroxisomal enzyme A (one of two), homodimer: database code for precursor TH11_RAT, 424 amino acids (43.77 kDa); also from yeast (precursor): database code THIK_YEAST, 417 amino acids (44.68 kDa); mitochondrial enzyme, homotetramer: database code THIM_RAT, 397 amino acids (41.82 kDa).

acetyl-CoA carboxylase EC 6.4.1.2; *systematic name*: acetyl-CoA:carbon-dioxide ligase (ADP-forming); a multienzyme complex involved in the formation of **malonyl-CoA**, the first step in fatty-acid biosynthesis. It catalyses a reaction between ATP, acetyl-CoA, and HCO_3^- to form ADP, orthophosphate, and malonyl-CoA. Biotin is a cofactor. In bacteria it is a heterohexamer of biotin-carboxyl-carrier-protein, biotin carboxylase, and a 2:2 complex of the two subunits of carboxyl transferase. Examples from *Escherichia coli*: biotin carboxylase (EC 6.3.4.14), which catalyses the reaction between ATP, biotin-carboxyl-carrier-protein, and CO_2 to form ADP, orthophosphate, and carboxybiotin-carboxyl-carrier-protein; database code ACCC_ECOLI, 449 amino acids (49.26 kDa); biotin-carboxyl-carrier-protein: database code BCCP_ECOLI, 156 amino acids (16.67 kDa); carboxyl transferase α -subunit, database code ACCA_ECOLI, 318 amino acids (35.07 kDa); carboxyl transferase β -subunit, database code ACCD_ECOLI, 304 amino acids (33.31 kDa). Biotin-carboxyl-carrier-protein is carboxylated by biotin carboxylase and the carbonyl group is then transferred to acetyl-CoA by carboxyl transferase, thus forming malonyl-CoA. In mammals the activity is part of a trifunctional enzyme that has in a single protein the activities carboxyl carrier protein, biotin carboxylase (EC 6.3.4.14), and acetyl-CoA carboxylase (EC 6.4.1.2). Example from rat: database code COAC_RAT, 2345 amino acids (264.89 kDa).

[acetyl-CoA carboxylase] kinase EC 2.7.1.128; an enzyme that catalyses the phosphorylation by ATP of [acetyl-CoA carboxylase] with release of ADP. This phosphorylation step is one of the regulatory mechanisms for acetyl-CoA carboxylase, causing that enzyme to dissociate from an active polymeric form to an inactive monomeric form.

[acetyl-CoA carboxylase] phosphatase EC 3.1.3.44; an enzyme that catalyses the hydrolysis of phosphate from [acetyl-CoA carboxylase] phosphate. It reverses the phosphorylation catalysed by [acetyl-CoA carboxylase] kinase.

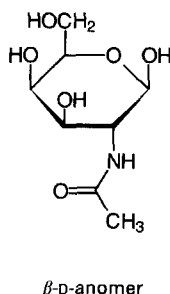
acetyl coenzyme A *abbr.*: acetyl-CoA; a derivative of **coenzyme A** in which the sulfhydryl group is acetylated. Originally termed 'active acetate', it is an important metabolite, derived from pathways such as glycolysis, fatty-acid oxidation, and degradative metabolism of some amino acids. It is further metabolized by the **tricarboxylic-acid cycle** and represents a key intermediate in lipid and terpenoid biosynthesis and other anabolic reactions.



N-acetylcysteine or **N-acetyl-L-cysteine** a thiol-protecting agent used intravenously as an antidote in acetaminophen poisoning. It acts by enhancing glutathione synthesis, thereby increasing the capacity for detoxification and excretion of acetaminophen as a **mercapturic acid**. Methionine can be similarly used. It also has **mucolytic** properties, and is used in aiding the isolation of mycobacteria from sputum.



N-acetylgalactosamine *symbol*: D-GalpNAc; *abbr.*: NAGA; the D isomer, 2-acetamido-2-deoxy-D-galactopyranose, is a common structural unit of oligosaccharides, such as the blood-group substances and O-linked glycoproteins, in which the sugar is in glycosidic linkage to a protein or serine residue, or, in the case of the blood-group substances, to a lipid hydroxyl group. The reactant in synthetic reactions is UDP-N-acetylgalactosamine, which is formed by epimerization of N-acetylglucosamine.

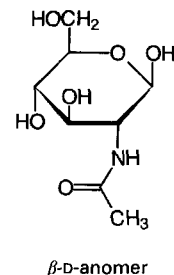


N-acetylgalactosamine-4-sulfatase EC 3.1.6.12; *other names*: arylsulfatase B; chondroitinsulfatase; chondroitinase; an enzyme that hydrolyses the 4-sulfate groups of the N-acetyl-D-galactosamine 4-sulfate units of chondroitin sulfate and dermatan sulfate. It is a lysosomal enzyme involved in the degradation of proteoglycans, which accumulate in Maroteaux-Lamy syndrome (**mucopolysaccharidosis VI**), a storage disease resulting from a deficiency of the enzyme. Example (precursor) from human: database code ARSB_HUMAN, 533 amino acids (59.62 kDa).

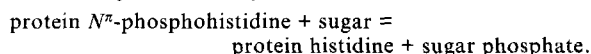
N-acetylglucosamine-6-sulfatase EC 3.1.6.4; *other names*: chondroitinsulfatase; chondroitinase; galactose-6-sulfate sulfatase; an enzyme that hydrolyses the 6-sulfate groups of the N-acetyl-D-galactosamine 6-sulfate units of chondroitin sulfate, and also the D-galactose 6-sulfate units of keratan sulfate. It is a lysosomal enzyme involved in the degradation of proteoglycans. Keratan sulfate and chondroitin 6-sulfate accumulate in Morquio A syndrome, a storage disease resulting from a deficiency of the enzyme. Example (precursor) from human: database code GA6S_HUMAN, 522 amino acids (57.96 kDa).

N-acetylglactosaminidase EC 3.2.1.53; either of two types of enzyme that catalyse the hydrolysis of respectively α- and β-linked terminal nonreducing N-acetyl-D-galactosamine residues. Both are lysosomal enzymes. Deficiency of α-N-acetylglactosaminidase is associated with a storage disease in which sialyloligosaccharides are found in urine.

N-acetylglucosamine *symbol*: D-GlcNAc; *abbr.*: NAG; the D isomer, 2-acetamido-2-deoxy-D-glucopyranose, is a common structural unit of plant glycoproteins and of many animal and bacterial glycoproteins. It is often the terminal sugar of an oligosaccharide moiety of a glycoprotein, linked glycosidically to the amide nitrogen of a protein asparagine residue. The acetyl group is introduced in a reaction between acetyl-CoA and glucosamine 6-phosphate to give N-acetylglucosamine 6-phosphate, which undergoes a mutase reaction to form N-acetylglucosamine 1-phosphate, from which UDP-N-acetylglucosamine is formed. The latter is the reactant in pathways of oligosaccharide synthesis and is also a precursor of N-acetylneuraminic acid (*see* **sialic acid**).



N-acetylglucosamine phosphotransferase EC 2.7.1.69; *systematic name*: protein-N^γ-phosphohistidine: sugar N-pros-phosphotransferase; *other name*: enzyme II of the phosphotransferase system; an enzyme that catalyses the reaction:



It is a component of the phosphoenolpyruvate-dependent sugar phosphotransferase system, a major carbohydrate active-transport system; the phosphoryl group from phosphoenolpyruvate is transferred to phosphoryl carrier protein HPR by enzyme I, and from phospho-HPR to the sugar by enzyme II. It is an integral membrane protein. Example from *Escherichia coli*: database code PT2N_ECOLI, 648 amino acids (68.27 kDa).

N-acetylglucosamine-6-sulfatase EC 3.1.6.14; an enzyme that catalyses the hydrolysis of sulfate groups of N-acetyl-D-glucosamine 6-sulfate units of heparan sulfate and keratan sulfate. It is a lysosomal glycoprotein. A deficiency is associated with the storage disease **mucopolysaccharidosis III**. Example (precursor) from human: database code GL6S_HUMAN, 552 amino acids (62.01 kDa).

N^γ-(β-N-acetylglucosaminyl)-L-asparaginase EC 3.5.1.26; *other names*: aspartylglucosylamine deaspartylase; aspartylglucosylaminase; glycosylasparaginase; an enzyme that catalyses the hydrolysis of N^γ-(β-N-acetyl-D-glucosaminyl)-L-asparagine, released from glycoproteins, to N-acetyl-β-glucosaminylamine and L-aspartate. A deficiency of the enzyme results in the lysosomal storage disease, **aspartylglucosaminuria**, in which there is an accumulation of the enzyme's substrate. Example (precursor) from human: database code ASPG_HUMAN, 346 amino acids (37.15 kDa).

N-acetylglucosaminyl transferase any of various glycosyltransferase enzymes within the subclass EC 2.4.1 that transfer an N-acetylglucosaminyl residue from UDP-N-acetylglucosamine to an oligosaccharide, and which are important in oligosaccharide synthesis. An example is EC 2.4.1.144, β-1,4-mannosyl-glycoprotein β-1,4-N-acetylglucosaminyltransferase; *other name*: N-glycosyl-oligosaccharide-glycoprotein N-acetylglucosaminyltransferase III. It catalyses the addition of N-acetylglucosamine in β(1-4) linkage to the β-linked mannose of the trimannosyl core of N-linked sugar chains. It is a type II membrane protein of the Golgi stack. Example from rat: database code GNT3_RAT, 536 amino acids (61.69 kDa). *See also* **N-acetylglucosamine synthase**, **lipopolysaccharide**, **N-acetylglucosaminyltransferase**, **UDP-N-acetylglucosamine-dolichyl phosphate-N-acetylglucosamine phosphotransferase**.

β⁶-N-acetylglucosaminyltransferase *see* **β-1,3-galactosyl-D-glycosyl-glycoprotein β-1,6-N-acetylglucosaminyltransferase**.

N-acetylglutamate synthase (*abbr.* AGS) *see* **amino-acid N-acetyltransferase**.

acetylglutamic acid the L isomer, N-acetyl-L-glutamic acid, is a key intermediate in ornithine formation in bacteria and plants. It is converted to N-acetyl-L-glutamic γ-semialdehyde, from which N-acetyl-L-ornithine is formed in a transamination reaction. It activates **carbamoyl-phosphate synthase** (ammonia).

β -N-acetylhexosaminidase

nia). It is formed from acetyl-CoA and glutamate by the action of amino acid *N*-acetyltransferase, EC 2.3.1.1.

β -N-acetylhexosaminidase EC 3.2.1.52; *other names*: β -hexosaminidase; hexosaminidase; an enzyme that catalyses the hydrolysis of terminal nonreducing *N*-acetyl-D-hexosamine residues in *N*-acetyl- β -D-hexosaminides. Example from human: database code HEXA_HUMAN, 529 amino acids (60.62 kDa).

N-acetylglucosamine synthase EC 2.4.1.90; *systematic name*: UDPgalactose-*N*-acetyl-D-glucosamine 4- β -D-galactosyltransferase; *other names*: *N*-acetylglucosamine β -(1 $\rightarrow$ 4)-galactosyltransferase; UDPgalactose-*N*-acetylglucosamine β -D-galactosyltransferase. An enzyme, located in the rough endoplasmic reticulum, that catalyses a reaction between UDPgalactose and *N*-acetyl-D-glucosamine to form UDP and *N*-acetylglucosamine. In humans the enzyme also has the activity of β -N-acetylglucosaminyl-glycopeptide β -1,4-galactosyltransferase (EC 2.4.1.38); *other names*: glycoprotein 4- β -galactosyltransferase; thyroid galactosyltransferase; UDPgalactose-glycoprotein galactosyltransferase. It catalyses a reaction between UDPgalactose and *N*-acetyl- β -D-glucosaminylglycopeptide to form UDP and β -D-galactosyl-1,4-*N*-acetyl- β -D-glucosaminylglycopeptide. α -Lactalbumin is an allosteric regulator and converts this activity to **lactose synthase**. Database code NALS_HUMAN, 400 amino acids (44.22 kDa).

N-acetylmuramoyl-L-alanine amidase *see* autolysin.

N-acetylneuraminic acid *see* sialic acid.

acetyloleoylglycerol *see* oleoylacylglycerol.

acetylsalicylic acid *see* aspirin.

ACG 1 a codon in mRNA for L-threonine. 2 *abbr.* for acycloguanosine (*see* acyclovir).

Ach *symbol* for the arachidoyl (i.e. eicosanoyl) group.

Δ_2 Ach *symbol* for the (all-*Z*)-eicosa-8,11-dienoyl group; *see* eicosadienoic acid.

Δ_3 Ach *symbol* for the (all-*Z*)-eicosa-5,8,11-trienoyl group; *see* eicosatrienoic acid.

Δ_4 Ach *symbol* for the arachidonoyl (i.e. (all-*Z*)-eicosa-5,8,11,14-tetraenoyl) group; *see* arachidonoyl.

ACH *abbr.* for acetylcholine.

Achaete-Scute complex *see* AS-C protein.

A chain 1 the shorter polypeptide chain of insulin, containing 21 amino-acid residues. 2 the heavy chain (H chain) of an immunoglobulin.

achatin-1 an endogenous neuroexcitatory tetrapeptide, Gly-D-Phe-Ala-Asp, isolated from the ganglia of the giant African snail, *Achatina fulica*: database code ACHI_ACHFU.

AChe *abbr.* for acetylcholinesterase.

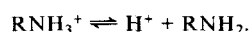
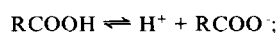
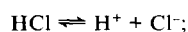
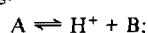
achiral not chiral. —achirality *n*.

achlorhydria an inability to secrete gastric acid. It is a disorder, probably autoimmune, that is linked with pernicious anemia.

achromic point the point in time during the action of amylase on starch at which the reaction mixture no longer gives a colour with iodine, i.e. the reaction has proceeded to the point when the starch has all been degraded at least as far as **achroödextrins**.

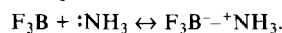
achroödextrin any **dextrin** that is small enough not to give a colour with iodine.

acid 1 in the Brønsted-Lowry concept, a molecular species having a tendency to lose a hydron forming a conjugate base, e.g.



2 in the Lewis concept, a substance capable of accepting from

a base an unshared pair of electrons, which then form a covalent chemical bond, e.g.



acid anhydride any compound formed by the elimination of the elements of water from the acidic groups of two acids, e.g. acetic anhydride (two acetic acid molecules) or acetyl phosphate (one molecule each of acetic and phosphoric acids).

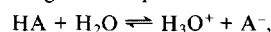
acid-base balance term descriptive of the hydrogen-ion status of the blood, the mechanisms that regulate it, and the causes of its deviation from normal.

acid-base catalysis catalysis of a chemical reaction in which either an acid or a base mediates the formation of a reactive intermediate.

acid-base titration a titration in which either acid or base is added to a solution and the progress of the titration is followed by pH measurements, either electrometrically or with the use of pH indicators.

acid carboxypeptidase *see* cysteine-type carboxypeptidase.

acid dissociation constant or **acidity constant** *symbol*: K_a ; the thermodynamic equilibrium constant for the dissociation of an **acid**. For a dilute solution of a weak acid, HA, dissociating in water according to the equilibrium:



$$K_a = (a_{H_3O^+} \times a_{A^-})/a_{HA}$$

where *a* is the **activity** of the species designated by the subscripts. The activity of the water has been omitted from the equation since it may be taken as unity for a dilute aqueous solution. K_a is a measure of the strength of the acid, i.e. of its ability to donate hydrons to water. *Compare* **basic dissociation constant**. *See also* **pK**.

acid dye a **dye** containing an anionic acidic organic group that binds to and stains positively charged macromolecules.

acidemia or (*esp. Brit.*) **acidaemia** (*archaic*) a condition in which there is excessive acidity (i.e. increased hydrogen-ion concentration, lowered pH) of the blood. *Compare* **acidosis**, **alkalemia**.

acid-fast bacillus any bacterium able to resist decolorization by mineral acids after the application of specific basic aniline dyes; this property is possible due to the presence in these organisms of **mycolic acid**, together with a semipermeable membrane that allows the passage of the stain but not of the decolorizing acid.

acid β -glucosidase *see* glucosylceramidase.

acid hydrolase any **hydrolase** enzyme that is active in mildly acidic conditions (pH 5–6); often found in **lysosomes**.

acidic 1 of, relating to, containing, or characteristic of an **acid**. *Compare* **basic** (def. 1). 2 having an acid reaction in water; of or relating to an aqueous solution having a pH < 7.0. *Compare* **basic** (def. 2).

acidic amino acid any amino acid containing more potentially anionic groups than potentially cationic groups. All such amino acids have a net negative charge at neutral pH. Examples are aspartic and glutamic acids.

acid mucopolysaccharide any of a group of related heteropolysaccharides, found widely distributed in animal connective tissues, that contain *N*-acetylated hexosamine in its characteristic repeating disaccharide unit. They include chondroitin, **chondroitin sulfates**, **dermatan sulfates**, **hyaluronic acid** (*see* **hyaluronate**), and **keratan sulfates**.

acid number or **acid value** the mass, in milligrams, of potassium hydroxide required to neutralize the free fatty acid in one gram of fat; a measure of the mass of free acid in the sample.

acidophilic 1 staining readily with **acid dyes**. 2 (of organisms) preferring or thriving in a relatively acid environment.

acidosis a clinical condition in which excess acid or a base deficit tends to cause increased hydrogen-ion concentration (i.e. lowered pH) in the blood.

acidosome a non-lysosomal vesicle found in the ciliate proto-



zoan *Paramecium*. The organelle is involved in acidification of digestive phagocytic particles through fusion.

acidotropic seeking an acid environment. The term is used e.g. in connection with Ser/Thr protein kinases that require Glu or Asp as part of the recognition site. The resulting Ser(P) or Thr(P) then acts as an acidic residue, extending the site with the result that further Ser or Thr residues are phosphorylated.

acid phosphatase *abbr.* (in clinical biochemistry): ACP; EC 3.1.3.2; *systematic name*: orthophosphoric monoester phosphohydrolase (acid optimum); *other names*: alkaline phosphomonoesterase; phosphomonoesterase; glycerophosphatase; a lysosomal enzyme (except in red cells). It catalyses the hydrolysis of orthophosphoric monoester to an alcohol and orthophosphate. Zinc and magnesium are cofactors. It is present in high concentrations in the prostate gland, and is also present in red cells, platelets, bone, liver, and spleen. Its measurement in blood may be of use clinically in monitoring progress in cases where prostatic cancer has metastasized, but not where cancer is confined to the prostate, being elevated in only about 30% of cases. Normal range in human plasma 4–11 IU L⁻¹. Example: human lysosomal acid phosphatase precursor: database code PPAL_HUMAN, 423 amino acids (48.29 kDa).

acid proteinase an older name for enzymes of the sub-subclass **aspartic endopeptidase**, EC 3.4.23. It was suggested by their characteristic low optimum pH.

acid sphingomyelinase *see* sphingomyelin phosphodiesterase.

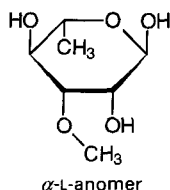
acinus (*pl.* acini) 1 a saclike structure that forms the terminal part of a gland. It comprises a cluster of cells surrounding a small duct. 2 one of the collection of small drupes making up an aggregate fruit such as a raspberry. —**acinar** *adj.*

ackee or **akee** a tree, *Blighia sapida*, native to tropical Africa and widely cultivated in the West Indies, especially Jamaica, for its fruit, the fleshy aril of which is edible when cooked and forms an important item of local diet. Unripe fruits contain toxic amounts of **hypoglycin** and can cause **Jamaican vomiting sickness**.

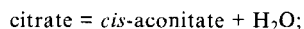
Acm *abbr.* for acetamidomethyl-.

AcNeu (formerly) *symbol* for N-acetylneuraminic acid (*see* sialic acid).

acofriose 6-deoxy-3-O-methylmannose; the L enantiomer is a component of some **cardiac glycosides**.

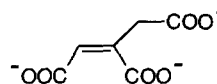


aconitase EC 4.2.1.3; *other name*: aconitate hydratase; *systematic name*: citrate (isocitrate) hydro-lyase. A hydrolase enzyme that catalyses the reaction:



it also reversibly converts isocitrate into *cis*-aconitate + H₂O. An iron-sulfur protein, it removes H_R from the *pro-R*-CH₂-COOH group of citrate (*see* **citrate** for structure). Under kinetic conditions in which it forms isocitrate from citrate the product is (1*R*,2*S*)-1-hydroxypropane-1,2,3-tricarboxylate ((2*R*,3*S*)-isocitrate). Example from pig: database code NRL_5ACN, 754 amino acids (82.60 kDa); 3-D structure known.

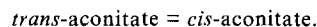
aconitate the *cis* isomer, (Z)-prop-1-ene-1,2,3-tricarboxylate, is an intermediate in the conversion of citrate to isocitrate in the tricarboxylic-acid cycle, by the action of **aconitase**.



cis-aconitate

aconitate hydratase *see* **aconitase**.

aconitate Δ-isomerase EC 5.3.3.7; *systematic name*: aconitate Δ²-Δ³-isomerase. An enzyme that catalyses the reaction:



acoustic gene transfer a method of transforming (typically plant) cells by using ultrasound.

ACP *abbr.* for 1 acyl carrier protein. 2 (in clinical biochemistry) acid phosphatase.

εAcp (formerly) *symbol* for a residue of the ε-amino acid ε-caproic acid, now known as 6-aminoheptanoic acid. εAhx is the preferred symbol.

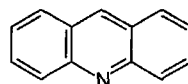
acquired immune deficiency syndrome *abbr.*: AIDS; defined by the Centers for Disease Control in the USA as a reliably diagnosed disease that is at least moderately indicative of an underlying immune deficiency or any other cause of reduced resistance reported to be associated with that disease. In humans it is a fatal condition (unless a successful therapy can be found), commonly associated with infection by **HIV**.

acquired immunity or **adaptive immunity** immunity (active or passive, humoral or cellular) that is established during the life of an individual, as contrasted with innate or natural immunity. Such immunity is specific for the inducing agent and is marked by an enhanced response on repeated encounters with that agent. The key features are memory (*see* **memory cell**) and specificity.

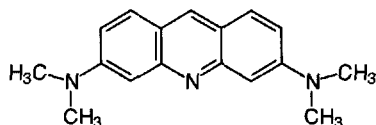
acquired tolerance 1 (in immunology) tolerance to an antigen that is established during the life of an individual. Immunological tolerance can (rarely) be produced in an adult animal by prolonged injection of massive doses of antigen. The tolerance persists as long as antigen persists in the animal. *See also* **self tolerance**. 2 (in pharmacology) tolerance (generally to psychoactive compounds) that develops on prolonged or repeated drug administration. It can be either **pharmacokinetic** usually by increased drug metabolism, or **pharmacodynamic**. *See* **tachyphylaxis**, **tolerance**.

acrasin a chemotactic substance produced by the myxamoebae of the cellular slime mould *Dictyostelium discoideum*, now identified as cyclic AMP.

acridine the parent compound of a series of derivatives e.g. 3,6-diaminoacridine (proflavin), that are potent mutagens. Some acridines are found in coal tar. They induce **frameshift mutations** during the replication of DNA by binding to DNA and distorting the double helix, causing additional bases to be incorporated or bases to be omitted. They are used as topical antiseptics and antimalarial agents.



Acridine Orange 3,6-bis(dimethylamino)acridine; a dye used as a probe of nucleic acids in microscopy and related techniques. When illuminated under UV light it yields a green (DNA) or reddish-orange (RNA) fluorescence.



Acridine Orange

acriflavin a mixture of the **acridine** derivatives 3,6-diamino-10-methylacridinium chloride (about 65%) and **proflavin**.

acrocentric describing a chromosome in which the **centromere** is very close to one end.

acrolein 3-propenal; $\text{CH}_2=\text{CH}-\text{CHO}$; an unstable flammable liquid with a pungent odour that irritates the eyes and mucosae. It polymerizes (especially under light) to form a plastic solid.

acrolein test a qualitative test for the presence of glycerol, either free or esterified, based upon its oxidative dehydration to acrolein when heated with solid potassium hydrogen sulfate.

acromegaly a chronic disease marked by the gradual enlargement of the bones of the hands, feet, head, and chest with thickening of the skin, lips, and vocal chords. It is caused by excessive secretion of, or increased sensitivity to, **somatotropin**, and is often due to a tumour of the somatotrope cells of the pituitary.

acrosin EC 3.4.21.10; a trypsin-like serine endopeptidase. A major proteinase of mammalian spermatozoa, synthesized in a zymogen form, proacrosin, and stored in the **acrosome**. It comprises a heavy chain (catalytic) and a light chain linked by two disulfide bonds; it is not inhibited by α_1 -antitrypsin. It catalyses the hydrolysis of Arg-|-Xaa and Lys-|-Xaa bonds, with preferential cleavage in the order Arg-|-Xaa $\gg$ Lys-|-Lys $\gg$ Lys-|-Xaa. Example (precursor) from human: database code ACRO_HUMAN, 421 amino acids (45.75 kDa).

acrosomal process a long thin actin-containing spike produced from the head of certain types of sperm when they make contact with the egg at fertilization. It is seen in sea urchins and other marine invertebrates having eggs surrounded by a thick gelatinous coat.

acrosome a structure in the head of a spermatozoon that contains acid hydrolases concerned in the breakdown of the outer membrane of the ovum during fertilization. It lies anterior to the nucleus just beneath the plasma membrane. *See also* **acrosin**.

acrylamide or **acrylamide monomer** the trivial name for propenamide, $\text{CH}_2=\text{CH}-\text{CONH}_2$; a water-soluble solid that is highly toxic and irritant, and readily polymerizes under the action of UV light or chemical catalysts into **polyacrylamide**.

ACTH *abbr.* for adrenocorticotropic(h)ic hormone (**corticotropin**).

1-24 ACTH *abbr.* for tetracosactrin.

actidione *see* **cycloheximide**.

actin a major protein constituent of the thin filaments of muscle and of the **microfilaments** found in practically all eukaryotic cells – it comprises 5–10% of the protein of such cells (*see* **actin filament**). In solutions of low ionic strength, actin is a globular 42 kDa monomer called **G-actin**. At physiological ionic strengths, G-actin polymerizes into a fibrous form called **F-actin**, which resembles two strings of beads wound round each other. F-actin is a helix of actin monomers with a helix diameter of about 7 nm, the structure repeating at intervals of 36 nm along the helix axis. When a solution of actin is mixed with a solution of the muscle protein myosin a complex called **actomyosin** is formed and the viscosity of the solution increases markedly. This increase in viscosity is reversed by the addition of ATP, which acts to dissociate the actomyosin complex. It is thought that the force of muscle contraction arises from an interaction of actin, myosin, and ATP. Example, α -cardiac actin

(human): database code ACTC_HUMAN, 377 amino acids (41.97 kDa); seven motifs.

actin-binding domain a structural feature found in **actin-binding proteins**.

actin-binding protein any of several proteins that associate with either actin monomers or actin filaments in cells and modify their properties. Many of these proteins are found in the cell cortex, an actin-rich layer just below the plasma membrane. Examples include **dystropin**, **profilin**, **spectrin** and **ankyrin**, **fimbrin** and α -actinin (*see* **actinin**), **filamin**, **gelsolin**, and the **myosins**. The term is sometimes applied specifically to **filamin**.

actin depolymerizing factor *see* **destrin**.

actin filament a two-stranded helical polymer of the protein **actin**. Actin filaments form the thin filaments of muscle and also the **microfilaments** of the cytoskeleton of eukaryotic cells. Hence they are a major component of the contractile apparatus of skeletal muscle, and one of the three types of protein filament that form the cytoskeleton, the others being **microtubules** and **intermediate filaments**. The filaments, comprising polymerized globular actin molecules, appear as flexible structures with a diameter of 5–9 nm. They are organized into a variety of linear bundles, two-dimensional networks, and three-dimensional gels. In the cytoskeleton they are most highly concentrated in the cortex of the cell just beneath the plasma membrane.

actinic describing electromagnetic radiation, especially higher frequencies of visible light and UV radiation, capable of initiating photochemical reactions.

actinidine a monoterpenoid alkaloid occurring in *Actinidia polygama*.

actinin a minor protein constituent of muscle, found to be concentrated in both the Z line and the I band. Two components of actinin have been identified: α -**actinin**, F-actin cross-linking protein, is a dimer of ≈ 200 kDa with an action similar to that of **actinogelin**. Example from human: database code AACT_HUMAN, 892 amino acids (102.86 kDa); and β -**actinin**, a dimer of ≈ 70 kDa, is similar in action to **gelsolin**.

actinogelin a protein factor that effects Ca^{2+} -sensitive gelation of actin filaments. It was first obtained from Ehrlich ascites tumour cells, and has a molecular mass of 100–120 kDa. α -Actinin (*see* **actinin**) has a similar action.

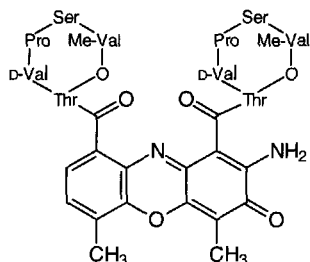
actinoid or **actinide** any member of the series of 15 metallic elements with proton numbers 89 (actinium) to 103 (lawrencium) inclusive that occur together in group 3 and period 7 of the IUPAC **periodic table**; sometimes the term is restricted to the 14 elements following actinium. Actinoid is now the preferred name. All actinoids are radioactive, and those of proton number 93 or greater are artificial. They are all electropositive, and their chemical properties are similar, due usually to the filling of an inner electron subshell (5f) progressively across the series. Like the **lanthanoids**, they thus represent a series of inner transition elements.

actinomycin any of a large group of antibiotics isolated from various species of *Streptomyces* bacteria and characterized by having a substituted phenoxazine ring linked to two cyclic **heterodetic** peptides. The principal member of the group is **actinomycin D** (also called actinomycin C_1 , actinomycin IV), which, at low concentrations, inhibits transcription without appreciably affecting DNA replication in both prokaryotic and eukaryotic cells. Actinomycin D binds tightly to duplex DNA thereby preventing it from being an effective template for RNA synthesis. Spectroscopic and hydrodynamic studies of complexes of actinomycin D and DNA suggest that the phenoxazine ring of actinomycin is intercalated between neighbouring base pairs in DNA. Other conformational studies indicate that actinomycin D recognizes the base sequence GpC in DNA.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyenvltd.com.



actinomycin D

actinomyosin (formerly) the contractile system in muscle comprising **actin** and **myosin**. Compare **actomyosin**.

action potential the localized change of electrical potential across a nerve fibre or muscle membrane that marks the position of an impulse as it is propagated. It is caused by sodium ion fluxes across the membrane resulting from transitory opening of **sodium channels**.

action spectrum a graph or table showing the relative efficiencies of different frequencies of radiation in causing a chemical or biochemical reaction, e.g. in photosynthesis or gas exchange. Efficiency, or **quantum yield**, is given by the measurable effect divided by the intensity of the (monochromatic) incident radiation.

activate 1 to make or render active, reactive, or capable of action. 2 to make radioactive. —**activated** *adj.*; **activation** *n.*

activated alumina particles of alumina, aluminium oxide, that have been rendered more adsorbent by heating strongly.

activated amino acid an amino-acid residue in an aminoacyladenylate molecule. The amino acid reacts with ATP to form aminoacyl-AMP in a reaction catalysed by an aminoacyl-tRNA synthetase. It is the first metabolic step in the biosynthesis of a polypeptide. The same enzyme catalyses the transfer of the amino acid to an ester link with the 3'-terminal hydroxyl of tRNA.

activated complex term that may be used to denote the assembly of atoms at the transition state of a chemical reaction.

activation 1 the action or process of rendering an atom, molecule, or other substance reactive or more reactive, whether physicochemically, chemically, or biochemically. 2 the process of rendering material artificially radioactive: radioactivation. See **radioactivation analysis**. 3 the initial changes in an ovum during fertilization, covering the period from first contact with a sperm to dissolution of the nuclear membranes. 4 the initial changes in the conversion of a spore to a vegetative cell. This can be effected by various agents or processes and may involve alteration of one of the spore's outer layers.

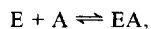
activation analysis an alternative term for **radioactivation analysis**.

activation energy see **Arrhenius equation**.

activation hormone a polypeptide hormone synthesized in the neurosecretory brain cells of insects. It regulates the functioning of the whole endocrine system, stimulating the secretory activity of other glands.

activator any substance that activates a chemical or enzymic reaction. The term was formerly used in biochemistry especially to describe metal-ion **cofactors** for enzymes.

activator constant symbol: K_A ; the equilibrium dissociation constant of the reaction of an enzyme with an activator. For example, in the reaction:



$$K_A = [EA]/[E][A],$$

where [EA], [E], and [A] are the concentrations of enzyme-activator adduct, free enzyme, and free activator respectively.

active acetaldehyde (sometimes) the α -hydroxyethyl derivative, on C-2 of the thiazolidine ring, of thiamine diphosphate.

active acetate (formerly) an alternative name for **acetyl coenzyme A**.

active C₁ *abbr.* for active one-carbon.

active centre all the features of primary, tertiary, and quaternary structure of an enzyme – including the **active site** – that are required for substrate binding, specificity, and catalysis.

active-enzyme centrifugation a technique for determining the sedimentation and diffusion coefficients of an enzyme-substrate complex. A thin lamella of an enzyme solution is layered onto a substrate solution in an ultracentrifuge cell and, on rotation, as the enzyme molecules sediment in a band through the substrate solution they catalyse the enzymic reaction, the progress of which is observed optically (e.g. by spectrophotometry). The required coefficients are then calculated from either the rate of appearance of the product of the reaction or the rate of disappearance of the substrate. The centrifugation may be performed on impure enzyme preparations and at the very low concentrations of enzyme used in kinetic studies.

active fatty acid (formerly) an alternative term for an **acyl-CoA** derivative.

active formaldehyde (formerly) an alternative term for N^5,N^{10} -methylene tetrahydrofolate.

active immunity or **adaptive immunity** a type of immunity resulting from the stimulation of the immune response by an immunogen.

active immunization stimulation of the reactivity of the immune system of an individual towards an immunogen by administration of the immunogen, thereby producing a state of active immunity.

active methionine or **active methyl** name sometimes used for **S-adenosylmethionine**.

active methyl 1 an alternative name for **active methionine**. 2 one form of **active one-carbon** (e.g. 5-methyltetrahydrofolate).

active one-carbon any of the one-carbon units carried on tetrahydrofolate and concerned in a wide variety of biosynthetic reactions. They include the methyl-, methylene-, formimino-, formyl-, and methenyl- substituents.

active site 1 the general region of an enzyme molecule containing the **catalytic residues** identified with the binding and reaction of substrate(s). It includes those amino-acid residues that are, in the enzyme-substrate complex, either **contact amino acids**, i.e. those that at some point are only one bond distance removed from some point of the substrate molecule, or **auxiliary amino acids**, i.e. those that are not in such intimate physical contact with the substrate but nonetheless play a definite role in the action of the enzyme. See also **contributing amino acid**. 2 (sometimes) the portion of a peptide hormone responsible for its biological activity. Isolated fragments containing the active site may show some activity, but may function less efficiently than the intact hormone. See also **message sequence**.

active site-directed irreversible inhibitor a custom-made inhibitor of a given enzyme. Such inhibitors are typically trifunctional molecules containing: (1) a functional group able to bind to the enzyme's active site; (2) a nonpolar part that can interact with a nonpolar region on the enzyme, serving to align the inhibitor; and (3) an alkylating group capable of reacting with a susceptible group on the enzyme and irreversibly forming a covalent bond with it.

active sulfate an alternative name for 1 **adenosine-5'-phosphosulfate**. 2 3'-phosphoadenosine-5'-phosphosulfate.

active transport any energy-dependent process by which molecules or ions are transported across membranes against a chemical potential gradient. Compare **facilitated diffusion**, **simple diffusion**.

activin one of two gonadal glycoproteins related to **transforming growth factor- β** (the other is **inhibin**); present in two forms in

activity

human gonads, it exists as a dimer of inhibin β_A or β_B chains linked by disulfide bonds. Activin A is a dimer of β_A chains; activin AB is a dimer of β_A and β_B chains. A potent selective stimulator of FSH secretion by the anterior pituitary gland, not via GnRH receptors, it modulates induction of hemoglobin accumulation and proliferation of erythroid progenitor cells in human bone marrow culture and is important in embryonic axial development.

activity 1 the natural or normal functioning of an enzyme, hormone, inhibitor, or other agent; or the intensity with which such an agent functions. *See also* **enzymic activity**. 2 the number of nuclear **transformations** (def. 5) that occur or that can be detected in a given quantity of radioactive material in unit time (*see also* **becquerel**, **curie**). This term is also often used in this sense for **radioactivity** and, loosely, for the radioactive material itself or its emitted radiation; *see also* **specific activity**. 3 or **relative activity** *symbol*: a ; the apparent or effective concentration of a chemical substance as judged by the behaviour of the substance in a standard state. The activity of an entity B is defined by

$$a_B = \exp[(\mu_B - \mu_B^\circ)/RT]$$

where μ_B and μ_B° are respectively the chemical potential and standard chemical potential of entity B, R is the gas constant, and T the thermodynamic temperature. 4 **absolute activity** *symbol*: λ ; the absolute activity of an entity B is defined by

$$\lambda_B = \exp(\mu_B/RT),$$

where μ_B is the chemical potential of entity B, R is the gas constant, and T is the thermodynamic temperature. *See also* **activity coefficient**.

activity coefficient *symbol*: γ ; the ratio of the **activity** (def. 3) of a component of a solution to its concentration. When expressed on an amount concentration basis it is denoted γ_c , when on a molality basis γ_m , and when on a mole fraction basis γ_x . *See also* **mean ionic activity coefficient**.

activity stain any reagent that develops a colour when acted on by a particular enzyme. Such a reagent is used to detect the presence of an enzyme in a gel or paper electrophoresis strip, in which it differentially stains the enzyme protein in question but not other proteins.

actobindin a monomeric protein capable of binding two molecules of monomeric (i.e. G-) actin. Example from *Acanthamoeba castellanii*: database code ACTO_ACACA, 88 amino acids (9.53 kDa).

actomyosin *see* **actin**.

ACU a codon in mRNA for L-threonine.

acumentin an actin-modulating protein, Ca^{2+} -nondependent, isolated from rabbit alveolar macrophages.

acute-phase protein any of the non-antibody proteins that show raised plasma concentrations soon after the onset of infection or tissue injury in homoiothermic animals. They include **complement** (def. 3) proteins, **C-reactive protein**, **fibrinogen** and other coagulation proteins, and **interferon**.

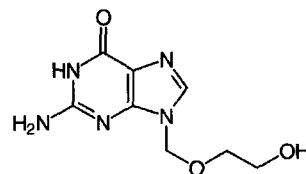
acute transfection the short-term infection of cells with DNA.

acyclic describing an organic compound devoid of a ring of atoms.

acycloguanosine an alternative name for **acyclovir**.

acyclovir or **acycloguanosine** 9-(2-hydroxyethoxymethyl)guanine; an antiviral agent widely used in the treatment of human herpes infections. It is selectively phosphorylated by herpesvirus-induced thymidine kinase and the phosphorylated compound is a potent inhibitor of herpesvirus-induced DNA polymerase. One proprietary name is **Zovirax**.

1-acylglycerol-3-phosphate O-acyltransferase



acyclovir

acyl *generic name* for any group formally derived by removal of a hydroxyl group from the acid function of an organic acid. Examples include any group of general structure R-CO- derived from a carboxylic acid; R-SO_2- from a sulfonic acid; and R-PO(OH)- from a phosphonic acid. However, unless the context indicates otherwise the term refers to such groups derived from carboxylic acids.

acyl-ACP dehydrogenase *see* **enoyl-[acyl-carrier protein] reductase (NADPH, A-specific)**.

acyl-activating enzyme *see* **long-chain-fatty-acid-CoA ligase**.

acylation the process of introducing an acyl group into a compound by substitution for a hydrogen atom. —**acylated** *adj.*

acyl carrier protein *abbr.*: ACP; any of the relatively small acidic proteins that are associated with the fatty acid synthase system of many organisms, from bacteria to plants. They contain one 4'-phosphopantetheine prosthetic group bound covalently by a phosphate ester bond to the hydroxyl group of a serine residue. The sulfhydryl group of the 4'-phosphopantetheine moiety serves as an anchor to which acyl intermediates are (thio)esterified during fatty-acid synthesis. In animals the fatty acid synthase system is a 500 kDa polyfunctional enzyme containing two identical chains, each with one ACP unit. *Escherichia coli* ACP, a separate single protein, contains 77 amino-acid residues (8.85 kDa); the phosphopantetheine group is linked to serine 36. Example (ACP-1 precursor) from *Arabidopsis thaliana*: database code ACP1_ARATH, 137 amino acids (15.04 kDa).

acyl-CoA *generic name* for any derivative of **coenzyme A** in which the sulfhydryl group is in thioester linkage with a fatty-acyl group.

acyl-CoA reductase EC 1.2.1.50; *recommended name*: long-chain-fatty-acyl-CoA reductase; *systematic name*: long-chain-aldehyde:NADP⁺ oxidoreductase (acyl-CoA-forming). An enzyme that catalyses a reaction between a long-chain aldehyde, CoA, and NADP⁺ to form long-chain acyl-CoA and NADPH. It is part of the bacterial **luciferase** system. Example: *see* **Lux proteins**.

acyl-CoA synthetase *see* **long-chain-fatty-acid-CoA ligase**.

acyl enzyme any intermediate formed in an enzymic acyl-transfer reaction in which a group at the active site of the enzyme is acylated.

acylglycerol 1 any mono-, di-, or triester of glycerol with (one or more) fatty acids (termed respectively monoacylglycerol, diacylglycerol, or triacylglycerol); formerly known as mono-, di-, or triglyceride. 2 acylglycerols (*plural*), also denotes any mixture of mono-, di-, and/or triacylglycerols whatever its degree of complexity, including any comprising a **neutral fat**.

2-acylglycerol O-acyltransferase EC 2.3.1.22; *other names*: acylglycerol palmitoyltransferase; monoglyceride acyltransferase; an enzyme that catalyses the acylation by acyl-CoA of 2-acylglycerol to form diacylglycerol and CoA. It is an enzyme involved in the **monoacylglycerol pathway** of triacylglycerol biosynthesis.

1-acylglycerol-3-phosphate O-acyltransferase

EC 2.3.1.51; *systematic name*: acyl-CoA:1-acyl-*sn*-glycerol-3-phosphate 2-O-acyltransferase. An enzyme that catalyses the acylation by acyl-CoA of 1-acyl-*sn*-glycerol 3-phosphate (lysophosphatidic acid) to form 1,2-diacyl-*sn*-glycerol 3-phosphate.



phate (phosphatidic acid) with release of CoA. Acylated acyl carrier protein can also act as substrate. In animals, the enzyme specifically transfers unsaturated fatty acyl groups, a preference that contributes to the high proportion of unsaturated fatty acids on the 2-position of phosphoglycerides. Example from *Escherichia coli*: database code PLSC_ECOLI, 245 amino acids (27.42 kDa).

1-acylglycerophosphocholine O-acyltransferase EC 2.3.1.23; *other name*: lysolecithin acyltransferase; an enzyme that catalyses the acylation by acyl-CoA of 1-acyl-*sn*-glycero-3-phosphocholine to form 1,2-diacyl-*sn*-glycero-3-phosphocholine and CoA. It is involved in the resynthesis of phosphatidylcholine in situations where the latter has been degraded to lysophosphatidylcholine.

acylhomoserine *see* **O-succinylhomoserine**.

acyl migration any intramolecular rearrangement reaction in which, under certain conditions, an acyl group moves from one functional group to another, which may be of the same or of a different kind. **N → O acyl migration** (also termed **N → O acyl shift**, **N → O acyl transfer**, or **N → O peptidyl shift**) is a reaction that can occur in polypeptides and proteins subjected to hydrolysis by concentrated acid at room temperature or in the presence of an acid chloride (e.g. POCl₃), whereby the β-hydroxyl group of a serine or threonine residue displaces the amino group of the same residue from its amide (= peptide) linkage with the carboxyl group of the adjacent residue, thus forming an acid-sensitive β-ester linkage. The reaction may be used to advantage in partial acid hydrolysis of polypeptides and proteins. In the presence of alkali the ester reverts to the original amide by **O → N acyl migration**, a reaction that can also occur during the **Edman degradation** of a peptide. An acyl group, e.g. acetyl or maleyl, that happens to be present on the β-hydroxyl group of a serine or threonine residue, moves to the free amino group of such a residue when it becomes N-terminal, so preventing the next cycle of the degradation.

N-acylneuraminatase glycohydrolase *see* **sialidase**.

acylphosphatase EC 3.6.1.7; *systematic name*: acylphosphate phosphohydrolase. An enzyme that catalyses the hydrolysis of acyl phosphate to fatty-acid anion and orthophosphate. Example from chicken: database code ACYM_CHICK, 102 amino acids (11.27 kDa); three motifs.

acyl-protein synthase *see* **myelin-proteolipid O-palmitoyltransferase**.

acyl shift *see* **acyl migration**.

acylsphingosine deacylase *see* **ceramidase**.

N-acylsphingosine galactosyltransferase EC 2.4.1.47; an enzyme that catalyses the formation of the cerebroside D-galactosylceramide from UDPgalactose and N-acylsphingosine with release of UDP.

acyltransferase any enzyme of sub-subclass EC 2.3.1. Such enzymes transfer acyl groups, forming either esters or amides, by catalysing reactions of the type:

acyl-carrier + reactant = acyl-reactant + carrier.

For example if acyl = acetyl, carrier = CoA, and reactant = choline, the reaction is that of **choline acetyltransferase**.

AD (*in clinical biochemistry*) *abbr.* for alcohol dehydrogenase (preferred to ADH because of possible confusion with anti-diuretic hormone).

Ada or **ADA** *abbr.* for N-(2-acetamido)iminodiacetic acid; [(carbamoylmethyl)imino]diacetic acid; a **Good buffer substance**, pK_a (20 °C) = 6.6.

Adair equation an equation describing the oxygen saturation curve of hemoglobin:

$$R = \frac{K_1[p_{O_2}] + 2K_1K_2[p_{O_2}]^2 + 3K_1K_2K_3[p_{O_2}]^3 + 4K_1K_2K_3K_4[p_{O_2}]^4}{4(1 + K_1[p_{O_2}] + K_1K_2[p_{O_2}]^2 + K_1K_2K_3[p_{O_2}]^3 + K_1K_2K_3K_4[p_{O_2}]^4)}$$

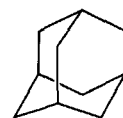
where *R* is the fractional saturation of hemoglobin with oxygen, *K*₁, *K*₂, *K*₃, and *K*₄ are the stepwise intrinsic association constants for the four oxygen-binding sites, and *p*_{O₂} is the partial pressure of oxygen.

For a generalized form of this equation *see* **Bjerrum formation function**. [Defined by Gilbert Smithson Adair (1896–1979), British biophysicist, in 1925.]

adamalysin EC 3.4.24.46; *other name*: proteinase II; a metalloendopeptidase that catalyses the cleavage of Phe¹-Val², His⁵-Leu⁶, His¹⁰-Leu¹¹, Ala¹⁴-Leu¹⁵, Leu¹⁵-Tyr¹⁶, and Tyr¹⁶-Leu¹⁷ in the insulin B-chain. Zn²⁺ and Ca²⁺ are cofactors. It is a snake venom proteinase whose natural substrate(s) are protease inhibitor(s), such as the **serpins**. Example from the Eastern diamondback rattlesnake, *Crotalus adamanteus*: database code ADAM_CROAD, 203 amino acids (23.05 kDa).

1-adamantanamine or **α-adamantanamine** *see* **amantadine**.

adamantane tricyclo[3.3.1.1^{3,7}]decane; a substance isolated from petroleum and said to have virostatic activity. *See also* **amantadine**.



adaptation (*in biology*) **1** any change in an organism's structure or function that allows it better to deal with its environmental conditions. **2** the mechanism whereby bacterial cells that approach cells of the opposite mating type, but fail to mate, can recover and continue dividing.

adapted gradient in **gradient elution**, a form of gradient in which the composition of the eluate is adjusted, on the basis of information obtained in a trial separation, to be optimal at each moment of the elution, so that the optimum resolution between the desired components of the sample is obtained.

adaptin a major coat protein of clathrin-coated vesicles. It functions as a component of the adaptor complexes that link clathrin to the receptors being taken into the cell in the coated vesicles, recognizing a motif of four amino acids (FRXY) in the cytoplasmic domain of the receptor. Example from rat: database code ADA5_RAT, 435 amino acids (49.60 kDa).

adaptive describing changes in an organism, induced by environmental factors, that tend to increase its viability.

adaptive enzyme *see* **inducible enzyme**.

adaptive immunity *an alternative name for* **active immunity**.

adaptor **1** any of various devices useful for joining together two or more parts, otherwise incompatible, such as electrical connectors or pieces of glassware. **2** any synthetic single- or double-stranded oligodeoxynucleotide useful in recombinant DNA technology for joining two incompatible cohesive ends of restriction fragments. *Compare* **linker**. **3** or **adaptor molecule** the molecule that was postulated as carrying the amino acid to the (messenger) RNA in the **adaptor hypothesis** of protein synthesis, now known to be a tRNA molecule.

adaptor hypothesis the hypothesis, first postulated by **Crick**, that during protein biosynthesis each amino-acid residue is carried to the RNA template by its appropriate small adaptor RNA molecule, and that the adaptor is the part that fits on to the messenger RNA. The hypothesis was subsequently confirmed by the discovery of **transfer RNA**.

ADCC *abbr.* for antibody-dependent cell-mediated cytotoxicity.

Addison's disease a disease due to deficiency of production of cortisol and other cortical steroids. It results from atrophy of the adrenal cortex, a condition having a variety of causes. [Described by Thomas Addison (1793–1860), British physician and endocrinologist, in 1849.]

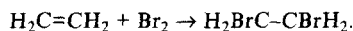
addition **1** (*in chemistry*) the formation of an **adduct**. **2** an **addition reaction**.

addition compound *see* **adduct**.

addition reaction or **addition** any organic chemical reaction involving the combination of two or more substances to form a single product in which there are more groups attached to carbon atoms than there were in the original reactants. Such



reactions thus involve a net reduction of bond multiplicity in one of the reactants, as in the example:



address 1 that part of the information contained within the amino-acid sequence of a hormonal polypeptide or its biosynthetic precursors that determines the receptor-specific affinity of the hormone. *Compare message* (def. 4). 2 (in computer technology) a label, name, or number identifying a device or a location in a memory.

address sequence or **address region** that segment of the amino-acid sequence of a polypeptide hormone or prohormonal polypeptide in which the address of the hormone resides and through which the hormone is considered to bind to its specific receptor. *Compare message sequence*.

adducin a membrane skeleton protein (heterodimer) that interacts with a junctional complex that links **spectrin** assemblies. The complex in the red blood cell consists of **tropomyosin** and **actin** with band 4.1 (see **band**). Adducin probably binds to band 4.1. Examples from human erythrocyte: adducin α , database code S18207, 737 amino acids (80.88 kDa); adducin β , database code S18208, 726 amino acids (80.76 kDa).

adduct 1 any new chemical species, AB, formed by direct combination of two separate chemical species, A and B, in such a way that there is no change in **connectivity** of atoms within the moieties A and B. This term is preferred to **complex**, which is less explicit. 2 (formerly) the product of a reaction between molecules occurring in such a way that the original molecules or their residues have their long axes parallel to one another.

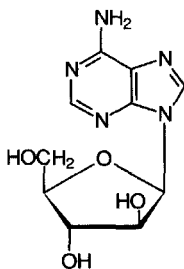
Ad symbol for a residue of the purine base adenine (alternative to A).

ADE2 a gene that encodes a multifunctional protein with activities for **phosphoribosylaminoimidazole carboxylase** and **phosphoribosylaminoimidazolesuccinocarboxamide synthase**. It has been studied as a model of position effect in gene expression. The gene is expressed in all cells if in its normal position near the middle of the chromosome, but if moved experimentally to the end of the chromosome it is silenced. The lack of the enzyme produces a red pigmentation in these colonies.

adenine symbol: A or Ade; 6-aminopurine; 1H-purin-6-amine; a purine base. It is one of the five main bases found in nucleic acids and a component of numerous important derivatives of its corresponding ribonucleoside, adenosine. It can exist in a tautomeric imino form.



adenine arabinoside other names: vidarabine, ara-A; 9- β -D-arabinofuranosyladenine; an antibiotic produced by *Streptomyces* bacteria that acts as an antiviral agent, causing mispairing of purines and pyrimidines in nucleic-acid synthesis.



9- β -D-form

adenine nucleotide translocase see **ADP/ATP carrier protein**.

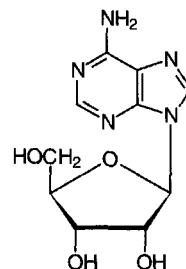
adenine phosphoribosyltransferase abbr.: APRT; EC 2.4.2.7; other names: AMP pyrophosphorylase; transphosphoribosidase; an enzyme that catalyses the formation of AMP from 5-phospho- α -D-ribose 1-diphosphate and adenine with release of pyrophosphate. It is an enzyme of purine metabolism that salvages adenine released by degradative enzymes, converting it back to a nucleotide. Example: from human: APT_HUMAN, 179 amino acids (19.45 kDa).

adeno+ or (before a vowel) **aden+** comb. form meaning of, pertaining to, or like a gland; glandular; found in glands.

adenohypophysis the glandular anterior lobe of the hypophysis (or **pituitary gland**). It produces corticotropin, gonadotropin, lipotropin, somatotropin, thyrotropin, and other hormones. *Compare neurohypophysis*. —**adenohypophyseal** or **adenohypophysial** adj.

adenoma any benign tumour formed by the multiplication of the epithelial cells that form the ducts and acini of glandular organs. The meaning has been broadened to include the benign tumours that arise from the solid masses of epithelium that form some of the endocrine glands. Adenomas often accurately reproduce the tissues from which they are derived and produce a secretion identical with or similar to that produced by the normal glandular tissue.

adenosine symbol: A or Ado; adenine riboside; 9- β -D-ribofuranosyladenine; a ribonucleoside found widely distributed in cells of every type as the free nucleoside and in combination in nucleic acids and various nucleoside coenzymes. It is a potent regulator of physiological transmission in both central and peripheral nervous systems, where it activates specific receptors (see **adenosine receptor**). It can inhibit or stimulate the release of a number of neurotransmitters, including acetylcholine, β -aminobutyrate, catecholamines, excitatory amino acids, and 5-hydroxytryptamine. The effect depends on whether the receptor involved inhibits adenylate cyclase (and thus inhibits release) or stimulates it.



adenosine aminohydrolase see **adenosine deaminase**.

adenosine 2',3'-(cyclic)phosphate see **adenosine phosphate**.

adenosine 3',5'-(cyclic)phosphate see **adenosine 3',5'-phosphate**.

adenosine deaminase EC 3.5.4.4; systematic name: adenosine aminohydrolase; an enzyme that catalyses the hydrolysis of adenosine to inosine and NH_3 . It is involved in the degradation of adenosine, the inosine thereby formed being then converted to hypoxanthine and thence to uric acid. The enzyme is also found on the outer surface of T cells, and a form that acts on double-stranded RNA (dsRNA) converts adenosines to inosines within dsRNA. Example from mouse, 3-D structure with 6-hydroxy-1,6-dihydropurine ribonucleoside: database code NRL_1ADA, 349 amino acids (39.62 kDa).

adenosine diphosphatase see **apyrase**.

adenosine 5'-diphosphate symbol: Ado5'PP or ppA; the recommended name for adenosine diphosphate (abbr.: ADP); 5'-diphosphoadenosine; 5'-adenyl phosphate; adenosine 5'-trihydrogen diphosphate; a universally distributed nucleotide that occurs both in the free state and as a component

adenosinediphosphoglucose

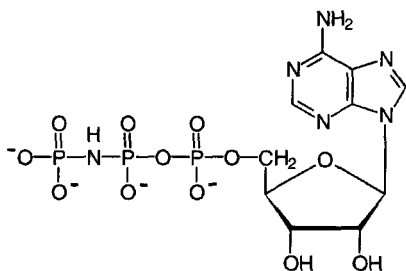
of certain **nucleotide coenzymes**. It is a metabolic precursor of adenosine 5'-triphosphate, ATP, and is the product of many enzyme reactions in which ATP is hydrolysed by an **adenosine triphosphatase** or in which the terminal phosphoric residue of ATP is transferred to another organic compound by a **kinase** (including **adenylate kinase**).

adenosinediphosphoglucose or **adenosine (5')diphospho(1)- α -D-glucose** *symbol*: AdoPPGlc or Ado-5'PP-Glc or A5'pp1Glc; the alternative recommended names for adenosine diphosphate glucose (*abbr.*: ADPG or ADP-Glc or ADPglucose); adenosine 5'-(α -D-glucopyranosyl diphosphate); α -D-glucopyranosyl 5'-adenyl phosphate; a **nucleoside-diphosphosugar** in which the distal phosphoric residue of adenosine 5'-diphosphate is in glycosidic linkage with glucose. ADPglucose is synthesized from glucose 1-phosphate and adenosine 5'-triphosphate, ATP, through the action of glucose 1-phosphate adenylyltransferase (EC 2.7.7.27). It is an intermediate in the incorporation of glucosyl groups into starch in plants and into storage glucans in bacteria.

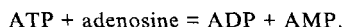
adenosinediphosphoribose or **adenosine(5')diphospho(5)- β -D-ribose** *symbol*: AdoPPRib or A5'pp5Rib or (Rib5)ppA; the alternative recommended names for adenosine diphosphate ribose (*abbr.*: ADP-Rib or ADPRibose); adenosine 5'-(β -D-ribofuranose 5-diphosphate); a **nucleoside diphosphosugar** in which the explicit sugar moiety is a second residue of D-ribose and is in ester linkage with the distal phosphoric residue of adenosine 5'-diphosphate atypically at C-5 instead of at C-1. ADP-ribose is synthesized from D-ribose 5-phosphate and ADP through the action of ribose-5-phosphate adenylyltransferase (EC 2.7.7.35), releasing orthophosphate. In addition, free ADP-ribose is released from nicotinamide-adenine dinucleotide, NAD⁺, on hydrolytic cleavage of the glycosylamine bond to nicotinamide through the action of NAD⁺ nucleosidase (EC 3.2.2.5). *See also* **cyclic adenosinediphosphoribose**.

adenosinediphosphoribosyl *abbr.*: ADPRibosyl; the glycosyl group formally derivable from adenosinediphosphoribose by loss of the anomeric (i.e. C-1) hydroxyl group from its distal ribose moiety. The ADPRibosyl group of nicotinamide-adenine dinucleotide may be utilized for **ADP-ribosylation**.

adenosine 5'-[β , γ -imido]triphosphate *symbol*: AdoPP[NH]P or p[NH]ppA; the recommended name for β , γ -imidoadenosine 5'-triphosphate (*abbr.*: ATP[β , γ -NH]); 5'-adenyl imidodiphosphate; 5'-adenyl iminodiphosphonate (*abbr.*: AMP-PNP); adenosine (5' $\rightarrow$ O³)-1,2- μ -imidotriphosphate; a synthetic analogue of adenosine triphosphate, in which the oxy group between the latter's intermediate and terminal phosphorus atoms is replaced by an imido group. It competitively inhibits ATP-dependent enzymes, including mitochondrial ATPase.



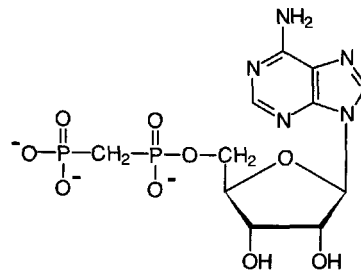
adenosine kinase EC 2.7.1.20; an enzyme that catalyses the reaction:



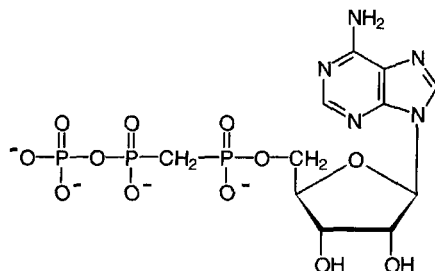
adenosine 5'-[α , β -methylene]diphosphate *symbol*: AdoP[CH₂]P or p[CH₂]pA; the recommended name for α , β -methyleneadenosine 5'-diphosphate (*abbr.*: ADP[α , β -CH₂]); 5'-adenyl methylphosphonate (*abbr.*: AMP-CP); adeno-

adenosine 5'-[β , γ -methylene]triphosphate

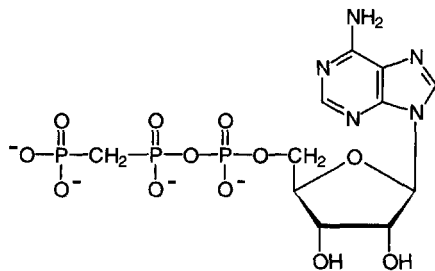
sine (5' $\rightarrow$ O¹)-1,2- μ -methylenediphosphate; a synthetic analogue of adenosine 5'-diphosphate, ADP, in which the oxy group between the latter's two phosphorus atoms is replaced by a methylene group. It inhibits 5'-nucleotidase obtained from rat-heart membranes.



adenosine 5'-[α , β -methylene]triphosphate *symbol*: AdoP[CH₂]PP or pp[CH₂]pA; the recommended name for α , β -methyleneadenosine 5'-triphosphate (*abbr.*: ATP[α , β -CH₂]); 5'-adenyl methylenediphosphate (*abbr.*: AMP-CPP); adenosine (5' $\rightarrow$ O¹)-1,2- μ -methylenetriphosphate; a synthetic analogue of adenosine 5'-triphosphate, ATP, in which the oxy group between the latter's innermost and intermediate phosphorus atoms is replaced by a methylene group. It competitively inhibits rat-liver adenylyl cyclase.



adenosine 5'-[β , γ -methylene]triphosphate *symbol*: AdoPP[CH₂]P or p[CH₂]ppA; the recommended name for β , γ -methyleneadenosine 5'-triphosphate (*abbr.*: ATP[β , γ -CH₂]); 5'-adenyl methylenediphosphonate (*abbr.*: AMP-PCP); adenosine (5' $\rightarrow$ O¹)-1,2- μ -methylenetriphosphate; a synthetic analogue of adenosine 5'-triphosphate, ATP, in which the oxy group between the latter's intermediate and terminal phosphorus atoms is replaced by a methylene group. It inhibits the binding of ouabain to Na⁺/K⁺-transporting ATPase, the sodium pump.



adenosine monophosphate

adenosine monophosphate *abbr.*: AMP; *an alternative name for any adenosine phosphate*, but in particular for adenosine 5'-phosphate, especially when its distinction from adenosine (5') diphosphate and adenosine (5')-triphosphate requires emphasis.

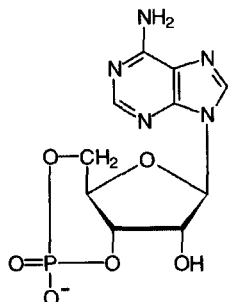
adenosine phosphate *symbol*: AdoP; *adenosine monophosphate (abbr.: AMP); any phosphoric monoester or diester of adenosine*. There are three monoesters – adenosine 2'-phosphate, adenosine 3'-phosphate, and **adenosine 5'-phosphate** – and two diesters – adenosine 2',3'-phosphate and **adenosine 3',5'-phosphate** – although adenosine 5'-phosphate is the ester commonly denoted (the locant being omitted if no ambiguity may arise). Adenosine 2'-phosphate (*symbol*: Ado2'P) is also named **adenosine 2'-monophosphate (abbr.: 2'AMP) or 2'-adenylic acid or adenylic acid a**, and adenosine 3'-phosphate (*symbol*: Ado3'P) is also named **adenosine 3'-monophosphate (abbr.: 3'AMP) or 3'-adenylic acid or adenylic acid b or (formerly) yeast adenylic acid**.

adenosine 2'-phosphate *see* **adenosine phosphate**.

adenosine 2',3'-phosphate *see* **adenosine phosphate**.

adenosine 3'-phosphate *see* **adenosine phosphate**.

adenosine 3',5'-phosphate *symbol*: Ado-3',5'-P; *the recommended name for cyclic adenosine 3',5'-monophosphate (abbr.: cyclic AMP or cAMP); adenosine 3',5'-cyclophosphate; adenosine 3',5'-(cyclic)phosphate; a monophosphoric diester of adenosine*. It is a universally distributed key metabolite, produced by the action of **adenylate cyclase** on adenosine 5'-triphosphate, ATP. The first compound to be named a **second messenger**, it mediates many effects in signal transduction pathways. It was first identified as a heat-stable activator of glycogen phosphorylase kinase, and is now known also to activate cyclic-AMP-dependent protein kinase and to regulate numerous other enzymic activities or physiological processes.

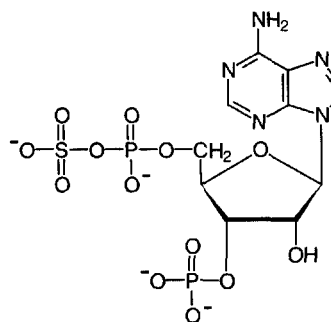


adenosine 5'-phosphate or 5'-adenylic acid or 5'-phospho-adenosine or 5'-O-phosphoadenosine *symbol*: Ado5'P; *alternative recommended names for adenosine 5'-monophosphate (abbr.: 5'AMP); adenosine 5'-(dihydrogen phosphate); adenylic (mono)ribonucleotide or (formerly) muscle adenylic acid*. (The locant is commonly omitted if there is no ambiguity as to the position of phosphorylation.) It is a metabolic regulator, biosynthesized from inosine 5'-phosphate, 5'-inosinic acid; it is formed also by pyrophosphatase-catalysed cleavage of (inorganic) diphosphate from ATP.

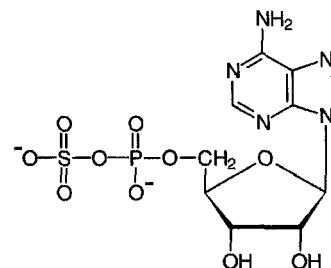
adenosine 3'-phosphate 5'-phosphosulfate or 3'-phospho-adenosine 5'-phosphosulfate *symbol*: PAdoPS; *abbr.*: PAPS; *alternative recommended names for active sulfate; 3'-phospho-5'-adenylyl sulfate; adenosine 3'-phosphate 5'-P-phosphato-sulfate; 3'-phospho-5'-adenylic sulfuric monoanhydride; a naturally occurring mixed anhydride*. It is synthesized from **adenosine 5'-phosphosulfate** by phosphorylation with ATP through the action of adenylylsulfate kinase (EC 2.7.1.25). It is an intermediate in the formation of a variety of sulfo compounds in biological systems. For example, in animals it is involved in sulfate transfer in the formation of sulfatides and in

adenosine 5'-β-thiodiphospho

the synthesis of chondroitin sulfate and other sulfated polysaccharides, while in bacteria, by a process analogous to that involving adenosine 5'-phosphosulfate in plants, adenosine 3'-phosphate 5'-phosphosulfate interacts with reduced thioredoxin to yield adenosine 3',5'-bis(phosphate) and sulfite; the latter can then undergo further reduction to sulfide, from which cysteine may be synthesized.



adenosine 5'-phosphosulfate *symbol*: AdoPS; *abbr.*: APS; *the recommended name for 5'-adenylyl sulfate; adenosine 5'-P-phosphatosulfate; 5'-adenylic sulfuric monoanhydride; a naturally occurring mixed anhydride*. It is synthesized from ATP and (inorganic) sulfate by the action of sulfurylase, sulfate adenylyltransferase (EC 2.7.7.4), and is an intermediate in the formation of **adenosine 3'-phosphate 5'-phosphosulfate**. In plants, by a process analogous to that involving **adenosine 3'-phosphate 5'-phosphosulfate** in bacteria, adenosine 5'-phosphosulfate can undergo reduction to yield sulfide, which may then be utilized for cysteine synthesis.

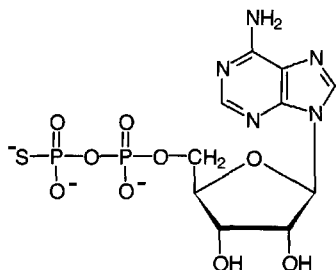


adenosine receptor or P₁ purinoceptor one of three types, A₁, A₂, and A₃ of membrane protein that bind adenosine or its analogues; they are **seven-transmembrane-domain proteins**. The selective agonist for A₁ is N⁶-cyclopentyladenosine; that for A₂ is (2-*p*-carboxyethyl)phenylamino-5'-*N*-carboxamidoadenosine. Binding of agonist to A₁ causes inhibition of **adenylate cyclase**, opening of K⁺ channels, and inhibition of Ca²⁺ channels. Activation of A₂ brings about stimulation of adenylyl cyclase. Activation of A₃ causes stimulation of adenylyl cyclase. In some cases, particularly where A₁ receptors are involved, inositolphospholipid turnover may be stimulated. Examples (all from human): A₁ receptor: database code AAIR_HUMAN, 326 amino acids (36.15 kDa); A₂A receptor: database code AA2A_HUMAN, 412 amino acids (44.71 kDa); A₃ receptor: database code AA3R_HUMAN, 318 amino acids (36.18 kDa). *See also* **purinoceptor**.

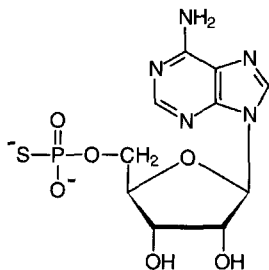
adenosine 5'-β-thiodiphosphate *symbol*: AdoPP[S] or [S]ppA; *abbr.*: ADP[S] or ADP-β-S; *the recommended name for adenosine 5'-[β-thio]diphosphate; adenosine 5'-(2-thiodiphosphate); adenosine (5'→O₂)-1-thiodiphosphate; a synthetic*

adenosine 5'-thiophosphate

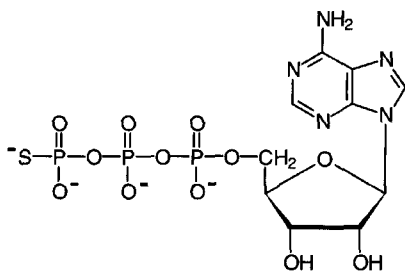
analogue of adenosine 5'-diphosphate, ADP, in which an oxygen atom of its terminal phosphoric residue is replaced by a sulfur atom. It acts variously as a substrate or an inhibitor of ADP-dependent systems.



adenosine 5'-thiophosphate *symbol:* AdoP[S] or [S]pA; *the recommended name for* adenosine 5'-[α-thio]monophosphate (*abbr.:* AMP-S); a synthetic analogue of adenosine 5'-phosphate, AMP, in which an oxygen atom of its phosphoric residue is replaced by a sulfur atom. It acts variously as a substrate or an inhibitor of AMP-dependent systems.



adenosine 5'-γ-thiotriphosphate *symbol:* AdoPPP[S] or [S]pppA; *abbr.:* ATP[S] or ATP-γ-S; *the recommended name for* adenosine 5'-[γ-thio]triphosphate; adenosine (5'→O³)-1-thiotriphosphate; a synthetic analogue of adenosine 5'-triphosphate, ATP, in which an oxygen atom of its terminal phosphoric residue is replaced by a sulfur atom. It can variously substitute for ATP or act as an inhibitor of ATP-dependent systems.



adenosinetriphosphatase *abbr.:* ATPase; **1** EC 3.6.1.3; ATP phosphohydrolase; an enzyme catalysing the hydrolysis of ATP to ADP and orthophosphate. **2** any of various enzymes listed separately as EC 3.6.1.32 to EC 3.6.1.38. Some function as exchangers or transporters for Na⁺, K⁺, or Ca²⁺ (see **calcium-transporting ATPase**, **sodium-potassium ATPase**); myosin ATPase (EC 3.6.1.32) is involved in muscle contraction. **H⁺**-

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

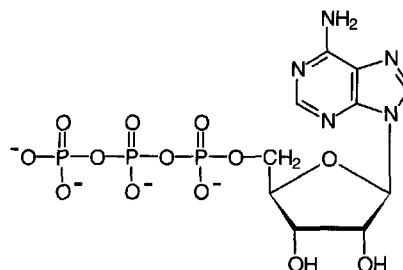
Huynguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynguyen@huynguyentld.com.

S-adenosylmethionine

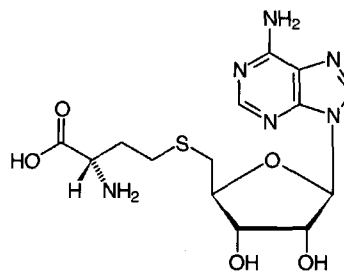
transporting ATP synthase (also called mitochondrial or chloroplast ATPase) (EC 3.6.1.34) is involved in ATP synthesis.

adenosine 5'-triphosphate *symbol:* Ado5'PPP or pppA; *the recommended name for* adenosine triphosphate (*abbr.:* ATP); 5'-triphosphoadenosine; 5'-adenylyl diphosphate; adenosine 5'-(tetrahydrogen triphosphate); a universally important coenzyme and enzyme regulator. It is formed from adenosine 5'-diphosphate by oxidative phosphorylation in coupled mitochondria, by photophosphorylation in plants, and by substrate-level phosphorylation. Reactions in which it participates are often driven in the direction leading to hydrolysis of ATP. The chemical energy so released may be utilized in active transport; it may be converted to mechanical energy (e.g. for muscular contraction, movement of cilia, etc.), to light energy (for bioluminescence), or to electrical energy (in electric fish); or it may be released as heat. ATP also participates in numerous synthetic reactions by the transfer to other metabolites of a phosphoric or a diphosphoric residue, of an adenosyl residue, or of an adenylyl residue.



adenosyl any chemical group formed by the loss of a 2'-, a 3'-, or a 5'-hydroxyl group from the ribose moiety of adenosine. *Compare* **adenylyl**.

S-adenosylhomocysteine the L enantiomer, S-(5'-adenosyl)-L-homocysteine (*symbol:* AdoHcy; *abbr.:* SAH), S-(5'-deoxyadenosine-5'-yl)-L-homocysteine, is formed from **S-adenosylmethionine**. It is a strong inhibitor of S-adenosylmethionine-mediated methylation reactions and is cleaved to adenosine and homocysteine.



S-adenosylmethionine the L enantiomer, active methionine, S-(5'-adenosyl)-L-methionine (*symbol:* AdoMet; *abbr.:* SAM), S-(5'-deoxyadenosine-5'-yl)-L-methionine, is an important intermediate in one-carbon metabolism, the methionine's methyl group (activated by the adenosyl moiety bonded through sulfonium) being donated to an acceptor molecule by transmethylation and S-adenosylhomocysteine being produced. It is important also as an intermediate in the production of ethylene from L-methionine in plants, being cleaved to 5'-methylthioadenosine and L-aminocyclopropane-1-carboxylic acid.



late. The latter compound is then fragmented under aerobic conditions to ethylene, formate, ammonium, and carbon dioxide.

adenosylmethionine decarboxylase EC 4.1.1.50; *systematic name*: S-adenosyl-L-methionine carboxy-lyase. An enzyme of polyamine (and hence **trypanothione**) biosynthesis that catalyses the decarboxylation of S-adenosyl-L-methionine to (5'-deoxyadenosin-5'-yl)(3-aminopropyl)methylsulfonium salt; pyruvate acts as a cofactor. The product participates in reactions in which its 3-aminopropyl group is transferred to **putrescine** to form **spermidine**, and a second 3-aminopropyl group is then transferred to spermidine to form **spermine**. In most cases its subunits (α and β) are derived from a single proenzyme. Example from yeast: database code DCAM_YEAST, 396 amino acids (46.18 kDa); α , 1-87; β , 88-end.

S-adenosylmethionine synthetase *see* **methionine adenosyltransferase**.

adenovirus any of a group of non-enveloped icosahedral viruses, the Adenoviridae, containing linear double-stranded DNA. They affect mammals or birds (usually being specific to one or a few closely related host species) and are often associated with disease of the respiratory tract.

adenyl a misnomer for **adenyllyl** or **adenylate**.

adenylate 1 either the monoanion or the dianion of adenylic acid. 2 any mixture of the acid and its anions. 3 any salt or ester of adenylic acid.

adenylate cyclase or **adenyllylcyclase** or (sometimes, incorrectly) **adenyl cyclase** EC 4.6.1.1; *systematic name*: ATP pyrophosphate-lyase (cyclizing); a phosphorus-oxygen lyase enzyme that catalyses the elimination of a moiety of diphosphate from a molecule of adenosine 5'-triphosphate (ATP) to form **adenosine 3',5'-phosphate** (cyclic AMP). It thereby is involved as an effector of signal transduction and other fundamental regulatory mechanisms. Its distribution is very wide, as also is its structural variation and the variation of its properties among the different forms. In mammals it is a membrane-bound glycoprotein, the various isoforms being in the mass range 115 to 180 kDa. They all have a predicted general structure indicating two structurally related cytoplasmic domains, relatively well conserved among the isoforms, having homology to the guanylate cyclase catalytic domain. One of them bears an ATP-binding site. In addition they have two membrane-spanning domains, each of which contains six membrane spans, the sequence differing significantly among the isoforms. Types I to VI have been described. Type I, 1134 amino acids, is brain-specific, Ca^{2+} /calmodulin activated, and inhibited by G-protein $\beta\gamma$ subunits. Types II (1088 amino acids) and IV (1064 amino acids) are **calmodulin** insensitive and activated by $\beta\gamma$ subunits; type II is found in brain and olfactory tissue, type IV is widely distributed except in testis. Type III, from olfactory sensory neurons, is calmodulin sensitive and not regulated by $\beta\gamma$ subunits. Types V and VI form a subgroup that is widely distributed and not stimulated by $\beta\gamma$ subunits. *Saccharomyces cerevisiae* enzyme (2026 amino acids) has a weak structural relationship with the mammalian enzyme. It is positively regulated by the *RAS1* and *RAS2* gene products, but this property is not shared by a recombinant enzyme from *Schizosaccharomyces pombe*. A plant gene has been cloned, the deduced protein sequence having no similarity to prokaryotic counterparts, but showing striking similarity to the catalytic region of *S. cerevisiae* adenylyl cyclase, and with the cytoplasmic domains of bovine adenylyl cyclase and two mammalian guanylate cyclases. Soluble adenylyl cyclases are found in some bacterial toxins. Example from *Escherichia coli*: database code CYAA_ECOLI, 848 amino acids (97.43 kDa). Example from human: database code HSADCYCL, 476 amino acids (53.37 kDa).

adenylate energy charge (or **energy charge**) a measure of the phosphorylating power of an adenylyl pool, equal to one half the average number of anhydride-bound phosphoric groups per adenosine moiety present in the pool. It may be defined in

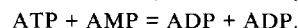
terms of the concentrations of AMP, ADP, and ATP in the pool and expressed by the quotient:

$$([\text{ATP}] + 0.5[\text{ADP}])/([\text{ATP}] + [\text{ADP}] + [\text{AMP}]).$$

Compare **phosphorylation state ratio**.

adenylate isopentenyltransferase EC 2.5.1.27; *other name*: cytokinin synthase; an enzyme that catalyses the reaction between Δ^2 -isopentenyl diphosphate and AMP to form (N^6 - Δ^2 -isopentenyl)adenosine 5'-monophosphate, an intermediate in the formation of **zeatin** riboside, and pyrophosphate. Example from *Agrobacterium vitis* plasmid pTIS4: database code S30106, 236 amino acids (25.81 kDa).

adenylate kinase *abbr.* (in clinical biochemistry): AK; EC 2.7.4.3; *systematic name*: ATP:AMP phosphotransferase; *other name*: myokinase. An enzyme that catalyses the reaction:



Example, adenylyl kinase isoenzyme 1 (human): database code KADI_HUMAN, 194 amino acids (21.65 kDa), five motifs.

adenylate pool or **adenylate system** the total amount of AMP, ADP, and ATP in a cell, tissue, or organism.

adenyl cyclase a misnomer for **adenylate cyclase**.

adenylic acid the trivial name for any phosphoric monoester of adenosine. The position of the phosphoric residue on the ribose moiety of a given ester may be specified by a prefixed locant; *see* **adenosine phosphate**, **adenosine**. However, 5'-adenylic acid is the ester commonly denoted, its locant usually being omitted if no ambiguity may arise. 5'-Adenylic acid is also an alternative recommended name for **adenosine 5'-phosphate**.

adenyllyl the adenosine[mono]phospho group; the acyl group derived from adenylic acid (*see* **adenosine 5'-monophosphate**).

Compare **adenosyl**.

adenyllylate to introduce adenylyl groups into a compound generally through the action of an adenylyltransferase.

—**adenyllylated** *adj.*; **adenyllylation** *n.*

adenyllyl cyclase a misnomer for **adenylate cyclase**.

adenyllyl sulfate *see* **adenosine 5'-phosphosulfate**

adenyllylsulfate kinase EC 2.7.1.25; *other name*: APS kinase; an enzyme that synthesizes 3'-phosphoadenosine-5'-phosphosulfate from ATP and adenylylsulfate with release of ADP. Example from yeast: KAPS_YEAST, 202 amino acids (23.03 kDa). *See* **phosphoadenosine-5'-phosphosulfate**.

adenyllyltransferase *generic name* for any of a number of enzymes within sub-subclass EC 2.7.7, nucleotidyltransferases, that are specific for the transfer of an adenylyl group from a donor (usually adenosine triphosphate, ATP) to an acceptor (such as a nucleotide, a polynucleotide, a protein, a sugar phosphate, or sulfate); e.g. FMN adenylyltransferase (EC 2.7.7.2), ribose-5-phosphate adenylyltransferase (EC 2.7.7.35).

ADH *abbr. for*: 1 antidiuretic hormone. 2 (in clinical biochemistry) alcohol dehydrogenase (preferred form: AD).

adhalin or **DAG2** a dystrophin-associated glycoprotein of skeletal muscle sarcolemma that is specifically deficient in severe childhood autosomal recessive muscular dystrophy. It has a 17-residue signal sequence and one transmembrane domain, and there are two sites for N-linked glycosylation. Example from rabbit: database code OCDAGII, 387 amino acids (42.45 kDa).

adherens junction a **cell junction** in which the cytoplasmic face of the plasma membrane is attached to actin filaments.

adhesion (in pathology) the abnormal union of surfaces or parts, usually due to the formation of fibrous tissue following inflammation.

adhesion molecules molecules expressed on the surface of a cell that mediate the adhesion of the cell to other cells or to the extracellular matrix. They bind to receptors, classed collectively as **integrins**. Adhesion molecules are grouped into classes such as **selectins**; the immunoglobulin superfamily, which includes **ICAM**, **MadCAM**, **NCAM**, **PECAM**, and **VCAM**; the



adiabatic

cadherins; and **CD44**. Adhesion molecules play a part in morphogenesis (e.g. cadherins) and in the treatment of inflammation and wounds (e.g. selectins and the immunoglobulin superfamily).

adiabatic describing any thermodynamic process that occurs with neither gain nor loss of heat between the system and its surroundings. — **adiabatically** *adv.*

adiabatic calorimeter a calorimeter in which the temperature of the outer jacket is kept as close as possible to that of the inner compartment so that heat losses from the latter are minimized.

adiabatic system (*in thermodynamics*) any geometrically defined volume that does not exchange thermal energy with its surroundings. *Compare* **closed system**, **open system**.

adipocyte a cell of adipose tissue; a fat cell or lipocyte of an animal.

adipogenesis the formation of fat or adipose tissue.

adipokinetic fat-mobilizing; lipotropic.

adipokinetic hormone *abbr.*: AKH; an alternative name for **lipotropin**.

adipose tissue fat or fatty tissue. *See also* **brown adipose tissue**, **white adipose tissue**.

adipsin **1** another name for complement factor D. **2** a protein of the serine endopeptidase family secreted into serum by adipocytes. It is expressed abundantly in adipocytes and sciatic nerve. There are two forms, of M_r 37 000 and 44 000.

adiuretin (*sometimes*) an alternative name for **antidiuretic hormone**.

adjuvant **1** (*in immunology*) any substance or mixture of substances that increases or diversifies the immune response to an antigen. *See also* **Freund's adjuvant**. **2** (*in pharmacology*) any remedy or drug that assists or modifies the action of other remedies or drugs.

adjuvant peptide an alternative name for **muramyl-dipeptide**.

ad lib without restraint. Short for the Latin *ad libitum*, meaning literally 'at pleasure'.

A-DNA *abbr.* for A-form of DNA.

Ado symbol for a residue of the ribonucleoside adenosine (alternative to A).

AdoHcy symbol for S-adenosylhomocysteine.

AdoMet symbol for S-adenosylmethionine.

adonitol a former name for **ribitol**.

adonose a former name for **ribulose**.

AdoP symbol for any adenosine phosphate.

Ado2'P symbol for adenosine 2'-phosphate.

Ado-2',3'-P symbol for adenosine 2',3'-phosphate.

Ado3'P symbol for adenosine 3'-phosphate.

Ado-3',5'-P symbol for adenosine 3',5'-phosphate.

Ado5'P symbol for adenosine 5'-diphosphate (alternative to pAdo).

AdoP[CH₂]P symbol for adenosine 5'-[α,β -methylene]diphosphate (alternative to p[CH₂]pA).

AdoP[CH₂]PP symbol for adenosine 5'-[α,β -methylene]triphosphate (alternative to pp[CH₂]pA).

Ado5'PP symbol for adenosine 5'-diphosphate (alternative to ppA).

AdoPP[CH₂]P symbol for adenosine 5'-[β,γ -methylene]triphosphate (alternative to p[CH₂]ppA).

AdoPPGlc or **Ado-5'PP-Glc** symbol for adenosinediphosphoglucose (alternatives to A5'pp1Glc).

AdoPP[NH]P symbol for adenosine 5'-[β,γ -imido]triphosphate (alternative to p[N]ppA).

Ado5'PPP symbol for adenosine 5'-triphosphate (alternative to pppA).

AdoPP[S] symbol for adenosine 5'- β -thiodiphosphate (alternative to [S]ppA).

AdoP[S] symbol for adenosine 5'-thiophosphate (alternative to [S]pA).

AdoPPP[S] symbol for adenosine 5'- γ -thiotriphosphate (alternative to [S]pppA).

AdoPPRib symbol for adenosine diphosphoribose (alternative to pRibppA).

to A5'pp5Rib or (Rib5)ppA).

AdoPS symbol for adenosine 3'-phosphate 5'-phosphosulfate.

adoptive immunity immunity conferred by transfer of immunologically active cells from one individual to another.

ADP *abbr.* for adenosine 5'-diphosphate.

ADPase *abbr.* for adenosine diphosphatase (*see* **apyrase**).

ADP,ATP carrier protein or **adenine nucleotide translocase** an integral membrane protein of the inner mitochondrial membrane, responsible for the transport of ADP and ATP across the membrane. It exchanges one molecule of ADP³⁻ for one molecule of ATP⁴⁻, and is thought to be driven by the proton gradient established across the inner membrane. It is a homodimer, each chain having three homologous domains. Different types occur in different mammalian tissues but all are related. Example, muscle isoform I (human): database code ADT1_HUMAN, 298 amino acids (33.03 kDa).

ADP[α,β -CH₂] *abbr.* for α,β -methyleneadenosine 5'-diphosphate; i.e. adenosine 5'-[α,β -methylene]diphosphate.

ADPG *abbr.* for adenosinediphosphoglucose; ADP-Glc is preferred.

ADP-Glc or **ADPglucose** (*preferred*) *abbr.* for adenosinediphosphoglucose; *see also* **ADPG**.

ADP-Rib or **ADPRibose** *abbr.* for adenosinediphosphoribose.

ADP-ribose (*cyclic*) *see* **cyclic adenosine 5'-diphosphoribose**.

ADPRibosyl *abbr.* for the adenosinediphosphoribosyl group.

ADP-ribosylation the transfer of one or more **adenosine-diphosphoribosyl** (*abbr.* ADPRibosyl) groups from nicotinamide-adenine dinucleotide, NAD⁺, to a protein through the action of an **ADP-ribosyltransferase**. In eukaryotic cells the transfer occurs particularly to the α subunits of **G proteins**, and then is stimulated by treatment with certain bacterial toxins such as **cholera toxin** or **pertussis toxin**. Linear poly(ADPRibosyl) attachments, which may consist of up to 50 ADPRibose units, are found on nuclear proteins and also on some cytoplasmic proteins. In the stepwise generation of such attachments an incoming ADPRibosyl group forms a (12') glycosidic bond to the nucleosidic ribose moiety of the most recently attached ADPRibose unit. ADP-ribosylation occurs also in cells of *Escherichia coli* infected with bacteriophage T4, where single ADPRibosyl groups become attached to the host's RNA polymerase. — **ADP-ribosylated** *adj.* *See also* **ADP-ribosyltransferase**, **NAD(P)⁺-arginine ADP-ribosyltransferase**.

ADP-ribosylation factor a protein that acts as activator for **ADP-ribosyltransferase**. Example (human and bovine): **cholera toxin** activator: database code ARF1_HUMAN, 181 amino acids (20.67 kDa).

ADP-ribosyltransferase EC 2.4.2.30; *recommended name*: NAD⁺ ADP-ribosyltransferase; *systematic name*: NAD⁺: poly(adenine-diphosphate-D-ribose)-acceptor ADP-D-riboseyl-transferase; *other names*: poly(ADP) polymerase; poly(adenosine diphosphate ribose) polymerase; NAD⁺ ADP-ribosyltransferase (polymerizing); poly(ADP-ribose) synthetase; *abbr.*: PARP or ADPRT. An enzyme that catalyses a reaction between NAD⁺ and (ADP-D-ribose)_n-acceptor to form nicotinamide and (ADP-D-ribose)_{n+1}-acceptor. The enzyme acts on a number of nuclear proteins and thereby regulates events in differentiation and cell proliferation. The presence of DNA is necessary, and zinc is a cofactor. Example from *Bos taurus*: database code PPOL_BOVIN, 1015 amino acids (113.36 kDa).

ADP[S] or **ADP- β -S** *abbr.* for adenosine 5'- β -thiodiphosphate.

adren+ or **adreno+** *comb. form* denoting **1** the **adrenal gland(s)**. **2** **epinephrine** or a related catecholamine.

adrenal **1** pertaining to or produced by the **adrenal glands**. **2** the adrenal gland itself.

adrenal androgen any of the C₁₉ steroid hormones produced in the cortex of the adrenal gland, including androstenedione and testosterone.

adrenal cortex *see* **adrenal gland**.

adrenal cortical hormone or **adrenocortical hormone** any (cor-



adrenalectomize

tico)steroid hormone elaborated and secreted by the cortex of the **adrenal gland**.

adrenalectomize or **adrenalectomise** to carry out adrenalectomy. — **adrenalectomized** or **adrenalectomised** *adj.*

adrenalectomy surgical removal of one or (usually) both adrenal glands.

adrenal gland or (less commonly) **suprarenal gland** an endocrine organ in vertebrates. There is a single pair in mammals, one near each kidney; in other vertebrates there may be multiple adrenal glands. The gland has two components: an inner medulla, derived from the neural crest, that biosynthesizes and secretes epinephrine and norepinephrine; and an outer cortex, derived from the coelom, that is concerned in the biosynthesis and secretion of steroid hormones. The cortex consists in turn of three histologically defined zones: an outer **zona glomerulosa**, the cells of which are responsible for the biosynthesis of aldosterone and deoxycorticosterone; an intermediate **zona fasciculata**; and an inner **zona reticularis**. The cells of the zona fasciculata and zona reticularis are responsible for the biosynthesis of the glucocorticoid cortisol and the androgens dehydroepiandrosterone and androstenedione.

adrenaline *see* **epinephrine**.

adrenal medulla *see* **adrenal gland**.

adrenal medullary hormone any catecholamine hormone elaborated and secreted by the adrenal medulla. *See* **adrenal gland**.

adrenergic **1** describing a nerve or other cell, or cell receptor (*see* **adrenoceptor**) that is activated by epinephrine, norepinephrine, or an epinephrine-like substance. **2** any nerve that acts by releasing epinephrine, norepinephrine, or an epinephrine-like substance from its nerve ending. *See also* **cholinergic**, **GABAergic**, **noradrenergic**, **peptidergic**, **purinergic**, **serotonergic**. — **adrenergically** *adv.*

adrenergic receptor an alternative term for **adrenoceptor**.

β-adrenergic-receptor kinase *abbr.*: **βARK**, **βAR kinase** or **BARK**; EC 2.7.1.126; an enzyme that phosphorylates specifically only the agonist-occupied form of the β-adrenoceptor and closely related receptors at the C terminus. It appears to be important in mediating rapid agonist-specific (homologous) desensitization. Its cDNA codes for a precursor protein of 689 amino acids, database code **ARK1_HUMAN** (79.58 kDa). The purified protein migrates as a single band of 80 kDa on electrophoresis. It has a protein kinase catalytic domain that has sequence similarity to protein kinase C and cyclic AMP-dependent protein kinase.

adrenic acid *see* **docosatetraenoic acid**

adreno+ *see* **adren+**.

adrenoceptor or **adrenergic receptor** or **adrenoreceptor** or **adrenotrophic receptor** any receptor on an effector cell that is activated by epinephrine or related catecholamines. Structurally they are all of the seven-transmembrane-domain type. Adrenoceptors may be classified phenomenologically into different types according to their sensitivities to agonists and antagonists.

α Adrenoceptors have a relative order of agonist potency: epinephrine ≥ norepinephrine > isoprenaline (isoproterenol); a relative order of antagonist potency: phentolamine >> propranolol. They are associated with stimulatory effects such as vasoconstriction and contraction of the iris, nictitating membrane, urinary bladder, seminal vesicles, and vas deferens and with relaxation of propulsive smooth muscle in the gut. In some species they mediate stimulation of gluconeogenesis and hepatic glycogenolysis. There are two groups, α₁ and α₂. α₁ Receptors act through the phosphoinositide/Ca²⁺ second messenger system and are of four subtypes: α_{1A}, norepinephrine > epinephrine, 560 amino acids (rat); α_{1B}, norepinephrine = epinephrine, 515 amino acids (rat); α_{1C}, norepinephrine = epinephrine (different antagonist sensitivity from α_{1B}), 466 amino acids (bovine); α_{1D}, 560 amino acids (rat). α_{2A} Receptors all inhibit the formation of cyclic AMP; they also open K⁺ channels. *Bản đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách Sinh học, xin liên hệ:*

nels and inhibit Ca²⁺ channels; 450 amino acids (human). α_{2B} Receptors inhibit Ca²⁺ channels; 450 amino acids (human). α_{2C} Receptors have no effect on ion channels; 461 amino acids (human). Examples: α_{1A} receptor, database code **A1AA_HUMAN**, 572 amino acids (60.46 kDa); α_{2A} receptor (subtype c10), database code **A2AA_HUMAN**, 450 amino acids (48.96 kDa).

β Adrenoceptors have actions that can be ascribed to the activation of adenylate cyclase. They may be divided phenomenologically into three classes: (1) β₁ adrenoceptors, in which the relative order of agonist potency is isoprenaline > norepinephrine ≥ epinephrine and the relative order of antagonist potency is practolol > propranolol. They are associated with cardiac stimulation and glycogenolysis, lipolysis in white adipose tissue, and calorogenesis in brown adipose tissue; (2) β₂ adrenoceptors, in which the relative order of agonist potency is isoprenaline > epinephrine > norepinephrine and the relative order of antagonist potency is propranolol > practolol; they are associated with skeletal muscle glycogenolysis, promotion of secretion of glucagon and insulin, vasodilation, and bronchodilation; and (3) β₃ adrenoceptors with agonist potency norepinephrine > epinephrine. Examples from human: β₁, database code **B1AR_HUMAN**, 477 amino acids (51.16 kDa); β₂, database code **B2AR_HUMAN**, 413 amino acids (46.50 kDa); β₃, database code **B3AR_HUMAN**, 408 amino acids (43.47 kDa). *See also* **β-adrenergic receptor kinase**, **β-arrestin**.

adrenoceptor kinase *see* **β-adrenergic receptor kinase**.

adrenocortical of, pertaining to, or derived from the cortex of the **adrenal gland**, e.g. adrenocortical hormone. *See* **adrenal androgen**, **corticosteroid**.

adrenocorticotrophic hormone or **adrenocorticotrophic hormone** *abbr.*: **ACTH**; an alternative name for **corticotropin**.

adrenocorticotrophic hormone-releasing factor an alternative name for **corticotropin releasing hormone**.

adrenocorticotropin or **adrenocorticotrophin** an alternative name for **corticotropin**.

adrenodoxin or (sometimes) **adrenoredoxin** a ferredoxin isolated from adrenal-cortex mitochondria that acts as an electron carrier in hydroxylase systems acting on steroids. It transfers electrons from adrenodoxin reductase to **cholesterol monooxygenase**. Example from *Bos taurus*: database code **ADX1_BOVIN**, 186 amino acids (19.76 kDa); residues 1–58 are the mitochondrial transit sequence, 59–186 adrenodoxin. *See also* **NADPH:adrenodoxin oxidoreductase precursor**. Compare **putidaredoxin**.

adrenodoxin reductase *see* **NADPH:adrenodoxin oxidoreductase precursor**.

adrenomedullary of, pertaining to, or derived from the medulla of the **adrenal gland**.

adrenomedullin a hypotensive peptide that may function as a hormone in circulation control. Example from human: database code **ADML_HUMAN**, 185 amino acids (20.40 kDa).

adrenoreceptor a variant spelling of **adrenoceptor**.

adrenoredoxin (sometimes) a variant spelling of **adrenodoxin**.

adrenosterone androst-4-ene-3,11,17-trione; a hormone with weak androgenic effect, originally called Reichstein's substance G.

adrenergic receptor an alternative name for **adrenoceptor**.

adrex or **adx** *abbr.* for adrenalectomized.

adriamycin former (generic) name for **doxorubicin**.

adseverin another name for **scinderin**.

adsorb to undergo or elicit **adsorption**. — **adsorbable** *adj.*; **adsorbability** *n.*

adsorbate a substance that is adsorbed to the surface of another substance from either a gas or a liquid phase.

adsorbent **1** capable of **adsorption**. **2** a solid that adsorbs another substance from either a gas phase or a liquid phase.

adsorptiochromism the colour change that sometimes ac-



adsorption

companies adsorption of organic compounds onto inorganic substances, e.g. onto alumina.

adsorption 1 any process in which a gas, liquid, or solute adheres to the exposed surfaces of a material, especially a solid, with which it is in contact. In **physical adsorption** the adhesion is through van der Waals forces of interaction, whereas in **chemisorption** (or chemical adsorption) the adhesion is through formation of weak chemical bonds. *Compare* **absorption**. 2 (in immunology) the nonspecific attachment of an antigen (or antibody) onto the surfaces of red cells or inert particles so that the antibody (or antigen) to it may be detected by agglutination of the cells or particles. *Compare* **immunosorption**. 3 (in microbiology) the process of attachment of a phage or other virus to a cell.

adsorption chromatography any form of **chromatography** in which separation of the components of a mixture is based mainly on differences between the adsorption affinities of the components for the surface of an active solid.

adsorption coefficient in any adsorption equilibrium (of a substance from a solution), the mass of adsorbed substance per unit mass of adsorbent divided by the concentration of the substance in solution. It has the dimensions of reciprocal concentration. *Compare* **distribution coefficient**, **partition coefficient**.

adsorption isotherm any plot of the amount of solute adsorbed by an adsorbent (or of ligand bound by, e.g., a macromolecule, often expressed as the **saturation fraction**) versus the concentration of the free solute (or ligand), at constant temperature. *Compare* **Langmuir adsorption isotherm**.

adx *see* **adrex**.

AESF *abbr. for* aminoethylbenzenesulfonyl fluoride.

+aemia *see* **+emia**.

aequorin a Ca^{2+} -dependent **photoprotein** responsible for luminescence by oxidation of the chromophore coelenterazine. It is obtainable from the hydrozoan jellyfish *Aequorea forskalea*. In the absence of any other cofactors or oxygen it emits light on the addition of calcium ions, making it useful for the determination of free calcium ions. Example, aequorin I from *Aequorea victoria*: database code AEQ1_AEQVI, 196 amino acids (22.49 kDa).

aerate 1 to supply with or expose to air. 2 to pass air through a liquid. —**aeration** *n*.

aerobe any organism or class of organisms that can grow in the presence of dioxygen. **Facultative aerobes** are also capable of growing in the absence of dioxygen, whereas **obligate** (or strict) **aerobes** have an absolute requirement for dioxygen. *Compare* **anaerobe**.

aerobic 1 describing conditions in which gaseous or dissolved dioxygen is present. 2 describing an organism or process that requires or is able to use dioxygen. 3 of or produced by an aerobe. *Compare* **anaerobic**.

aerolysin a channel-forming protein secreted by the human pathogen *Aeromonas hydrophila*. The cytolytic toxin is a dimer but forms stable heptameric structures that insert into lipid bilayers to produce well-defined channels, leading to destruction of the membrane permeability barrier and osmotic lysis. Example (precursor) from *A. hydrophila*: database code AERA_AERHY, 493 amino acids (54.28 kDa).

aerosol a colloidal dispersion of solid particles or liquid droplets in air or another gas.

Aet *symbol for* the aminoethyl group, $-\text{CH}_2-\text{CH}_2-\text{NH}_2$.

aetiology *see* **etiology**.

afamin a mammalian serum protein similar to albumin, α -fetoprotein, and vitamin D-binding protein. Example from human: database code HUMAFAMIN, 599 amino acids (68.99 kDa).

afferent 1 conveying inwards to a part, organ, or centre, as of a blood vessel, nerve, or duct. 2 an afferent part, e.g. an afferent blood vessel or afferent nerve. *Compare* **efferent**.

affine the quality of having affinity.

affinity 1 chemical attraction; the tendency of a chemical substance to combine with, bind to, or dissolve in other chemical

substances. 2 any measure of such chemical attraction. 3 denoting biomolecular interaction that exhibits specificity.

affinity adsorbent any biospecific adsorbent used in affinity chromatography.

affinity chromatography a general chromatographic method that may, in principle, be used to isolate either of the components of a reversibly reacting chemical system provided that one component can be coupled to an insoluble matrix through a covalent linkage. The other component can then be bound to the immobilized component and the system eluted with a buffer that liberates the bound component. The technique has been applied to the separation of various substances, including enzymes, substrates, antigens, antibodies, nucleic acids, and even whole living cells. Pure antibodies can be prepared by this means: the antigen is covalently coupled to the dextran beads in the chromatography column, the antibody-containing solution is run into the column in neutral buffer, the specific antibody binds to the antigen, and the antibody is subsequently released with a buffer of high or low pH or with a denaturing reagent. The technique can also be used to isolate antigen. *See also* **affinity-elution chromatography**, **dye-ligand chromatography**. —**affinity-chromatographic** *adj*.

affinity constant or **binding constant** *an alternative name for association constant* (especially in relation to the binding of and/or to macromolecules, as in antigen-antibody, hormone-receptor, and enzyme-inhibitor reactions).

affinity cytochemistry a technique for detecting the distribution of specific cell-surface receptors. An easily visible (or electron-dense) material conjugated with a reagent specific for a particular cell-surface receptor is allowed to react with the cells in question and is then detected by light or electron microscopy.

affinity electrode a type of electrode useful for assaying specific proteins. It comprises a metal (e.g. titanium) wire whose surface has been oxidized and then covalently attached to a ligand capable of interacting biospecifically and reversibly with the protein in question. Binding of the complementary protein to the electrode results in a measurable change in electric potential relative to that given by a reference electrode. The latter is prepared in a similar way but lacks the specific ligand.

affinity electrophoresis or **affinoelectrophoresis** a type of **electrophoresis** in which the support medium contains an agent, immobilized by entrapment or by covalent linkage, that interacts specifically and selectively with certain of the components of the mixture to be analysed, thereby altering the electrophoretic mobility of those components.

affinity-elution chromatography a technique in which a compound that is nonspecifically bound to the matrix of a chromatographic column is specifically eluted by binding to a ligand in the eluting solvent. **Biospecific-elution chromatography** is a variant of this technique.

affinity gel any gel that serves as an **affinity matrix**.

affinity-isolated *an alternative term for* **affinity-purified**.

affinity label an active-site-directed irreversible inhibitor of an enzyme, antibody, or other protein. It is a chemically reactive compound that resembles a substrate or other specific ligand and bonds covalently to the active site or specific site on the protein. The affinity labelled groups can then be identified by **fingerprinting** and thus reveal the composition at the active site. It is sometimes termed **Trojan horse inhibitor**.

affinity matrix or **affinity support** any supporting material to which the biospecific reagent is attached in **affinity chromatography**.

affinity precipitation the precipitation of an enzyme by a homo- or hetero-bifunctional derivative of its coenzyme or/and a substrate or inhibitor. An example is the precipitation of **lactate dehydrogenase** by $N^{2'}$ -adipodihydrazido-bis- $(N^6$ -carboxymethyl-NAD), a reactive derivative of its coenzyme, NAD.

affinity-purified or **affinity-isolated** describing a specified sub-



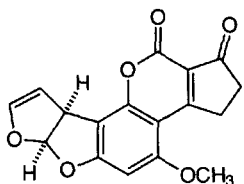
affinity support

stance, usually a biological macromolecule such as an antigen or antibody, that has been purified (or isolated) by a technique such as **affinity chromatography** or **affinity electrophoresis**, thus implying high purity.

affinity support an alternative term for **affinity matrix**.

affinoelectrophoresis an alternative term for **affinity electrophoresis**.

aflatoxin any of a group of related and highly toxic secondary metabolites (**mycotoxins**) produced by strains of the moulds *Aspergillus flavus* or *A. parasiticus*, together with further metabolites of these mycotoxins. Their main structural feature is a fused coumarin-bis(dihydrofuran) ring system. The most important are aflatoxins B₁, B₂, G₁, and G₂ — so designated from the colour of their fluorescence (B, blue; G, green). The mycotoxins are produced naturally by the moulds growing on various seed crops, especially groundnuts (peanuts), and certain cereals (e.g. maize) during storage under moist conditions. They are acutely toxic and highly carcinogenic to many species of animals (probably including humans), and are responsible for turkey X disease. The main organ affected is the liver; aflatoxin B₁ is the most potent hepatocarcinogen known. The term is derived as a contraction of *A. flavus* toxin.



aflatoxin B₁

AFLP *abbr.* for amplification fragment length polymorphism. The use of the **polymerase chain reaction** to amplify DNA in the study of **restriction fragment length polymorphism**.

A form 1 A form of DNA (*abbr.*: A-DNA or DNA-A); the molecular conformation adopted by fibres of the sodium salt of duplex DNA at a relative humidity of 75% or less. It consists of a right-handed double helix containing about 11 nucleotide residues per turn, with the planes of the base pairs inclined at about 70° to the axis of the helix. Unlike the **B form** of DNA it has a large hole (≈0.8 nm diameter) at the axis and a very deep major groove. *See also* **C form**, **Z form**. 2 A form of RNA (*abbr.*: A-RNA); the molecular conformation of double-stranded regions of RNA that is favoured at low-salt concentrations and moderate temperatures; it resembles the A form of DNA. (*see def.* 1).

AFP *abbr.* for α -fetoprotein.

Ag 1 *abbr.* for antigen. 2 *symbol* for silver.

AGA 1 a codon in non-mitochondrial mRNA for L-arginine. 2 a codon in human mitochondrial mRNA for chain termination. 3 *abbr.* for N-acetylglutamic acid.

agamic reproduction *see* **asexual reproduction**.

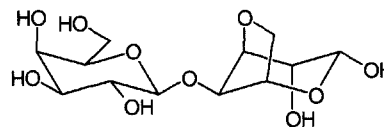
agamogeny the development of a new individual from a single cell. *Compare* **vegetative reproduction**.

agar or **agar-agar** a complex sulfated galactan extracted from certain seaweeds, especially *Gelidium* and related genera. The two main components are **agarose** and **agarpectin**. Agar forms an aqueous gel suitable for the solidification of microbiological culture media and for use as a support medium in zone electrophoresis or (immuno)diffusion techniques. It is not metabolized by most organisms. The gelling temperature varies from about 25 to 35 °C for different types of preparation. The gel is then stable to about 90 °C.

agar-diffusion method a method for determining the sensitivity of a microorganism to an antimicrobial drug. The zone of growth inhibition is measured around a ditch, hole, or a fil-

ter-paper disk containing the drug and located on an agar culture medium seeded with the microorganism in question.

agarobiose 4-O- β -D-galactopyranosyl-3,6-anhydro-L-galactose; a **disaccharide** that forms the basic unit of **agarose**.



agarobiose

agarpectin a charged component of **agar** consisting of a mixture of polysaccharides containing D-galactose, 3,6-anhydro-L-galactose, and monoesterified sulfuric acid units. Samples from some algae may contain other components such as pyruvate, L-arabinose, and D-glucuronate units.

agarose an essentially uncharged component of **agar** comprising a ≈120 kDa alternating carbohydrate polymer. It consists of -3)D-Galp-(β 1-4)-3,6An-L-Galp(α 1- repeating units, containing small amounts of ionized sulfate and pyruvate groups. It is widely used as a matrix in zone and immunoelectrophoresis, immunodiffusion, and gel filtration and affinity chromatography. *See also* **agarobiose**.

AGC a codon in mRNA for L-serine.

agenized flour *see* **methionine sulfoximine**.

AGG 1 a codon in non-mitochondrial mRNA for L-arginine. 2 a codon in human mitochondrial mRNA for chain termination.

agglutinate 1 to adhere or cause to adhere. 2 to cause the clumping of cells, particles, etc., or to undergo such clumping.

agglutination 1 the act or process of adhering. 2 the process in which suspended cells or other **antigen** coated particles clump together when **antibody** is added due to an antigen-antibody reaction. 3 the result of such a process. *See also* **agglutinin**.

agglutinin an **antibody** that has the ability to agglutinate the corresponding antigen, e.g. when the antigen is present on the surface of a suspended cell or other particle.

agglutिनogen an **antigen**, usually particulate, that stimulates the production of **agglutinin**.

aggrecan the major proteoglycan of cartilage. It has approximately one **glycosaminoglycan** chain for every 20 amino-acid residues, and a total mass of about 3 MDa. It assembles with glycosaminoglycans, especially hyaluronan, into aggregates several micrometres in diameter. Example (of protein precursor) from human juvenile cartilage chondrocyte: database code HUMAGPRO, 2316 amino acids (238.97 kDa). *See also* **versican**.

aggregate 1 formed of separate units or particles collected into a whole or into larger units. 2 an assemblage or sum of many separate units or particles. 3 to form or be formed into a single body or larger body.

aggregation 1 the process of forming an **aggregate** (*def.* 2). 2 a cluster or group of particles held together into larger units.

aggregation number the number of monomers of an amphipathic substance that form a micelle under any particular conditions.

aggressin a diffusible nontoxic substance produced by a microorganism that promotes the invasive power of the microorganism in the host.

aglucon or (*formerly*) **aglucone** an **aglycon** derived from a glucoside (i.e. where the sugar moiety was a glucose residue).

aglycon or **aglucon** or (*formerly*) **aglycone** or **aglucone** the part

of any **glycoside** that remains after the sugar moiety has been chemically or enzymically removed.

agmatine 1-amino-4-guanidobutane; a putative endogenous ligand for **imidazoline receptors** synthesized from arginine by the enzyme arginine decarboxylase (EC 4.1.1.19). In some invertebrates e.g. the sponge *Geodia gigas* and some cephalopods, the guanidine group can undergo phosphorylation to phosphoagmatine, which acts as a **phosphagen**.

agnotobiotic pertaining to the growth of organisms of a single species in the presence of one or more other species of which at least one is unknown. *Compare* **syxenic**.

+agogue or (*US*) **+agog** suffix denoting an agent that elicits or enhances the secretion of the indicated substance. [from the Greek *agōgos*, leading, drawing forth.] **→+agogic adj.**

agonist any ligand, especially a drug or hormone, that binds to receptors and thereby alters the proportion of them that are in an active form, resulting in a biological response. A conventional agonist increases this proportion, whereas an **inverse agonist** reduces it. *See also* **full agonist**, **partial agonist**. *Compare* **antagonist**. **→agonistic adj.**

Agranoff's turtle a zoomorphic mnemonic for avoiding the confusion that has tended to occur over the numbering of the carbon atoms of *myo*-inositol and its chiral derivatives, especially when their Haworth projections are converted into diagrams representing their normal chair conformations. The chair conformation of the parent compound is likened to a turtle, with its body corresponding to the C₆ ring, its coplanar limbs and tail representing the five equatorial hydroxyl groups, and its erect head the axial hydroxyl group. The turtle is viewed from above; then, for derivatives named as members of the D series, numbering of the turtle's appendages proceeds counterclockwise, commencing with its right front paw (i.e. its *dextro* 'hand'), which is designated 1D; its head becomes 2D, its left front paw is 3D, and so on around its body. Conversely, numbering of the L series of derivatives commences with the left front paw (the *laevo* 'hand'), designated 1L, and then proceeds clockwise, the head becoming 2L, and so on. [After its originator Bernard William Agranoff (1926–), who devised it in 1978.]

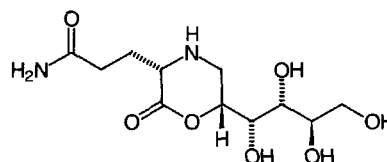
agrin a component of the synaptic basal lamina that causes aggregation of acetylcholine receptors and acetylcholinesterase on the surface of muscle fibres of the neuromuscular junction. It occurs in embryonic nervous tissue and muscle, especially during early development. At least five different forms arise by alternative splicing; they differ in their acetylcholine receptor clustering activity. An EGF-like domain is present. Example (precursor) from rat: database code AGRI_RAT, 1959 amino acids (208.41 kDa).

Agrobacterium a genus of Gram-negative aerobic rod-shaped soil bacteria. Most strains can initiate formation of galls in plants (*see* **crown-gall disease**). A feature of the infection of plants by *Agrobacterium* species is that the bacterium can subvert the host plant tissue to produce amino acids known as **opines**, which the bacterium can use as an energy, carbon, and nitrogen source. The type of opine produced is determined by the bacterial strain. Continued presence of the bacteria is not necessary for transformation of the cells of the host plant. Transformation by *Agrobacterium tumefaciens* has been shown to be due to large plasmids (140–235 kb). These are known as tumour-inducing or Ti plasmids, and they have been exploited in the production of **transgenic** plants.

agrochemical a chemical that is used in agriculture or horticulture, especially as a biocide, fertilizer, etc.

agroclavine 8,9-didehydro-6,8-dimethylergoline; a nonpeptide **ergot** alkaloid.

agropine an **opine**; a rare amino-acid derivative that is produced by a certain type of crown-gall tumour. The genes responsible for its synthesis are part of the T-DNA from a **Ti plasmid**. *See also* **crown-gall disease**.



agropine

AGU a codon in mRNA for L-serine.

Ah *abbr. for* aromatic hydrocarbon; the name of a genetic locus in higher organisms that governs biological responses to some aromatic hydrocarbons. The proteins it encodes include cytochrome P450 1A1, a liver microsomal monooxygenase that oxidizes a variety of unrelated compounds, including xenobiotics such as the environmental carcinogens, benzo[a]pyrene and 3-methylcholanthrene. This enzyme is induced by these compounds and, more potently, by polychlorinated biphenyls such as 2,3,7,8-tetrachlorodibenzo-*p*-dioxin (TCDD), acting through the **Ah receptor**. The 5'-flanking region of the gene for this enzyme contains several short sequence motifs known as **xenobiotic response elements** (XREs).

ahnak a human gene that encodes a giant protein of ≈700 kDa with a large internal domain (≈4300 amino acids) of highly conserved repeated sequences, much of which are 128 amino-acid residues in length and contain a heptad repeat. The ahnak protein is found in human nuclei and is of unknown function. [From Hebrew *Ahnak*, giant.]

Ah receptor *abbr. for* aryl hydrocarbon receptor; a protein encoded by the **Ah** gene that binds a number of aryl hydrocarbons and mediates their biochemical and toxic effects. It is activated when it binds a ligand, and then translocates from the cytoplasm to the nucleus, where it is believed to enhance gene transcription by binding to the **xenobiotic-response element** sequence. It contains a basic helix-turn-helix motif; the DNA-binding form is a heterodimer of this protein and **ARNT**. Example (precursor) from human: database code AHR_HUMAN, 808 amino acids (91.62 kDa).

εAhx or (formerly) **εAcp** symbol for a residue of the ε-amino acid 6-aminoheptanoic acid (formerly known as ε-aminocaproic acid).

αHyl symbol for a residue of the α-amino acid L-allohydroxylysine; *threo*-5-hydroxy-L₅-lysine.

AIB *abbr. for* 2-aminoisobutyric acid.

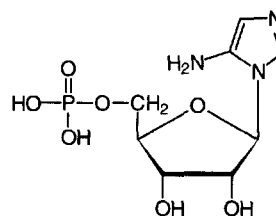
AICAR *abbr. for* 5-amino-4-imidazolecarboxamide ribonucleotide (an intermediate in **purine biosynthesis**).

AICAR transformylase *see* **phosphoribosylaminoimidazolecarboxamide formyltransferase**.

AIDS *abbr. for* acquired immunodeficiency syndrome.

alle (**Alle**) symbol for a residue of the α-amino acid L-alloisoleucine, (2*S*,3*R*)-2-amino-3-methylpentanoic acid.

AIR *abbr. for* for 5-aminoimidazole ribonucleotide, an intermediate in **purine biosynthesis**.



AIR carboxylase *see* **phosphoribosylaminoimidazole carboxylase**.

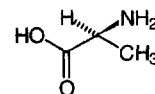
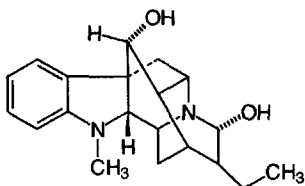
Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

air-lift bioreactor

air-lift bioreactor a bioreactor in which the reaction medium is kept mixed and gassed by the introduction of air or another gas at the base of a column-like reactor.

ajmaline ajmalan-17,20-diol; an alkaloid from the roots of *Rauwolfia serpentina*. It is used clinically as an antihypertensive and antiarrhythmic, having the effect of normalizing heart rhythm.



L-alanine

AK (in clinical biochemistry) *abbr.* for adenylate kinase.

AKAP *abbr.* for A-kinase anchor protein or cAMP-dependent protein kinase anchor protein; a protein that anchors the kinase to cytoskeletal and/or organelle-associated proteins, targeting the signal carried by cyclic AMP to specific intracellular effectors. The N-terminal region, which is highly basic, is required for interaction with **calmodulin**. Example (human): database code AK79_HUMAN, 427 amino acids (47.02 kDa).

akaryotic describing a cell without a nucleus.

akee a variant spelling of **ackee**.

AKH *abbr.* for adipokinetic hormone.

A-kinase an enzyme that phosphorylates target proteins in response to a rise in intracellular **cyclic AMP**. *See* **cyclic AMP-dependent protein kinase**.

Akt the mouse or human homologue of *v-akt*, the oncogene of the transforming retrovirus AKT8. *v-akt* encodes a serine/threonine protein kinase that contains an SH2 domain. Database code KAKT_MLVAT, 501 amino acids (57.87 kDa).

al+ prefix denoting an acyclic monosaccharide or monosaccharide derivative. *See also* **aldehyde**.

+al suffix denoting an unbranched acyclic mono- or dialdehyde.

Al symbol for aluminium.

Ala symbol for a residue of the α -amino acid L-alanine, 2-aminopropanoic acid (alternative to A).

β Ala symbol for a residue of the β -amino acid β -alanine, 3-aminopropanoic acid.

ALA or **dALA** *abbr.* for δ -aminolevulinate.

Ala AT or **ALAT** (in clinical biochemistry) (formerly) *abbr.* for alanine aminotransferase. ALT is preferred.

ALA dehydratase *see* **porphobilinogen synthase**.

alamethicin a linear ionophorous antibiotic polypeptide containing a high proportion of 2-methylalanine residues and blocked at both ends. *See also* **peptaibophols**.

Ac-2-MeAla-L-Pro-2-MeAla-L-Ala-2-MeAla-L-Ala-L-Gln-2-MeAla-L-Val-2-MeAla-Gly-L-Leu-2-MeAla-L-Pro-L-Val-2-MeAla-2-MeAla-L-Glu-L-Gln-L-phenylalaninol

alanical the aldehyde obtained by reduction of the carboxyl group of alanine.

alaninate 1 alanine anion, the anion $\text{CH}_3\text{-CH}(\text{NH}_2)\text{-COO}^-$. 2 any salt containing alanine anion. 3 any ester of alanine.

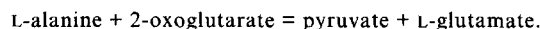
alanine the trivial name for α -aminopropionic acid; 2-aminopropanoic acid; $\text{CH}_3\text{-CH}(\text{NH}_2)\text{-COOH}$; a chiral α -amino acid. L-alanine (*symbol*: A or Ala), (S)-2-aminopropanoic acid, is a coded amino acid found in peptide linkage in proteins; codon: GCA, GCC, GCG, or GCU. In mammals it is a nonessential dietary amino acid and is glucogenic. Residues of D-alanine (*symbol*: D-Ala or DAla), (R)-2-aminopropanoic acid, are found in cell-wall peptidoglycans of various bacterial species and in other materials, e.g. **cyclosporin**.

β -alanine *symbol*: β Ala; the trivial name for β -aminopropionic acid; 3-aminopropanoic acid; $\text{H}_2\text{N-CH}_2\text{-CH}_2\text{-COOH}$; an achiral β -amino acid. It occurs free in brain and in combination in pantothenate (and hence in coenzyme A) and in the isopeptides anserine and carnosine, but it is not a constituent of proteins. It has an inhibitory effect on the central nervous system.

alanine cycle *see* **glucose-alanine cycle**.

β -alanine-oxoglutarate aminotransferase *see* **4-aminobutyrate transaminase**.

alanine transaminase *abbr.* (in clinical biochemistry): ALT; EC 2.6.1.2; *systematic name*: L-alanine:2-oxoglutarate aminotransferase; *other names*: glutamic-pyruvic transaminase; glutamic-alanine transaminase. An enzyme that catalyses the reversible reaction:



Widely distributed in all tissues and organisms, it is a pyridoxal-phosphate enzyme. Example, cytoplasmic isoform: database code ALAT_HUMAN, 495 amino acids (54.38 kDa). In clinical chemistry its assay alongside aspartate transaminase (AST) normally present in plasma in higher concentrations, may be useful as an indicator of liver damage, as levels of ALT are higher in liver than those of AST, in contrast to other tissues. In hepatitis, plasma levels of ALT may exceed those of AST.

alaninium alanine cation, $\text{CH}_3\text{-CH}(\text{NH}_3^+)\text{-COOH}$.

alanino the alkylamino group, $\text{CH}_3\text{-CH}(\text{COOH})\text{-NH-}$, derived from alanine.

alaninol the alcohol obtained by the successive reduction of the carboxyl group of alanine.

alanyl the acyl group, $\text{CH}_3\text{-CH}(\text{NH}_2)\text{-CO-}$, derived from alanine.

alarmone any signal molecule that serves to reorient a cell's economy in response to stress. Such molecules include ppGpp, which is produced, e.g., in microorganisms, in response to growth-rate limitation caused by amino-acid stress and acts to correct this in various ways, and diadenosine tetraphosphate (AppppA), which stimulates proliferation when DNA replication is halted at a replication fork.

ALA synthetase *see* **5-aminolevulinate synthase**.

ALAT *See* **Ala AT**.

albinism an inherited metabolic disorder in which there appears to be a complete or nearly complete failure to form melanin. It was first listed in 1908 by the British physician Archibald Garrod (1857–1936) in his lecture entitled 'Inborn Errors of Metabolism'. In some cases it is due to a deficiency of 'tyrosinase' (**monophenol monooxygenase**), but other types are known in which this enzyme is present, and for which the precise cause is unknown.

albumin any of a group of globular proteins that are soluble in distilled water and solutions of half-saturated ammonium sulfate, but insoluble in fully saturated ammonium sulfate solutions. **Serum albumin**, the major protein of serum, has a good binding capacity for water, Ca^{2+} , Na^+ , K^+ , fatty acids, hormones, bilirubin, and drugs. Its main function is the regulation of the colloidal osmotic pressure of blood; it has structural similarity to α -fetoprotein and **vitamin D binding protein**. Example (precursor) from human: database code ALBU_HUMAN, 609 amino acids (69.29 kDa). *See also* **endosperm albumin**, **lactalbumin**, **ovalbumin**, **serum albumin**. [The name origi-

Bản đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.



albuminoid

nates from Latin *albus*, white, as it exists in nearly pure form in egg-white, from which it was easily obtained by the ancients.]

albuminoid 1 of or relating to an **albumin**. 2 *an alternative name for scleroprotein*.

albuminuria the presence of excessive amounts of protein in the urine. It is usually a sign of renal disease.

albuterol *see* **salbutamol**.

Alcalase the proprietary name for a proteolytic enzyme preparation obtained by fermentation of a strain of *Bacillus subtilis*. It is characterized by stability and activity at high temperatures and alkaline pH values, and is used in enzymic laundering and various industrial processes.

alcapton 2,5-dihydroxyphenylacetic acid. *See* **alcaptonuria**, **homogentisate**.

alcaptonuria or **alkaptonuria** a rare hereditary metabolic disease, usually due to an autosomal recessive gene, in which the enzyme homogentisic acid oxidase (homogentisate 1,2-dioxygenase, EC 1.13.11.5) is virtually absent. This leads to the inability to metabolize further the homogentisate produced by the normal metabolism of phenylalanine and tyrosine. Homogentisate therefore accumulates in the body and is excreted in the urine, which gradually turns dark as homogentisate is oxidized to a melanin-like product – a process speeded up at an alkaline pH.

alcohol 1 any of a class of alkyl compounds containing a hydroxyl group. An alcohol is called primary, secondary, or tertiary according to whether the carbon atom bearing the hydroxyl group is itself attached to one, two, or three other carbon atoms. 2 specifically, ethyl alcohol, ethanol. *See also* **absolute alcohol**. —**alcoholic** *adj.*, *n.*

alcohol dehydrogenase any of the nicotinamide-nucleotide-linked dehydrogenase enzymes of groups EC 1.1.1.1 (NAD⁺) or EC 1.1.1.2 (NADP⁺) that catalyse the oxidation of alcohols to aldehydes. These are zinc proteins. Also EC 1.1.1.71, alcohol dehydrogenase [NAD(P)⁺]; *other name*: retinal reductase; EC 1.1.99.8, alcohol dehydrogenase (acceptor), a quinoprotein. Examples: alcohol dehydrogenase I from *Saccharomyces cerevisiae*; database code ADH1_YEAST, 347 amino acids (36.66 kDa); alcohol dehydrogenase 6 (human); database code ADH6_HUMAN, 368 amino acids (39.03 kDa).

alcoholic fermentation the conversion of glucose to ethanol by a group of reactions that are characteristic of yeast.

alcohol oxidase *abbr.*: AOX or **methanol oxidase** *abbr.*: MOX; EC 1.1.3.13; a flavoprotein (FAD) enzyme that catalyses the oxidation by dioxygen of a primary alcohol to the corresponding aldehyde. It is important in methylotrophic yeasts, in which it is the first enzyme in the pathway for utilization of methanol, converting this to formaldehyde, thence to CO₂.

alcohololysis the solvolysis of a covalent derivative of an acid by a reaction in which one of the products combines with the H atom of an alcohol's hydroxy group and the other product with the alcohol's alkoxy group.

aldaric acid any dicarboxylic acid formed by oxidation of both terminal groups of an **aldose** to carboxyl groups. There are three tetraric acids (D-, L-, and *meso*-tartaric acids), four pentaric acids (D- and L-arabinaric acids, xylaric acid [*meso*], and ribaric acid [*meso*]), and ten hexaric acids (D- and L-glucaric acids, D- and L-idaric acids, D- and L-mannaric acids, D- and L-talaric acids, allaric acid [*meso*], and galactaric acid [*meso*]).

aldehyde 1 *an alternative name for the formyl group*, —CHO. 2 any organic compound with the formula R—CH=O. *Compare* **ketone**. —**aldehydic** *adj.*

aldehyde dehydrogenase any enzyme that catalyses the oxidation of an aldehyde group to a carboxylic acid. Many examples are known, acting specifically on a wide range of substrates. Those acting on aliphatic aldehydes are divided into four classes. Class 1, or Ald C, are tetrameric cytosolic enzymes, and include aldehyde dehydrogenase (NAD⁺), EC 1.2.1.3; *systematic name*: aldehyde:NAD⁺ oxidoreductase; an

enzyme of wide specificity. Example from human: database code DHAC_HUMAN, 500 amino acids (54.73 kDa). This EC designation also includes Class 2 enzymes, or Ald M, which are tetrameric mitochondrial enzymes. Class 3, or Ald D, are dimeric cytosolic enzymes, and include aldehyde dehydrogenase [NAD(P)⁺], EC 1.2.1.5; *systematic name*: aldehyde:NAD(P)⁺ oxidoreductase. Example from human: database code DHAP_HUMAN, 453 amino acids (50.35 kDa).

aldehyde ferredoxin oxidoreductase an enzyme that catalyses the reversible oxidation of aldehydes to the corresponding carboxylic acid, using ferredoxin as an electron acceptor. Containing tungsten, molybdopterin molecules, and Fe₄S₄ clusters, the enzyme has remarkable thermostability. Its 3-D structure is known.

aldehyde- *abbr.*: al-; prefix designating an uncyclized monosaccharide or monosaccharide derivative. It is used in (semi)systematic nomenclature to stress the acyclic nature of such compounds. *Compare* **keto-**.

aldimine any **imine** that is an analogue of an aldehyde; the general structure is RCH=NR where R may be any organyl group or H. *Compare* **ketimine**.

alditol any polyhydric alcohol derived from the acyclic form of a **monosaccharide** by reduction of its aldehyde or keto group to an alcoholic group. Older names for these compounds include glycitol and sugar alcohol.

aldo- prefix indicating aldehydic; e.g. aldohexose.

aldoketose any monosaccharide derivative containing both a (potential) aldehydic carbonyl group and a (potential) ketonic carbonyl group.

aldol 1 any organic compound that is both an aldehyde and an alcohol, especially where the two functions are separated by two linked carbon atoms in accordance with the general structure: HO—C—C—CH=O. 2 *the trivial name for acetaldehyde* (3-hydroxybutanal) a compound formed by the self-condensation of two acetaldehyde molecules.

aldolase 1 *a generic name for nearly every enzyme of the subclass EC 4.1.2, the aldehyde-lyases, enzymes catalysing aldol condensations and their reversal*. 2 *abbr. (in clinical biochemistry)*: ALS; *common name for the enzyme fructose-bisphosphate aldolase*; EC 4.1.2.13; *systematic name*: D-fructose-1,6-bisphosphate D-glyceraldehyde-3-phosphate-lyase; *other name*: fructose-1,6-bisphosphate triosephosphate-lyase. An enzyme that catalyses the reversible fission of D-fructose 1,6-bisphosphate to glyceralone phosphate and D-glyceraldehyde 3-phosphate. It acts also on (3S,4R)-ketose 1-phosphates. In mammals three isoenzymes are known: aldolase A in muscle; aldolase B in liver, kidney, and small intestine; and aldolase C in brain. All three isoenzymes catalyse the reversible splitting of both D-fructose 1,6-bisphosphate and D-fructose 1-phosphate, but with different relative limiting rates; liver aldolase B shows approximately the same limiting rate with both substrates, while aldolase A and aldolase C show, respectively, a 50-fold and a 10-fold lower maximal activity with fructose 1-phosphate than with fructose 1,6-bisphosphate. Each is of ≈160 kDa and is composed of four subunits, which are immunologically distinct in the three isoenzymes. Examples, aldolase A from *Mus musculus*: database code ALFA_MOUSE, 363 amino acids (39.22 kDa); aldolase from *Pisum sativum*: database code ALF1_PEA, 357 amino acids (38.44 kDa). Three different but related aldolase genes (probably formed from a single ancestral gene) have been identified; in humans the aldolase-A gene is in chromosome 16, the aldolase-B gene is present as a single copy in chromosome 9, and the aldolase-C gene is in chromosome 17. *See also* **hereditary fructose intolerance**.

aldol condensation the base-catalysed addition reaction of two aldehydes or an aldehyde and a ketone to form an **aldol** (def. 1).

aldonic acid a monocarboxylic acid having a chain of three or more carbon atoms and formally derived from an **aldose** by oxidation of the aldehydic group.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

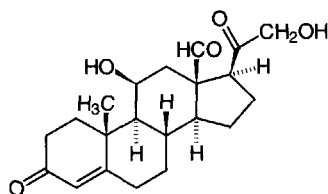
aldopentose

aldopentose any of the eight possible five-carbon-atom aldoses.

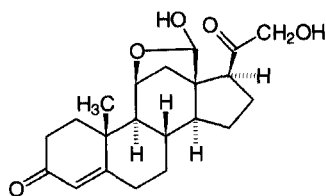
aldopyranose any aldose in the pyranose form.

aldose a monosaccharide in which the (potential) carbonyl group is terminal (i.e. aldehydic). The term is frequently modified to indicate the number of carbon atoms in the chain, as in aldotriose, aldohexose, etc.

aldosterone 11 β ,21-dihydroxy-3,20-dioxopregn-4-en-18-al; the most powerful naturally occurring mineralocorticoid hormone, originally named **electrocortin** by its discoverers Sylvia Simpson and James Tait in 1954. The name was changed to aldosterone after its structure had been determined by Reichstein in 1953. It exists in solution as an equilibrium mixture of the aldehyde and the hemiacetal formed with the hydroxyl at position 11. Produced by the cells of the zona glomerulosa of the adrenal cortex, its main action is to increase reabsorption of sodium ions by the distal renal tubules and thus to regulate water and electrolyte metabolism. The biosynthesis of aldosterone is stimulated by angiotensin II (see **angiotensin**, **renin-angiotensin system**), **corticotropin**, and increased potassium-ion concentrations. 11-Deoxycorticosterone is converted into aldosterone by **aldosterone synthase**. Aldosterone exerts its effects through binding to cytosolic receptors. The resulting complex is transferred to the nucleus, where it acts to stimulate production of mRNA for sodium transport proteins. The action is thus on a timescale of many minutes.



aldehyde form



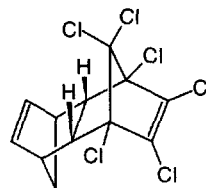
hemiacetal form

aldosterone synthase EC 1.14.15.4; *recommended name*: steroid 11 β -monooxygenase; *other names*: cytochrome P450 11B2; steroid 18-hydroxylase; P450C18. A heme-thiolate enzyme that converts 11-deoxycorticosterone to aldosterone, by successive hydroxylations at the 11 β and 18 positions. Reduced adrenodoxin is a reactant, being converted to oxidized adrenodoxin. Example, human (precursor): database code CPN2_HUMAN, 503 amino acids (57.58 kDa).

aldotetrose any of the four possible four-carbon-atom aldoses.

aldotriose either D- or L-glyceraldehyde; see **aldose**.

aldrin 1,2,3,4,10,10-hexachloro-1,4,4a,5,8,8a-hexahydro-1,4:5,8-dimethanonaphthalene; an organochlorine compound formerly used as an insecticide, but now banned in certain countries on account of its toxicity.



aldrin

aleuron or **aleurone** 1 granules of insoluble protein found in plant cells, especially in the seeds of cereals, where they appear to act as storage material. 2 the single layer of cells on the outside of the endosperm of cereal seeds containing such granules. Translocated **gibberellins** stimulate the cells to synthesize α -amylase, which is secreted into the endosperm to hydrolyse the stored starch.

aleuroplast any colourless plastid or leukoplast storing protein. Aleuroplasts are found in plant cells, particularly seeds.

alexin or **alexine** former term for **complement**; hence **alexinated** means treated with complement.

ALG abbr. for anti-lymphocyte globulin.

alga (pl. **algae**) any of a group of simple eukaryotic photosynthetic protists. They may be unicellular or multicellular, and are generally aquatic. The prokaryotes formerly called blue-green algae are now known as **cyanobacteria**.—**algal** *adj.*

algicide any chemical agent that selectively kills algae.

algin the sodium salt of **alginic acid**

alginate lyase EC 4.2.2.3; *recommended name*: poly(β -D-mannuronate) lyase; *systematic name*: poly(β -D-1,4-mannuronide) lyase. An enzyme that catalyses a reaction resulting in eliminative cleavage of polysaccharides containing β -D-mannuronate residues to give oligosaccharides with 4-deoxy- α -L-erythrohex-4-enopyranuronosyl groups at their ends. Example from *Pseudomonas aeruginosa*: database code PSEALGL, 367 amino acids (40.78 kDa).

alginic acid a hydrophilic polysaccharide ($\approx$ 240 kDa) occurring in brown algae (brown seaweeds), especially the Californian giant kelp (*Macrocystis pyrifera*) and horsetail kelp (*Laminaria digitata*). Sodium alginate is used commercially as an emulsifier and thickener in foodstuffs, pharmaceuticals, etc. It has an interrupted structure of stretches of α 1-4-linked α -L-gulopyranosyluronic acid residues, stretches of β 1-4-linked β -D-mannopyranosyluronic acid residues, and stretches where both uronic acids occur in alternating sequence. Similar glycans are secreted by some bacteria.

algorithm any systematic mathematical procedure that allows a problem to be solved in a finite number of steps. *Compare heuristic, stochastic.*

alicyclic describing an organic compound derived from a saturated cyclic hydrocarbon. *Compare aliphatic, aromatic.*

aliphatic describing organic compounds in which the carbon atoms form open (noncyclic) chains. *Compare alicyclic, aromatic.*

aliquant one of a number of unequal parts of a whole.

aliquot one of number of equal parts of a whole; often used loosely, and erroneously, for a sample.

alizarin 1,2-dihydroxyanthraquinone; the **aglycon** of **ruberythric acid**.

alkalemia or (esp. Brit) **alkalaemia** a condition in which there is increased alkalinity (i.e. lowered hydrogen-ion concentration and hence raised pH) of the blood. *Compare acidemia, alkalosis.*

alkali any of a class of bases that neutralize acids and are themselves neutralized by acids, and form caustic and/or corrosive aqueous solutions. The term is applied in particular to hy-

alkali metal

dioxides of the **alkali metals**, though the term is often extended to other substances with similar though weaker properties.

alkali metal any element of group 1 of the IUPAC **periodic table**; the group comprises lithium (Li), sodium (Na), potassium (K), rubidium (Rb), cesium (Cs), and francium (Fr).

alkaline **1** having the properties of an **alkali**; **basic**. **2** describing an aqueous solution having a pH > 7.

alkaline earth any metallic element belonging to group 2 of the IUPAC **periodic table**; the group comprises beryllium (Be), magnesium (Mg), calcium (Ca), strontium (Sr), barium (Ba), and radium (Ra).

alkaline phosphatase EC 3.1.3.1; *systematic name*: orthophosphoric monoester phosphohydrolase (alkaline optimum); *other names*: alkaline phosphomonoesterase; phosphomonoesterase; glycerophosphatase. An enzyme that catalyses the hydrolysis of orthophosphoric monoester to an alcohol and orthophosphate. The cofactors are zinc and magnesium. In human plasma the level is raised in the last trimester of pregnancy. Plasma levels may also be greatly elevated in cases of Paget's disease of bone, osteomalacia, and cirrhosis or biliary obstruction; levels may become moderately elevated in other types of bone disease. The assay of tissue-specific isoenzymes increases the value of these estimations; e.g. the placental isoenzyme is elevated in plasma in testicular seminoma. The normal range in human plasma is 30–90 IU L⁻¹. Example from intestine (bovine): database code PPBI_BOVIN, 44 amino acids (4.77 kDa); example from *Escherichia coli* (precursor): database code PPB_ECOLI, 471 amino acids (49.38 kDa).

alkaline tide the slight increase in plasma and urine pH that occurs after meals. It is believed to be due to withdrawal of hydrogen ions from the blood during the formation of gastric HCl.

alkali reserve or **alkaline reserve** a term formerly used to denote the bicarbonate-CO₂ extractable from blood plasma.

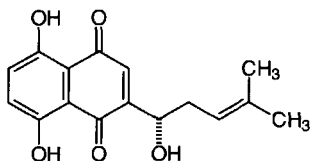
alkaloid any member of a broad group of nitrogen-containing basic organic compounds present in various dicotyledonous plants and some fungi. Although only about 5% of the world's plant species have so far been examined, they have yielded over 2000 different alkaloids. Heterocyclic alkaloids derived from amino acids are termed **true alkaloids**. Alkaloids with and without heterocyclic rings and not derived from amino acids are termed **pseudoalkaloids**; in these the carbon skeleton is usually isoprenoid derived. Alkaloids often have marked and specific pharmacological activity.

alkalosis a clinical condition in which total base excess or total acid deficit has the potential to cause decreased hydrogen-ion concentration (i.e. raised pH) in the blood, in the absence of compensating biochemical and physiological changes.

alkane any saturated aliphatic hydrocarbon compound.

alkanet the plant, *Alkanna tinctoria*, and its root, which contains the pigment **alkannin**.

alkannin (S)-5,8,-dihydroxy-2-(1-hydroxy-4-methyl-3-pentenyl)-1,4-naphthalenedione; a red pigment derived from the root of the alkanet plant, and used as a dye in cosmetics and food. The (+)(R) enantiomer is also a natural product, **shikonin**, and the racemate is known as shikalkin.



alkaptonuria a variant spelling of **alcaptonuria**.

alkatriene any unsaturated aliphatic hydrocarbon containing two (conjugated or unconjugated) double bonds.

alkatrienyl any group derived from an alkatriene by removal of a hydrogen atom.

alkenyl any group derived from an alkene by removal of a hydrogen atom.

1-alkenyl-2-acylglycerol cholinephosphotransferase EC 2.7.8.22; an enzyme that catalyses the formation of the plasmalogen, plasmenylcholine (i.e. 1-alkenyl-2-acylglycerophosphocholine), from CDPcholine and 1-alkenyl-2-acylglycerol with release of CMP.

1-alkenylglycerophosphocholine O-acyltransferase EC 2.3.1.104; an enzyme that catalyses the formation of the plasmalogen, plasmenylcholine (i.e. 1-alkenyl-2-acylglycerophosphocholine) from acyl-CoA and 1-alkenylglycerophosphocholine with release of CoA.

alkenylglycerophosphocholine hydrolase EC 3.3.2.2; *other name*: lysoplasmalogenase; a phospholipase involved in the turnover of **plasmalogens**. It catalyses the hydrolysis of 1-(1-alkenyl)-sn-glycero-3-phosphocholine to an aldehyde and sn-glycero-3-phosphocholine.

1-alkenylglycerophosphoethanolamine O-acyltransferase EC 2.3.1.121; an enzyme that catalyses the formation of the **plasmalogen**, 1-alkenyl-2-acyl-glycerophosphoethanolamine, from acyl-CoA and 1-alkenylglycerophosphoethanolamine with release of CoA.

alkenylglycerophosphoethanolamine hydrolase EC 3.3.2.5; a phospholipase enzyme involved in the turnover of **plasmalogens**. It catalyses the hydrolysis of 1-(1-alkenyl)-sn-glycero-3-phosphoethanolamine to an aldehyde and sn-glycero-3-phosphoethanolamine.

alkyl any group derived from an alkane by the removal of one hydrogen atom. Alkyl groups are often designated by the symbol R.

alkylating agent any of a group of compounds that react with another compound so as to introduce an alkyl group into the second compound.

alkylation the process of replacing a hydrogen atom in a compound by an alkyl group.

1-alkylglycerophosphocholine O-acetyltransferase EC 2.3.1.67; an enzyme that catalyses the formation of 1-alkyl-2-acetyl-sn-glycero-3-phosphocholine (see **platelet-activating factor**) from acetyl-CoA and 1-alkyl-sn-glycero-3-phosphocholine with release of CoA. See also **1-alkylglycerophosphocholine O-acyltransferase**.

1-alkylglycerophosphocholine O-acyltransferase EC 2.3.1.63; an enzyme that catalyses the formation of the **ether lipid**, 1-alkyl-2-acyl-sn-glycero-3-phosphocholine, from acyl-CoA and 1-alkyl-sn-glycero-3-phosphocholine with release of CoA.

O⁶-alkylguanine alkyltransferase see **O⁶-methylguanine**.

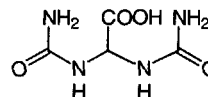
alkyne any unsaturated aliphatic hydrocarbon compound containing one or more triple bonds.

alkynyl any group derived from an **alkyne** by the removal of one hydrogen atom.

All symbol for alllose.

all+ see **allo+**.

allantoic acid or **diureidoacetate** the end product of **purine** metabolism in mammals and some fish, formed from **allantoin** by a hydrolytic reaction. Most fish metabolize allantoic acid further to urea and glyoxalate. It is widely distributed in plants as an important source of stored nitrogen.



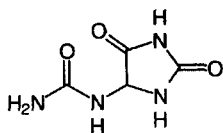
allantoin (2,5-dioxo-4-imidazolidinyl)urea; 5-ureidohydantoin; an intermediate or end product of purine catabolism, formed from uric acid by urate oxidase. In certain animal groups it is converted to allantoic acid.

allele or **allelomorph** any of the forms of the same **gene** that occur at the same **locus** on a homologous chromosome but differ in their effects.



allele-specific oligonucleotide

fer in base sequence. Two or more alleles are said to be **allelic** or **allelomorphic** to each other, and if more than two alleles exist in a population, the locus is said to show **multiple allelism**.



allantoin

allele-specific oligonucleotide *abbr.*: ASO; an oligonucleotide that is constructed with a DNA sequence homologous to a specific allele. Two ASOs can be made so that they differ in sequence at only one nucleotide base, thereby distinguishing a mutant allele with a point mutation from its corresponding wild-type allele.

allelic exclusion the process by which a cell uses either the gene from its maternal chromosome or the one from the paternal chromosome, but not both. It seems to occur only in genes that encode antibodies and T-cell receptors. Individual B-lymphocytes display allelic exclusion of their heavy and light IgG genes.

allelomorph another term for **allele**.

allelopathic agent any plant excretory product that may be autotoxic or affect neighbouring plants, such as salicylate in *Quercus falcata*.

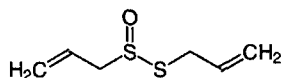
allelozyme any of two or more variants of a particular enzyme (with similar catalytic properties) whose amino-acid sequences are encoded in allelic structural genes (*see allele*); a class of **isoenzyme**. *Compare multilocus enzyme*.

allergen any antigen that stimulates an allergic reaction, inducing a type I hypersensitive reaction. *See allergy, hypersensitivity*.

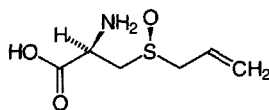
allergic of, relating to, or involving **allergy**.

allergy or **allergic reaction** **1** a state of altered (usually increased) reactivity of the body to foreign material. **2** **hypersensitivity**.

allicin 2-propene-1-sulfinothioic acid *S*-2-propenyl ester; an antibacterial compound derived from garlic (*Allium sativum*) and having an odour of garlic.



alliin 3-(2-propenylsulfinyl)-L-alanine; *S*-allyl-L-cysteine sulfide; a component of garlic (*Allium sativum*) and other plants that develops an odour of garlic following the action of alliinase (alliin lyase; EC 4.4.1.4).



allitol a *meso*-hexitol; derived formally by reduction of the aldehyde group of either D- or L-allose.

allo+ or (before a vowel) **all+** *comb. form* **1** denoting other, dissimilar, different. *Compare hetero+* (def. 1), *homo+* (def. 1), *iso+* (def. 1), *xeno+* (def. 1). **2** referring to a dissimilar genome. *Compare homo+* (def. 2), *hetero+* (def. 2), *iso+* (def. 2), *xeno+* (def. 2). **3** referring to an isomeric form of an enantiomer of a

compound that has more than one pair of enantiomers. *Compare iso+* (def. 3), *noro+* (def. 1).

allo+ *prefix* denoting the configuration of a set of four (usually) contiguous >CHOH groups, as in D- or L-allose. *See monosaccharide*.

alloantigen an antigen that is part of an animal's self-recognition system, e.g. **major histocompatibility complex** molecules. When injected into another animal, they trigger an immune response aimed at eliminating them. *Compare alloreactivity*.

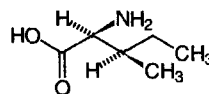
allocystathionine *see cystathionine*.

allogeneic or **allogenic** describing cells, tissues, organisms, etc. that are of different genetic constitution. *Compare heterogenic, homologous*.

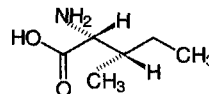
allograft any tissue graft between allogeneic individuals; i.e. from a donor of one genotype to a host of another genotype.

allohydroxylysine *see hydroxylysine*.

alloisoleucine *trivial name* for α -amino- β -methylvaleric acid; (2*R**,3*S**)-2-amino-3-methylpentanoic acid, CH₃-CH₂-CH(CH₃)-CH(NH₂)-COOH; an α -amino acid with two chiral centres. Because molecules of alloisoleucine possess a second chiral centre, at C-3, in addition to the chiral centre at C-2 common to molecules of all α -amino acids other than glycine, the enantiomers L-alloisoleucine (*symbol*: alle), (2*S*,3*R*)-2-amino-3-methylpentanoic acid, and D-alloisoleucine (*symbol*: D-alle), (2*R*,3*S*)-2-amino-3-methylpentanoic acid, are diastereoisomeric with those of **isoleucine**, (2*R**, 3*R**)-2-amino-3-methylpentanoic acid. L-Alloisoleucine does not occur in peptide linkage in proteins. Residues of D-alloisoleucine are found in the peptidolipid of the actinomycete, *Nocardia esteroides*, and various members of the actinomycin group of antibiotics contain (usually) two residues per molecule.

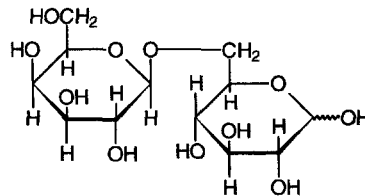


L-alloisoleucine



D-alloisoleucine

allolactose *trivial name* for β -D-galactopyranosyl-(1 $\rightarrow$ 6)-D-glucopyranose; an isomer of lactose and the natural intracellular inducer of the **lac operon** in *Escherichia coli*.



allomerism a state of similarity in the crystalline structures of substances of different chemical composition. —**allomerous** *adj.*

allometric **1** differing in relative growth rates. **2** of or relating to allometry.

allometry **1** the study of the growth of parts of an organism



allomone

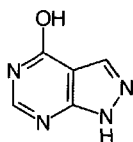
relative to the growth of the whole organism. **2** change in the proportion of any part of an organism occurring during growth.

allomone any chemical substance produced or acquired by an organism that, when it contacts an individual of another species in the natural context, evokes in the receiver a behavioural or developmental reaction adaptively favourable to the transmitter. *Compare* **kairomone**, **pheromone**.

allomorphism variability in the crystalline structure of certain substances having the same chemical composition.

allophycocyanin a **phycobiliprotein** present in small amounts in red algae and cyanobacteria.

allopurinol 4-hydroxypyrazolo[3,4-*d*]pyrimidine; a constitutional isomer of hypoxanthine and a competitive inhibitor of xanthine oxidase (EC 1.2.3.2). It is a synthetic drug used in the treatment of gout and other conditions characterized by raised blood urate content. Patients with gout excrete less urate and more xanthine and hypoxanthine than normal.



alloreactivity the T-cell response to non-self MHC molecules.

all-or-none describing a chemical reaction, or a pharmacological or physiological response, that either occurs to the fullest extent or not at all, depending on the conditions.

allose *symbol*: All; *allo*-hexose; an aldohexose that differs from glucose only in the configuration of the hydroxyl on C-3.

allosteric describing protein molecules that are assumed to possess two or more stereospecifically different, nonoverlapping receptor sites. One of these, the **active site**, binds the substrate; the other, the **allosteric site**, is complementary to the structure of another, or the same, metabolite, the **allosteric effector**, which binds specifically and reversibly. The formation of the enzyme–allosteric effector complex does not activate or deactivate a reaction involving the effector itself; it is assumed only to bring about a discrete reversible alteration of the molecular structure of the protein, called **allosteric transition**. This modifies the properties of the active site, changing one or several of the kinetic parameters that characterize the biological activity of the protein. *See also* **Monod–Wyman–Changeux model**.

allosteric activation any activation of an enzyme by a positive **allosteric effector**.

allosteric constant *symbol*: *L*; the equilibrium constant for the transition between two forms of an **allosteric** protein in the absence of ligand: $L = T_0/R_0$, where T_0 and R_0 are the concentrations of the less affine T-form and the more affine R-form of the protein, respectively. *See* **Monod–Wyman–Changeux model**.

allosteric effect an effect that arises when the reaction of ligands with one site of any polyvalent molecule affects the reaction of ligand(s) at one or more other sites as a result of conformational changes. The reactions may or may not both be reversible; when they are, and represent equilibria, they are **reciprocal effects** or **linked functions**.

allosteric effector a specific ligand that binds to the allosteric site of a protein with different affinities in the two states of the protein (R and T) in the **Monod–Wyman–Changeux model** of allosteric transition. An effector may be described as positive if it has an activating effect, or as negative if it is inhibitory. *See* **allosteric**. *Compare* **autosteric effector**.

allosteric enzyme designating any enzyme that: (1) contains a site topologically distinct from the active site; (2) shows sigmoid kinetics; (3) occurs at a branch point in a metabolic pathway; and (4) obeys the concerted model for allosteric proteins. Since all these properties may sometimes, but not always, go together, the term is somewhat imprecise.

allosteric inhibition any inhibition of an enzyme by a negative **allosteric effector**.

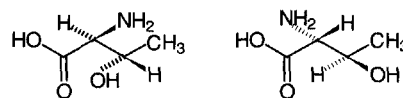
allosteric protein a protein that exhibits **allosterism**. *See also* **allosteric**.

allosteric site *see* **allosteric**.

allosteric transition *see* **allosteric**, **Monod–Wyman–Changeux model**.

allosterism or **allostery** a property of an enzyme or other macromolecule by which its principal biological reactivity is modified by the binding of an effector to a site other than the binding site of the principal reactant, thereby bringing about a conformational change in the macromolecule such as to alter its principal reactivity. *See* **allosteric**.

allothreonine *trivial name for* α -amino- β -hydroxybutyric acid; (2*R**,3*R**)-2-amino-3-hydroxybutanoic acid; $\text{CH}_3\text{--CH(OH)--CH(NH}_2\text{)--COOH}$; an α -amino acid with two chiral centres. Because molecules of allothreonine possess a second chiral centre, at C-3, in addition to the chiral centre at C-2 common to all α -amino acids other than glycine, the enantiomers L-allothreonine (*symbol*: aThr), (2*S*,3*S*)-2-amino-3-hydroxybutanoic acid, and D-allothreonine (*symbol*: D-aThr), (2*R*,3*R*)-2-amino-3-hydroxybutanoic acid, are diastereoisomeric with those of **threonine**, (2*R**,3*R**)-2-amino-3-hydroxybutanoic acid. [Note: The enantiomers of allothreonine may also be named semisystematically as derivatives of erythrose: L_g-allothreonine in amino-acid nomenclature is synonymous with 2-amino-2,4-dideoxy-L_g-erythronic acid in carbohydrate nomenclature, and D_s-allothreonine with 2-amino-2,4-dideoxy-D_g-erythronic acid (the subscript letters d or g being added to the configurational prefixes where there might be uncertainty regarding the reference centre of chirality; *see* **D/L convention**).] L-Allothreonine does not occur in peptide linkage in proteins; however, one residue per molecule is present in the peptide antibiotic telomycin. Residues of D-allothreonine have been found in the glycolipid and peptidolipid of actinomycetes, and in mycobacteria.



L-allothreonine

D-allothreonine

allotopic **1** of, relating to, or having the property of **allotopy**. **2** (*in genetics*) describing a mutation that imparts to one tissue a particular metabolic characteristic normally found in another tissue.

allotopy the phenomenon of the formation of a membrane–enzyme complex resulting in alteration of the properties of both enzyme and membrane. For example, a change in enzymatic activity is sometimes observed when a soluble enzyme is adsorbed on an interface.

allotrope any of the different physical forms in which a chemical element can exist; e.g. diamond, graphite, and fullerenes are allotropes of carbon.

allotropy or **allotrophy** or **allotropism** the phenomenon of a substance, especially an element, existing in more than one physical form (**allotrope**), usually in the same phase; e.g. the different crystalline forms of carbon or phosphorus, or the different molecular forms of dioxygen and ozone. *Compare* **polymorphism** (def. 2). —**allotropic** *adj.*

allotype an antigenically distinct variant of a protein or other antigen arising from intraspecies genetic differences. Each individual has a particular variant at each immunoglobulin gene locus, which will often differ from those in other individuals of the same species.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

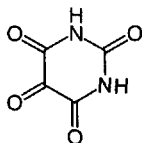
Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

allotypic specificity

allotypic specificity an antigenic specificity that is not the same in a given protein in all normal individuals of a species.

allotropy the property, possessed by certain proteins and other antigens, of existing in antigenically distinct forms or **allotypes**. Thus allotypes of the same protein, although not distinguished by usual physicochemical and chemical criteria, can elicit specific antibodies in some other members of the same species. They are those **idiotypes** that behave as **xenotypes** in other individuals of the same species.

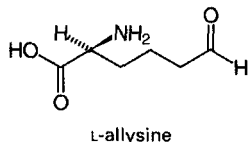
alloxan mesoxalylurea; 2,4,5,6-(1*H*,3*H*)-pyrimidinetrone; a compound used experimentally to induce diabetes through its action in selectively destroying pancreatic B-cells.



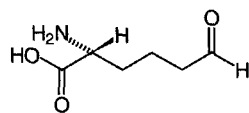
allozyme any enzyme variant produced by a particular allele.

allulose *trivial name (not recommended)* for **psicose**.

allysine 6-oxonorleucine; 2-aminoadipaldehydic acid; an α -amino acid not normally found in proteins but enzymically formed from lysine *in situ* as an intermediate in the creation of covalent cross-links between adjacent polypeptide strands during the maturation of certain connective-tissue proteins (e.g. collagen, elastin). The enzyme **lysyl oxidase** oxidizes the terminal amino-methylene groups of residues of either lysine or 5-hydroxylysine to aldehyde groups, forming residues of allysine or **hydroxyallysine** respectively. *See also* **desmosine**, **isodesmosine**, **lathyrism**, **noncoded amino acid**, **syndesine**.



L-allysine



D-allysine

almond emulsin *see* **1,3- α -L-fucosidase**.

ALP (*in clinical biochemistry*) *abbr.* for alkaline phosphatase.

alpha symbol: α (lower case) or A (upper case); the first letter of the Greek alphabet. For uses *see* **Appendix A**.

alpha cell an alternative term for **A cell**.

alpha-1 cell an alternative term for **D cell**.

alpha chain or **α chain** 1 the heavy chain of IgA immunoglobulin molecules. 2 one of the two types of polypeptide chain present in normal adult hemoglobin (Hb A) molecules.

alpha effect the enhancement of nucleophilicity that occurs when the atom adjacent to a nucleophilic site bears a lone pair of electrons. It has been invoked to explain the high toxicity of, e.g., hydroxylamine and the cyanide ion.

alpha-fetoprotein *see* **α -fetoprotein**.

alpha globulin or **α -globulin** any of a group of plasma proteins that migrate most anodically among the globulins in electrophoresis at pH values slightly above 7.

alpha helix or **α -helix** a helical, or spiral, configuration of a polypeptide chain in which successive turns of the helix are

held together by hydrogen bonds between the amide (peptide) links, the carbonyl group of any given residue being hydrogen-bonded to the imino group of the third residue behind it in the chain. This is the case for all of the carbonyl and amide groups of the peptide bonds of the main chain. The alpha helix has 3.6 residues per turn and the translation (def. 2), or pitch, along the helical axis is 1.5 Å per residue and 5.4 Å per turn. The helix may be left- or right-handed, the latter being much more common. It is one of the two basic elements of the **secondary structure** adopted by a polypeptide chain within the hydrophobic core of a globular protein (the other being **beta strand**). The alpha helix was first described by Pauling and Corey in 1951, based on model building. *See also* **gamma helix**.

alpha hemolysis or **α -hemolysis** a type of hemolysis, characterizing certain strains of streptococci, in which a greenish tinge occurs around the bacterial colonies in a blood-agar plate. *Compare* **beta hemolysis**.

alphanumeric describing any coding system or data set that provides for letters, numbers, and other symbols such as punctuation. [From 'alphabetic' and 'numeric'.]

alpha oxidation or **α -oxidation** a series of reactions occurring in plant tissue by which a free fatty acid of chain length ranging from C₁₈ to C₁₃ is oxidatively degraded with the simultaneous release of a molecule of CO₂ from the carboxyl group and the formation of a free fatty acid containing one carbon atom fewer. NAD⁺ is the only cofactor required.

alpha particle or **α particle** a fast-moving positively charged helium nucleus, ⁴He²⁺, emitted in the decay of certain **radionuclides**.

alpha radiation radiation consisting of fast-moving alpha particles.

alpha ray a directed stream of alpha particles.

alpha receptor *see* **adrenoceptor**.

ALS *abbr.* for 1 antilymphocyte serum. 2 (*in clinical biochemistry*) (fructose-bisphosphate) aldolase, EC 4.1.2.13.

Alt symbol for altrose.

ALT (*in clinical biochemistry*) *abbr.* for alanine transaminase.

alternate pathway *see* **alternative pathway** (def. 2).

alternating axis of symmetry *see* **symmetry**.

alternating current *abbr.*: a.c. or AC; an electric current that varies with time in a sinusoidal manner. The abbreviation a.c. may also be applied to a voltage of varying polarity.

alternative pathway 1 a misleading term sometimes applied to any of the metabolic pathways by which glucose is broken down in animals, especially the **pentose (phosphate) pathway**. 2 (or **alternate pathway**) a pathway by which **complement** components C3–C9 are activated without a requirement for C1, C2, or C4.

alternative splicing the occurrence of alternative patterns of splicing of a primary RNA transcript of DNA to produce different mRNAs. After excision of **introns**, selection may determine which **exons** are spliced together to form the mRNA. An example is the production of mRNA for 19S IgM, which is secreted, and 7S IgM, which is inserted into the lymphocyte membrane, both products being coded for by the same gene, but the mRNA for 7S IgM having fewer of the original exons.

Altman, Sidney (1939–), Canadian-born US biochemist; Nobel Laureate in Chemistry (1989) jointly with T. R. Cech 'for their discovery of catalytic properties of RNA'.

altro+ prefix denoting a particular configuration of a set of four (usually) contiguous >CHOH groups, as in D- or L-altrose. *See* **monosaccharide**.

altrose symbol: Alt; **altro-hexose**; an aldohexose that differs from glucose in the configuration of the hydroxyl groups on C-2 and C-3.

alum-precipitated toxoid any preparation in which a toxoid is adsorbed onto an adjuvant containing an aluminium compound, e.g. aluminium hydroxide gel or an alum. The term is applied especially to diphtheria toxoid.

alum precipitation a technique in which a soluble immunogen is converted into particulate form by mixing with a solution of

Alu sequence

an alum (e.g. aluminium potassium sulfate) and adjusting the pH to near 6.5. The antigen is adsorbed on the aluminium hydroxide precipitate so formed, which acts as an adjuvant in immunization.

Alu sequence or **AluI sequence** or **Alu repeat** or **AluI repeat** any of various base sequences dispersed in human, rat, and mouse (and perhaps other) genomes of ≈ 0.3 kbp and spaced at approximately 5 kbp. Each Alu sequence comprises two similar 130 bp boxes (**Alu left** and **Alu right**). The name derives from the presence in such sequences of sites for the restriction enzyme *AluI*.

alveolus (*pl. alveoli*) any small pit, cavity, or saclike dilatation, such as an air cell of the lungs, an **acinus** of a compound gland, or a tooth socket. — **alveolar** *adj.*

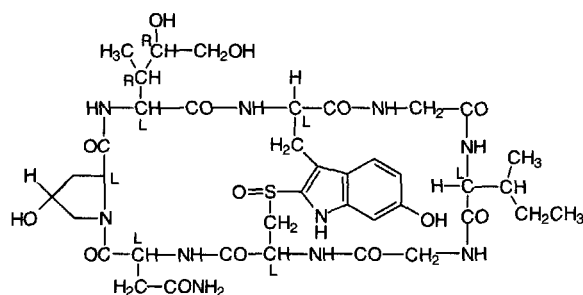
Alzheimer's disease or **Alzheimer's dementia** or **Alzheimer sclerosis** a genetically and phenotypically complex syndrome of progressive cognitive failure. It is the most common cause of late-life dementia (senile dementia) in developed nations, but early onset before the age of 65 (presenile dementia) may also occur. All cases develop numerous extracellular deposits of the **β -amyloid peptide** in the brain, and almost all have intraneuronal bundles of abnormal filaments composed of highly phosphorylated forms of the microtubule-associated **tau protein**. Missense mutations in the chromosome 21 gene coding for the β -amyloid peptide precursor have been described, and a chromosome 14q gene coding for a 467-amino-acid protein with seven putative membrane-spanning domains has been linked with the early-onset disease. Inheritance of the E₄ allele of the gene encoding **apolipoprotein E** is one of the main risk factors for late-onset Alzheimer's disease. [After Alois Alzheimer (1864–1915), German neurologist who in 1907 described the condition of a patient referred to as Auguste D.]

Am *symbol for* americium.

Amadori rearrangement an acid- or base-catalysed chemical rearrangement reaction in which an *N*-substituted aldose-amine is converted into the corresponding *N*-substituted 1-amino-1-deoxy-2-ketose. It occurs, e.g., in the **Maillard reaction**, in the reaction of carbohydrates with phenylhydrazine, in the formation of hemoglobin A_{1c}, and in a step in tryptophan biosynthesis – isomerization of *N*-(5'-phosphoribosyl)-anthranilic acid into 1-(α -carboxyphenylamino)-1'-deoxyribose-5'-phosphate.

Amanita toxin *see* **amatoxin**.

amanitin any of various **amatoxins**, especially α -, β -, γ -, and ϵ -amanitin. These occur, together with other **amatoxins**, in the highly poisonous agaric commonly called the death-cap fungus or deadly agaric (*Amanita phalloides*), and in some related species.



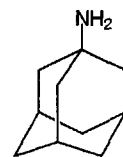
α -amanitin

amantadine *trivial name for* 1-adamantanamine; 1-aminoadamantane; C₁₀H₁₇N; a polycyclic antiviral agent that is especially effective as a prophylactic against influenza A virus. It appears not to affect viral attachment to cells but to block penetration or uncoating. *See also* **adamantane**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.



amantadine

amatoxin any member of a family of closely related bicyclic octapeptides present in the highly poisonous agaric commonly called the death-cap fungus or deadly agaric (*Amanita phalloides*), and in some related species. **α -Amanitin** is the best known member and the chief toxic constituent of *A. phalloides*. The majority of amatoxins are highly toxic, though much slower in action than the congeneric and less potent phallotoxins. For example, the ingestion by an adult man of 10 mg α -amanitin (the quantity in a single fruit body of *A. phalloides*) leads over the course of several days to death in the absence of treatment. The common elements of the molecular structure of amatoxins include residues of *trans*-4-hydroxy-L-proline, (usually mono- or dihydroxy-)L-isoleucine, and the bisfunctional amino acid (usually 6'-hydroxy-)L,L-tryptathionine *S*-oxide (which forms a sulfoxide bridge across the molecule; *see* structure at **amanitin**), together with two residues of glycine. α -Amanitin binds very tightly ($K_{\text{diss}} = 10^{-8}$ M) to eukaryotic RNA polymerase II, thereby blocking the formation of mRNA precursors and hnRNA. It also binds, but rather less tightly ($K_{\text{diss}} = 10^{-6}$ M), to eukaryotic RNA polymerase III, thereby blocking the formation of tRNA and 5S RNA. It has little action on RNA polymerase I. Hence the synthesis of RNA and protein is markedly inhibited in eukaryotic cells but not in prokaryotic cells.

amber 1 a fossil resin. 2 *see* **amber mutant**.

amber codon or **amber triplet** *symbol:* UAG; one of the three terminator codons or **nonsense codons** in an mRNA molecule.

Amberlite *proprietary name for* various cation- and anion-exchange resins.

amber mutant or **amber** a T4 bacteriophage carrying an **amber mutation**. *See also* **ochre mutant**, **opal mutant**.

amber mutation a temperature-sensitive conditionally lethal mutation, capable of occurring in almost any gene of a bacterial virus, that causes the synthesis of a particular polypeptide chain to terminate prematurely. It is due to the alteration of one codon in the corresponding messenger RNA to an **amber codon**, i.e. a termination codon.

amber suppressor any of a number of mutations in *Escherichia coli* that suppress an **amber codon** in mRNA thereby allowing insertion of one of several alternative amino acids into a polypeptide at that site. The mutations result in changed anticodons of the tRNA molecules, consequent on single base substitutions in their DNA coding sequences.

ambi+ or **ambo+** *comb. form* denoting on both sides, both. *Compare* **amphi+**. [From Latin *ambo*, both.]

ambient of or relating to the immediate environment.

ambiquitous describing an enzyme that has reversibly convertible soluble and membrane-bound forms. — **ambiquity** *n*.

ambo+ a variant form of **ambi+**.

ambo- *prefix* designating chiral compounds in which both of the possible molecular configurations are present at an indicated chiral centre or residue but not necessarily in equimolar proportions. Such situations may arise especially with chiral compounds obtained by partial chemical synthesis from chiral precursors in asymmetric reactions. Examples: the amino-acid sequence Phe-**ambo**-Ala-Leu, representing a mixture of Phe-Ala-Leu and Phe-D-Ala-Leu; and the trivial name 4'-**ambo**-8'-**ambo**- α -tocopherol, representing a synthetic α -tocopherol in which C-2 has the natural, (*R*)-, configuration and C-4' and



amboceptor

C-8' have both (*R*)- and (*S*)-configurations though not necessarily in equimolar proportions. *Compare* **DL**-, **rac**-.

amboceptor (formerly) an alternative term for hemolytic antibody. [Named by Ehrlich, who pictured hemolysis as an antibody having specific receptors both for erythrocytes and for complement, thereby serving to link the two together.]

ameba an alternative US spelling of **amoeba**.

amelogenin the predominant protein in dental enamel during early development. Example (bovine): database code AMEG_BOVIN, 170 amino acids (19.24 kDa).

amidorphin an opioid peptide corresponding to residues 104–129 of bovine proenkephalin A, but which is amidated at its C terminus.

Ames test *see* **Salmonella mutagenesis test**.

amethopterin another name for **methotrexate**.

amidase EC 3.5.1.4; acylamide amidohydrolase; an enzyme that catalyses the hydrolysis of a monocarboxylic acid amide to monocarboxylate and NH_3 .

ω -amidase EC 3.5.1.3; ω -aminodicarboxylate amidohydrolase; an enzyme that catalyses the hydrolysis of an ω -amidodicarboxylic acid to dicarboxylate and NH_3 .

amidate to convert any oxy acid into its corresponding amide. —**amidated** *adj.*; **amidation** *n*.

amide any compound containing one, two, or three acyl groups attached to a nitrogen atom. An amide may be derived formally or actually by condensation of an oxy acid with ammonia or a primary or secondary amine. Amides derived from carbon acids may be termed **carboxamides**, those from sulfonic acids **sulfonamides**, etc. Examples include: acetamide, methanamide, $\text{CH}_3\text{-CO-NH}_2$; sulfanilamide, 4-aminobenzenesulfonamide, $\text{H}_2\text{N-C}_6\text{H}_4\text{-SO}_2\text{-NH}_2$; and diethylformamide, formdiethylamide, $\text{H-CO-N(C}_2\text{H}_5)_2$. The term includes **imide** and **peptide**.

amidination the process or reaction of introducing an amidino group into a chemical entity.

amidine any compound of the type R-C(=NH)-NH_2 .

amidino the monovalent group -C(=NH)-NH_2 .

amidino+ *comb. form* denoting the presence of an amidino group.

***N*-amidinoglycine** *see* **glycocytamine**.

amido+ *comb. form* denoting the presence of an amide group.

Amido Black *see* **Naphthalene Black**.

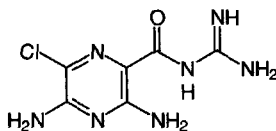
amidolysis any cleavage of an amide (including a peptide) to the parent oxy acid and ammonia or an amine.

amidophosphoribosyltransferase EC 2.4.2.14; *systematic name*: 5-phosphoribosylamine:pyrophosphate phospho- α -D-ribosyltransferase (glutamate-amidating); *other names*: glutamine phosphoribosylpyrophosphate amidotransferase; phosphoribosyldiphosphate 5-amidotransferase. An enzyme that catalyses the formation of 5-phospho- β -D-ribosylamine from L-glutamine, 5-phospho- α -D-ribose 1-diphosphate, and H_2O with release of L-glutamate and pyrophosphate. This is an early reaction in purine biosynthesis. Example from *Escherichia coli*: database code PUR1_ECOLI, 504 amino acids (56.38 kDa).

amidorphin an opioid peptide corresponding to residues 104–129 of bovine proenkephalin A, but which is amidated at its C terminus.

amidotransferase *see* **amidophosphoribosyltransferase**, **glutamine amidotransferase**.

amiloride *N*-amidino-3,5-diamino-6-chloropyrazinecarboxamide; a potent and specific inhibitor of Na^+ entry into cells. It inhibits Na^+ , K^+ -ATPase at low concentrations. It is used as a potassium-sparing diuretic.



amiloride-binding protein *see* **amine oxidase (copper-containing)**.

amine any organic compound that is weakly basic in character and contains an **amino** or a substituted amino group. Amines are called primary, secondary, or tertiary according to whether one, two, or three carbon atoms are attached to the nitrogen atom. *See also* **quaternary ammonium compound**.

amine oxidase (copper-containing) EC 1.4.3.6; *other names*: diamine oxidase; diamino oxyhydrase; histaminase; an enzyme that catalyses the oxidation by dioxygen of an amino group to an aldehyde, with release of ammonium and hydrogen peroxide. 6-Hydroxydopa and copper are cofactors. Example, human amiloride-sensitive amine oxidase (copper-containing) precursor (amiloride-binding protein; *abbr.*: ABP): database code ABP_HUMAN, 751 amino acids (85.27 kDa); this enzyme binds **amiloride**.

amine oxidase (flavin-containing) *see* **monoamine oxidase**.

aminergic describing a receptor that is activated by a specific (biogenic) amine. The term is usually applied to nerves that act by releasing a monoamine (norepinephrine, dopamine, or 5-hydroxytryptamine) from their nerve endings. *Compare* **adrenergic**, **cholinergic**, **dopaminergic**, **peptidergic**, **purinergic**, **serotonergic**.

amino the chemical group -NH_2 in an organic molecule. It is basic in character and formally derived by the removal of a hydrogen atom from ammonia.

amino+ *comb. form* denoting the presence of an amino group.

amino acid any organic acid containing one or more amino substituents. The term is usually restricted to amino, especially α -amino, derivatives of aliphatic carboxylic acids, but it can also include β -amino derivatives.

amino-acid accepting RNA *see* **transfer RNA**.

amino-acid *N*-acetyltransferase EC 2.3.1.1; *other name*: *N*-acetylglutamate synthase (*abbr.*: AGS); an enzyme that catalyses the formation of *N*-acetyl-L-glutamate (*abbr.*: AGA) from acetyl-CoA and L-glutamate with the release of CoA. This is the enzyme responsible for the formation of AGA as an obligate activator of **carbamoyl-phosphate synthase** (def. 1), the first step in the **ornithine-urea cycle**. Example from *Escherichia coli*: database code ARG1_ECOLI, 443 amino acids (49.14 kDa). For another example, *see* **glutamate *N*-acetyltransferase**.

amino-acid analyser any instrument or apparatus for the automated analysis of mixtures of amino acids. The amino acids are separated individually and quantified spectrophotometrically.

amino-acid arm the base-paired segment of **transfer RNA** containing both the 3' and the 5' ends of the RNA molecule, on which the specific amino acid is attached to the 3'-hydroxyl at the 3' end.

amino-acid oxidase any enzyme that catalyses the oxidation by dioxygen of the amino group of an α -amino acid to an oxo acid with release of ammonium and hydrogen peroxide. L-amino-acid oxidase, EC 1.4.3.2, is specific for L enantiomers and D-amino-acid oxidase, EC 1.4.3.3, is specific for D enantiomers. Both enzymes have FAD as coenzyme. Examples: L-amino-acid oxidase from *Neurospora crassa*: database code OXLA_NEUCR, 695 amino acids (76.80 kDa); D-amino acid oxidase (human) a peroxisomal enzyme: database code OXDA_HUMAN, 347 amino acids (39.37 kDa).

amino-acid residue that part of any amino-acid molecule that is present when the amino acid is combined in a (poly)peptide, i.e. the amino acid less the atoms (a hydrogen atom or/and a hydroxyl group) that have been removed during the formation of a peptide bond or bonds.

amino-acid sequence or **primary structure** the sequential order of amino-acid residues in a polypeptide or protein. It is usually given from the end carrying an α -amino group not in peptide linkage, i.e. from the N-terminal end.

amino-acid site a less correct term for **aminoacyl site**.

amino acid-tRNA ligase *other name*: aminoacyl-tRNA syn-

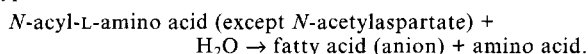


thetase. Any of the sub-subclass EC 6.1.1 of enzymes that catalyse the formation of aminoacyl-tRNA from ATP, amino acid, and tRNA with release of pyrophosphate and AMP. Each amino acid has its cognate amino acid-tRNA ligase and tRNA acceptor species; e.g. in the case of glutamate-tRNA ligase (EC 6.1.1.17), the tRNA is tRNA^{Glu} and the product is Glu-tRNA^{Glu}. The enzymes are complicated. Aminoacyl adenylate is normally an intermediate and the initial aminoacyl-tRNA product usually involves 2'-substitution with subsequent rearrangement. Specificity is not perfect but the enzymes have one or more mechanisms for correcting mischarged aminoacyl-tRNAs, which occur with structurally similar amino acids (e.g. Ile-tRNA^{Val}). Structurally the enzymes belong to several classes. Example of type I: tyrosine-tRNA ligase from *Bacillus stearothermophilus* (EC 6.1.1.1); homodimer, database code *SYT_BACST*, 419 amino acids (47.25 kDa). The 3-D structure of this, along with its mutants and complexes with substrates, is known; it has a large helical content and a characteristic ATP-binding motif.

amino aciduria a condition in which, on a normal dietary intake, amino acids appear in the urine. It may be due to renal defects or an inherited disease of amino-acid metabolism. See **Hartnup disease**, **maple syrup urine disease**.

aminoacyl the acyl group, R-CH(NH₂)-CO-, formed by removal of the hydroxyl group from the α -carboxyl group of an (unknown or unspecified) α -amino acid.

aminoacylase 1 EC 3.5.1.14; an *N*-acyl amino-acid aminohydrolase; an enzyme that catalyses any reaction of the general type:



Example (human): database code *ACY1_HUMAN*, 408 amino acids (45.88 kDa). 2 EC 3.5.1.15; *recommended name*: asparatoacylase; an enzyme that hydrolyses *N*-acetylaspargate to acetate and aspartate. Example (human): database code *ACY2_HUMAN*, 313 amino acids (35.73 kDa).

aminoacyl site or **A site** or (correctly) **aminoacyl-tRNA site** the site on a **ribosome** to which the incoming aminoacyl-tRNA is bound during protein synthesis. In bacteria this involves **ribosomal proteins** L1, L5, L7/L12, L20, L30, L33, both 16S and 23S rRNA (L7 and L12 differ in acetylation only).

aminoacyltransferase any enzyme of sub-subclass EC 2.3.2 that transfers aminoacyl groups, forming either an ester or an amide.

aminoacyl-tRNA the form in which amino acids are transported to the ribosomes during protein synthesis. The amino acid is esterified through its carboxyl group to the 3'-hydroxyl group of the 3'-terminal adenosine of a molecule of **transfer RNA**.

aminoacyl-tRNA synthetase an alternative name for **amino acid-tRNA ligase**.

α -aminoadipate 2-aminohexanedioate; an intermediate in the synthesis of lysine (see **aminoadipic pathway**). It is a competitive inhibitor of **glutamate receptors**.

aminoadipic pathway a metabolic pathway for the biosynthesis of lysine that occurs in some algae, *Euglena*, and fungi. It involves a two-carbon extension of 2-oxoglutarate to 2-oxoadipate (acetyl-CoA addition to form homocitrate, thence *cis*-homoaconitate, homoisocitrate, and oxidation to 2-oxoadipate), transamination to 2-aminoadipate, reaction of the latter with ATP to yield 5-adenylyl-2-aminoadipate, and further reductive reactions to 2-aminoadipic-5-semialdehyde, saccharopine, and finally L-lysine. Aminoadipate is also involved in lysine catabolism in mammals.

amino alcohol any aliphatic organic molecule or moiety containing both an amino and a hydroxyl substituent.

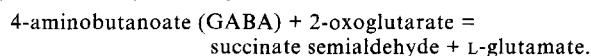
***p*-aminobenzoic acid** *abbr.*: PABA; 4-aminobenzoic acid; a compound present in yeast as a factor in the B complex of vitamins, and a structural component of folic acid and related compounds. It is a substrate for the bacterial enzyme, dihy-

dropteroate synthase, EC 2.5.1.15, competitive inhibitors of which (e.g. **sulfonamides**) are effective antibacterial agents. It is used in sunscreen preparations. *De novo* synthesis is through the intermediate chorismate formed by the **shikimate pathway**. It is formed in a test of pancreatic function, the **PABA test**.

γ -aminobutyrate receptor see **GABA receptor**.

γ -aminobutyrate shunt or **GABA shunt** a variation of the **tricarboxylic-acid cycle** in which α -oxoglutarate is converted to L-glutamate either by transamination or by reductive amination. The glutamate is decarboxylated to form γ -aminobutyrate, which can re-enter the tricarboxylic-acid cycle after being oxidized to succinate. This metabolic pathway occurs in brain tissue, presumably to form γ -aminobutyric acid, and is also prominent in green plants.

4-aminobutyrate transaminase EC 2.6.1.19; *other names*: γ -amino-*N*-butyrate transaminase; GABA transaminase; β -alanine-oxoglutarate aminotransferase; an aminotransferase enzyme important in the metabolism of **γ -amino-*n*-butyric acid** (*abbr.*: GABA). It catalyses the reaction:



Example from yeast: database code *GATA_YEAST*, 471 amino acids (52.90 kDa).

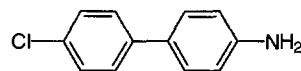
α -amino-*n*-butyric acid 2-aminobutanoic acid; a component of **ophthalmic acid**.

γ -amino-*n*-butyric acid *abbr.*: GABA; 4-aminobutanoic acid, H₂N-CH₂-CH₂-CH₂-COOH; an amino acid not found in proteins but occurring principally in the central nervous system, where it is an important inhibitory **neurotransmitter**. It is also found in some plants. See also **GABA receptor**.

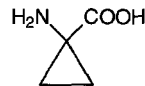
ϵ -aminocaproic acid *symbol*: ϵ AcP; (formerly) an alternative name for **6-aminohexanoic acid**.

7-aminocephalosporanic acid a compound obtained by the acid hydrolysis of **cephalosporin C** and used in the manufacture of semisynthetic cephalosporins.

4-amino-4'-chlorodiphenyl *abbr.*: pCPA; an irreversible inhibitor of **tryptophan 5-hydroxylase** an enzyme in the pathway for synthesis of 5-hydroxytryptamine.



1-aminocyclopropane-1-carboxylic acid a key intermediate in the conversion of L-methionine into ethylene in higher plants. It is also a potent and selective ligand for the glycine modulatory site of the NMDA receptor (see **glutamate receptor**). It is formed from *S*-adenosyl-L-methionine by 1-aminocyclopropane-1-carboxylate synthase, EC 4.4.1.14.



aminodeoxysugar or **amino sugar** or **amino-saccharide** or **amino-monosaccharide** any monosaccharide or monosaccharide derivative in which an alcoholic hydroxyl group has been replaced by an amino group.

aminoethoxyvinylglycine *abbr.*: AVG; an inhibitor of ethylene formation in higher plants. It is produced by *Streptomyces* spp. It acts at the level of 1-aminocyclopropane-1-carboxylate synthase (see **1-aminocyclopropane-1-carboxylic acid**).

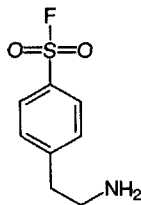
aminoethyl *symbol*: Aet; the -CH₂-CH₂-NH₂ group.

aminoethylbenzenesulfonyl fluoride *abbr.*: AEBSF; an inhibitor of serine endopeptidases. It is both soluble and stable in aqueous solutions, and is less toxic (LD₅₀ 2.8 g kg⁻¹) than

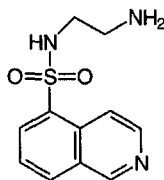


***N*-(2-aminoethyl)isoquinoline-5-sulfonamide**

phenylmethylsulfonyl fluoride, but with the same action and similar potency. Compare organophosphate.



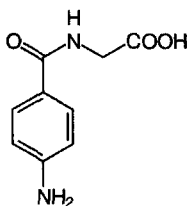
***N*-(2-aminoethyl)isoquinoline-5-sulfonamide** or **H9** a selective inhibitor of casein kinase I, protein kinase A, and, less potently, protein kinase C.



aminoglycoside antibiotic any member of a group of broad-spectrum antibiotics, of similar toxicity and pharmacology, that contain an aminodeoxysugar, an amino- or guanidino-substituted inositol ring, and one or more residues of other sugars. The group includes **streptomycin**, **neomycin**, framycetin, **kanamycin**, paromomycin, and **gentamicin**.

6-aminohexanoic acid *symbol*: εAhx; a compound that acts as an inhibitor of **streptokinase** and **urokinase** and prevents the synthesis of **plasmin**.

para-aminohippuric acid *abbr.*: PAH; *N*-(4-aminobenzoyl)glycine; it is used in renal clearance tests. Its clearance exceeds that of inulin, indicating that it is actively secreted by renal tubular cells. It shares this transport system with penicillin and competitively inhibits penicillin secretion. PAH clearance is taken as a measure of effective renal plasma flow and the inulin:PAH clearance ratio as the filtration fraction, normally about 18%.



α-amino-3-hydroxy-5-methylisoxazolepropionic acid receptor or **AMPA receptor** an excitatory amino acid receptor coupled directly to a transmembrane cation channel. It was formerly known as the quisqualate receptor. It is a type of **glutamate receptor**.

aminoimidazole carboxamide ribonucleotide *see* **AICAR**.

aminoimidazole ribonucleotide *see* **AIR**.

amino-imino tautomerism a form of **tautomerism** in which the amino and the imino forms of a compound are in equilibrium, the change corresponding to the migration of a hydrogen atom. Compare **keto-enol tautomerism**.

α-aminoisobutyric acid α-methylalanine, 2-amino-2-methylpropanoic acid, (CH₃)₂C(NH₂)COOH; a non-metabolizable amino-acid analogue used in metabolic and other studies.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

δ-aminolevulinate or (*esp. Brit.*) **δ-aminolaevulinate** *abbr.*: ALA, δALA, or DALA; 5-amino-4-oxopentanoate; H₂N-CH₂-CO-[CH₂]₂-COO⁻; a compound formed in animals and bacteria from glycine and succinyl-CoA (*see* **5-aminolevulinate synthase**); in plant chloroplasts it is formed from glutamate (*see* **glutamate-1-semialdehyde 2,1-aminomutase**). It is an intermediate in **tetrapyrrole** biosynthesis.

5-aminolevulinate synthase EC 2.3.1.37; *systematic name*: succinyl-CoA:glycine C-succinyltransferase (decarboxylating); *other names*: δ-aminolevulinate synthase; δALA synthetase. The first and regulatory enzyme in the porphyrin biosynthetic pathway. It catalyses a reaction between succinyl-CoA and glycine to form 5-aminolevulinate, CoA, and CO₂. Pyridoxal phosphate is a cofactor. Example from human erythroid type (precursor): database code HEM0_HUMAN, 587 amino acids (64.62 kDa).

aminolysis 1 any chemical reaction in which an amine reacts analogously to ammonia in **ammonolysis**. 2 any hydrolytic deamination reaction in which an amino group is replaced by a hydroxyl group.

2-amino-4-methylhex-4-enoic acid a nonprotein amino acid found in plants, e.g. *Aesculus californica* (California buckeye).

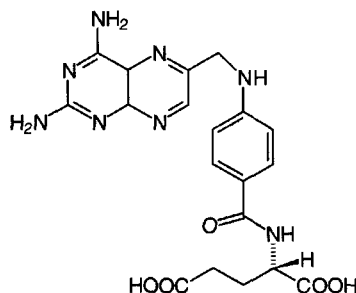
amino-monosaccharide an alternative name for **aminodeoxysugar**.

6-aminopenicillanic acid *abbr.*: 6-APA; a compound formally derived by the removal of the acyl side chain from **penicillin**. It is obtained industrially from fermentations in the absence of side-chain precursors but more usually by removal of the side chain using an amidase from *Escherichia coli* as an immobilized support. It is a very important intermediate for the manufacture of semisynthetic penicillins.

aminopeptidase any enzyme of sub-subclass EC 3.4.11, α-aminoacylpeptide hydrolases, that hydrolyse N-terminal amino-acid residues from oligopeptides or polypeptides.

4-aminophthalhydrazide *see* **luminol**.

aminopterin 4-aminofolic acid; a potent inhibitor of dihydrofolate reductase (*K*_i < 10⁻⁹ M). It is used as an antineoplastic agent in the treatment of acute leukemia and choriocarcinoma. It blocks nucleotide biosynthesis and thus also DNA biosynthesis, and has a similar action to **amethopterin**.



amino-saccharide an alternative name for **aminodeoxysugar**.

amino-terminal determination the analysis of a peptide or a protein to determine the nature of the amino-acid residue at the N terminus. The free terminal α-amino group is substituted with a reagent such as 2,4-dinitrofluorobenzene (Sanger's reagent), dansyl chloride, or phenylisothiocyanate (Edman's reagent). The substituted terminal residue is then removed by hydrolysis and identified by, e.g., chromatography.

amino terminus or **amino terminal** an alternative name for **N terminus** (of a polypeptide chain when the α-amino, or α-imino group is free). Compare **carboxyl terminus**.

aminotransferase or **transaminase** any enzyme, of the large sub-subclass EC 2.6.1, that catalyses the reversible transfer of a nitrogenous group, usually an amino group, to an acceptor, generally a 2-oxo acid. Pyridoxal phosphate is a coenzyme, the

amitosis

intermediate pyridoxamine phosphate being formed in the reaction. *See also* **transamination**.

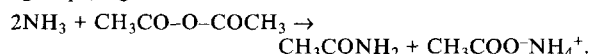
amitosis division of a cell nucleus by a process other than mitosis.

ammonification of nitrate *see* **nitrate reduction**.

ammonium **1** the NH_4^+ ion, formed by hydration of a molecule of ammonia. It is formed from N_2 by root-nodule bacteria (*Rhizobium* spp.) in leguminous plants, and is an excretory product in **ammonotelic** animals. In higher animals it is toxic in elevated concentrations, and for excretion is converted to uric acid in **uricotelic** animals, and to urea in **ureotelic** animals. **2** the chemical group $-\text{NH}_3^+$.

ammonium sulfate $(\text{NH}_4)_2\text{SO}_4$; a salt much used in the purification of proteins, which vary in their solubility in ammonium sulfate solutions. An early step in the purification of a protein is thus often an ammonium sulfate precipitation.

ammonolysis any chemical reaction in which a covalent bond is cleaved by the action of ammonia, one or more of the hydrogen atoms of the ammonia being replaced by other atoms or groups; e.g.



ammonotelic describing animal groups (e.g. crustaceans, marine mussels, octopods) in which ammonium is the principal end-product of the degradation of nitrogen-containing compounds.

amniocentesis the process of withdrawing a sample of amniotic fluid from the **amnion** by means of a hollow needle. It is particularly useful in the diagnosis of the status of a fetus with respect to proteins or DNA but, in the human, sampling can only be carried out during the second trimester of pregnancy.

amnion a membranous sac, filled with amniotic fluid, that surrounds and protects the embryo in higher animals. It is developed in reptiles, birds, and mammals but not in amphibians or fish.

amoeba or (*US*) **ameba** (*pl.* **amoebae** or **amebae**) any of a genus (*Amoeba*) of naked rhizopod protozoans in the order Amoebozoa, characterized by a thin pellicle and frequent alterations in shape due to the constant formation and retraction of pseudopodia. —**amoebic** or (*US*) **amebic** *adj.*

amoeboid or (*US*) **ameboid** **1** of, pertaining to, or reminiscent of an amoeba or amoebae. **2** describing cells, etc. that move by forming pseudopodia.

amorph a mutant allele that is inactive and fails to produce a measurable effect. It may act as a genetic block to normal biosynthesis.

amorphous **1** lacking definite shape or form. **2** noncrystalline; devoid of real or apparent crystalline form.

amount concentration *see* **concentration**.

amount of substance or **chemical amount** *symbol:* n ; a fundamental physical quantity – one of the seven SI base physical quantities – proportional to the number of specified elementary entities of a substance (i.e. atoms, molecules, ions, radicals, electrons, etc., or any specified group(s) of such particles). The SI base unit of the amount of substance is the **mole**. The proportionality factor is the same for all substances, and is the reciprocal of the **Avogadro constant**, N_A . It is given by: $n_B = N_B/N_A$, where n_B and N_B are the amount of substance B and the number of entities of substance B, respectively. The terms 'amount of substance' and 'chemical amount' may often usefully be abbreviated to 'amount'.

amount-of-substance concentration *see* **concentration**.

amp *abbr.* for ampere.

AMP *abbr.* for adenosine monophosphate; i.e. adenosine phosphate.

2'AMP *abbr.* for adenosine 2'-monophosphate; i.e. adenosine 2'-phosphate.

3'AMP *abbr.* for adenosine 3'-monophosphate; i.e. adenosine 3'-phosphate.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

5'AMP *abbr.* for adenosine 5'-monophosphate; i.e. adenosine 5'-phosphate.

AMPA receptor *abbr.* for DL- α -amino-3-hydroxy-5-methylisoxazolepropionic acid; *see* **α -amino-3-hydroxy-5-methylisoxazolepropionic acid receptor**.

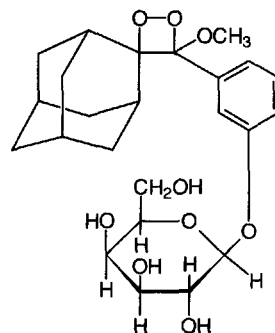
AMP-CP *abbr.* for 5'-adenylyl methylenephosphonate; i.e. adenosine 5'-[α,β -methylene]diphosphate.

AMP-CPP *abbr.* for 5'-adenylyl methylenediphosphate; i.e. adenosine 5'-[α,β -methylene]triphosphate.

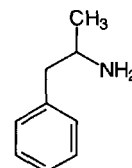
ampere *symbol:* A; *abbr.:* amp; the SI base unit of electric current. It is defined as that constant current, which if maintained in two straight parallel conductors of infinite length, of negligible circular cross-section, and placed 1 metre apart in a vacuum, would produce between these conductors a force equal to 2×10^{-7} newton per metre of length. [After André Marie Ampère (1775–1836), French physicist.]

amperometric titration any titration in which the end point is detected by measuring the electric current flowing when a potential is applied between two electrodes in a solution.

AMPGD *abbr.* for 3-(4-methoxyspiro)-1,2-dioxetane-3,2'-tricyclo[3.3.1.1]decan-(4-yl)phenyl- β -D-galactopyranoside; a chemiluminescent substrate for β -galactosidase (*see* **chemiluminescence**). It is used to detect the activity of the β -galactosidase gene when the latter is employed as a reporter gene in transgenic plants, etc.



amphetamine (*R,S*)1-phenyl-2-aminopropane; ($\pm$)- α -methylphenethylamine; racemic desoxynorephedrine; a drug that has potent effects by stimulating the release of monoamines, inhibiting **monoamine oxidase**, and blocking neuronal reuptake of monoamines from the synapse. It is noted for its marked CNS stimulant actions. *See also* **dextroamphetamine**.



amphi+ or (*before a vowel*) **amph+** *prefix* **1** denoting on both sides of, of both kinds. **2** denoting around.

Amphibia a class of anamniote vertebrate animals in the super-class Tetrapoda. They are characterized by having gills during the larval stage; these are typically replaced by lungs in the adults. The skin is moist, glandular, and scaleless.

amphibian **1** a member of the class Amphibia. **2** adapted for life both on land and in water; amphibious.

amphibolic describing an enzyme or metabolic pathway that is concerned in both **anabolism** and **catabolism**. —**amphibolism** *n.*

amphile *an alternative name for amphiphile*.

amphion or **amphion** any molecule that contains ionic groups of opposite charge, not necessarily in equal numbers; an ion-



amphipath

ized **ampholyte**. Compare **dipolar ion**. —**amphionic** or **ampho-ionic** *adj.*

amphipath *an alternative name for amphiphile.*

amphipathic or **amphiphilic** describing a molecule, such as a glycolipid or phospholipid, that contains both hydrophobic (i.e. lipophilic, nonpolar) and hydrophilic (i.e. aquaphilic, polar) parts; displaying **amphipathy**; being an **amphiphile**.

amphipathic helix an **alpha helix** with a specific topography consisting of opposite polar and nonpolar forces and a regular charge distribution. It is proposed as a basic structural element of the lipid-associating domains of apolipoproteins in very-low-density lipoproteins and high-density lipoproteins.

amphipathy or **amphiphilicity** the property (of a molecule, substance, etc.) of having affinity for both lipoidal and aqueous media; the extent or the degree to which a substance is amphipathic.

amphiphile or **amphile** or **amphipath** a substance whose molecules have an affinity for both aqueous and nonaqueous media; an **amphipathic** substance. [From Greek *amphi-*, of both kinds, + *philos*, loving.]

amphiphilic *an alternative term for amphipathic.*

amphiphysin an acidic protein that is localized in many synapses of avian and mammalian nervous systems and also expressed in adrenal gland and anterior and posterior pituitary tissues. It is an autoantigen in paraneoplastic stiff-man syndrome, a generalized progressive muscular hypertonia. Example from human: database code HSAMPHI, 695 amino acids (76.17 kDa).

amphiprotic describing a substance that can either gain or lose one or more protons.

amphiregulin a bifunctional transmembrane glycoprotein with a C-terminal EGF-like domain. It inhibits growth of several human carcinoma cells in culture and stimulates proliferation of human fibroblasts and certain other tumour cells. It is induced by phorbol acetate. Example from human: database code AMPR_HUMAN, 252 amino acids (27.86 kDa).

amphioion *a variant spelling of amphion.*

Ampholine *the proprietary name for a series of mixtures of carrier ampholytes*, which between them span the pH range 2.5–11. They consist of polyamino-polycarboxylic acids formed by reaction between acrylic acid and various polyethylene-polyamines. Each mixture contains a large number of ampholyte species of slightly differing isoelectric points. The mean and maximum apparent molecular masses are approximately 400 and 500 respectively. They are used in **isoelectric focusing**. They set up a pH gradient through the electrophoretic medium, and components being separated equilibrate to a point in the medium corresponding to their isoelectric point.

ampholyte any amphoteric electrolyte; *see also* **carrier ampholyte**, **amphion**.

ampholyte-displacement chromatography a technique for the fractionation of protein mixtures by displacement chromatography on an ion exchanger using a mobile phase containing a mixture of amphoteric substances of closely spaced isoelectric points (i.e. carrier ampholytes). Separation of the components is thought to involve specific interaction between the proteins and the carrier ampholytes as well as competition between them for sites on the ion exchanger.

amphoteric describing a molecule that can react as acidic towards strong bases and as basic towards strong acids, e.g. an amino acid.

amphotericin B Fungizone; a polyene macrolide antifungal antibiotic produced by *Streptomyces* spp. It is useful against many mycotic infections, acting as a membrane-active lytic agent. At appropriate concentrations it permeabilizes cells, creating pores formed of five to ten molecules of antibiotic in association with cholesterol. These pores allow the passage of

low M_r substances. Because it is active only against sterol-containing membranes, mitochondria are not affected. It interacts preferentially with ergosterol, giving it a high affinity for plant membranes; hence its value as a fungicide. *See also* **polyene antibiotic**.

ampicillin a semisynthetic antibiotic of the **penicillin** type, and having a $C_6H_5CH(NH_2)-$ side chain.

ampicillin-resistant *symbol:* Ap^r ; describing a cell, culture, organism, etc. that shows resistance to the lethal effects of the antibiotic **ampicillin**. This is a β -lactam antibiotic and resistance is often mediated by a class of enzymes called **β -lactamases**, which are secreted either into the periplasmic space of Gram-negative bacteria or into the medium of Gram-positive bacteria. The cloning vector pBR322 contains an ampicillin-resistance gene.

ampicillin-sensitive *symbol:* Ap^s ; describing a cell, culture, organism, etc. that shows sensitivity to the antibiotic ampicillin.

amplification **1** the act, process, or result of amplifying. **2** the increase in the strength of an electromagnetic, chemical, or acoustic signal effected by an amplifier. **3** the production of additional copies of a stretch of genomic DNA, as a result of DNA either added to cells or originating in the chromosomes. These copies may result in tandem arrays in either an extra-chromosomal or chromosomal location. This can result during selection of cells for resistance to certain agents; e.g. methotrexate blocks folate metabolism and may lead to amplification of dihydrofolate reductase genes. **4** the increase in the amount of a DNA with a specific sequence resulting from the **polymerase chain reaction**.

amplifier **1** an electronic device used to increase the strength of an electromagnetic signal; especially such a device used to increase the strength of an acoustic signal. **2** a combination of a receptor and various enzymes used to increase the strength of a chemical signal, as in a **cascade sequence**.

amplify to enlarge or to increase in strength or effect, especially an electromagnetic, acoustic, or chemical signal.

amplimer the amplified product of a **polymerase chain reaction**.

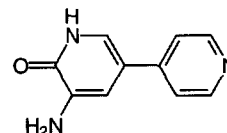
amplitude **1** magnitude; greatness of extent. **2** *symbol:* a ; the maximal displacement from zero or from a mean position of an oscillation or a curve.

AMP-PCP *abbr. for* 5'-adenylyl methylenediphosphonate; i.e. adenosine 5'-[β,γ -methylene]triphosphate.

AMP-PNP *abbr. for* 5'-adenylyl imidodiphosphonate; i.e. adenosine 5'-[β,γ -imido]triphosphate.

AMP-S *abbr. for* adenosine 5'-[α -thio]monophosphate; i.e. adenosine 5'-thiophosphate.

amrinone 5-amino-(3,4'-bipyridin)-6(1*H*)-one; an inhibitor of cardiac (type III) phosphodiesterase, resulting in increased cardiac contractility and vasodilatation. It is useful in the treatment of heart failure.



AMS (*in clinical biochemistry*) *abbr. for* α -amylase.

amu or **AMU** *symbol for* atomic mass unit. *See also* **unified atomic mass unit**.

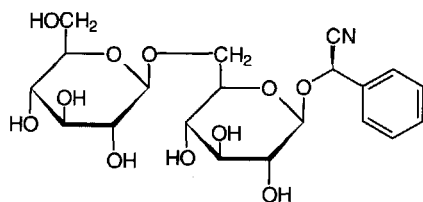
amunine *another name for* corticotropin-releasing factor.

amygdalin (–)-D-mandelonitrile β -D-gentiobioside; a **cyanogenic** glycoside derived biogenically from phenylalanine and found principally in oil of bitter almonds, although it is present in other plants belonging to the Rosaceae.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



amygdalin

amygdalose former name for **gentiobiose**.

amyl former name for the **pentyl** group.

amyl+ see **amyl+**.

amylase any enzyme able to hydrolyse *O*-glucosyl linkages in starch, glycogen, and related polysaccharides (see **α -amylase**, **β -amylase**).

α -amylase EC 3.2.1.1; *systematic name*: 1,4- α -D-glucan glucohydrolase. An enzyme that catalyses the endohydrolysis of 1,4- α -D-glucosidic linkages with the production of mainly α -maltose from amylose, but also limit dextrins from **glycogen** and **amylopectin** since it cannot catalyse hydrolysis of 1,6 linkages. It is typical of animal cells (especially pancreas), and saliva. It has five motifs. The plasma level of α -amylase is raised in acute pancreatitis, and assay of the pancreas-specific isoenzyme aids diagnosis. Normal levels in human plasma are $<300 \text{ IU L}^{-1}$, (this is very method-dependent). Example, α/β -amylase precursor from *Bacillus polymyxa*: database code AMYB_BACPO, 1196 amino acids (130.75 kDa). The protein is cleaved proteolytically to form the α (551 amino acids) and β (419 amino acids) enzymes.

β -amylase EC 3.2.1.2; *systematic name*: 1,4- α -D-glucan maltohydrolase; *other names*: saccharogen amylase; glycogenase. An enzyme that removes successive maltose units from the non-reducing ends of carbohydrate chains; the anomeric group undergoes inversion so that the major product is β -maltose. In some organisms it shares the same biosynthetic precursor as α -amylase. β -Amylases are characterized by their formation as insoluble zymogens during the process of seed maturation, and are also found in bacteria. Example from *Bacillus cereus* (precursor): database code AMYB_BACCE, 544 amino acids (61.33 kDa).

amylin or **islet amyloid polypeptide** (abbr.: IAPP) a 37-residue peptide, sequence KCNTATCATQRLANFLVHSSNNFG AILSTNVGSNTY-NH₂, isolated from pancreatic islet amyloid deposits of an insulinoma and from amyloid-rich pancreases of type 2 diabetic patients; it is found in pancreatic cells and secreted with insulin. Structurally it closely resembles **calcitonin gene-related peptide** (CGRP), and has 43% and 46% amino-acid sequence homology with human α - and β -CGRP respectively. Like CGRP it is C-terminally amidated and has an N-terminal disulfide bridge linking cysteines 2 and 7. It opposes the action of glycogen synthase in skeletal muscle and may modulate insulin action. The peptide inhibits insulin-stimulated glucose metabolism in muscle but not in adipocytes. Example, human precursor: database code IAPP_HUMAN.

amyl+ or (before a vowel) **amyl+** comb. form denoting starch.

amyl-1,6-glucosidase EC 3.2.1.33; *other name*: dextrin 6- α -D-glucosidase; an enzyme that catalyses the endohydrolysis of 1,6- α -D-glucoside linkages at points of branching in chains of 1,4-linked α -D-glucose residues. Example (human) also has glycogen debranching enzyme (4- α -glucanotransferase) activity: database code GDE_HUMAN, 1515 amino acids (172.30 kDa).

amyloid 1 a waxy eosinophilic translucent material deposited in tissues in **amyloidosis**. Originally thought to be polysaccharide, it is now known to be fibrils made of protein molecules of various origins with a twisted β -pleated sheet structure. It is insoluble, and exhibits a green fluorescence when stained with Congo

Red and viewed through crossed Nicol prisms. Amyloid fibres are relatively inert and resistant to solution in physiological solvents and to proteolytic enzymes. 2 pertaining to or characterized by the production of amyloid. 3 any non-heterogeneous substance resembling starch. 4 like or containing starch.

amyloid A protein or **serum amyloid A** (abbr.: SAA or AA) any of various plasma proteins synthesized by liver and induced by **cytokine** stimulation. The extracellular accumulation in various tissues of amyloid proteins results in secondary amyloidosis in human and other species. These deposits are highly insoluble and resistant to proteolysis. They disrupt tissue structure and compromise function. Four motifs are characteristic of SAAs. Example, SAA-1 (human) precursor: database code SAA1_HUMAN, 122 amino acids (13.43 kDa).

amyloidosis any of various pathological states in which the deposition of **amyloid** (def. 1) occurs in tissues. It is often associated with chronic infections and immunocyte dyscrasias. In the absence of medical intervention amyloid accumulates implacably, causing pressure atrophy of the affected tissue(s). Death subsequently results from interference with the normal physiological processes of affected vital organs. The descriptive term β -fibrillosis has been suggested for amyloidosis, after the beta-pleated sheet structure of the amyloid fibrils. However, the term is not commonly used. See also **amyloid A protein**.

β -amyloid peptide a glycoprotein associated with Alzheimer's disease and derived from a precursor, A4, capable of several forms through alternative splicing. It has three motifs. Database code (for human A4): A4_HUMAN, 770 amino acids (86.85 kDa).

amylolysis the process of converting starch into soluble oligo- and monosaccharides, especially by enzymic action. —**amylolytic** adj.

amylopectin a polydisperse highly branched **glucan** composed of chains of D-glucopyranose residues in $\alpha(1\rightarrow4)$ glycosidic linkage. The chains are joined together by $\alpha(1\rightarrow6)$ glycosidic linkages. A small number of $\alpha(1\rightarrow3)$ glycosidic linkages and some 6-phosphate ester groups also may occur. The chains are of similar structure to those of glycogen but with fewer and longer branches. Those in amylopectin have been found variously to contain 24 to 30 glucose residues per nonreducing end-group. The size of the molecule varies greatly according to source and method of measurement, with average M_r values of 10^6 to 10^8 reported. Amylopectin occurs in starch, commonly as a mixture with about one-third its mass of **amylose**. However, some starches, notably the so-called waxy starch of maize (*Zea mays*), are almost pure amylopectin. Starch granules serve as the principal cellular store of glucose in most plants.

amylopectin 6-glucanohydrolase see **α -dextrin endo-1,6- α -glucosidase**.

amylopectinosis see **glycogen disease**.

amyloplast a colourless **plastid** or **leukoplast** in which starch is stored. It occurs in various plant tissues, e.g. cotyledons, endosperm, and storage organs such as potato tubers.

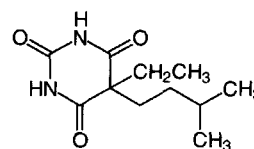
amylopsin former term for **pancreatic α -amylase**.

amylose a form of starch comprising a long unbranched **glucan** with $\alpha(1\rightarrow4)$ glycosidic linkages.

amylsucrase an alternative name for **sucrase** (def. 3).

amyl-(1,4-1,6)-transglycosylase see **branching enzyme**.

Amytal proprietary name for the barbituric sedative 5-ethyl-5-isoamylbarbituric acid, 5-ethyl-5-isopentylbarbituric acid. It inhibits the **electron-transport chain** between flavoproteins and ubiquinone.



Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



an+

an+ or (*before a consonant*) **a+** prefix indicating not, without.

+an suffix used in forming the semisystematic name of a parent chemical structure from the trivial name of the key compound to which the parent structure is related if the parent structure contains a heteroatom and is saturated e.g. glucosan, any internal anhydride of glucose.

An symbol for anhydro+.

anabiosis a return to life after apparent death; a state of suspended animation induced in an organism by, e.g., desiccation or freezing, and reversed by rehydration or warming.

anabolic pertaining to, involving, or exhibiting anabolism.

anabolic pathway any sequence of reactions involving anabolism; i.e. one in which energy is taken up. Compare catabolic pathway. See also amphibolic.

anabolic steroid any of various synthetic steroids that stimulate protein production *in vivo* by mimicking the action of androgens. Drug design aims to minimize masculinizing effects when used in female patients.

anabolism the energy-requiring part of metabolism in which simpler substances are transformed into more complex ones, as in growth and other biosynthetic processes. Compare catabolism.

anabolite any product of an anabolic pathway.

anabolize or **anabolise** to effect anabolism.

anaemia a variant spelling (*esp. Brit.*) of anemia.

anaerobe any organism that can grow in an atmosphere devoid of dioxygen. A facultative anaerobe is also capable of growing in a dioxygen-containing atmosphere whereas an obligate anaerobe (or strict anaerobe) can only grow in the absence of dioxygen.

anaerobic 1 describing an organism that is able to grow in the absence of dioxygen. 2 of or produced by an anaerobe. 3 (of an atmosphere) devoid of dioxygen. 4 (of a process) characterized by the absence of dioxygen.

anaerobic fermentation any of several catabolic pathways used by living organisms to extract chemical energy from various organic substances in the absence of dioxygen. Compare anaerobic respiration.

anaerobic glycolysis the metabolic pathways by which glucose is degraded anaerobically to lactate or ethanol. See glycolysis.

anaerobic indicator any indicator of the attainment and maintenance of anaerobic conditions in an anaerobic jar. It may be based on a dye that is decolorized in anaerobic conditions or consist of a culture of a strict anaerobe in one of the dishes in the jar.

anaerobic jar or **McIntosh & Fildes jar** an apparatus used to hold culture dishes or tubes under anaerobic conditions. It usually consists of an upright jar with an airtight lid; the air is evacuated and replaced by dihydrogen, which combines with any remaining dioxygen, the reaction being catalysed by a palladium catalyst. [After James McIntosh (1882–1948) and (Sir) Paul (Gordon) Fildes (1882–1971), British bacteriologists who first described the apparatus in 1916.]

anaerobic respiration the metabolic processes by which organisms degrade organic compounds in the absence of dioxygen to yield energy. They make use of inorganic oxidizing agents or accumulate reduced coenzymes. Compare anaerobic fermentation.

anaerobiosis life in the absence of dioxygen or air.

anagenesis the progressive or 'upward' evolution of a species; any evolutionary process that results in biological improvement. Compare cladogenesis. —**anagenetic** *adj.*; **anagenetically** *adv.*

anaglyph a stereoscopic moving or still picture made up of two superimposed images of the same object, taken from slightly different angles and reproduced in complementary colours (usually red and bluish green) so as to produce a three-dimensional effect when viewed through spectacles having one lens of each colour.

AnalaR proprietary name for any analytical (chemical) reagent.

that conforms to the tests and purity standards in 'AnalaR' Standards for Laboratory Chemicals (1977) AnalaR Standards Ltd.

analbuminemia or (*esp. Brit.*) **albuminaemia** a rare inherited disease in which there are very low plasma albumin levels but few symptoms.

analogous 1 having or showing similar or corresponding attributes. 2 (*in biology*) describing parts, organs, or tissues that have a similar function but differ in structure or evolutionary origin.

analogue or (*esp. US*) **analog** a chemical compound that is structurally similar to another. Such compounds can be used as probes to investigate specificity of, e.g. enzyme activity.

analogue enrichment or **analog enrichment** enrichment of microorganisms capable of co-metabolic degradation of organic pollutants by application of biodegradable analogues of the pollutant to the microbial ecosystem. See co-metabolism.

analyser or (*US*) **analyzer** 1 the Nicol prism of a polariscope or polarimeter that receives the light and indicates the degree and/or direction of rotation of its plane of polarization. Compare polarizer. 2 an instrument for routinely performing chemical analyses, often automatically and/or on a succession of samples. See amino-acid analyser, autoanalyzer, sequenator.

analysis (*pl. analyses*) 1 the process of determining the nature of the chemical constituents of a mixture or of the chemical groups present in a compound (qualitative analysis), or the process of determining the amounts of the constituents of a mixture (quantitative analysis). 2 the results of such an analysis. 3 the branch of mathematics concerned with the concepts of algebra and calculus. —**analytic** or **analytical** *adj.*

analysis of covariance a statistical method for determining the extent of the variance of one variable that is due to variability in some other variable.

analysis of variance a statistical method for segregating the experimentally observed variance between two populations into portions traceable to specific sources.

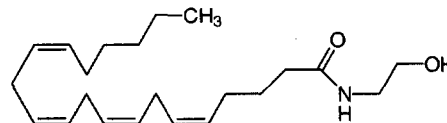
analyte a component to be measured in an analysis (*def.* 1).

analytical chemistry the branch of chemistry concerned with analysing materials by chemical (or physical) methods.

analytical ultracentrifuge an ultracentrifuge equipped with optical systems designed to determine concentration distributions at any time during the measurement. It is used in determinations of sedimentation coefficients and sedimentation equilibrium.

anamnesic reaction any immunological response in which a second or subsequent exposure to an antigen causes a greater and more rapid reaction than that elicited by the initial exposure. It is a manifestation of immunological memory.

anandamide arachidonoyl ethanolamide; a specific endogenous agonist that serves as a neurotransmitter of the cannabinoid receptor. It is formed in neurons by phosphodiesterase cleavage of *N*-arachidonoyl-phosphatidylethanolamine, and is probably then released to the extracellular space. A series of analogues such as *N*-linoleoyl ethanolamide and *N*-palmitoyl ethanolamide are formed simultaneously from appropriate precursors, but their role remains unclear. It may also be formed by condensation of arachidonate with ethanolamine. It is inactivated by reuptake into neurons and hydrolysis by *N*-acyl-ethanolamine amidohydrolase to ethanolamine and arachidonate.



anaphase a stage in mitosis or meiosis during which the two

anaphylactic shock

sister chromatids of each chromosome separate and move towards opposite poles of the spindle.

anaphylactic shock a severe generalized form of **anaphylaxis** in which there is widespread release of histamine and other vasoactive substances causing edema, constriction of the bronchioles, heart failure, circulatory collapse, and sometimes death.

anaphylactoid reaction an acute syndrome resembling **anaphylactic shock** though caused not by an immunological reaction but by injection of a variety of substances such as bee or snake venom.

anaphylatoxin any of a group of substances that mediate inflammation and are produced in serum during the fixation of complement, especially during the production of C3a and C5a. They cause mast cell degranulation and histamine release.

anaphylaxis extreme immunological sensitivity of the body or tissues to the reintroduction of an antigen. It is a form of **anamnesic reaction** and is accompanied by pathological changes in tissues or organs due to the release of pharmacologically active substances. —**anaphylactic adj.**

anaphylotoxin inactivator *see* **lysine(arginine) carboxypeptidase**.

anaplasia the reversion by a cell or tissue to a more primitive, embryonic, or undifferentiated form with loss of its characteristics and increased capacity for multiplication.

anaplerosis 1 the filling up of a deficiency; the repair or replacement of losses, especially of tissue. 2 (*in biochemistry*) the process in which an intermediate in a central metabolic pathway is replenished through a distinct **anaplerotic metabolic pathway**. This prevents the central pathway from ceasing to function because of withdrawal of the intermediate for some other metabolic purpose. [From Greek *ana*, again, back, up + *plerosin*, to fill.] —**anaplerotic adj.**

anaplerotic metabolic pathway any sequence of metabolic reactions that serves to maintain the operation of the central pathways of metabolism and which allows the operation and coordination of other metabolic pathways that enter and leave the central pathways.

anchorin CII *see* **annexin V**.

ancrod an enzyme, extractable from the venom of the Malayan pit viper, *Agkistrodon rhodostoma*, that produces unstable fibrin from fibrinogen by proteolysis. The resulting fibrinogen depletion results in an anticoagulant effect.

Andersen's disease an alternative name for type IV **glycogen disease**. [After Dorothy Hansine Andersen (1901–63), American physician.]

androgamone *see* **gamone**.

androgen any substance, natural or synthetic, that is able to stimulate the development of male sexual characteristics. Naturally occurring androgens are C₁₉-steroid hormones. They are produced especially by the testis (e.g. testosterone) and also by the adrenal cortex, ovary, and placenta. *Compare* **estrogen**. —**androgenic adj.**

androgen receptor any of the androgen-binding nuclear proteins that mediate the effects of androgens by regulating gene expression; they are **zinc-finger** proteins that bind discrete DNA sequences upstream of transcriptional start sites when the hormone-receptor complex is formed. The steroid-binding domain is in the C-terminal region. In humans, many variants are known that are associated with abnormalities, including testicular feminization syndrome, and complete or partial androgen insensitivity leading to external genitalia varying between female and nearly normal male. Example from human: database code ANDR_HUMAN, 919 amino acids (98.99 kDa).

androgen resistance *see* **testicular feminization**.

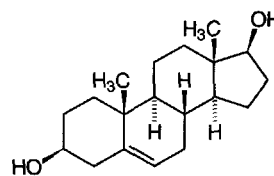
androstane the C₁₉ parent compound from which androgens are structurally derived.

androstenediol 5-androstene-3 β ,17 β -diol; a metabolite of dehydroepiandrosterone, from which it is formed by the action of 17 β -hydroxysteroid dehydrogenase, EC 1.1.1.51.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

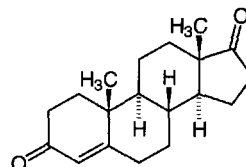
Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

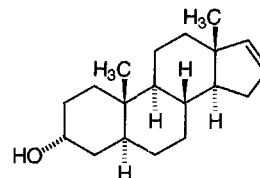


androstenediol

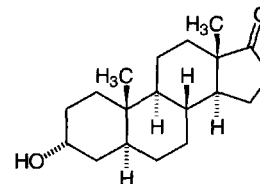
androstenedione 4-androstene-3,17-dione; a precursor of **testosterone** and **estrone**, formed in the testis and ovary from 17 α -hydroxyprogesterone by the action of 17 α -hydroxyprogesterone aldolase (17,20-lyase), EC 4.1.2.30, or from dehydroepiandrosterone by the action of 3 β -hydroxy- Δ^5 -steroid dehydrogenase, EC 1.1.1.145. It is also formed in adrenal cortex, from 17-hydroxypregnenolone by an isoenzyme of the 17,20-lyase. Androstenedione is secreted by the adrenals into the circulation and serves as a precursor for sex-hormone formation in peripheral tissues.



androst-16-en-3 α -ol 3 α -hydroxy-5 α -androst-16-ene; 'boar taint'; a major constituent of boar pheromone. It is also present in truffles, which may account for the ability of pigs to locate these underground delicacies.



androsterone 5 α -androstane-3 α -ol-17-one; a metabolite of **testosterone**, from which it is formed in the liver via androstenedione. The latter is converted to androstenedione by 3-oxo-5 α -steroid 4-dehydrogenase, EC 1.3.99.5. Androstenedione in turn is acted on by 3 α -hydroxysteroid dehydrogenase, EC 1.1.1.213, forming androsterone.



+ane suffix (*in systematic chemical nomenclature*) denoting 1 an alkane, e.g. pentane. 2 a saturated heterocyclic compound in which the heteroatom is other than nitrogen, e.g. thiolane. 3 a a hydride of boron, silicon, germanium, tin or lead, e.g. borane, silane; b a hydride of an unbranched chain of identical atoms of certain elements.

anemia or (*esp. Brit.*) **anaemia** any condition in which the blood has an abnormally low number of red cells (i.e. erythrocytes) or hemoglobin content, or is deficient in total volume.

There are many causes, including hypersensitivity reactions to (e.g.) drugs (**hemolytic anemia**) and imbalance of Fe^{2+} metabolism (**iron deficiency anemia**). **Pernicious anemia** is characterized by defective production of erythrocytes and the presence of megaloblasts in the bone marrow; sometimes it is accompanied by neurological changes. It is caused by deficiency of vitamin B_{12} due either to a dietary deficiency of the vitamin or to a failure to produce **intrinsic factor** required for its absorption from the gut.

anergy (*in immunology*) any deficiency in the immune response to an agent that normally induces a response.

aneuploid describing a nucleus or organism in which the chromosome number is not an exact multiple of the haploid number (n). In humans ($n = 23$) examples include the presence of an extra copy of a single chromosome, e.g. trisomy 21 (Down's syndrome), or the absence of a single chromosome, e.g. monosomy (as found in Turner's syndrome, 45,X). Compare **euploid**, **monosomic**, **nullisomic**, **trisomic**.

aneurin or **aneurine** a former name for **thiamine**.

Anfinsen, Christian Boehmer (1916–95), US protein chemist; Nobel Laureate in Chemistry (1972) 'for his work on ribonuclease, especially concerning the connection between the amino-acid sequence and the biologically active conformation' [prize shared with S. Moore and W. H. Stein].

angiogenesis the formation of blood vessels, whether during embryogenesis, tissue repair, or invasive growth of tumours.

angiogenin a **cytokine**, produced by fibroblasts, lymphocytes, and colon epithelial cells, that induces neovascularization and can serve as a substratum for endothelial and fibroblast cell adhesion. A monomeric protein, it is a member of the pancreatic ribonuclease family and specifically hydrolyses tRNAs, thus inhibiting protein synthesis. Example from *Mus musculus* (precursor): database code ANGI_MOUSE, 145 amino acids (16.23 kDa).

angiosperm a flowering plant; any member of the class Angiospermae, division Spermatophyta (seed plants). Compare **gymnosperm**.

angiotensin any of three related peptides, two of which regulate adrenal activity and also increase vasoconstriction and raise blood pressure. **Angiotensin I**, the decapeptide

H-Asp-Arg-Val-Tyr-Ile-His-Pro-Phe-His-Leu-OH,

is cleaved off **angiotensinogen** by the kidney enzyme **renin**; this peptide has no physiologically important action. **Angiotensin II**, the octapeptide formed from angiotensin I by removal of -His-Leu-OH by the action of **angiotensin converting enzyme**, is the most powerful pressor agent known, and the substance responsible for essential hypertension. It is also one of the most potent known stimulators of aldosterone production by the adrenal gland. It is rapidly destroyed by tissue peptidases, including **angiotensinase**. **Angiotensin III**, the heptapeptide formed by removal of the N-terminal Asp residue from angiotensin II (and the only one of its breakdown products to have activity), appears to be as active on the adrenal gland as angiotensin II and may also have significant vasoconstrictor activity.

angiotensinase an enzyme with proteinase activity that inactivates angiotensin II. Angiotensinase A is an alternative name for **glutamyl aminopeptidase**, EC 3.4.11.7, and angiotensinase C is another name for **lysosomal Pro-X carboxypeptidase**, EC 3.4.16.2.

angiotensin converting enzyme EC 3.4.15.1; *recommended name*: peptidyl dipeptidase A; *other names*: serum converting enzyme; peptidase P; peptidyl carboxypeptidase I; carboxycathepsin; kininase II. A Cl^- -dependent zinc-containing dipeptidase present in lung, the brush borders of the proximal renal tubule, the endothelium of vascular beds, and plasma. It catalyses the release of a C-terminal dipeptide, -Xaa-|-Xbb-Xcc, when neither Xaa nor Xbb is proline, hence its conversion of angiotensin I to angiotensin II. It also acts on bradykinin. It is a transmembrane glycoprotein; two types (testis and somatic).

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

arise by alternative splicing and different transcriptional start points. Example, somatic type precursor (human): database code ACE_HUMAN, 1306 amino acids (149.55 kDa).

angiotensin converting enzyme inhibitors compounds that prevent the synthesis of angiotensin II from its precursor, angiotensin I, e.g. **captopril**, **enalapril**. Such inhibitors are used in the treatment of cardiac failure and hypertension.

angiotensinogen a plasma α_2 -globulin formed in liver and the precursor of **angiotensin I**, **II**, and **III**; it is similar to the **serpins**. Example, precursor from human: database code ANGT_HUMAN, 485 amino acids (53.09 kDa).

angiotensinogenase *see* **renin**.

angiotensin receptor any of several membrane proteins that bind angiotensin and mediate its intracellular effects. Two types of receptor, AT_1 and AT_2 , have been identified on the basis of their response to drugs. The AT_1 receptor (previously known as AII_1 or AII_a) has high affinity for **losartan**, and lower affinity for CGP42112A; the reverse is true for AT_2 receptors. The potency of endogenous ligands is angiotensin II > angiotensin III. In peripheral tissues, AT_1 receptors, but not AT_2 receptors, interact with G-proteins, inhibit adenylate cyclase, and activate phospholipases C and A_2 . AT_2 receptors may stimulate a phosphotyrosine phosphatase leading to inhibition of guanylate cyclase. Sulfhydryl reducing agents inhibit the binding of angiotensin II to AT_1 receptors but enhance its binding to AT_2 . Both types are G-protein-coupled seven-transmembrane domain proteins. Examples: AT_1 receptor, database code AG2R_BOVIN, 359 amino acids (41.09 kDa); AT_2 receptor, database code AG22_MOUSE, 363 amino acids (41.37 kDa).

angle 1 the inclination between two lines or planes: **plane angle**. It is dimensionless and given in radians (*symbol*: rad), degrees (*symbol*: °; $1^\circ = \pi/180$ rad); minutes (*symbol*: ') $1' = \pi/10800$ rad; or seconds (*symbol*: ") $1'' = \pi/64800$ rad. 2 the (three-dimensional) vertex of a cone: **solid angle**. It is dimensionless but may be expressed in **steradians** (*symbol*: sr).

angle head or **angle rotor** any rotor for a centrifuge in which the tubes containing the liquid are at a fixed angle to the axis of rotation. Compare **swinging bucket rotor**.

ångström or **Ångström** *symbol*: Å; a unit of length equal to 10^{-10} metres. It is used in atomic measurements and for wavelengths of electromagnetic radiation, and is temporarily approved for use with SI units. [After Anders Jonas Ångström (1814–74), Swedish physicist noted for his work on spectroscopy.]

angular velocity *symbol*: ω ; the rate of rotation of an object expressed as its rate of motion through an angle about an axis. It is expressed in rad s^{-1} .

anhydrase *see* **carbonic anhydrase**.

anhydride a derivative of a substance that yields the substance when combined chemically with water.

anhydro+ *symbol*: An; *prefix* (in a chemical name) denoting the loss of the elements of water from within one molecule or residue. The positions from which removal has occurred may be indicated by a pair of locants, e.g. 2,5-anhydro-D-gluconic acid.

anhydrosialidase EC 3.2.1.138; *other names*: anhydronneuraminidase; sialidase L; an enzyme that catalyses the hydrolysis of α -sialosyl linkages in N-acetylneuraminic acid glycosides, releasing 2,7-anhydro- α -N-acetylneuraminic acid.

anhydrosugar any intramolecular ether formed by the elimination of water from two (indicated) alcoholic hydroxyl groups of a single molecule of a monosaccharide, whether aldose or ketose; i.e. any anhydromonosaccharide whose formation does not involve the reducing group.

anhydrous devoid of water, especially (of a substance or material) lacking water of crystallization or adsorbed water, or (of an environment) very dry. *See also* **dehydration**.

animal pole *see* **vegetal pole**.

animal protein factor a name formerly used for a growth factor for livestock, found to be present in meat and fish but not



anion

in feeds of plant origin. It was later shown to be identical with anti-pernicious anemia factor, i.e. vitamin B₁₂.

anion an ion with a net negative charge; during electrolysis of a solution anions migrate towards the **anode**. *Compare cation*. — **anionic** *adj.*

anion exchanger or **anion-exchange resin** any **ion-exchange resin** that carries positively charged groups, e.g. quaternary ammonium groups, and binds anions.

anion gap (*in clinical chemistry*), the difference, in meq L⁻¹, between the concentrations of serum sodium plus potassium and of the serum chloride plus bicarbonate; i.e.

$$\text{anion gap} = [\text{Na}^+ + \text{K}^+] - [\text{Cl}^- + \text{HCO}_3^-].$$

Normal values are 12 ± 2 meq L⁻¹. Variations in the anion gap provide a useful insight into the acid-base status of a subject.

anionic detergent a detergent in which the polar part of the molecule carries a negative charge. *Compare nonionic detergent*.

anisotropic describing a medium in which physical properties, such as the velocity of electromagnetic radiation transmission, electrical or heat conductivities, or compressibility, have different values when measured along axes in different directions. — **anisotropically** *adv.*; **anisotropy** or **anisotropism** *n.*

anisotropic motion rotation about a single axis, especially of a molecule or a **spin label**.

ankyrin or **band 2.1 (protein)** or **nexin** or **syndein** a 200 kDa cytoskeletal protein that attaches other cytoskeletal proteins to integral membrane proteins. It is bound tightly to the cytoplasmic surface of the human erythrocyte membrane, to which it attaches the cytoskeletal protein **spectrin**. Example from human: database code ANK1_HUMAN, 1880 amino acids (206.14 kDa). [After the Greek *ankyra*, anchor].

annealing 1 the renaturation of heat-denatured nucleic acids or proteins by slow cooling. 2 the formation of hybrid double-stranded nucleic-acid molecules by the slow cooling of a mixture of denatured, single-stranded nucleic acids. It is used to detect complementary regions of different nucleic-acid molecules. 3 the slow regulated cooling, especially of metals or glasses, to relieve the stresses caused by heating or other treatments. — **annealer** *n.*

annexin any of a ubiquitous family of Ca²⁺-dependent phospholipid-binding proteins. They have been isolated from organisms as diverse as mammals, *Drosophila melanogaster*, and *Hydra vulgaris*. Characteristically they have a low intrinsic affinity for Ca²⁺ (K_d 25–1000 μM), although annexin VI has $K_d \approx 1$ μM, but in the presence of phospholipid the affinity for Ca²⁺ increases up to 100-fold. The Ca²⁺- and phospholipid-binding properties that typify the family derive from a region of common primary structure (identity with any other member of the family is 40–60%) in a conserved 34 kDa C-terminal region (the 'core'); each different type of annexin has a unique N-terminal domain known as the 'tail'. The C-terminal region in all cases except one consists of four repeats of a ≈70 amino-acid sequence, in which there is a high homology region known as the **endonexin fold**, with a characteristic motif GXGTDE; the exception is annexin VI, which has eight repeats. The true physiological roles of the annexins remain unknown, but their main characteristics, exhibited variously by different annexins and reflected in the alternative names they acquired in the course of discovery, include: the ability to aggregate membranes and vesicles (all annexins); inhibition of phospholipase A₂ (especially I, but also II to VI); formation of ion channels (especially I and V to VII); anticoagulant properties (especially I, and VIII); and acting as substrates for tyrosine kinases (I and II) and protein kinase C (I, II, IV, and XII). Alternative names and database codes are given under the individual entries.

annexin I *other names*: lipocortin I; p35 (EGF receptor substrate); calpactin II; chromobindin 9; GIF; lipomodulin; macrocortin; renocortin. Example from rat: database code ANX1_RAT, 345 amino acids (38.65 kDa).

annexin II *other names*: calpactin I heavy chain; lipocortin II; p36 (pp60^{src} substrate); chromobindin 8; protein I; placental anticoagulant protein (PAP)-IV. Example from human: database code ANX2_HUMAN, 338 amino acids (38.43 kDa).

annexin III *other names*: lipocortin III; placental anticoagulant protein (PAP)-III; 35-α-calcimedlin; calphobindin III; inositol 1,2-cyclic phosphate hydrolase. Example from rat: database code ANX3_RAT, 324 amino acids (36.28 kDa).

annexin IV *other names*: endonexin I; protein II; 32.5K calelectrin; lipocortin IV; chromobindin 4; placental anticoagulant protein (PAP)-II; placental protein PP4-X; 35-β-calcimedlin; placental anticoagulant protein II. Example from human: database code ANX4_HUMAN, 318 amino acids (35.68 kDa).

annexin V *other names*: placental anticoagulant protein (PAP)-I; inhibitor of blood coagulation (IBC); lipocortin V; 35K cal-electrin; endonexin II; placental protein PP4; vascular anticoagulant (VAC)-α; 35-γ-calcimedlin; calphobindin I; anchoring CII. Example from human: database code ANX5_HUMAN, 319 amino acids (35.76 kDa).

annexin VI *other names*: p68; p70; 73K; 67K calelectrin; lipocortin VI; protein III; chromobindin 20; 67K calcimedlin; calphobindin II; synhibin. Example from human: database code ANX6_HUMAN, 672 amino acids (75.66 kDa).

annexin VII *another name for synexin*.

annexin VIII *other name*: vascular anticoagulant (VAC)-β. Example from human: database code ANX8_HUMAN, 327 amino acids (36.84 kDa).

annexin IX *other name*: *Drosophila melanogaster* annexin. Database code (fragment) from *D. melanogaster*: ANX9_DROME.

annexin X *other name*: intestine-specific annexin. Example from human: database code ANXI_HUMAN, 315 amino acids (35.29 kDa). A similar protein occurs in *Drosophila melanogaster*.

annexin XI *other names*: (sometimes, e.g. in rabbit) calcyclin-associated annexin (CAP); CAP-50. There are two variants (derived by alternative splicing). Example, variant I, bovine: database code ANXA_BOVIN, 503 amino acids (53.96 kDa).

annexin XII a member of the annexin family obtained from *Hydra attenuata* (*Hydra vulgaris*). It is of unknown function. Example from *H. vulgaris*: database code ANXC_HYDAT, 316 amino acids (35.07 kDa).

annihilation radiation electromagnetic radiation of wavelength 2.4 pm (energy 0.51 MeV) that is emitted as a pair of photons when a positron absorbed into matter reaches the end of its range and combines with an electron in mutual destruction.

anode 1 the electrode of an electrolysis cell or electrophoresis apparatus towards which anions of the electrolyte migrate under the influence of an electric field applied between the electrodes. 2 the positive electrode, pole, or terminal of such a cell or apparatus, or of a discharge tube, solid-state rectifier, or thermionic valve; the electrode by which electrons flow out of the device. 3 the negative pole or terminal of an electric (primary or secondary) cell or battery (of cells). *Compare cathode*. — **anodal** or **anodic** *adj.*; **anodally** or **anodically** *adv.*

anodic stripping voltammetry *abbr.*: ASV; an electrochemical procedure for estimating the concentration of metal ions in a solution. Exhaustive or partial (depending on the concentrations involved) deposition of the metal occurs on the cathode of an electrolytic cell at a constant applied electric potential (controlled potential deposition can be used for selectivity when mixtures are involved). The metal deposit is then anodically stripped by using a rapid sweep of voltage of opposite polarity and a peak-type current-voltage curve is obtained. The amount of metal deposited may be calculated coulometrically from the area under the peak, or by other means, and hence the concentration of metal ions in the initial solution may be determined.

anodic to during electrophoresis, migrating further towards



anolyte

the anode or less far towards the cathode than (the specified substance).

anolyte the electrolyte in immediate contact with the anode in isoelectric focusing.

anomalous dispersion 1 an X-ray crystallographic dispersion that occurs when the frequency of the X-rays used falls near an absorption frequency of an atom in the crystal. Anomalous dispersion techniques are frequently used in the determination of absolute configuration. 2 an optical rotatory dispersion that cannot be expressed by a simple one-term Drude equation.

anomer either of the two stereoisomers of the cyclic form of an aldose or a ketose, or an aldose or ketose derivative, dependent on the position of the free hydroxyl group belonging to the internal hemiacetal grouping. The anomer having the same configuration, in the Fischer projection, at the anomeric and the reference asymmetric carbon atom is designated α ; the other anomer is designated β . In correct usage in regard to carbohydrates, the anomeric and configurational descriptors must always be linked together; e.g. α -D-glucopyranose, not α -glucose.

anomeric 1 of or pertaining to an anomer. 2 or glycosidic of or pertaining to the free hydroxyl group belonging to the internal hemiacetal grouping in the cyclic form of a monosaccharide or monosaccharide derivative.

anomeric effect the tendency for an electronegative substituent at C-1 of a pyranose to assume the axial orientation, in contrast to the equatorial orientation predicted on steric grounds.

anorexia absence of desire for food.

anosmia loss of the sense of smell.

anoxia the absence or a deficiency of dioxygen. Compare hypoxia. —anoxic adj.

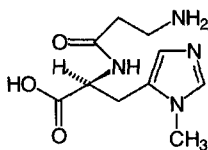
ANP abbr. for atrial natriuretic peptide. See natriuretic peptide.

ANPR abbr. for atrial natriuretic peptide receptor.

ANS abbr. for 1-anilino-8-naphthalene sulfonate; a common fluor used in extrinsic fluorescence studies of proteins.

ansamycin any antibiotic belonging to the rifamycin or the streptovaricin classes.

anserine *N*- β -alanil-*N*^ω-methyl-L-histidine; an isopeptide occurring in skeletal muscle and brain of some animals and humans (up to 20 mmol kg⁻¹). Compare carnosine.



Anson unit an old measure of the activity of a proteinase. It was based upon the extent to which a sample containing a particular proteinase can digest denatured hemoglobin, under standard conditions appropriate to the enzyme in question, with liberation of nonprecipitable material giving the same colour value as one millimole of tyrosine on reaction with a diluted Folin-Ciocalteu reagent. [After Mortimer Lewis Anson (1901–68), US protein chemist, who described it in 1958.]

ant+ a variant of anti+.

Antabuse proprietary name for disulfiram.

antagonism 1 opposition; counteraction. 2 the mutual opposition of an agonist and an antagonist. 3 the interference with, or inhibition of, the growth of an organism by one of another kind, as by competing for nutrients or producing an antibiotic substance.

antagonist 1 anything that antagonizes. 2 any agent, especially a drug or hormone, that reduces the action of another.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

agent, the **agonist**. Many act at the same receptor as the agonist. Such antagonists may be either surmountable or insurmountable, depending on prevailing conditions. Antagonism may also result from combination with the substance being antagonized (**chemical antagonism**), or the production of an opposite effect through a different receptor (**functional antagonism** or **physiological antagonism**), or as a consequence of competition for the binding site of an intermediate that links receptor activation to the effect observed (indirect antagonism). The term 'functional antagonism' is also used to describe a less well-defined category in which the antagonist interferes with other events that follow receptor activation. 3 a muscle that works in opposition to another (the agonist). See also **competitive antagonism**, **noncompetitive antagonism**. Compare **agonist**, **inverse agonist**, **synergist**. —antagonistic adj.; antagonistically adv.

antagonize or **antagonise** to act in opposition to; to counteract. —antagonized or antagonised adj.

antanamide a highly lipophilic homodetic cyclic decapeptide, cyclo(-Pro-Phe-Phe-Val-Pro-Pro-Ala-Phe-Phe-Pro-). It occurs in the death-cap fungus, *Amanita phalloides*, together with the **amatoxins** and **phallotoxins**. When administered to experimental animals (at 1 mg per kg of body weight), either before or at most a few minutes after a lethal dose of phallotoxin, it acts as an antidote to the toxin, apparently by competing for binding sites. *In vitro* it readily forms complexes with sodium and calcium ions, and can act as a sodium ionophore.

ANT-C abbr. for antennapedia complex; one of two major families of homeotic genes of *Drosophila* (the other being **BX-C**), mutation which affects head and thoracic segments.

ante+ prefix denoting before in time or position.

antergrade a variant of anterograde.

anteiso+ prefix denoting a branched-chain fatty acid having a methyl group at the *n*-3 position; e.g. 16-methyloctadecanoic acid, also named anteisonadecanoic acid.

antenna complex any of the light-harvesting pigment-protein complexes of photosynthetic organisms that funnel absorbed incident radiation to the photochemical reaction centres.

antennapedia complex see **ANT-C**.

anterograde or **antegrade** 1 moving forwards; extending towards the front. 2 moving along the axon of a nerve cell in a direction away from the cell body (as in anterograde transport of vesicles). Compare **retrograde**.

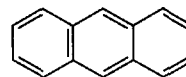
antheridiol a steroid hormone produced by water moulds (e.g. *Achlya* spp.) to regulate their sexual reproduction.

anthocyanidin any member of a group of water-insoluble red, blue, or violet polyhydroxylated 2-phenylbenzopyrylium compounds that occur as glycosides – anthocyanins – in plants. They comprise one class of the widespread group of flavonoids.

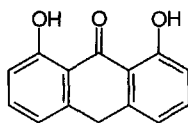
anthocyanin any member of a group of intensely coloured soluble glycosides of anthocyanidins that occur in plants. They are responsible for most of the scarlet, purple, mauve, and blue colouring in higher plants, especially of flowers.

antho-K amide L-3-phenyllactyl-Phe-Lys-Ala-NH₂; a neuropeptide from the sea anemone *Anthopleura elegantissima*.

anthracene a tricyclic aromatic compound, C₁₄H₁₀; it is a component of a number of **scintillation cocktails**.



anthralin 1,8-dihydroxyanthrone; a substance that inhibits **leukotriene** biosynthesis. It is used as an antipsoriatic drug.

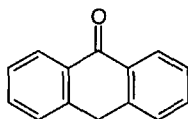


anthralin

anthranilate 2-aminobenzoate (anion, salt, or ester); an intermediate in the biosynthesis of tryptophan.

anthranilate synthase EC 4.1.3.27; *systematic name*: chorismate pyruvate-lyase (amino-accepting). An enzyme that catalyses the first step in tryptophan biosynthesis, in which chorismate and L-glutamine react to form anthranilate, pyruvate, and L-glutamate. It is a tetramer of two components, I and II: in the absence of II, I catalyses the formation of anthranilate using NH_3 rather than glutamine; II provides glutamine amidotransferase activity. Example from yeast: I database code TRPE_YEAST, 528 amino acids (59.70 kDa); II database code TRPG_YEAST, 484 amino acids (53.46 kDa).

anthrone 9(10*H*)-anthracenone; a compound used in the colorimetric determination of carbohydrates, especially hexoses, with which it gives a green colour.

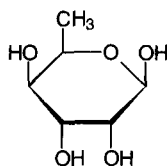


anti- or (sometimes before a vowel or *h*) **ant-** prefix 1 acting against, counteracting, or inhibiting. 2 opposite to (in direction). 3 converse or reverse of; complementary to.

anti- prefix (*obsolete*) denoting a *cis* isomer.

antiamoebin or **antiamebin** see **peptaibophol**.

antiarose the D isomer, 6-deoxy-D-gulose; occurs in glycosidic linkage in some **cardiac glycosides**.

 β -D-pyranose form

antibiotic 1 any of numerous substances of relatively low M_r produced by living microorganisms (and also certain plants) that are able selectively and at low concentrations to destroy or inhibit the growth of other organisms, especially microorganisms. Also included are the many semi- or wholly synthetic organic compounds with similar antimicrobial properties. Many are useful chemotherapeutic agents. 2 of or relating to an antibiotic.

antibody any glycoprotein of the **immunoglobulin** family that is capable of combining noncovalently, reversibly, and in a specific manner with a corresponding antigen. Antibodies frequently, but not always, counteract the biological activity of antigens. They are produced in higher animals by cells of the lymphoid cell series, especially plasma cells, in direct response to the introduction of immunogens or autocoupling haptens. Antibodies were originally identified as a class of serum proteins.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentid.com.

teins, and most are present in the γ -globulin fraction of the serum. In the ultracentrifuge most antibodies have a sedimentation coefficient of 7S, except for the pentameric IgM at 19S.

See also **cell-bound antibody**.

antibody-mediated hypersensitivity or **type II hypersensitivity** an immune response caused by the presence of antibody to cell-surface antigens and components of the extracellular matrix. These antibodies can sensitize the cells for antibody-dependent cytotoxic attack by **K cells** or **complement**-mediated lysis. Type II hypersensitivity is seen in the destruction of red cells in transfusion reactions and in hemolytic disease of the newborn.

anticholinesterase any agent, other than an antibody, that inhibits **cholinesterase** (def. 2). An example is **physostigmine**, used in the diagnosis and treatment of myasthenia gravis.

anticircular chromatography a technique of thin-layer chromatography in which the substances to be separated are applied just inside the perimeter of a circular thin-layer plate and are made to travel radially inwards by subsequent application of the developing solvent at the edge of the plate. It is very fast and is particularly useful for the resolution of substances whose R_f values approach unity. Compare **circular chromatography**.

anticlinal see **conformation** (def. 2).

anticoagulant 1 retarding or preventing coagulation. 2 any substance that retards or prevents coagulation, especially of blood or milk.

anticoagulate to treat (esp. blood) so as to prevent coagulation.

anticodon a triplet of nucleotides in any particular transfer RNA that is complementary to and therefore pairs with a given **codon** in messenger RNA. The pairing is antiparallel, 5'→3' in the messenger, 3'→5' in the anticodon. Thus the anticodon of 5'-ACG-3' is 3'-UGC-5'. However, to maintain the convention of writing sequences in the 5'→3' direction, the anticodon is thus written, but with a reverse arrow above: $\overleftarrow{\text{CGU}}$.

anticodon arm the part in the clover-leaf model of transfer RNA containing a loop bearing the anticodon.

anticooperativity negative **cooperativity** between binding sites on the same (macro)molecule by which binding of a ligand to one site makes the binding of a second ligand more difficult.

antidiuretic 1 describing any drug or other agent that decreases the excretion of urine. 2 an antidiuretic substance.

antidiuretic hormone *abbr.*: ADH; an alternative name (esp. in clinical chemistry) for **vasopressin**. It is so named from the hormone's most important biological action, viz. reduction of the rate of urine output by stimulation of the rate of water reabsorption by the kidney(s).

antidote a substance or other agent that limits or reverses the effects of a poison.

antifolate any of a class of drugs that interfere with the formation of tetrahydrofolate, and are used in the treatment of acute leukemia and choriocarcinoma. One of the best known is **methotrexate**.

antifreeze glycoprotein any of a group of glycoproteins found in the serum of certain cold-water fish, particularly nototheniids and chaenichthyids, which inhabit polar seas; e.g. *Trematomus borchgrevinki* and *Dissostichus mawsoni*. These glycoproteins depress the freezing point of aqueous solutions about 200–500 times as effectively as sodium chloride on a molar basis. They differ only in mass (10.5 kDa, 17 kDa, and 21.5 kDa) and are composed of repeating units of a diglycosyl tripeptide, Ala-Ala-Thr-*O*-disaccharide, in which the disaccharide consists of a β -galactosyl residue bound to an internal α -N-acetylgalactosaminyl residue.

antigen 1 (or **complete antigen**) any agent that, when introduced into an immunocompetent animal, stimulates the production of a specific antibody or antibodies that can combine with the antigen. It may be a pure substance, a mixture of substances, or particulate material (including cells or cell fragments). In this sense the term includes **agglutinin** and



antigen-antibody complex

allergen and is synonymous with **immunogen** (def. 1). **2** (or **incomplete antigen**) any substance that can combine with a specific antibody but is not an **immunogen** (def. 2), i.e. it is not by itself able to stimulate antibody production though it may do so if combined with a **carrier** (def. 8). In this sense the term includes **hapten**. — **antigenic adj.**

antigen-antibody complex or **immune complex** any specific macromolecular complex of antigen and antibody. It may be soluble, especially in **antigen excess**, or insoluble (as a precipitate) when antigen and antibody are present in optimal proportions. Components of **complement** may bind to antigen-antibody complexes. *See also* **immune complex-mediated hypersensitivity**.

antigen-binding capacity *abbr.*: ABC; the total amount of antibody in a preparation, e.g. serum, based on a determination of the amount of antigen bound by unit amount of the preparation.

antigen excess the condition when a mixture of antigen and antibody contains sufficient antigen to combine with all the combining sites of the specific antibody molecules, leaving some free uncombined antigenic determinants and resulting in the formation of soluble antigen-antibody complexes. This is the explanation for the type of 'precipitin' curve when increasing amounts of antigen are incubated with a fixed amount of antibody and the amount of precipitate is measured.

antigenic determinant any specific chemical structure within, and generally small in relation to, an antigen molecule that is recognizable by a combining site of an antibody, or T-cell or B-cell receptor, and at which combination takes place. It determines the specificity of the antibody-antigen reaction. Several different antigenic determinants may be carried by a single molecule of antigen. *See also* **epitope**.

antigenic drift any of the minor changes in the antigenicity of influenza virus that result from spontaneous mutation, with corresponding minor changes in the amino-acid sequence of the viral hemagglutinin. *Compare* **antigenic shift**.

antigenicity the capacity of an agent to stimulate the formation of specific antibodies to itself.

antigenic shift any of the major changes in the antigenicity of influenza virus that result from spontaneous mutation, with corresponding major changes in the amino-acid sequence of the viral hemagglutinin. *Compare* **antigenic drift**.

(antigen-independent) mitogenic factor a **lymphokine** that acts as a **mitogen** for lymphocytes.

antigen-presenting cell *abbr.*: APC; a cell, especially a macrophage or dendritic cell, that recognizes an antigen to be targeted for neutralization. It takes up the antigen and processes it, incorporating antigen fragments into its own membrane and presenting them in association with class II **major histocompatibility complex** molecules to T lymphocytes, which are then stimulated to mount a response.

antigibberellin any of the organic compounds that cause plants to grow with thick stems or with an appearance differing from that obtained with **gibberellin**. The effect can usually be reversed by gibberellin.

antiglobulin consumption test a serological method for demonstrating antibody (globulin) attachment to cells. It is based on showing that antiglobulin is removed from solution on addition of the cells in question.

antiglobulin test or **Coombs test** a test for the presence of incomplete **antibody** against red blood cells. It entails the addition of heteroantibody to immunoglobulin (i.e. antiglobulin) to cause the agglutination of red blood cells previously coated with nonagglutinating incomplete antibody. *Compare* **double-antibody method**. [After Robin Royston Amos Coombs (b. 1921), British immunologist and hematologist who (with others) described it in 1945.]

antihemophilic factor or **antihemophilic factor A** or **antihemophilic globulin** an alternative name for factor VIII; *see* **blood clotting factors**.

antihemophilic factor B an alternative name for factor IX; *see* **blood coagulation**.

antihemophilic factor C an alternative name for factor XI; *see* **blood coagulation**.

antihemophilic globulin an alternative name for **antihemophilic factor**.

antihistamine any drug or other agent that antagonizes an action of histamine on the body. Antihistamines are used in the treatment of immediate-type hypersensitivity.

antihormone any substance that acts to attenuate hormone-induced responses, regardless of the mechanism involved.

anti-idiotypic an antibody that reacts with **epitopes** on the V region of the T-cell receptor.

anti-inducer any compound that inhibits **operon** induction by competing for inducer binding to free repressor and stabilizing the repressor-operator complex.

antilipolytic 1 describing any agent that inhibits **lipolysis**. 2 an antilipolytic agent.

antilipotropic 1 describing any agent that is active metabolically in deflecting methyl groups from the synthesis of choline. 2 an antilipotropic agent.

antilogarithm *abbr.*: antilog; *symbol*: (decadic) $\lg^{-1}$; (napierian) $\ln^{-1}$; the number that is represented by a **logarithm**.

antilymphocyte globulin the globulin fraction of an **antilymphocyte serum**.

antilymphocyte serum *abbr.*: ALS; any serum containing antibodies to lymphocytes. It is used as a selective immunosuppressive agent.

anti-messenger DNA *abbr.*: amDNA; an alternative name for **complementary DNA**.

antimetabolite any substance that inhibits the utilization of a metabolite, e.g. by acting antagonistically through competition in a transport system or at a key enzyme site.

antimicrobial 1 describing a drug, antibiotic agent, physical process, radiation, etc. that is inimical to microbes. 2 any substance having antimicrobial properties.

antimicrobial spectrum a list of the types of microbes against which an antimicrobial agent is effective.

antimitotic 1 describing any agent that decreases the rate of mitosis. 2 an antimitotic agent.

antimonial 1 of, relating to, or containing antimony. 2 a drug containing antimony.

antimorph 1 a mutant gene producing an effect opposite to the wild-type gene at the same locus. *Compare* **allele**. 2 *obsolete term for an enantiomorph*.

antimutagen any agent (often a purine nucleoside) that can decrease the rate of induced or spontaneous mutation.

antimutator gene any mutant gene that decreases the mutation rate.

antimycin any member of a group of antibiotics produced by *Streptomyces* spp. They associate strongly with mitochondria and block the passage of electrons from cytochrome *b* to cytochrome *c*₁; as little as 7 pmol antimycin A₁ per gram of mitochondrial protein is effective. They are used experimentally to distinguish between events in the earlier and later parts of the electron-transport chain.

antineoplastic 1 describing any drug or other treatment that prevents or limits the growth of a neoplasm. 2 an antineoplastic agent.

antinuclear factor or **antinuclear antibody** any antibody that reacts with a component of the cell nucleus.

antioxidant or **antioxidant** any substance, often an organic compound, that opposes oxidation or inhibits reactions brought about by dioxygen or peroxides. Usually the antioxidant is effective because it can itself be more easily oxidized than the substance protected. The term is often applied to components that can trap **free radicals**, such as α -tocopherol (*see* **vitamin E**), thereby breaking the chain reaction that normally leads to extensive biological damage. The mechanism by which the free-radical scavenger is discharged of its free radical is not clear, but it is likely that ascorbate, glutathione, and,

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



antiparallel

ultimately, NAD⁺ and the **respiratory chain** are involved. Compounds such as di-*tert*-butyl-*p*-cresol act as antioxidants and are often added to protect labile compounds during storage or incubation. *See also* **quinhydrone**.

antiparallel 1 (*in biochemistry*) (describing of a pair of parallel linear structures, such as two polynucleotide or polypeptide chains, having directional polarity or asymmetry in opposite directions. **2** (*in quantum physics*) (of the spins of a pair of electrons occupying the same atomic or molecular orbital) described by different spin quantum numbers. *Compare* **parallel** (def. 2). —**antiparallelism** *n*.

antiperiplanar *see* **conformation**.

antipernicious anemia factor *former name for* **vitamin B₁₂**.

antiport or **countertransport** the process of coupled solute translocation in which two solutes equilibrate across an osmotic barrier, the translocation of one solute being coupled to the translocation of the other in the opposite direction. *Compare* **symport**.

antiporter any substance or structural feature, but usually a protein, that promotes the exchange diffusion of two specific substances across a membrane by **antiport**. The convention for denoting such a protein (e.g. a tetracycline-resistance protein) is tetracycline/H⁺ antiporter.

antirachitic 1 describing any agent that opposes or prevents the development of **rickets**. **2** any agent with antirachitic properties.

antirachitic factor or **antirachitic vitamin** *former term for* any substance with **vitamin D** activity.

antirepressor any gene product that acts to decrease repression by a repressor of gene expression.

antirestriction the process whereby a host bacterium is prevented from inactivating foreign duplex DNA; i.e. the prevention of **restriction**. It is commonly achieved by **host-controlled modification** of the DNA, but in certain cases a specific bacteriophage-coded **antirestriction protein** is produced that directly inhibits the host's **restriction endonuclease**.

antirestriction protein *see* **antirestriction**.

anti-RNA single-stranded RNA complementary to mRNA. *See* **antisense RNA**.

antiscorbutic 1 describing an agent that opposes or prevents the development of **scurvy**. **2** any agent with antiscorbutic properties.

antiscorbutic factor or **antiscorbutic vitamin** *former term for* any substance with **vitamin C** activity.

antisense DNA *see* **noncoding strand**.

antisense oligonucleotide an oligoribonucleotide, or an analogue of one, having properties similar to those of **antisense RNA**.

antisense RNA or **messenger-RNA-interfering complementary RNA** (*abbr.*: micRNA) an artificial single-stranded RNA molecule complementary in sequence to all or part of a molecule of messenger RNA or of some other specific RNA transcript of a gene. It can therefore hybridize with the specific RNA and so interfere with the latter's actions or reactions. *Compare* **complementary RNA**.

antisense strand *see* **noncoding strand**.

antiseptis the elimination of pathogenic and other microorganisms using chemical or physical methods; the promotion of **asepsis**.

antiseptic 1 of, relating to, or bringing about **antiseptis**. **2** any chemical agent used in antiseptis, especially one that can be safely applied to the skin.

antiserum (*pl.* antisera or (*esp. US*) antisera) serum that contains a high level of antibodies, usually antibodies against specific antigens.

antisigma factor a protein that interferes with the recognition of initiation sites by the **sigma factor** of DNA-dependent RNA polymerase.

antistasin the generic name for a family of potent inhibitors of factor Xa. Antistasins are isolated from leeches, and the name

was first given to a 119-amino-acid protein present in salivary gland of the Mexican leech *Haementeria officinalis*; the protein has 20 cysteine residues, all in disulfide bridges (probably characteristic of the family). It appears to present a **bait**-like structure to factor Xa, which cleaves it at Arg³⁴. The 3-D structure has been partially solved, indicating a reactive site in domain I, sequence CRKTC. Antistasins have been considered as possible chemotherapeutic anticoagulants but have strong metastatic activity. Example, **ghilanten** from the Amazonian leech *H. ghilltiani*: database code ANTA_HAEGH, 119 amino acids (13.30 kDa).

antistreptolysin-O *abbr.*: ASO; antibody formed against streptolysin-O. *See* **streptolysin**.

antisuppressor any mutation that diminishes the effects of a suppressor of gene expression.

antitemplate strand *see* **coding strand**.

antiterminator (or **antitermination factor**) a protein that prevents the termination of RNA synthesis. It is found as a regulatory device in, e.g., phage lambda, enabling a terminator to be masked from RNA polymerase so that distal genes can be expressed. Antiterminators are associated with switches from early to middle expression following induction (gene N product, database code REGN_LAMBD, 107 amino acids (12.24 kDa)) and from early to late expression (gene Q product, database code REGQ_LAMBD, 207 amino acids (22.45 kDa)). The N system is referred to under **nus**. Antitermination is an important mechanism in the reproduction of some bacteriophages, allowing RNA polymerase to read through the terminator to the immediate early genes of the phage and thus initiate production of the mature phage with consequent cell lysis.

antithrombin III or **heparin cofactor** or **factor Xa inhibitor** a 62 kDa protein inhibitor of serine proteinases, normally present in plasma, that reacts with factor IIa (thrombin), and also factors IXa, Xa, XIa, and XIIa to form inert 1:1 stoichiometric complexes. The formation of complexes is slow in the absence of heparin but very fast in its presence. Example from *Bos taurus*: ANT3_BOVIN, 433 amino acids (49.13 kDa).

antitoxin any antibody to a **toxin** of a (micro)organism that neutralizes the toxin both *in vitro* and *in vivo*.

α_1 -antitrypsin or **α_1 -proteinase inhibitor** a major α_1 -globulin, normally present in human plasma, that inhibits many serine proteinases, including trypsin but especially elastase, and is the archetypal **serpin**. Its concentration rises dramatically in response to infection or tissue injury (*compare* **acute-phase protein**). A 51 kDa glycoprotein, its prime physiological role is to inhibit elastase released by neutrophil leukocytes. In humans, its synthesis is controlled by an autosomal allelic system and over 20 different phenotypes have been classified, one of which is associated with emphysema of early onset; in particular, the so-called Z mutant (Glu $\rightarrow$ Lys at position 342) is associated with a decreased level of α_1 -antitrypsin in the blood. Smoking tends to cause oxidation of Met³⁵⁸, by increasing hydrogen peroxide formation by lung neutrophils, and this leads to a less active α_1 -antitrypsin and hence reduced inhibition of elastase. Excessive activity of elastase may damage lung tissue, resulting in emphysema. Thus smoking is associated with emphysema of early onset. Individuals who are homozygous for the Z mutant and who also smoke suffer decreased levels of a less active anti-elastase and have a high probability of developing emphysema. The protein has two chains; examples (human): chain A, database code NRL_7APIA, 339 amino acids (38.09 kDa); 3-D structure known; chain B, database code NRL_7APIB, 36 amino acids (4.13 kDa); 3-D structure known.

antitussive 1 relieving or suppressing coughing. **2** an agent with antitussive properties.

antivitamin any agent that antagonizes a specific vitamin. Sometimes it is a structural analogue of the vitamin and a competitive inhibitor of the vitamin's action.



antixerophthalmic factor

antixerophthalmic factor former term for any substance with vitamin A activity.

antioxidant a variant spelling of **antioxidant**.

antrin a decapeptide, isolated from rat gastric antrum. It has the sequence

Ala-Pro-Ser-Asp-Pro-Arg-Leu-Arg-Gln-Phe

and is identical to residues 25–34 of preprosomatostatin (see **somatostatin**).

anucleate or **anucleated** lacking a **nucleus**.

anucleolate lacking a **nucleolus**.

AOX abbr. for **alcohol oxidase**.

AP see **AP endonuclease**, **AP protein**.

6-APA abbr. for 6-aminopenicillanic acid.

apamin a neurotoxin comprising about 2% of the dry mass of the venom of the honey bee (*Apis mellifera*). It is a highly basic octadecapeptide amide, structure

CNCKAPETALCAKKCEEH-NH₂,

containing two disulfide bridges (between Cys residues 1 and 11, and 3 and 15), and is the smallest neurotoxic peptide known. Compare **melittin**.

APC abbr. for 1 activated protein C. 2 antigen-presenting cell.

APC the human gene for adenomatous polyposis coli protein. The gene is a tumour suppressor, and mutations are associated with familial adenomatous polyposis, Gardner's syndrome, etc. and contribute to colorectal cancer. The protein product is of unknown function and has serine phosphorylation, glycosylation, and myristoylation sites. It is concentrated in the basolateral region of crypt epithelial cells. Product database code: APC_HUMAN, 2843 amino acids (311.32 kDa).

Ape symbol for a residue of the α -amino acid L- α -aminovaleric acid, L-2-aminopentanoic acid.

AP endonuclease abbr. for apyrimidinic or apurinic nuclease; any of various DNA repair enzymes that make breaks at sites generated by DNA glycosylase. There are three types: (1) **deoxyribonuclease V**; (2) **exodeoxynuclease III** (endonuclease activity) see **deoxyribonuclease III**; and (3) DNA-(apurinic or apyrimidinic site) lyase, EC 4.2.99.18; other names: AP lyase; AP endonuclease class I; endodeoxyribonuclease (apurinic or apyrimidinic); deoxyribonuclease (apurinic or apyrimidinic); *Escherichia coli* endonuclease III; phage-T4 UV endonuclease; *Micrococcus luteus* UV endonuclease. They catalyse a reaction in which the C–O–P bond 3' to the apurinic or apyrimidinic site in DNA is broken by a β -elimination reaction, leaving a 3'-terminal unsaturated sugar and a product with a terminal 5'-phosphate. Example: endonuclease III from *E. coli*; this binds a Fe₄S₄ cluster, which is not important for the catalytic activity but which is probably involved in the proper positioning of the enzyme along the DNA: database code END3_ECOLI, 211 amino acids (23.53 kDa).

A peptide the octadecapeptide cleaved from the α chains of a fibrinogen molecule when it is converted into fibrin by the proteolytic action of thrombin. Compare **B peptide**.

aperture (in physics) 1 an opening in an optical or other similar instrument that controls the amount of radiation entering or leaving it. 2 the diameter of the refracting surface of a lens or reflecting surface of a mirror. See also **f-number**.

aphidicolin a potent antiviral and antimitotic agent, first isolated as an antibiotic from the fungus *Cephalosporium aphidicola*. It inhibits eukaryotic DNA polymerase.

Api symbol for D-apiose.

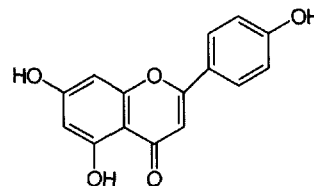
apical surface the surface of an epithelial cell that faces the lumen.

apigenin 4',5,7-trihydroxyflavone; the aglycon of **apiin**. It inhibits cyclooxygenase and estrogen synthase. See **D-apiose**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

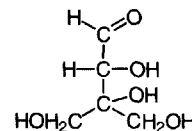
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



apigenin

apiin apigenin-7-apiosylglucoside, apioside; the name is derived from *Apium* (celery) from which apiin is isolated. See also **apigenin**, **apiose**.

apiose symbol: Api; the D isomer, 3-C-(hydroxymethyl)-D-glycero-aldotetrose is a branched carbon chain sugar that occurs widely in plants either in glycosides or polysaccharides. It is a component of **apiin**.



D-apiose

aplasia incomplete or deficient development of an organ or tissue.

aplastic 1 not showing normal growth or change in structure. 2 relating to or showing **aplasia**.

Apm symbol for a residue of the α -amino acid α -aminopimelic acid, 2-aminoheptanedioic acid.

A₂pm or **Dpm** symbol for the di(α -amino acid) α,ϵ -diaminopimelic acid, 2,6-diaminoheptanedioic acid.

APMSF *p*-amidinophenylmethylsulfonylfluoride; a specific irreversible inhibitor of trypsin-like serine endopeptidases, with greater inhibitory activity than **phenylmethylsulfonylfluoride**.

apo+ prefix denoting detached, separate. It is used especially of enzymes and other proteins. For example, apocboxylase is the protein part of the holoenzyme carboxylase, i.e. it lacks the covalently bound biotin residue(s). Apoferritin is a 450 kDa protein with 24 identical subunits, that combines with ferric hydroxide-phosphate to form **ferritin**. See also **apoenzyme**.

Apo(a) or **lipoprotein(a)** a serine endopeptidase (EC 3.4.21.-) related to **plasminogen**, having 37 repeats of plasminogen **kringle** domains and capable of cleaving **fibrinogen**. It is covalently linked (by disulfide bond) to apolipoprotein B-100. Its plasma concentration is related to the incidence of atherosclerosis. Example from human: database code APOA_HUMAN, 4577 amino acids (500.77 kDa).

apocrine 1 describing a type of secretion that entails loss of part of the cytoplasm of the secretory cell. 2 describing a gland made up of such cells. Compare **eccrine**, **holocrine**, **merocrine**.

apoenzyme the protein part of an enzyme that forms a catalytically active enzyme (holoenzyme) when combined with an activator, a coenzyme, or a prosthetic group. It determines the specificity of the catalytic system.

apoferritin see **ferritin**.

apolar 1 lacking poles, especially of nerve cells. 2 **nonpolar**.

apolar interaction any of the entropy-driven interactions between nonpolar parts of molecules when in polar solvents. It is important in maintaining the structure of protein molecules in polar solvents, in antigen-antibody interactions, in virus-protein and enzyme-subunit aggregation and disaggregation, and in determining enzyme specificity.



apolipophorin

apolipophorin a protein component of **lipophorin**. This protein increases the lipid-carrying capacity of lipophorin by covering the expanding hydrophobic surface resulting from diacylglycerol uptake. Example (precursor) apolipophorin IIIB from *Locusta migratoria*: database code APL3_LOCOMI, 179 amino acids (19.09 kDa).

apolipoprotein the protein component of any lipoprotein, but especially of plasma lipoproteins. The latter are designated by the abbreviation 'Apo', e.g. ApoA (for apolipoprotein A), ApoB, etc., and are subdivided into subtypes, e.g. ApoA-I for apolipoprotein A, subtype I. See following entries for individual types of apolipoprotein.

apolipoprotein A-I *abbr.*: ApoA-I; the major protein of plasma **high-density lipoprotein** (HDL), also found in chylomicrons. It is synthesized in the liver and small intestine, and participates in the reverse transport of cholesterol from tissues to the liver by promoting cholesterol efflux from tissues; it acts as a cofactor for **lecithin-cholesterol acyltransferase**. Deficiency of ApoA-I is associated with **Tangier disease**. Example (precursor) from human: database code APA1_HUMAN, 267 amino acids (30.74 kDa).

apolipoprotein A-II *abbr.*: ApoA-II; an apolipoprotein associated with **high-density lipoprotein**, which it stabilizes. It is a homodimer (disulfide linked). Example from human: database code APA2_HUMAN, 100 amino acids (11.62 kDa).

apolipoprotein A-IV *abbr.*: ApoA-IV; an apolipoprotein that is a major component of **high-density lipoprotein** and chylomicrons. It is involved in the catabolism of chylomicrons and very-low-density lipoprotein, being required for efficient activation of **lipoprotein lipase** by apolipoprotein C-II. It is a potent activator of **lecithin-cholesterol acyltransferase**. Example from human: database code APA4_HUMAN, 396 amino acids (45.32 kDa).

apolipoprotein B *abbr.*: ApoB; a major protein constituent of chylomicrons, very-low-density lipoprotein, and low-density lipoprotein. It exists in two forms, B-100 (amino acids 28-4563 of the sequence) and B-48 (amino acids 28-2179); the latter is of intestinal origin and derived from editing of mRNA to form an internal stop codon. Excessive plasma levels occur in familial **hypercholesterolemia**. Example from human: database code APB_HUMAN, 4563 amino acids (514.97 kDa).

apolipoprotein C-I *abbr.*: ApoC-I; a component of chylomicrons and very-low-density lipoprotein (VLDL), synthesized mainly in liver. It modulates the interaction of apolipoprotein E with β -migrating VLDL and inhibits binding of β -VLDL to low-density lipoprotein receptor-related protein. Example from human: database code APC1_HUMAN, 83 amino acids (9.32 kDa).

apolipoprotein C-II *abbr.*: ApoC-II; a component of very-low-density lipoprotein (VLDL) and an activator of several triacylglycerol lipases. It has a reversible association with plasma chylomicrons, VLDL, and high-density lipoprotein. Deficiency results in hypertriglyceridemia, xanthomas, and increased risk of pancreatitis and early atherosclerosis (hyperlipoproteinemia, type IB). Example from human: database code APC2_HUMAN, 101 amino acids (11.27 kDa).

apolipoprotein C-III *abbr.*: ApoC-III; an apolipoprotein that constitutes 50% of the protein fraction of very-low-density lipoprotein. It binds to sugar and sialic-acid residues and inhibits lipoprotein and hepatic lipases. Example from human: database code APC3_HUMAN, 99 amino acids (10.84 kDa).

apolipoprotein D *abbr.*: ApoD; an apolipoprotein that occurs in a macromolecular complex with **lecithin-cholesterol acyltransferase** and in high-density lipoprotein. It is a **lipocalin** of known 3-D structure, probably involved in the transport and binding of bilin. The homodimer shows wide tissue distribution. Example from human: database code APD_HUMAN, 189 amino acids (21.25 kDa).

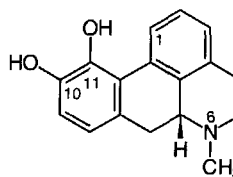
apolipoprotein E *abbr.*: ApoE; an apolipoprotein that constitutes 10-20% of very-low-density lipoprotein. It mediates

binding, internalization, and catabolism of lipoprotein particles and is a ligand for the low-density lipoprotein (ApoB/E) receptor and for the ApoE receptor (chylomicron remnant) of hepatic tissues. It is present in brain and may have a role in cholesterol transport. *O*-Glycoside and sialic-acid residues occur, but it is de-sialylated in plasma. It has nine allelic variants and two of these are associated with hyperlipoproteinemia type III. The 3-D structure is known; it contains a heparin-binding motif. Example from human: database code APE_HUMAN, 317 amino acids (36.11 kDa).

apolipoprotein H *abbr.*: ApoH; *other names*: β_2 -glycoprotein II precursor; APC inhibitor (*see* **activated protein C**). An apolipoprotein that binds to anions such as heparin and phospholipids. It may prevent activation of the intrinsic blood coagulation cascade by binding to phospholipids on the surface of damaged cells. *See also* **amyloid A protein**.

apolipoprotein receptor *see* **LDL receptor**, **VLDL receptor**.

apomorphine $6\alpha\beta$ -aporphine-10,11-diol; 5,6,6a,7-tetrahydro-6-methyl-4*H*-dibenzo[*de,g*]quinoline-10,11-diol; a synthetic opiate. The (*R*)-enantiomer is a potent dopamine receptor agonist, noted for its powerful emetic effects.



apoplast the nonliving part of plant tissue, external to the plasmalemma and composed of the cell walls, the intercellular spaces, and the lumen of dead structures such as xylem cells.

apoprotein the protein part of a **conjugated protein**.

apoptosis cell death may occur by accident, by cell **necrosis**, or by an intracellular controlled process characterized by a condensation and subsequent fragmentation of the cell nucleus during which the plasma membrane remains intact. The term apoptosis may be used broadly to encompass all forms of such normal or pathological cell death or may be confined to those processes involving morphological changes such as occur in normal animal development. A cascade of **caspases**, (**cysteine endopeptidases** that cleave at specific aspartic residues hence the name) is involved. In this cascade, caspases cleave procaspases to form additional caspases. Some of the caspases thus formed cleave other proteins, including deoxyribonucleases. [From **apo+** + Greek *ptosis*, falling.]

aporepressor the protein component of a complex that can act to inhibit gene expression. *See also* **repressor**.

apparent describing a physical quantity, especially an association constant, dissociation constant, equilibrium constant, etc., that has been determined under particular experimental conditions and hence is not the true (i.e. thermodynamic) constant. *See* **equilibrium constant**.

apparent exchangeable mass a term used in tracer kinetics in whole animals for a mass obtained by dividing the quantity of administered radioactivity that is retained by the specific activity of the tracer in the (blood) plasma. It equals the total exchangeable mass when the specific activities are equal throughout the system. *Compare* **compartment**.

apparent molecular weight *former name for* **apparent relative molecular mass**.

apparent relative molecular mass *relative molecular mass* that has been determined by a method that does not make allowance for any nonideality of the system.

apple domains sequences of 90 or 91 amino-acid residues that contain three highly conserved disulfide bonds linking the first and sixth, second and fifth, and third and fourth cysteine



applicator

residues present in each repeat. They can be drawn in the shape of an apple. Four are present in human kallikrein and in human coagulation factor XI. *See also* **Appendix E**.

applicator any device or instrument, such as a comb, rod, spatula, or tube, for applying samples in a reproducible manner to a small defined area, e.g. in chromatography, or for applying something to a particular part of the body.

approach to (sedimentation) equilibrium method or **Archibald method** a method for determining the relative molecular masses of (macro)molecules by sedimentation in an ultracentrifuge, by measuring the depletion of the macromolecules from the meniscus and their accumulation at the cell bottom. Its advantage is that only very short periods of sedimentation are required. [After William James Archibald (1912–)]

AP protein *abbr.* for activator protein, one of two **transcription factors**; AP1, a product of *c-jun* (*see* **JUN**), interacts with Fos (product of *c-fos*) to form a dimer. It harbours two functional domains, one responsible for DNA binding and one for binding to Fos, and includes a leucine zipper. Example from human: database code TAP1_HUMAN, 331 amino acids (35.63 kDa). AP2 interacts with inducible viral and cellular enhancer elements to stimulate transcription of selected genes. It contains the consensus binding sequence CCCAGGC and a helix-span-helix motif. Example from human: database code TAP2_HUMAN, 437 amino acids (48.01 kDa).

A₂pr or **Dpr** *symbol* for a residue of the α,β -diamino acid L- α,β -diaminopropionic acid; L-2,3-diaminopropanoic acid.

aprotic describing a substance, especially a solvent, that is unable to act as a proton acceptor or donor.

aprotinin a proteinase inhibitor derived from animal tissues. It is a basic polypeptide of known sequence of 58 amino-acid residues (≈ 6.5 kDa) that exists as a dimeric aggregate. It is stable in acid or neutral media and at high temperature, and it inhibits kallikrein, trypsin, chymotrypsin, plasmin, and, to a lesser extent, papain. *See also* **Trasylo**.

APRT *abbr.* for adenine phosphoribosyltransferase.

APS *abbr.* for adenosine 5'-phosphosulfate.

APT *abbr.* for alum-precipitated toxoid.

aptitude (*in microbiology*) the specific physiological state of a **lysogenic bacterium** that enables it, upon induction, to react so as to produce infectious bacteriophage particles.

apud or **APUD** *abbr.* for amine precursor uptake and decarboxylation; designating any of a series of cells that produce peptides and/or amines active as hormones or neurotransmitters, share a spectrum of cytochemical qualities, and appear in many cases to have a common embryological origin. They are thought to include the A, B, and D cells of the endocrine pancreas, the C cells of the thyroid, and the various hormone-producing cells of the pituitary and the gut.

apudoma any tumour of an **apud** cell, e.g. a gastrinoma, glucagonoma, insulinoma, or VIPoma.

apurinic acid a DNA derivative in which all the purine bases have been removed by selective hydrolysis, leaving the pyrimidine-deoxyribose bonds and the phosphodiester bonds of the backbone intact. It results from the gentle acid hydrolysis of DNA at pH 3.0.

apurinic endonuclease *see* **AP endonuclease**.

apyrase EC 3.6.1.5; *other names*: ATP-diphosphatase; adenosine diphosphatase (*abbr.*: ADPase); a Ca^{2+} -requiring enzyme that catalyses the hydrolysis of ATP to AMP and 2 orthophosphate. It also hydrolyses other nucleotides, especially ADP to AMP and orthophosphate. It was originally identified in plants, especially potatoes, but enzymes of this type have also been found in animals, e.g. as a surface enzyme on lymphoid cells identical to CD39, and as a secreted enzyme of the yellow-fever mosquito, *Aedes aegypti*. It is used in platelet incubations to destroy ADP. Example from *A. aegypti*: database code APY_AEDAE, 562 amino acids (62.73 kDa).

aprimidinic acid a DNA derivative from which the pyrimidine bases have been removed chemically, leaving the

purine-deoxyribose bonds and the phosphodiester bonds of the backbone intact.

aprimidinic endonuclease *see* **AP endonuclease**.

aq (*in chemistry*) *abbr.* for aqueous.

aqua+ *comb. form* 1 denoting acting on or reacting with water. 2 denoting presence of water as a neutral ligand in a coordination entity, e.g. aquacobalamin. The number of water molecules per entity is indicated by an approximate multiplying prefix, e.g. hexaquaachromium trichloride.

aquacide a substance that when added to a mixture removes water molecules, usually by forming a compound with the elements of water. —**aquacidal** *adj.*

aquaporin any of a class of **CHIPS** that are specific for the transfer of water. Example, specific for erythrocytes and kidney proximal tubule, from human: database code AQP1_HUMAN, 269 amino acids (28.49 kDa).

aqueous 1 of, having the characteristics of, or relating to water. 2 dissolved in or mixed with water. 3 or **aqueous humour** the fluid occupying the space between the crystalline lens and the cornea of the eye.

aquo+ *comb. form* 1 containing one or more water molecules in a coordination complex. 2 (*in chemical nomenclature*) derived from water.

aquo-ion a complex ion containing one or more water molecules held by coordination.

Ar *symbol* for 1 argon. 2 aryl.

Ara *symbol* for arabinose.

ara a plant gene related to **ras**.

araBAD an operon characterized in *Escherichia coli* and encoding L-ribulokinase, L-arabinose isomerase, and L-ribulose 5-phosphate 4-epimerase, three of the enzymes that metabolize arabinose. *See also* **araC**.

Arabidopsis thaliana a small Cruciferous plant (common wall cress) widely used in plant genetic research because of its small genome and very short generation time.

arabin+ a variant (*before a vowel*) of **arabino+**.

arabinan a branched homopolymer of L-arabinose found in plant pectins.

arabinaric acid *see* **aldaric acid**.

arabino+ or (*before a vowel*) **arabin+** *comb. form* indicating arabinose.

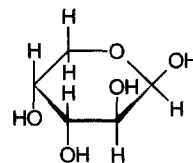
arabino- *prefix* (*in chemical nomenclature*) indicating a chemical compound containing a particular configuration of a set of three (usually) contiguous >CHOH groups, as in the acyclic form of D- or L-arabinose. *See* **monosaccharide**.

arabinofuranosylcytosine cytosine arabinoside; *see* **cytarabine**.

arabinogalactan any member of a subgroup of plant hemicelluloses that contain arabinose and galactose and are particularly abundant in larch.

arabinonucleoside the 1- β -D-arabino-*N*-furanoside of a purine or pyrimidine base. It is a structural analogue of a **riboside**.

arabinose *arabino*-pentose; a monosaccharide. L-Arabinose occurs free, e.g. in the heartwood of many conifers, and in the combined state, in both furanose and pyranose forms, as a constituent of various plant hemicelluloses, bacterial polysaccharides, etc. D-Arabinose occurs to some extent in plant glycosides and is a constituent of the **arabinonucleosides**.



β -D-arabinose

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

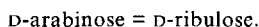
Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.



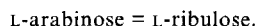
D-arabinose isomerase

D-arabinose isomerase EC 5.3.1.3; *systematic name*: D-arabinose ketol-isomerase. An enzyme that catalyses the reaction:



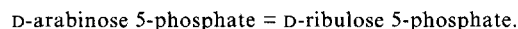
See also **L-arabinose isomerase**.

L-arabinose isomerase EC 5.3.1.4; *systematic name*: L-arabinose ketol-isomerase. An enzyme that catalyses the reaction:



It initiates the metabolism of L-arabinose. Example from *Escherichia coli*: database code ARAA_ECOLI, 500 amino acids (55.98 kDa). See also **araBAD**, **D-arabinose isomerase**.

arabinose-5-phosphate isomerase EC 5.3.1.13; *systematic name*: D-arabinose-5-phosphate ketol-isomerase. An enzyme that catalyses the reaction:



arabinoxide any glycoside derived from arabinose.

arabinoxylan any of various neutral polysaccharides that occur in association with acidic polysaccharides in cereal gums. They consist of a chain of (1→4)-linked β-D-xylan units with single α-L-arabinofuranosyl groups in (1→3) linkage.

arabitol or **arabinitol** *arabino*-pentitol; the pentitol formally derived by reduction of the aldehyde group of arabinose or lyxose. The D enantiomer is present in lichens and mushrooms.

araC the gene for the AraC protein, which acts as a regulator of expression of the *araBAD* operon and of *araC* itself. AraC acts as both a repressor (when intracellular arabinose levels are low), binding to *araO* operator sites, and an activator (in the presence of arabinose), when bound to the *araI* site, but only in the presence of cyclic AMP–CRP complex. It is a typical **helix-turn-helix** DNA-binding regulatory protein. Database code ARAC_ECOLI, 292 amino acids (33.35 kDa).

ara-C *abbr.* for arabinosylfuranosylcytosine.

arachain an unclassified proteinase obtained from the peanut, *Arachis hypogaea*.

arachidic acid the *trivial name* for eicosanoic acid; $\text{CH}_3\text{--}[\text{CH}_2]_{18}\text{--COOH}$; a constituent of arachis oil (from the peanut, *Arachis hypogaea*) and other oils. See also **arachidoyl**.

arachidonate 5-lipoxygenase EC 1.13.11.34; *other names*: 5-lipoxygenase; leukotriene- A_4 synthase; a **lipoxygenase** (def. 2) enzyme that catalyses the reaction of arachidonate with dioxygen to form (6E,8Z,11Z,14Z)-(5S)-5-hydroperoxyeicosa-6,8,11,14-tetraenoate. Iron is a cofactor. Example from human: database code LOX5_HUMAN, 673 amino acids (77.76 kDa).

arachidonate 12-lipoxygenase EC 1.13.11.31; *other name*: 12-lipoxygenase; a **lipoxygenase** (def. 2) enzyme that catalyses the reaction of arachidonate with dioxygen to form (5Z,8Z,10E,14Z)-(12S)-12-hydroperoxyeicosa-5,8,10,14-tetraenoate. Iron is a cofactor. Example from human: database code LOX2_HUMAN, 662 amino acids (75.45 kDa).

arachidonate 15-lipoxygenase EC 1.13.11.33; *other names*: arachidonate ω-6 lipoxygenase; 6-lipoxygenase; a **lipoxygenase** (def. 2) enzyme that catalyses the reaction of arachidonate with dioxygen to form (5Z,8Z,11Z,13E)-(15S)-15-hydroperoxyeicosa-5,8,11,13-tetraenoate. Iron is a cofactor. Example from human: database code LOX1_HUMAN, 661 amino acids (74.59 kDa).

arachidonic acid see **eicosatetraenoic acid**.

arachidonoyl *symbol*: $\Delta_4\text{Ach}$; the *trivial name* for (*all-Z*)-eicosa-5,8,11,14-tetraenoyl; $\text{CH}_3\text{--}[\text{CH}_2]_3\text{--}[\text{CH}_2\text{--CH=CH}]_4\text{--}[\text{CH}_2]_3\text{--CO--}$ (*all-Z* isomer); the acyl group derived from arachidonic (i.e. (*all-Z*)-eicosa-5,8,11,14-tetraenoic) acid, a polyunsaturated, unbranched, acyclic, aliphatic acid. It occurs naturally in membrane phospholipids, and is a biochemical precursor of **prostaglandins** and **leukotrienes**. It is also present in mosses, algae, and ferns, but not in higher plants. Distinguish from **arachidonyl** and **arachidoyl**.

arachidonyl the *trivial name* for the alkyl group, $\text{CH}_3\text{--}[\text{CH}_2]_3\text{--}[\text{CH}_2\text{--CH=CH}]_4\text{--}[\text{CH}_2]_3\text{--CH}_2\text{--}$, derived from the alkanol analogue of arachidonic acid. Distinguish from **arachidonoyl**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

arachidoyl or (formerly) **arachidyl** or **arachyl** *symbol*: Ach; the *trivial name* for eicosanoyl; $\text{CH}_3\text{--}[\text{CH}_2]_{18}\text{--CO--}$; the acyl group derived from arachidic (or eicosanoic or (formerly) arachic) acid, a saturated, unbranched, acyclic, aliphatic acid. It occurs naturally as triarachin in certain seeds, oils, and butter fat. Arachidyl is now the name for the alkyl group, $\text{CH}_3\text{--}[\text{CH}_2]_{18}\text{--CH}_2\text{--}$. Distinguish from **arachidonyl**.

arachidyl the *trivial name* for the alkyl group, $\text{CH}_3\text{--}[\text{CH}_2]_{18}\text{--CH}_2\text{--}$, the alkanol analogue of arachidic acid. See also **arachidoyl**.

arachin the major protein of the groundnut, *Arachis hypogaea*, comprising a 345 kDa globulin with 12 subunits. Not all groundnuts contain the same arachin; there exist at least two forms, designated A and B, having similar amino-acid compositions and general properties but differing in their subunit composition. Examples of subunits: database code ARA1_ARAHY, 176 amino acids (20.86 kDa); ARA5_ARAHY, 201 amino acids (22.22 kDa). See also **conarachin**.

Arber, Werner (1929–), Swiss microbiologist; Nobel Laureate in Physiology or Medicine (1978) jointly with D. Nathans and H. O. Smith 'for the discovery of restriction enzymes and their application to problems of molecular genetics'.

arbovirus *abbr.* for arthropod-borne virus (a term formerly used for **togavirus**).

arc a continuous curved line, as part of a circle or ellipse; e.g. the line of antigen–antibody precipitate obtained in **immuno-diffusion** or **immunoelectrophoresis**.

Archaea one of three primary kingdoms of cells, together with **Eukarya** and **Bacteria** (formerly **Eubacteria**), defined when cells are classified on the basis of rRNA sequence homologies. Archaea include all methanogens, extreme halophiles and most extreme thermophiles. Representative genera are: *Halo-bacterium*, *Methanobacterium*, *Sulfolobus*, *Thermoplasma*, and *Picrodictum* (growth optimum = 105 °C); about 40 others are known. —archaeal *adj.*

Archaeobacteria former name for **Archaea**.

Archibald method see **approach to (sedimentation) equilibrium method**.

Archimedes' principle the principle that when a body is partly or wholly immersed in a fluid there is an apparent loss of weight equal to the weight of the fluid displaced. [After Archimedes of Syracuse (?287–212 BC), Greek philosopher and applied mathematician.]

architectural gene a gene that determines the site of an enzyme within a cell.

A-region the presumptive initial oxidation site of a carcinogenic polycyclic aromatic hydrocarbon, on the terminal ring of the **bay region**, in the metabolic pathway leading to carcinogenesis (corresponding to what has sometimes been called the M-region). Compare **B-region**, **K-region**, **L-region**.

arenavirus any of a group of RNA viruses consisting of enveloped pleomorphic particles of 80–120 nm with helical nucleocapsids. The group includes Lassa virus and other viruses that spread to humans from rodents.

arene any monocyclic or polycyclic aromatic hydrocarbon.

arene oxide any epoxide formed by the addition of an atom of oxygen across any double bond of an **arene**.

ARF *abbr.* for ADP-ribosylation factor; a small monomeric cytosolic **GTPase** that, when bound to GTP, binds to the membranes of cells. It was originally discovered as a factor required for cholera toxin to ribosylate G_s , hence its name. Its structure is known and is highly conserved. It is a member of the **RAS** superfamily. Functions of ARF include regulation of membrane traffic in both endocytic and exocytic pathways, maintenance of organelle integrity, assembly of coat proteins (see **coatomer**), and activation of phospholipase D. See **phospholipase**.

Arg *symbol* for a residue of the α-amino acid L-arginine (alternative to R).

argentaffin cell a type of cell of the **apud** class, characterized by taking up silver stains with high affinity, without the addition of formaldehyde.

argentation chromatography

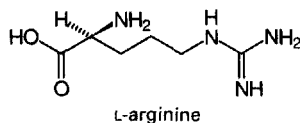
tion of reducing agent. [Named from Latin *argentum*, silver, and affinity.]

argentation chromatography any method of chromatography in which an adsorbent impregnated with silver nitrate is used to separate lipid classes according to the degree and geometry of their saturation. The technique may be used with either thin-layer chromatography or column chromatography, or in conjunction with gas-liquid chromatography.

arginase *abbr.* (in clinical biochemistry): ARS; *recommended name* for EC 3.5.3.1; *other names*: arginine amidinase; *systematic name*: L-arginine amidinohydrolase. An enzyme that catalyses the hydrolysis of L-arginine to L-ornithine and urea; it also acts on α -N-substituted L-arginines and canavanine. It is a key enzyme in the **ornithine-urea cycle**, occurring in high concentrations in the liver of ureotelic vertebrates and being absent or nearly so from the liver of uricotelic species. It has a very wide species distribution. In humans it is defective in the inherited disorder argininemia. Examples: from *Bacillus subtilis*, database code ARG1_BACSU, 296 amino acids (32.15 kDa); from *Rattus norvegicus*, database code ARG1_RAT, 323 amino acids (34.97 kDa).

arginate 1 arginine anion; $\text{H}_2\text{N}-\text{C}(=\text{NH})-\text{NH}-[\text{CH}_2]_3-\text{CH}(\text{NH}_2)-\text{COO}^-$. 2 any salt containing the arginine anion. 3 any ester of arginine.

arginine *trivial name* for N^5 -amidino-ornithine; α -amino- δ -guanidinopentanoic acid; 2-amino-5-guanidinopentanoic acid; $\text{H}_2\text{N}-\text{C}(=\text{NH})-\text{NH}-[\text{CH}_2]_3-\text{CH}(\text{NH}_2)-\text{COOH}$; a chiral α -amino acid. L-Arginine (*symbol*: R or Arg), (S)-2-amino-5-guanidinopentanoic acid, is a coded amino acid found in peptide linkage in proteins; codon: AGA (not in mitochondria of fruit fly or mammals) or AGG (not in mitochondria of mammals); CGA, CGC, CGG, or CGU. It is an essential dietary amino acid in rats and probably in young humans, and is glucogenic in mammals. D-Arginine (*symbol*: D-Arg or DArg), (R)-2-amino-5-guanidinopentanoic acid, is not known to occur naturally.



arginine (base) the ionic form of an arginine residue when the guanidino group is unprotonated.

arginine carboxypeptidase *see* lysine(arginine) carboxypeptidase.

arginine kinase EC 2.7.3.3; an enzyme that catalyses a reaction between ATP and L-arginine to form ADP and N^{ω} -phospho-L-arginine. The latter compound may play a role in invertebrate muscle analogous to that of **creatine phosphate** in vertebrate muscle. Example from *Homarus gammarus* (European lobster); database code KARG_HOMGA, 355 amino acids (39.81 kDa).

arginine phosphate *former name (not recommended) for phosphoarginine.*

arginine-urea cycle *see* ornithine-urea cycle.

[8-arginine]vasopressin or (*less correctly*) **arginine vasopressin** *abbr.*: AVP or [Arg⁸]vasopressin; the molecular form of **vasopressin** in which the variable eighth position in its amino acid sequence is occupied by an arginine residue. This is the form present in humans and most other mammals.

arginine vasotocin or [Arg⁸]oxytocin a pituitary peptide found in nonmammalian vertebrates with activity resembling both oxytocin and vasopressin.

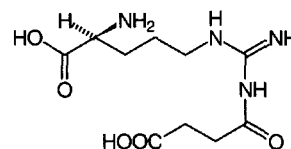
argininium 1 argininium(1+); the monocation of arginine. In theory, the term denotes any ion or mixture of ions formed from arginine and having a net charge of plus one, although in practice the predominant species is generally $\text{H}_2\text{N}-\text{C}(=\text{NH}_2^+)-$

$\text{NH}-[\text{CH}_2]_3-\text{CH}(\text{NH}_3^+)-\text{COO}^-$. 2 the systematic name for argininium(2+); the dication of arginine.

arginino 1 N^2 -arginino; the alkylamino group, $\text{H}_2\text{N}-\text{C}(=\text{NH})-\text{NH}-[\text{CH}_2]_3-\text{CH}(\text{COOH})-\text{NH}-$, derived from arginine by loss of a hydrogen atom from its α -amino group. 2 N^{ω} -arginino; the alkylguanidino group, $\text{HOOC}-\text{CH}(\text{NH}_2)-[\text{CH}_2]_3-\text{NH}-\text{C}(=\text{NH})-\text{NH}-$, derived from arginine by loss of a hydrogen atom from its ω -amino group.

argininosuccinase EC 4.3.2.1; *systematic name*: N-(L-argininosuccinate) arginine lyase; *recommended name*: argininosuccinate lyase. An enzyme of the **ornithine-urea cycle** that catalyses the hydrolysis of N-(L-arginino)succinate to fumarate and L-arginine. Example (human) homotetramer (deficiency in argininosuccinaciduria; very strong similarity to bird and reptile δ -crystallin); database code ARLY_HUMAN, 464 amino acids (51.78 kDa).

argininosuccinate 2-(N^{ω} -arginino)succinate; L-argininosuccinate is an intermediate in the **ornithine-urea cycle**, where it is synthesized by the enzymic condensation of L-citrulline with L-aspartate. *See* **arginosuccinate synthase**.



argininosuccinate synthase EC 6.3.4.5; *systematic name*: L-citrulline:L-aspartate ligase (AMP-forming); *other names*: argininosuccinate synthetase; citrulline-aspartate ligase. An enzyme of the **ornithine-urea cycle** that catalyses a reaction between ATP, L-citrulline, and L-aspartate to form L-argininosuccinate with release of AMP and pyrophosphate. A deficiency occurs in citrullinemia. Example (human) homotetramer; database code ASSY_HUMAN, 412 amino acids (46.37 kDa).

arginyl the acyl group, $\text{H}_2\text{N}-\text{C}(=\text{NH})-\text{NH}-[\text{CH}_2]_3-\text{CH}(\text{NH}_2)-\text{CO}-$, derived from arginine.

argon detector a sensitive detector used in gas chromatography that depends on the unique ionization properties of argon whereby its long-lived metastable excited atoms can transfer their energy of excitation by collision to other gas molecules of lower excitation potential. Thus argon can be used as the chromatographic carrier gas, the detector being an ionization chamber containing a source of ionizing radiation. Detectable changes of the level of ionization occur when the argon issuing from the column has another gas mixed with it. The detector was invented by James E. Lovelock. Its great advantage is that it does not depend on the chemical nature of the substances being separated.

[Arg⁸]vasopressin *abbr.* for [8-arginine]vasopressin.

arithmetic mean *see* mean.

arm a base-paired segment of the clover-leaf model of transfer RNA. There are several such arms: **amino-acid arm**, **anticodon arm**, **dihydrouridine arm**, **extra arm**, and **T ψ C arm**.

ARNT *abbr.* for aryl-hydrocarbon receptor nuclear translocator; *other name*: dioxin receptor; a protein that mediates translocation of the **Ah receptor** from the cytoplasm to the nucleus after ligand has been bound. Example from human: ARNT_HUMAN, 789 amino acids (86.64 kDa).

A-RNA *abbr.* for A form of RNA.

aro1 a fungal gene for aromatic amino-acid biosynthesis. It encodes a multifunctional protein for, e.g., **3-dehydroquinase**, **shikimate 5-dehydrogenase**, **shikimate kinase**, **EPSP synthase**. Example from *Emericella (Aspergillus) nidulans*; database code ARO1_EMENI, 1603 amino acids (174.89 kDa).

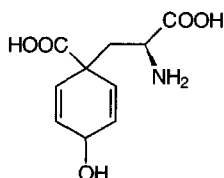
Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

aroA-H, aroL

aroA-H, aroL *Escherichia coli* genes for **chorismate** synthesis. **arogenate** or **pretyrosine** 3-(1'-carboxylato-4'-hydroxy-2',5'-cyclohexadienyl)alanine; an intermediate in phenylalanine and tyrosine biosynthesis in *Euglena gracilis* and some other microorganisms, and in plants. It is formed by transamination of **prephenate** and is subsequently decarboxylated and converted to tyrosine, or converted to phenylalanine by dehydration and decarboxylation.



aromatic describing any organic compound characterized by one or more planar rings, each of which contains (usually) three conjugated double bonds and $(4n + 2)$ delocalized pi-electrons, where n is a small integer. They undergo substitution reactions more readily than addition reactions. The simplest member of the class is benzene. The term was originally used to distinguish fragrant compounds from **aliphatic** compounds. —**aromaticity** *n*.

aromatic amino acid any amino acid containing an aromatic ring. In proteins these are phenylalanine, tryptophan, and tyrosine.

aromatic-L-amino acid decarboxylase EC 4.1.1.28; *systematic name*: aromatic-L-amino-acid carboxy-lyase; *other names*: dopa decarboxylase; tryptophan decarboxylase; hydroxytryptophan decarboxylase. An enzyme that catalyses the decarboxylation of L-tryptophan to tryptamine; pyridoxal 5'-phosphate is a coenzyme. The popular name derives from the fact that the enzyme also acts on 5-hydroxy-L-tryptophan to yield 5-hydroxytryptamine and on dihydroxy-L-phenylalanine (dopa) to yield **dopamine**. Example from human: database code DCD_HUMAN, 480 amino acids (53.83 kDa); it is a homodimer, and structurally similar to certain other eukaryotic decarboxylases.

ARP_a *abbr.* for assimilatory regulatory protein a, i.e. ferredoxin-thioredoxin reductase.

ARP_b *abbr.* for assimilatory regulatory protein b, i.e. **thioredoxin**.

arrestin or **S antigen** any protein of a family that includes the Ca^{2+} -binding protein of rod-cell outer segments. It binds to photoactivated-phosphorylated rhodopsin, thereby apparently preventing the transducin-mediated activation of phosphodiesterase. It has been implicated in autoimmune uveitis. Example from human: database code ARRS_HUMAN, 405 amino acids (45.00 kDa). β -Arrestins are similar in their C-terminal parts to α -transducin and other purine-nucleotide-binding proteins. β -Arrestins regulate β -adrenergic receptor function; they seem to bind phosphorylated β -adrenergic receptors, thereby significantly impairing their capacity to activate G_s proteins. Example (β -arrestin 2) from human: database code ARRC_HUMAN, 409 amino acids (45.99 kDa).

Arrhenius, Svante August (1859–1927), Swedish chemist and physicist; Nobel Laureate in Chemistry (1903) 'in recognition of the extraordinary services he has rendered to the advancement of chemistry by his electrolytic theory of dissociation'.

(Arrhenius) activation energy *symbol*: E_a or E_A ; *see* **Arrhenius equation**.

Arrhenius equation an equation representing the effect of temperature on the velocity of a chemical reaction. It may be expressed in the forms:

$$d \ln k / dT = E_a / RT^2 \quad (1)$$

or

$$d \ln k / d(1/T) = -E_a / R \quad (2)$$

or

$$k = Ae^{-E_a/RT} \quad (3)$$

where k is the rate constant for the reaction, E_a is its (**Arrhenius**) **activation energy**, R is the gas constant, T is the thermodynamic temperature, and A is a constant termed the frequency factor or the pre-exponential factor. Version 2 may be integrated to:

$$\ln k = \ln A - E_a / RT \quad (4).$$

Version 4 of the equation requires a plot of $\ln k$ against $1/T$, known as an **Arrhenius plot**, to be a straight line of slope $-E_a/R$, a relationship that has been shown to obtain for a considerable number of chemical reactions. *See also* **temperature coefficient of reaction**.

Arrhenius plot *see* **Arrhenius equation**.

ARS *abbr.* for 1 autonomously replicating sequence; any DNA sequence that can confer upon a plasmid the property of becoming a **replicon** in a particular host. The term is largely confined to fungal genetics. **2** (*in clinical biochemistry*) *abbr.* for arginase.

Arsenazo III 2,2'-(1,8-dihydroxy-3,6-disulfonaphthalene-2,7-bisazo)bisbenzenearsonic acid; an indicator substance used in the determination of low (micromolar) concentrations of free calcium ions. The free acid is purple; the calcium adduct is blue-violet.

arsenic *symbol*: As; a semi-metal element of group 15 of the IUPAC **periodic table**; atomic number 33; relative atomic mass 79.41. It exists in three forms: yellow, grey, and black. The yellow form, comparable to white phosphorus, is less stable than the other forms and is converted into the grey (metallic) form by heat. The black form is more stable than the yellow form but is metastable with respect to the grey form. Arsenic is required in trace amounts by some organisms, but generally it acts as an antimetabolite.

arsenical **1** of or containing arsenic. **2** a drug or other agent (e.g. an insecticide, rodenticide, or nerve gas) containing arsenic as an active principle.

artemesinin *see* **sesquiterpene lactone**.

arteriosclerosis a pathological condition in which there is a thickening and loss of elasticity of the walls of the arteries, which may become calcified. This interferes with the blood supply to various organs and tissues, resulting in impaired function. It normally, though not invariably, results from **atherosclerosis**.

Arthus reaction a **complement**-dependent hypersensitivity reaction caused when antigen reacts with precipitating antibody in the skin or other tissues, forming microprecipitates in and around the small blood vessels and, secondarily, damaging cells. In the **passive Arthus reaction** antibody is injected intravenously and the antigen applied locally; in the **reversed passive Arthus reaction** the opposite applies. An **active Arthus reaction** may occur when antigen is injected locally subsequent to previous injection of the same antigen. The inflammation that results is by type III mechanisms (*see* **hypersensitivity** (def. 2)). [Described by (Nicholas) Maurice Arthus (1862–1945), French physiologist and bacteriologist, in 1903.]

articulin any of a family of cytoskeletal proteins of the **epiplasm** of flagellate and ciliate organisms. They form filaments and larger sheets or tubes. They contain repeated valine- and proline-rich motifs with the consensus VPVP_ _V_ _ _V_.

aryl any univalent organic radical derived from an **arene** by loss of one hydrogen atom.

arylene any bivalent organic group derived from an **arene** by loss of two hydrogen atoms.

arylsulfatase any of a group of enzymes that catalyse the hydrolysis of a phenol sulfate to a phenol and sulfate. Example from *Escherichia coli*: database code ASLA_ECOLI, 551 amino acids (60.65 kDa). Arylsulfatases A, B, and C are com-

As

mon names for **cerebroside-sulfatase**, **N-acetylgalactosamine-4-sulfatase**, and **steryl-sulfatase** respectively.

As symbol for arsenic.

ASA dehydrogenase *see* **aspartate-semialdehyde dehydrogenase**.

ASAT (*in clinical biochemistry*) *abbr.* for aspartate aminotransferase. *See* **aspartate transaminase**.

AS-C *abbr.* for Achaete-Scute complex. *See also* **AS-C protein**.

ascarylose 3,6-dideoxy-L-mannose; 3,6-dideoxy-L-arabinohexose; a component of glycolipids found in eggs of the nematode, *Ascaris*, and in the bacterium, *Yersinia pseudotuberculosis*, type V. For the D enantiomer *see* **tyvelose**.

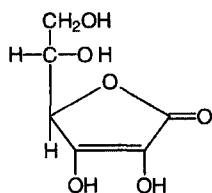
ascending boundary the solute boundary that moves upwards in one arm of the cell during electrophoresis in a **Tiselius apparatus**.

ascending chromatography any of the techniques of **chromatography** in which the mobile phase moves upwards over the solid phase.

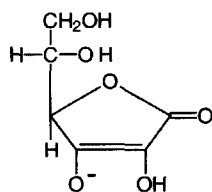
ascites the accumulation of serous fluid in the peritoneal cavity. The presence of ascites may be due to the growth of a tumour elsewhere in the body, the ascitic fluid then containing a suspension of single tumour cells. Administration of a **hybridoma** results in the formation of **monoclonal antibodies** which can be recovered from the ascitic fluid. —**ascitic** *adj.*

Ascoli test a precipitin test used in the serological diagnosis of anthrax. The antigens are extracted with boiling saline and detected by a ring test with immune serum.

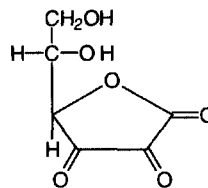
ascorbic acid a carbohydrate-like compound, containing six carbon atoms per molecule, the L enantiomer of which is found in fruit and vegetables. It has vitamin C activity in humans (most other primates can synthesize ascorbic acid) and guinea pigs. Humans are among the few higher animals that are unable to synthesize L-ascorbic acid, owing to a deficiency of **L-gulonolactone oxidase**, the enzyme catalysing the terminal step in L-ascorbic acid synthesis. The molecule contains an ene-diol group, one hydroxyl group of which is acidic (pK_a 4.04) and ionizes to give an anion ascorbate; the other ene-diol hydroxyl acts as a reducing group ($E^{\circ} = +0.058$ V, pH 7), and is oxidized on conversion to dehydroascorbate. Ascorbate is required together with ferric ion as a cofactor in the oxidation of prolyl residues in collagen to hydroxyprolyl residues and in other reactions.



ascorbic acid



ascorbate ion



dehydroascorbic acid

ascospore a (usually sexually derived) haploid spore formed within an **ascus**.

AS-C protein *abbr.* for Achaete-Scute complex protein; any of a family of proteins encoded by the Achaete-Scute complex, a gene cluster first discovered in *Drosophila* but with analogues in mammals and other organisms. They are involved in the determination of the neuronal precursors in the peripheral and central nervous systems. The T3, T4, and T5 proteins label strongly in the presumptive stomatogastric nervous system, while T8 is more prominent in the presumptive procephalic lobe. They resemble other members of the **MYC** family of helix-turn-helix transcription factors. Example, AS-C protein T4 (*Drosophila melanogaster*): database code AST4_DROME, 345 amino acids (38.08 kDa).

ascus (*pl.* **asci**) a sac-like fruiting body formed in ascomycete fungi, e.g. yeasts and *Neurospora*; it contains (typically eight) **ascospores** when mature.

+ase suffix denoting an enzyme; it is generally attached to a root indicating the substrate and/or the nature of the reaction catalysed. [From **diastase**.]

asemantic molecule any molecule that is not produced by an organism and therefore does not express any of the information that the organism contains. Absorbed molecules, although not derived from information contained within the organism, may nevertheless offer information about the organism relating to its absorption mechanisms. *See* **semantide**. *See also* **episemantic molecule**.

asepsis 1 the state of being **aseptic**. 2 the methods of making or keeping something aseptic. *Compare* **antiseptics**.

aseptic sterile; free from pathogenic microorganisms. *Compare* **antiseptic** (def. 2).

aseptic technique any technique incorporating measures that prevent the contamination of cultures, sterile media, etc. and/or the inappropriate infection of persons, animals or plants, by extraneous microorganisms.

asexual reproduction or **agamic reproduction** the development of a new individual from either a single cell or a group of cells without any sexual process. The term includes **agamogeny** and **vegetative reproduction**.

Ash protein another name for **Grb2**.

asialo+ *comb. form* denoting the absence from a compound of sialic-acid residues, e.g. after their deliberate enzymic removal from a naturally occurring **sialo-** (def. 2) compound.

A side the side of the nicotinamide ring of NAD(P)H from which projects the *pro-R* hydrogen atom (known as H_A) at the 4 position. *Compare* **B side**. *See also* **pro-R, pro-S convention**.

A site *abbr.* for aminoacyl-tRNA site (of a ribosome).

Asn symbol for a residue of the α -amino acid L-asparagine (alternative to N).

AsnC the asparagine biosynthetic operon regulatory protein, the product of *asnC*. In *Escherichia coli* it is the activator of *asnA* expression, the autogenous regulator of *asnC* expression, and the repressor of *gidA* (a gene of *E. coli* associated with glucose-inhibiting division) at the post-transcriptional level. It is a **helix-turn-helix** (HTH) protein and the prototype for AsnC HTH family of regulatory proteins; database code ASNC_ECOLI, 152 amino acids (16.87 kDa).

ASO *abbr.* for antistreptolysin-O; *see* **streptolysin**.

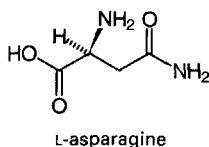
Asp

Asp symbol for a residue of the α -amino acid L-aspartic acid (alternative to D).

asparaginase EC 3.5.1.1; *other names*: L-asparaginase; *systematic name*: L-asparagine amidohydrolase; an enzyme that catalyses the hydrolysis of L-asparagine to L-aspartate and NH_3 . It is an effective antileukemic drug when injected into the bloodstream, its action being to deprive fast-growing tumour cells of the exogenous asparagine they require for rapid growth. Its clinical usefulness is limited because it causes damage to tissues with low asparagine synthetase activity. Several isoenzymes are found in several bacteria. Example, L-asparaginase I from *Escherichia coli*, a homotetramer: database code ASG1_ECOLI, 338 amino acids (37.08 kDa).

asparaginate 1 asparagine anion; the anion, $\text{H}_2\text{N}-\text{CO}-\text{CH}_2-\text{CH}(\text{NH}_2)-\text{COO}^-$, derived from asparagine. 2 any salt containing the asparagine anion. 3 any ester of asparagine.

asparagine *trivial name* for the β -amide of aspartic acid: aspartic 4-amide; α -aminosuccinamic acid; 2-amino-3-carbamoylpropanoic acid, $\text{H}_2\text{N}-\text{CO}-\text{CH}_2-\text{CH}(\text{NH}_2)-\text{COOH}$; a chiral α -amino acid. L-Asparagine (*symbol*: A or Asn; (*formerly*) also Asp(NH_2)), (S)-2-amino-3-carbamoylpropanoic acid, is a coded amino acid found in peptide linkage in proteins; codon: AAC or AAU. It is a nonessential dietary amino acid in mammals and is glucogenic. One residue per molecule of D-asparagine (*symbol*: D-Asn or DAsn), (R)-2-amino-3-carbamoylpropanoic acid, may occur in the antibiotic bacitracin A as an alternative to one of D-aspartic acid. *Compare* **isoasparagine**.



asparagine synthase (glutamine-hydrolysing) EC 6.3.5.4; *other name*: asparagine synthetase (glutamine-hydrolysing). An enzyme that catalyses a reaction between ATP, L-aspartate, and L-glutamine to form AMP, pyrophosphate, L-asparagine, and L-glutamate. Example from asparagus: database code ASN1_ASPOF, 589 amino acids (65.97 kDa); it contains a glutamine amidotransferase motif. *Compare* **aspartate-ammonia ligase**.

asparagino 1 N^2 -asparagino; the alkylamino group, $\text{H}_2\text{N}-\text{CO}-\text{CH}_2-\text{CH}(\text{COOH})-\text{NH}-$, derived from asparagine by loss of a hydrogen atom from its α -amino group. 2 N^4 -asparagino; the acylamino group, $\text{H}_2\text{N}-\text{CH}(\text{COOH})-\text{CH}_2-\text{CO}-\text{NH}-$, derived from asparagine by loss of a hydrogen atom from its amide group.

asparaginy the acyl group, $\text{H}_2\text{N}-\text{CO}-\text{CH}_2-\text{CH}(\text{NH}_2)-\text{CO}-$, derived from asparagine. *Compare* **isoasparaginy**.

aspartame N-L- α -aspartyl-L-phenylalanine methyl ester; a low-calorie artificial sweetener, about 160 times sweeter than sucrose in aqueous solution. Its sweetness is lost in cooking due to hydrolysis of the methyl ester. It is to be avoided in patients with phenylketonuria.

aspartase EC 4.3.1.1; *recommended name*: aspartate ammonia-lyase; *other name*: fumaric aminase. An enzyme that catalyses the hydrolysis of L-aspartate to fumarate and NH_3 . Example from *Escherichia coli*, homotetramer: database code ASPA_ECOLI, 478 amino acids (52.30 kDa).

aspartate 1 aspartate(1-) or hydrogen aspartate; the mono-anion derived from aspartic acid. In theory, the term denotes any ion or mixture of ions formed from aspartic acid and having a net charge of -1, although the species $^-\text{OOC}-\text{CH}_2-\text{CH}(\text{NH}_3^+)-\text{COO}^-$ generally predominates in practice. 2 *the systematic name* for aspartate(2-); the dianion derived from aspartic acid. 3 any salt containing an anion of aspartic acid. 4 any ester of aspartic acid.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

aspartate aminotransferase *see* **aspartate transaminase**.

aspartate-ammonia ligase EC 6.3.1.1; *other name*: asparagine synthetase; an enzyme that catalyses the reaction of ATP with L-aspartate and NH_3 to form AMP, pyrophosphate, and L-asparagine. Example from *Escherichia coli*: database code ASNA_ECOLI, 330 amino acids (36.61 kDa). *Compare* **asparagine synthase (glutamine-hydrolysing)**.

aspartate carbamoyltransferase *see* **aspartate transcarbamylase**.

aspartate kinase EC 2.7.2.4; *other name*: aspartokinase; an enzyme that catalyses the formation of 4-phospho-L-aspartate from ATP and L-aspartate with release of ADP. The reaction is the first step in the biosynthesis of lysine, methionine, and threonine in plants and bacteria. In *Escherichia coli*, the pathway for the synthesis of each amino acid is independently controlled by regulation of isoenzymes specific for each pathway. Example from yeast: database code AK_YEAST, 527 amino acids (58.04 kDa). *See* **homoserine dehydrogenase** for another example.

aspartate-semialdehyde dehydrogenase EC 1.2.1.11; *systematic name*: L-aspartate-4-semialdehyde:NADP⁺ oxidoreductase (phosphorylating); *other name*: ASA dehydrogenase; an enzyme that, although reversible, normally catalyses the formation of L-aspartate-4-semialdehyde, orthophosphate, and NADP⁺ from L-4-aspartylphosphate and NADPH. Its product is a component of a pathway involved in the biosynthesis of lysine and methionine, and can be converted either into dihydropicolinate (a precursor of lysine) or into homoserine (a precursor of methionine). Example from *Escherichia coli*: DHAS_ECOLI, 367 amino acids (39.97 kDa).

aspartate transaminase *abbr.* (in clinical biochemistry): AST; EC 2.6.1.1; *systematic name*: L-aspartate:2-oxoglutarate aminotransferase; *other names*: transaminase A; glutamic-oxaloacetic transaminase; glutamic-aspartic transaminase. An enzyme that catalyses a reaction between L-aspartate and 2-oxoglutarate to form oxaloacetate and L-glutamate. It is a pyridoxal-phosphate enzyme, very widely distributed, performing a pivotal role in amino-acid metabolism. In mammals this includes the transfer of amino groups to aspartate, which then transfers them into the **urea cycle** through **argininosuccinate synthase**. Very high plasma concentrations of AST (more than 100 times normal) are found in cases of severe tissue damage including acute hepatitis, crush injuries, and tissue hypoxemia. In jaundice and myocardial infarction the levels reach 10 to 20 times normal; in myocardial infarction AST starts to rise about 12 h after the infarct, reaching a peak at 24 to 36 h, then declining over 2 to 3 days. The normal level in human plasma is in the range 10–50 IU L⁻¹ (method-dependent; assumed presence of vitamin B₆ in the assay). Examples: (1) mitochondrial enzyme, homodimer: database code (precursor) AATM_HUMAN, 430 amino acids (47.42 kDa); (2) cytoplasmic isoform, homodimer: database code AATC_HUMAN, 412 amino acids (46.08 kDa).

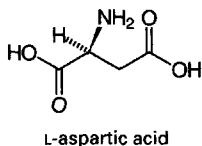
aspartate transcarbamylase *abbr.*: ATCase; EC 2.1.3.2; *recommended name*: aspartate carbamoyltransferase; *other names*: carbamylaspartotranskinase. An enzyme that catalyses a reaction between carbamoyl phosphate and L-aspartate to form N-carbamoyl-L-aspartate with release of orthophosphate. In many eukaryotes it is an activity of the **CAD protein**. In *Escherichia coli*, operon *pyrBI* codes for catalytic and regulatory chains; database code ECOPYRBI, catalytic chain 311 amino acids (34.37 kDa). There are six motifs and the 3-D structure of the enzyme in several states is known, e.g. database code NRL_1AT1A. The protein has a very high α -helical content. Carbamoyl aspartate is an important component of the pathway for *de novo* synthesis of pyrimidines.

aspartic denoting a residue of aspartic acid, whether or not the β -carboxyl group is protonated.

aspartic acid *trivial name* for α -aminosuccinic acid; 2-aminobutanedioic acid, $\text{HOOC}-\text{CH}_2-\text{CH}(\text{NH}_2)-\text{COOH}$; a chiral α -amino acid, L-Aspartic acid (*symbol*: D or Asp), (S)-



2-aminobutanedioic acid, is a coded amino acid found in peptide linkage in proteins; codon: GAC or GAU. In mammals it is a nonessential dietary amino acid and is glucogenic. Residues of D-aspartic acid (*symbol*: D-Asp or DAsp), (*R*)-2-aminobutanedioic acid, are found in the cell-wall material of bacteria of various species, and one residue per molecule may occur in the antibiotic bacitracin A as an alternative to one of D-asparagine. It is also formed by racemization of L-aspartic acid residues in long-lived proteins such as crystallins, dentine, and myelin.



aspartic amide 1 aspartic 1-amide; an alternative name for isoasparagine. 2 aspartic 4-amide; an alternative name for asparagine.

aspartic endopeptidase or (formerly) **acid proteinase** or **aspartyl protease** or **carboxyl protease** any enzyme of the subclass EC 3.4.23, comprising endopeptidases having a pH optimum below 5 by virtue of an aspartic residue being involved in the catalytic process. The group includes cathepsin D, chymosin, the pepsins, and renin.

asparto the alkylamino group, $\text{HOOC-CH}_2\text{-CH(COOH)-NH-}$, derived from aspartic acid.

aspartoacylase *see* **aminoacylase**.

aspartokinase *see* **aspartate kinase**; *see also* **aspartokinase/homoserine dehydrogenase**.

aspartokinase/homoserine dehydrogenase a multifunction enzyme, characterized in *Escherichia coli*, that catalyses two steps in the pathway from aspartate to diaminopimelate and lysine, to methionine, and to threonine and isoleucine. It is allosterically regulated by threonine. Example from *E. coli*: database code AK1H_EC01, 820 amino acids (89.02 kDa); homotetramer; the first 149 amino acids form the aspartokinase domain, while the last 350 residues form the homoserine dehydrogenase domain.

aspartoyl the diacyl group, $\text{-CO-CH}_2\text{-CH(NH}_2\text{)-CO-}$, derived from aspartic acid.

α -aspartyl aspart-1-yl; the α -monoacyl group, $\text{HOOC-CH}_2\text{-CH(NH}_2\text{)-CO-}$, derived from aspartic acid.

β -aspartyl or (formerly) **isoaspartyl** aspart-4-yl; the β -monoacyl group, $\text{HOOC-CH(NH}_2\text{)-CH}_2\text{-CO-}$, derived from aspartic acid.

β -aspartyl-N-acetylglucosaminidase EC 3.2.2.11; *systematic name*: 1- β -aspartyl-N-acetyl-D-glucosaminylamine L-asparaginohydrolase. An enzyme that catalyses the hydrolysis of 1- β -aspartyl-N-acetyl-D-glucosaminylamine to N-acetyl-D-glucosamine and L-asparagine. Example from *Flavobacterium meningosepticum*: database code A42259, 339 amino acids (37.16 kDa). *See also* **aspartylglucosaminuria**.

aspartylglucosaminuria a metabolic disorder with autosomal recessive inheritance; it results in progressive mental retardation due to lysosomal accumulation of aspartylglucosamine (2-acetamido-1- β -L-aspartamido-1,2-dideoxy- β -D-glucose), which is excreted in abnormally high amounts in the urine. This compound is commonly found in several glycoproteins in normal individuals. The first sign of the condition is usually a delay in speech development (1–4 years). The enzymic effect is a deficiency of lysosomal aspartylglucosaminidase (N^4 -(β -N-acetylglucosaminyl)-L-asparaginase, EC 3.5.1.26).

aspartylglucosylamine deaspartylase *see* **N^4 -(β -N-acetylglucosaminyl)-L-asparaginase**.

β -aspartyl phosphate or 4-phospho-L-aspartate or aspartyl- β -

phosphate an intermediate in the biosynthesis of lysine, threonine, and methionine in bacteria and plants. It is formed by **aspartate kinase**. It is converted to β -aspartyl semi-aldehyde, which is converted by the action of **homoserine dehydrogenase** to homoserine, a precursor of threonine or methionine. Alternatively addition of pyruvate to β -aspartyl phosphate leads to formation of dihydropicolinate, a precursor of lysine. A β -aspartyl phosphate residue is formed by phosphorylation of an aspartyl residue of $\text{Na}^+\text{,K}^+$ -ATPase during the transport process.

aspartyl protease an older name for **aspartic proteinase**.

β -aspartyl semi-aldehyde $\text{OHC-CH}_2\text{-CH(NH}_2\text{)-COOH}$; *see* **β -aspartyl phosphate**.

aspartyl shift or $\alpha \rightarrow \beta$ -**aspartyl shift** an intramolecular rearrangement reaction involving aspartic-acid residues that can occur during chemical synthesis or degradation of peptides. The β -carboxyl group of an aspartic residue displaces the α -carboxyl of the same residue from its amide (= peptide) linkage to the amino group of the adjacent amino-acid residue, thus forming an acid-sensitive β -amide (= isopeptide) linkage. Conditions must favour ionization of the β -carboxyl group. The reaction may be used to advantage in the partial hydrolysis of polypeptides and proteins by very dilute acid, but may occur to disadvantage during base-catalysed hydrolytic removal of protecting groups in the course of peptide synthesis.

Aspergillus a genus of filamentous fungi containing species of industrial and genetic importance. *A. niger* and *A. oryzae* are used in the production of citric acid, industrial enzymes, and fermented foods. The sexual species *A. nidulans* has been an important research tool in both biochemical and mitochondrial genetics. *A. flavus* is a source of **aflatoxins**.

asperlicin a naturally occurring antagonist of the peptide **cholecystokinin**, produced by various strains of *Aspergillus allicus*.

aspirin or **acetylsalicylic acid** 2-(acetyloxy)benzoic acid; an irreversible inhibitor of **prostaglandin-endoperoxide synthase**, through covalent acetylation of the α -amino group on the terminal serine active site. It is used as a drug for its antithrombotic activity and nonsteroidal anti-inflammatory and associated actions (antipyrexia, analgesia). *See also* **salicylic acid**.

Asp(NH₂) (formerly) *symbol* for L-asparagine.

Asp-NH₂ *symbol* for L-isoasparagine, an α -amino acid.

Asp(OMe) *symbol* for β -methyl aspartate; *O*⁴-methyl hydrogen aspartate.

Asp-OMe *symbol* for α -methyl aspartate; *O*¹-methyl hydrogen aspartate.

assay *n.* 1 the determination of the activity, potency, strength, etc. of a substance, either on an absolute basis or in comparison with that of a standard preparation. 2 the determination of the relative amount(s) of one or more components of a mixture, or of the degree of purity of a substance. 3 to carry out such a determination. *See also* **bioassay**, **immunoassay**, **microbiological assay**, **radioassay**.

assimilation the absorption of simple foodstuffs (or of the products of their digestion) and their use in the biosynthesis of complex constituents of the organism; sometimes restricted to **photosynthesis**. —*assimilatory adj.*

assimilatory nitrate reduction *see* **nitrate reduction**.

assimilatory regulatory protein a *abbr.*: ARP_a; *obsolete name* for ferredoxin-thioredoxin reductase, obtained from chloroplasts.

assimilatory regulatory protein b *abbr.*: ARP_b; *obsolete name* for thioredoxin, obtained from chloroplasts.

association a reversible union between two chemical entities, whether alike or different, to form a more complex substance. *Compare* **dissociation** (def. 1).

association constant *symbol*: K_{ass} or K_{a} ; the **equilibrium constant** for the reversible formation of a complex chemical compound from two or more simpler entities; the reciprocal of the **dissociation constant**, K_{diss} . Sometimes an **apparent association**



assortment

constant, K'_{ass} , constrained with respect to stated values of certain variables (e.g. pH), is determined (see **equilibrium constant**). For an association of the type: $A + B \rightleftharpoons AB$ the (concentration) association constant is given by:

$$K_{\text{ass}} = [AB]/([A] \times [B]) = 1/K_{\text{diss}}$$

See also **affinity constant**, **off-rate**, **on-rate**, **stability constant**.

assortment or **reassortment** during meiosis, the distribution to opposing cell poles of the two members of each pair of chromosomes at anaphase I and of the two members of each pair of chromatids at anaphase II.

AST (in clinical biochemistry) *abbr.* for aspartate transaminase.

astacin EC 3.4.24.21; *other name*: *Astacus* proteinase; crayfish small-molecule proteinase; an enzyme that catalyses the hydrolysis of peptide bonds in substrates containing five or more amino acids, preferentially with Ala in the P1' and Pro in the P2' positions (see **peptidase P-sites**). Zinc is a cofactor. Example, including 3-D structure, from the European fresh water crayfish: database code NRL_1AST, 200 amino acids (22.58 kDa).

astatin any of a subfamily of zinc metalloendopeptidases of unallocated number within sub-subclass EC 3.4.24, including **tolloid** (*Drosophila*) and **bone morphogenetic protein 1** (human), involved in differentiation.

astatine *symbol*: At; a halogen element of group 17 of the IUPAC **periodic table**; atomic number 85. It occurs as short-lived isotopes of mass numbers 215, 216, and 218 resulting from β -decay of other radioactive elements. Astatine-211, an α -emitter of half-life 7.5 h, results from α -bombardment of bismuth; it has potential in radiotherapy.

astaxanthin 3,3'-dihydroxy- β,β -carotene-4,4'-dione; a carotenoid pigment found mainly in animals (e.g. crustaceans, echinoderms) but also occurring in plants. It can occur free (as a red pigment), as an ester (e.g. the dipalmitate), or as a blue, brown, or green chromoprotein.

Aston, Francis William (1877–1945), British experimental physicist; Nobel Laureate in Chemistry (1922) 'for his discovery, by means of his mass spectrograph, of isotopes in a large number of non-radioactive elements, and for his enunciation of the whole-number rule'.

astral microtubule see **prometaphase**.

Astrup technique a micro technique for the rapid determination of the acid-base status of a sample of blood. The pH of the blood as taken is measured and the remainder of the sample split into two and each part equilibrated with a different CO_2/O_2 mixture of known $p(\text{CO}_2)$ and their pH values measured. These last two pH values and the corresponding $p(\text{CO}_2)$ values are plotted on a **Siggaard-Andersen nomogram** and the points joined by a straight line. The $p(\text{CO}_2)$ of the original sample may be obtained from the point on this line corresponding to the pH of the blood as taken. Furthermore, the **standard bicarbonate value** of the blood may be obtained from the point where the straight line intersects with a scale on the $p(\text{CO}_2)$ axis at 5.33 kPa (40 mmHg).

ASV *abbr.* for anodic stripping voltammetry.

Asx *symbol* for a residue of either of the α -amino acids L-aspartic acid or L-asparagine when the state of amidation is uncertain (alternative to B).

asymmetric synthesis any synthesis of a compound that produces only one of two possible enantiomers of the product. This is usually the case for enzymic but not for nonenzymic reactions.

asymmetry absence of **symmetry**. A particular example of its meaning in biochemistry relates to the different phospholipid composition of the inner and outer leaflets of a lipid bilayer, said to confer asymmetry of composition. —**asymmetric** *adj.*

asymmetry potential the usually small potential across a membrane electrode, such as a glass electrode, when both sides are immersed in identical solutions. It is probably due to slight imperfections in the membrane.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynguyen@huynguyentld.com.

asynapsis absence of **synapsis** (def. 1); i.e. the failure of homologous chromosomes to pair during meiosis.

At *symbol* for astatine.

atactic characterized by or showing irregularity in the spatial arrangement of the parts of a molecule.

atactic polymer any regular polymer in which the molecules display essential randomness with regard to the configurations at all main-chain sites of steric isomerism.

+ate suffix 1 designating any anion (including mixtures of anion and free acid), salt, or ester of any **oxoacid** (def. 1) other than an +ous acid. 2 denoting the product of a process, e.g. distillate, hydrolysate.

ATF *abbr.* for activating (or activator) transcription factor; any of various transcription factors for eukaryotic RNA polymerase II promoters that bind to **CRE**. Example, ATF-A and ATF-A- δ (human): these form a dimer; they have different sequences (amino acids 114–134 missing in δ), and possess a leucine zipper; database code ATFA_HUMAN, 483 amino acids (54.70 kDa).

(A + T)/(G + C) ratio the ratio of the sum of the adenine plus thymine bases to the sum of the guanine plus cytosine bases in a DNA molecule or preparation. The ratio is to some extent a characteristic of an individual type of DNA molecule. Compare **Chargaff's rules**.

athero-ELAMs endothelial leukocyte-adhesion molecules, i.e. **selectins**, expressed in atherosclerotic plaques.

atheroma a condition or process affecting blood vessels in which plaques form beneath the intima (i.e. inner lining). These start as infiltration by lipid-containing cells but later become invaded by fibrous tissue or become calcified. It is associated with hypercholesterolemia and defects in the **apolipoprotein B receptor** —**atheromatous** *adj.*

atherosclerosis a degenerative condition affecting arteries in which there is hyperplasia of the outer coat and fatty degeneration of the middle coat of the arteries due to **atheroma**.

aThr *symbol* for a residue of the α -amino acid L-allothreonine, (2S,3S)-2-amino-3-hydroxybutanoic acid.

atmosphere 1 the envelope of gas surrounding the Earth. 2 any local gaseous environment. 3 *symbol*: atm; a non-SI unit of pressure equal to 101.325 kPa; the pressure that will support a column of mercury 760 mm high at 273.15 K, sea-level, and latitude 45°.

atom a unit of matter consisting of a single **nucleus** surrounded by one or more orbital **electrons**. The number of electrons is normally sufficient to make the atom electrically neutral; adding or removing electrons converts the atom into a negative or positive ion, but this is regarded as a state of the same atom since the atom is characterized by its nucleus.

atomic absorption spectrophotometry a quantitative version of **atomic absorption spectroscopy** in which the absorption of electromagnetic radiation by the ground-state atoms in the flame is measured photoelectrically and the concentration of the element in question calculated by reference to the absorption observed with solutions of the elements of known concentration.

atomic absorption spectroscopy an analytical technique, based on the observation of the absorption of light energy by atoms, in which an atomic vapour of an element is formed by aspiration of a solution of the element into a flame, or otherwise, so that the large majority of the element's atoms remain in the non-emitting ground state. These ground-state atoms absorb electromagnetic radiation of their own specific resonance frequency, to an extent that is proportional to the density of the elemental atoms in the flame.

atomic mass constant *symbol*: m_u ; a fundamental constant equal to the mass of a carbon-12 atom divided by 12; it is equal to the **unified atomic mass unit**.

atomic mass unit *abbr.* (formerly): amu; see **unified atomic mass unit**.

atomic number an older alternative name for **proton number**.

atomic orbital the volume containing all the points within a

atomic radius

free atom at which the Schrödinger wave-function of an electron is of appreciable magnitude.

atomic radius the distance between the centre of the nucleus and the outermost electron shell of an atom. *Compare van der Waals radius.*

atomic weight *abbr.*: at. wt.; *former name for relative atomic mass.*

atomic weight unit *abbr.*: awu; one-twelfth of the mean mass of the neutral atoms of naturally occurring carbon.

atomizer or **atomiser** an instrument for breaking up a solution into a spray of fine droplets.

atom percent the proportion of any nuclide in a mixture expressed as a numerical percentage of all the atoms of that element present irrespective of their nuclidic masses.

atom percent excess a measure of the abundance of a stable nuclide in a sample expressed in terms of the excess, in **atom percent**, over that naturally present. It is used to express the extent of enrichment or dilution of substances labelled with stable isotopes.

atopy or **atopic allergy** the tendency of some individuals to develop immediate hypersensitivity reactions to **allergens**. It is probably hereditary.

ATP *abbr.* for adenosine 5'-triphosphate.

ATPase *abbr.* for adenosine triphosphatase.

ATP-diphosphatase *see* **apyrase**.

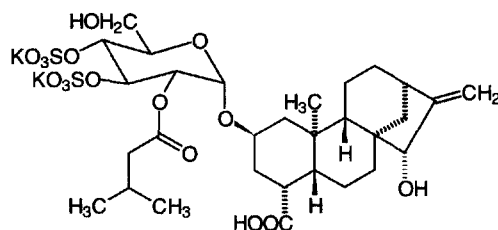
ATP[α,β -CH₂] *abbr.* for α,β -methyleneadenosine 5'-triphosphate; i.e. adenosine 5'-[α,β -methylene]triphosphate.

ATP[β,γ -CH₂] *abbr.* for β,γ -methyleneadenosine 5'-triphosphate; i.e. adenosine 5'-[β,γ -methylene]triphosphate.

ATP[β,γ -NH] *abbr.* for β,γ -imidoadenosine 5'-triphosphate; i.e. adenosine 5'-[β,γ -imido]triphosphate.

ATP(S) *abbr.* for adenosine 5'- γ -thiotriphosphate.

ATP synthetase *see* **H⁺-transporting ATP synthase**.



atractyloside

atractyloside an extremely toxic glycoside, obtained from the rhizomes of the Mediterranean thistle *Atractylis gummifera*, that produces hypoglycemia and convulsions in animals. It inhibits oxidative phosphorylation, in particular the translocation of mitochondrial adenine nucleotides. Carrier-bound adenine nucleotides are apparently replaced by atractyloside owing to the latter's higher affinity for a common translocation site. *Compare bongkreik acid.*

A-transferase EC 2.4.1.40; *recommended name*: fucosylgalactose α -N-acetylgalactosaminyltransferase; *other names*: fucosylglycoprotein α -N-acetylgalactosaminyltransferase, [histoblood group] A-transferase; an enzyme that can catalyse a reaction between UDP-N-acetyl-D-galactosamine and glycoprotein α -L-fucosyl-(1,2)-D-galactose to form UDP and glycoprotein N-acetyl- α -D-galactosaminyl-(1,3)-[L-fucosyl-(1,2)]-D-galactose. It thus catalyses the addition of N-acetylgalactosamine to the H antigen of the **ABH antigen** system, forming the A antigen. This and the **B-transferase** protein are products of alleles of the **ABO** gene, differing in only four residues.

atrial of or pertaining to an **atrium**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

atrial natriuretic peptide *see* **natriuretic peptide**.

atrial natriuretic peptide receptor *see* **natriuretic peptide receptor**.

atrioactivase a specific serine protease, present in bovine cardiac atria, that catalyses processing of the precursor of **natriuretic peptide** by selective cleavage at Arg⁸⁹.

atriopeptins biologically active peptides isolated from mammalian cardiac atria. Atriopeptin I (I) has 21 amino acids, with the structure



(3–19 disulfide link); atriopeptin II (II) is the same with additional Phe-Arg residues at the C terminus; atriopeptin III is the same as II but with an additional Tyr residue at the C terminus. I relaxes intestinal smooth muscle but not vascular smooth muscle, while II relaxes both. *Compare natriuretic peptide*.

atrium (*pl.* atria) or (formerly) **auricle** 1 either of the two chambers of the heart that lie above the ventricles. They receive blood from the afferent veins and pump it into the ventricles. 2 any of various anatomical chambers that receive the openings of other cavities.

atrolysin B EC 3.4.24.41; *other names*: *Crotalus atrox* metalloendopeptidase B; hemorrhagic toxin B (*abbr.*: HT-B); a snake-venom enzyme that catalyses the cleavage of His⁵-|-Leu⁶, His¹⁰-|-Leu¹¹, Ala¹⁴-|-Leu¹⁵, Tyr¹⁶-|-Leu¹⁷, and Gly²³-|-Phe²⁴ of the insulin B-chain. The reaction is identical to the cleavage of insulin B-chain by **atrolysin C**. The enzyme also cleaves Ser-|-Xaa bonds in glucagon. It is a zinc-requiring enzyme.

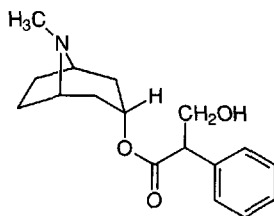
atrolysin C EC 3.4.24.42; *other names*: *Crotalus atrox* metalloendopeptidase C; hemorrhagic toxin C and D; a snake-venom enzyme that catalyses the cleavage of His⁵-|-Leu⁶, His¹⁰-|-Leu¹¹, Ala¹⁴-|-Leu¹⁵, Tyr¹⁶-|-Leu¹⁷, and Gly²³-|-Phe²⁴ of the insulin B-chain. With small-molecule substrates it prefers hydrophobic residues at the P2' position and small residues such as Ala or Gly at the P1 position (*see* **peptidase P-sites**). It is a zinc-requiring enzyme. Example from *C. atrox* (the Western diamondback rattlesnake): database code HRTD_CROAT, 203 amino acids (23.22 kDa).

atrolysin E EC 3.4.24.44; *other names*: *Crotalus atrox* metalloendopeptidase E; hemorrhagic toxin E; a snake-venom enzyme that catalyses the cleavage of Asn³-|-Gln⁴, Ser⁹-|-His¹⁰, and Ala¹⁴-|-Leu¹⁵ bonds in the insulin B-chain and Tyr⁴-|-Gln⁵ and Ala⁸-|-Ser⁹ in the A chain. It cleaves type IV collagen at Ala²⁵⁸-|-Gln²⁵⁹ in α_1 -collagen and at Gly¹⁹¹-|-Leu¹⁹² in α_2 -collagen. It is a zinc-requiring enzyme. Example from *C. atrox* (the Western diamondback rattlesnake): database code HRTE_CROAT, 478 amino acids (53.58 kDa).

atrolysin F EC 3.4.24.45; *other names*: *Crotalus atrox* metalloendopeptidase F; hemorrhagic toxin F; a snake-venom enzyme that catalyses the cleavage of Val²-|-Asn³, Gln⁴-|-His⁵, Leu⁶-|-Cys⁷, His¹⁰-|-Leu¹¹, Ala¹⁴-|-Leu¹⁵, and Tyr¹⁶-|-Leu¹⁷ bonds in the insulin B-chain. It is a zinc-requiring enzyme.

atrophy diminution in the size of a tissue or organ after full normal development has been attained. It is often a result of nutritional deficiency and/or decreased functional activity.

atropine ($\pm$)-hyoscyamine; tropyl DL-tropate; 1 α H,5 α H-tropan-3-ol ($\pm$)-tropate a tropane alkaloid obtained from deadly nightshade, *Atropa belladonna*; and other Solanaceae. The racemic mixture is a tertiary ammonium compound that acts as a **muscarinic receptor** antagonist. It is used as an anticholinergic, mydriatic, etc. *See also* **hyoscine**.



atropine

attachment proteins *general term* for proteins that are involved in binding other proteins to cell structures, e.g. **vinculin**, or α -actinin (*see* **actinin**), which bind actin to the plasma membrane.

attachment site *see* **active site** (def. 2).

A-T tailing homopolymer tailing of two types of duplex DNA molecule or fragment by the successive addition of adenine deoxynucleotide residues (A) to the 3' ends of molecules of one type and the complementary addition of thymine deoxynucleotide residues (T) to the 3' ends of molecules of the second type.

attenuance *symbol*: D ; the logarithm to the base 10 of the ratio of the **radiant powers** of the incident radiation, P_0 , and the transmitted radiation, P ; thus:

$$D = \lg(P_0/P) = \lg T^{-1},$$

where T is the **transmittance**. Attenuance reduces to **absorbance** if the incident beam is only either transmitted or absorbed, but not reflected or scattered.

attenuated virus a virus that has become less pathogenic following passage in culture outside its natural host, or by the use of physical or chemical means.

attenuator a sequence in DNA that is located between the **operator** and the gene for the first protein in an **operon**. It can cause transcription termination. The operon for the synthesis of **tryptophan** in bacteria is an example of an operon that contains an attenuator.

atto+ *symbol*: a; *prefix* used with SI units denoting 10^{-18} times.

attractant or **chemoattractant** a substance that a motile cell or organism tends to move towards; i.e. one that elicits positive **chemotaxis**. For a responsive bacterium, attractants can be either nutritive or nonmetabolizable. An example is **cytotaxin**.

Au *symbol* for gold.

AUA 1 a codon in mRNA, excluding that of mammalian mitochondria, for L-isoleucine. 2 a codon in mammalian mitochondrial mRNA for L-methionine (and for *N*-formyl-L-methionine, chain initiation, in some species).

AUC a codon in mRNA for L-isoleucine.

AUG a codon in mammalian mitochondrial mRNA, and in non-mitochondrial mRNA the only codon for L-methionine and usually for *N*-formyl-L-methionine, chain initiation.

Auger effect a process by which an orbital electron in an atom passes from an excited to a lower energy level. The X-ray so produced collides with, and ejects, another orbital electron, known as an **Auger electron**, with energy in the X-ray range. [After Pierre Victor Auger (1899–1993), French physicist.]

auracyanin a small blue copper-containing bacterial outer membrane glycoprotein (one Cu^{2+} ion per protein) that donates electrons to cytochrome c_{554} . Example from *Chloroflexus aurantiacus*: database code AUB1_CHLAU, 154 amino acids (15.53 kDa).

aureomycin 7-chlortetracycline. *See* **tetracycline**.

auriculin either of two fragments of atrial **natriuretic peptide**; auriculin A is the 4–27 and B the 4–28 fragment.

Auriculin a *proprietary name* for atrial natriuretic peptide.

aurosoma an artificially induced organelle occurring in animal

cells cultured in the presence of gold. It is an electron-dense lysosomal body containing gold particles.

Australia antigen an *alternative name* for hepatitis B antigen. *See* **hepatitis**.

aut+ a *variant form* of **auto+**.

autacoid any druglike principle that is produced in or can be extracted from the organs of internal secretion. Autacoids include both excitatory substances (**hormones**) and inhibitory substances (**chalone**s).

auto+ or (*before a vowel*) **aut+** *comb. form* 1 self; one's own. 2 occurring within oneself or itself. 3 acting by itself; automatic; spontaneous.

autoagglutination the tendency of a suspension of bacteria or other cells to agglutinate spontaneously in a simple medium not containing an **agglutinin** or other similar reagent.

autoallergy an *alternative term* for **autoimmunity**.

autoanalyser or (*esp. US*) **autoanalyzer** any instrument for effecting analyses automatically.

autoantibody an **antibody** produced in an animal that reacts with a constituent of the animal's own tissues, which thus acts as a 'self' antigen or **autoantigen**.

autoantigen an **antigen** that is a normal constituent of an individual and has the capacity to produce an **immune response** in the same individual in certain circumstances.

autocatalysis the catalysis of a chemical or biochemical reaction by one of the reaction products. —*autocatalytic adj.*

autochthonous derived from that part of the body where found; native. *Compare* **autologous**.

autoclave an instrument, used for sterilizing culture media, instruments, etc., that consists of an airtight chamber which can be filled with, or surrounded by, steam at high pressure.

autocoupling hapten a reactive compound of low molecular mass that, when injected into an animal, reacts with tissue antigens to form hapten-antigen compounds. These then lead to the formation of antibodies to the **hapten**.

autocrine describing an agent that acts on the cell in which it is produced.

autofluorogram an *alternative name* for **fluorogram** (when produced by **fluorography** (def. 2)).

autofluorography an *alternative term* for **fluorography** (def. 2).

autogenesis or **autogeny** an *alternative term* for **abiogenesis**. — *autogenetic adj.*

autogenous 1 self-produced. 2 originating within, especially within one's own body.

autogenous regulation the mechanism by which a protein directly controls the expression of its own structural gene.

autograft any graft or transplant of the subject's own tissue(s).

autoimmunity or **autoallergy** a state of immunological reactivity against constituents of the subject's own tissues.

autoimmunization or **autoimmunisation** the process of inducing **autoimmunity**.

autologous derived from the same organism. *Compare* **autochthonous**.

autolysate or (*esp. US*) **autolyzate** a product of **autolysis**, e.g. a suspension of broken cells.

autolysin EC 3.5.1.28; *other name*: *N*-acetylmuramoyl-L-alanine amidase; an enzyme that catalyses the hydrolysis of the link between *N*-acetylmuramoyl residues and L-amino-acid residues in certain bacterial cell-wall glycopeptides. The enzyme is required for cell separation, cell-wall turnover, etc. Example from *Staphylococcus aureus*: database code ALYS_STAAU, 481 amino acids (53.76 kDa). *See also* **lysostaphin**.

autolysis the process of self-digestion that occurs in plant and animal tissues after death of the organism or following separation of tissue from the rest of the organism. It is due to the action of the tissue's own enzymes. —*autolytic adj.*

autolysosome an **autophagic vacuole**.

automutagen any **mutagen** formed in an organism as a normal or abnormal metabolic product that is mutagenic in the same organism.



autonomic nervous system

autonomic nervous system *abbr.*: ANS; those parts of the **peripheral nervous system** and **central nervous system** that govern homeostasis in vertebrates and are generally independent of voluntary control.

autonomous existing or able to exist independently; used, e.g., of a tumour cell that is free of host control or of an episome or plasmid that replicates independently of the chromosome.

autophagic vacuole an enlarged **lysosome** containing subcellular structures such as mitochondria; a secondary lysosome.

autophagosome a **phagosome** containing endogenous cellular debris. *Compare heterophagosome.*

autophagy a process, apparently ubiquitous among eukaryotes, by which cells digest parts of their own cytoplasm. *Compare heterophagy. —autophagic adj.*

autophosphorylation the phosphorylation by a protein of one of its own residues. It is important in signal transduction by growth hormone receptors (*see, e.g., epidermal growth factor receptor*), but occurs in the case of other proteins, e.g. caldesmon.

autoprothrombin I an alternative name for factor VII; *see blood coagulation.*

autoprothrombin II an alternative name for factor IX; *see blood coagulation.*

autoprothrombin IIA an alternative name for protein C.

autopsy or **necropsy** the examination of a dead body, commonly to determine the cause of death or the presence of disease processes; a post-mortem (examination). *Compare biopsy.*

autoradiogram an **autoradiograph**, especially of a chromatogram or electrophoretogram.

autoradiograph or **autoradiogram** the image produced by **autoradiography**.

autoradiography a method by which radioactive material can be localized in, e.g., a tissue, cell, a cell part, or molecule, or in a chromatogram or electrophoretogram. The sample containing a radioactive substance is put in direct contact with a thick layer of specially prepared photographic emulsion. The radiation emitted in the decay of the radionuclide(s) activates individual silver halide grains in the emulsion rendering them susceptible to conversion to metallic silver grains by a photographic developer. After fixation the silver grains may be located either visually or by microscopy.

autosomal of, relating to, or being an **autosome**.

autosomal disease disease that results from an abnormality affecting the **autosomes**.

autosome any **chromosome** except the **sex chromosomes**. Hence the genes carried by the autosomes show **autosomal linkage** (or **autosomal inheritance**) according to the assortment of their respective autosomes to gamete cells during meiosis. Humans have 22 pairs of autosomes in addition to a pair of sex chromosomes.

autosteric effector any **effector** that binds to, and exerts its effect at, part of or immediately adjacent to the **active site** of an **allosteric** protein. It is distinguished from an **allosteric effector** operating at a distant site.

autotroph or **lithotroph** any organism that requires only simple inorganic substances to fulfil its nutritional requirements and for which gaseous or dissolved carbon dioxide is the sole source of carbon for the synthesis of cellular constituents. The term often includes any microorganism for which trace amounts of certain substances, e.g. vitamins, must also be supplied. Autotrophs may be subdivided into **chemoautotrophs** (or **chemolithotrophs**) and **photoautotrophs** (or **photolithotrophs**) according, respectively, to whether they are **chemotrophs** or **phototrophs**. *Compare heterotroph. —autotrophic adj.; autotrophy n.*

AUU a codon in mRNA for L-isoleucine.

auxiliary amino acid any amino-acid residue in the **active site**

(*def. 1*) of an enzyme that is more than one bond distance removed from the substrate molecule in the enzyme–substrate complex. *Compare contact amino acid.*

auxin any of a group of plant hormones, produced by regions of actively dividing and enlarging cells, that regulate aspects of plant growth. They include indole-3-acetic acid and indole-3-acetonitrile.

auxochrome any saturated atomic grouping (e.g. -OH, -CH₃, -NH₂, -Cl, -SH) that, when attached directly to a **chromophore**, shifts the selective absorption of light in the direction of longer wavelengths and enhances the intensity of absorption.

auxostat in industrial fermentation, a **chemostat** in which the **dilution rate** can be varied, normally in response to some parameter such as growth rate or pH, so that growth rate is more readily maximized.

auxotroph any strain of microorganism (alga, bacterium, or fungus) that differs from the **wild-type** by requiring a supply of one or more growth factors. *Compare prototroph. —auxotrophic adj.*

auxotrophy the condition of being an **auxotroph**.

average an alternative term for (arithmetic) **mean**.

average life *see mean life.*

average molar mass or (formerly) **average molecular weight** any of a series of numeric characteristics of a macromolecular system that are useful for assessing the extent of its polydispersity. The three most important are the **number-average molar mass**, M_n , the **mass-average molar mass**, M_m , and the **Z-average molar mass**, M_z . These three characteristics are given by the expressions:

$$M_n = \sum n_i M_i / \sum n_i \text{ (kg mol}^{-1}\text{)}$$

$$M_m = \sum n_i M_i^2 / \sum n_i M_i \text{ (kg mol}^{-1}\text{)}$$

$$M_z = \sum n_i M_i^3 / \sum n_i M_i^2 \text{ (kg mol}^{-1}\text{)}$$

where n_i is the amount of substance of species i and M_i is the molar mass of species i . Measurement of a **colligative property**, e.g. osmotic pressure, or end-group analysis, gives M_n ; light scattering, dielectric dispersion, or fluorescence depolarization measurements give M_m ; and data obtained by equilibrium ultracentrifugation will yield either M_m or M_z depending on how they are treated. For polydisperse or heterogeneous systems, $M_z > M_m > M_n$, whereas for monodisperse systems these three characteristics are equal (i.e. equal to the relative molecular mass, M_r , of the single species).

average molecular weight former name for **average molar mass**.

average radius of gyration an alternative term for **mean radius of gyration**.

average relative molecular mass former name for **average molar mass**.

avermectin B₁ an alternative name for **abamectin**.

avidin a 66 kDa glycoprotein comprising four essentially identical subunits, each of which consists of a single polypeptide chain of 128 amino-acid residues. Isolated from the white of the eggs of birds and amphibia, it binds extremely strongly to **biotin** and produces biotin deficiency when fed to experimental animals. Example from *Gallus gallus* (precursor): database code AVID_CHICK, 152 amino acids (16.77 kDa); residues 1–24 form the signal.

avidity 1 the tendency of an antibody to form more-or-less stable complexes with a (macromolecular) antigen; a measure of this tendency. 2 an alternative term for **affinity**.

avirulent lacking **virulence**.

avitaminosis a condition resulting from deficiency of one or more **vitamins**. The deficient vitamin may be specified, as in avitaminosis A, B, etc.

Avogadro constant or (formerly) **Avogadro's number** symbol:



avogram

L or N_A ; the number of molecular entities in one mole of any chemical species. It is a fundamental physical constant, of value $6.022\,136\,7(36) \times 10^{23} \text{ mol}^{-1}$. Compare **Loschmidt constant**.

Avogadro number or **Avogadro's number** the numerical value of the **Avogadro constant**.

Avogadro's hypothesis or **Avogadro's law** the principle that equal volumes of all ideal gases, at the same temperature and pressure, contain equal numbers of molecules. [After Lorenzo Romano Amadeo Carlo Avogadro, Count of Quaregna and Cerreto (1776–1836), Italian physicist and chemist who formulated it in 1811.]

avogram see **dalton**.

AVP *abbr.* for arginine vasopressin.

axenic pertaining to the growth of organisms of a single species in the absence of living organisms or living cells of any other species (also, strictly, in the absence of living cells of the organism itself except those in the intact living organism and its gametes). —**axenity** *n.*; **axenically** *adj.*

axenize or **axenise** to render axenic.

axenite an organism grown axenically.

axial 1 of, relating to, or being an **axis** (def. 1 or 2); located on or close to such an axis. 2 (in **stereochemistry**) *Symbol:* *a*. See **conformation**.

axial ratio the ratio of the length of the major axis to that of the minor axis of a prolate ellipsoid of rotation, a model often used for a protein or other macromolecule in solution.

axis 1 a straight line about which a body or three-dimensional figure, e.g. a crystal, is considered to rotate. 2 one of the reference lines of a co-ordinate system. 3 the second vertebra of the neck in higher vertebrates.

axis of rotation see **rotation axis**.

axis of symmetry see **symmetry**, **axis of**.

axokinin a heat-stable phosphoprotein, M_r 56 000, present in flagella and cilia of vertebrate and invertebrate species. Its cyclic AMP-dependent phosphorylation is required for initiation and maintenance of flagellar motility.

axon the long process of a neuron that conducts nerve impulses, usually away from the cell body to the terminals which are the site of storage and release of neurotransmitter. Compare **dendrite**.

axonal transport the directed transport of organelles and molecules along the axon of a neuron. It can be anterograde (i.e. outwards from the cell body) or retrograde (i.e. towards the cell body).

axoneme the bundle of microtubules and associated proteins that forms the core of cilia and flagella in eukaryotic cells and is responsible for their movements.

axoplasm the cytoplasm of the axon of a neuron, especially after it has been extruded from the axon.

aza+ *prefix* 1 (in **systematic chemical nomenclature**) denoting the replacement of a designated carbon atom (together with one associated hydrogen atom) by a nitrogen atom. 2 (in **trivial chemical names**) denoting presence of one or more nitrogen atoms.

azaserine *trivial name* for serine diazoacetate; *O*-diazoacetyl-L-serine, $N_2=CH-CO-O-CH_2-CH(NH_2)-COOH$; an antibiotic substance produced by *Streptomyces* spp. It is an inhibitor of amidotransferase enzymes and interferes with purine synthesis; it also retards the growth of transplantable animal neoplasms and is a specific inhibitor of L-glutamate ferredoxin oxidoreductase (transaminating), EC 1.4.7.1, in ammonium assimilation in plants.

azathioprine 6(1-methyl-4-nitro-5-imidazolyl) mercaptopurine: 6-[(1-methyl-4-nitroimidazol-5-yl)thio]purine; a derivative of **mercaptopurine**, into which it is converted *in vivo*. It is useful as an immunosuppressive and antimetabolic agent. *One proprietary name:* Imuran.

azelaic acid 1,7-heptanedicarboxylic acid, nonanedioic acid; a substance that acts as a substitute for biotin in the growth of a number of microorganisms and is proposed as an intermediate in biotin biosynthesis from oleic acid.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

azeotrope a mixture of two or more liquids that distills without change in composition and has a constant boiling temperature.

azide 1 any organic compound containing the monovalent azido group, $-N=N^+=N^-$. 2 the ion N_3^- , derived from hydrazoic acid, HN_3 . This ion blocks the electron transport chain by reacting with the ferric form of cytochrome aa_3 and thus acts as an inhibitor of respiration. 3 any salt or ester of hydrazoic acid.

azidothymidine see **AZT**.

azo the bivalent group $-N=N-$. It participates in noncyclic covalent linkage in organic molecules forming **azo compounds** or **diazo compounds**.

azocoll an insoluble bright-red collagen-rich proteinaceous material prepared by dyeing powdered cowhide or kangaroo-tail tendon with a dye prepared from tetrazolized benzidine, disodium 2-naphthol-3,6-disulfonate and sodium acetate. It is digested by most types of proteinase, the consequent solubilization of dye forming the basis of a simple nonspecific technique for the assay of proteolytic activity.

azo compound any organic compound with the general formula $R-N=N-R'$. Such compounds thus contain the bivalent azo group in covalent linkage, each nitrogen atom being linked to a carbon atom and replacing one hydrogen atom of each of the parent molecules RH and $R'H$. The latter may be the same or different, aliphatic or aromatic; e.g. azomethane, $CH_3-N=N-CH_3$; naphthalene-2-azobenzene, $C_{10}H_7-N=N-C_6H_5$. Compare **diazo compound**.

azoferredoxin or **protein II** a very oxygen-sensitive iron-sulfur protein that is a component of the **nitrogenase** enzyme system. It is a dimer of identical ≈ 30 kDa peptide chains. Each dimer contains two iron atoms, four S^{2-} groups, and 12 titratable thiol groups. In nitrogenase it is believed to be the electron carrier responsible for the reduction of molybdenum in **molybdoferredoxin**.

azomethine strictly, any **imine** having the more limited general structure $RR'C=NR$ where R and R' are any organyl groups, but often extended to include any compound where R' is H , the term then being considered to be synonymous with **Schiff base**.

azoprotein any of the modified proteins formed by coupling an aromatic diazo compound to one or more of the tyrosine residues of a protein, e.g. azoalbumin, azocasein.

azotemia *former term* for the accumulation of abnormally large concentrations of nitrogenous compounds, especially urea, in the blood.

Azotobacter a genus of Gram-negative strictly aerobic soil bacteria, all members of which can effect **nitrogen fixation**.

azotoflavin a somewhat larger than normal **flavodoxin** that is believed to act as a one-electron carrier between ferredoxin and azoferredoxin during nitrogen fixation in some bacteria, e.g. *Azotobacter*.

AZT *abbr.* for azidothymidine; 3'-azido-3'-deoxythymidine; *other names:* zidovudine; Retrovir; an analogue of thymidine, the phosphorylated form of which is an inhibitor of **reverse transcriptase** in **retroviruses**; it also terminates DNA synthesis. AZT undergoes phosphorylation in human T-cells to a nucleoside 5'-triphosphate, which competes with thymidine triphosphate and serves as a chain terminating inhibitor of HIV reverse transcriptase. It is used clinically to treat patients with HIV infection and AIDS.

azurin any member of a group of brilliant blue copper-containing proteins of low M_r found in some bacteria and thought to transfer electrons to cytochrome oxidase. Example from *Pseudomonas fluorescens* biotype B: database code AZUR_PSEFB, 128 amino acids (13.78 kDa).

azurophil(e) or **azurophilic** staining readily with blue aniline dyes.

azurophil(e) granule or **primary granule** a blue-staining lysosome of neutrophil leukocytes.

Bb

b *symbol for 1 molality. 2 one of two van der Waals coefficients. 3 breadth.*

B *symbol for 1 boron. 2 a residue of either of the α -amino acids L-aspartic acid or L-asparagine when the state of amidation is uncertain (alternative to Asx). 3 a residue of an incompletely specified base in a nucleic acid sequence that may be cytosine or guanine or either thymine (in DNA) or uracil (in RNA). 4 the ribonucleoside (5-)-bromouridine (alternative to BrUrd). 5 the B blood group. See **ABO system**.*

B *symbol for 1 Napierian absorbance (see **absorbance**). 2 magnetic flux density (bold italic).*

-B *conformational descriptor designating the boat conformation of a six-membered ring form of a monosaccharide or monosaccharide derivative. Locants of ring atoms that lie on the side of the structure's reference plane from which the numbering appears clockwise are indicated by superscripts preceding the letter, and those that lie on the other side of the reference plane by subscripts following the letter; e.g. methyl 2,6-anhydro- α -D-altropyranoside-^{2,5}B. See also **conformation** (under *cyclic compounds*).*

B_{max} *symbol for the amount of drug required to saturate a specific population of receptors in a membrane sample and hence an index of the density of receptor binding sites in the sample. It is usually derived from Scatchard analysis (see **Scatchard plot**) or from nonlinear regression analysis of data from a saturation binding assay.*

Ba *symbol for barium.*

Baa helices *abbr. for basic amphiphilic α helices; α helices that contain an array of basic amino-acid residues (His, Lys, or Arg) on one side and hydrophobic residues on the other. They are found, e.g., in the calmodulin-binding regions of many proteins.*

Babes-Ernst body or **Babes-Ernst granule** *a volutin granule seen in *Corynebacterium diphtheriae*. [After Victor Babes (1854–1926), Romanian bacteriologist, and Paul Ernst (1859–1937), German pathologist.]*

BAC *abbr. for bacterial artificial chromosome, based on the F factor (**F plasmid**) of *Escherichia coli*. Compare **YAC**.*

Bacillaceae *a family of eubacteria that includes the genera *Bacillus* and *Clostridium*.*

bacillary or **bacillar** *1 of, relating to, caused by, or containing bacilli. 2 an alternative term for **bacilliform** (def. 1).*

bacilliform *1 rod-shaped. 2 resembling a bacillus.*

bacillus (*pl. bacilli*) *any straight rod-shaped bacterial cell.*

Bacillus *a genus of large rod-shaped Gram-positive eubacteria belonging to the family Bacillaceae. Its members are aerobic or facultatively anaerobic, spore-bearing organisms. *B. subtilis* has become widely established as a vehicle for genetic engineering and many cloning vectors are available. *B. thuringiensis* synthesizes a toxin that is active against insects; different strains of the bacterium produce different forms of the toxin specific for different insect species.*

bacitracin *a cyclic antibiotic polypeptide complex produced by *Bacillus subtilis* and *B. licheniformis*. Commercial bacitracin is a mixture of at least nine bacitracins, mostly bacitracin A. Bacitracin A contains, in addition to several L amino acids, a number of D amino acids. Bacitracin inhibits bacterial cell wall synthesis by blocking the dephosphorylation of undecaprenyl diphosphate (undecaprenyl pyrophosphate) to the monophosphate form; it similarly inhibits the dephosphorylation of dolichyl diphosphate thereby blocking the formation of the core oligosaccharides of glycoproteins.*

backbone *1 (in biochemistry) the sequence of bonded atoms of a polymer to which the side groups and/or the side chains are attached. 2 (in zoology) the vertebral column or spine.*

backcross *1 any cross of a first-generation hybrid, F₁, with*

one parent or an individual genetically identical to one of the two parents. 2 the offspring of such a mating.

back-flush *an alternative term for **back-wash**.*

background *any intrusive sound or electronic signal registered by a detector but not coming from the source being measured, e.g. the radioactivity registered by a counter in the absence of a radioactive sample.*

back mutation *an alternative term for **reverse mutation**.*

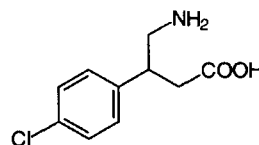
back scattering *an alternative term for **backward scattering**.*

back titration *an indirect titration procedure in which a measured excess of the reagent is added and the amount remaining after reaction with the analyte is titrated back to an end-point, the proportion consumed in the reaction being obtained by difference.*

backward scattering or **back scattering** *any scattering of radiation in a direction towards the source of the incident radiation.*

back-wash or **back-flush** *to pass a fluid through a chromatographic column in the direction opposite to that of the eluent flow. It is useful, e.g., to relieve bed-compression and to remove entrained gas.*

baclofen *γ -amino- β -(*p*-chlorophenyl)butyric acid; 3-(amino-methyl)-3-(4-chlorophenyl)propanoic acid; a 4-chlorophenyl derivative of γ -aminobutyric acid (GABA) that acts as a selective agonist for the **GABA_A receptor** and inhibits the release of other neurotransmitters in the central nervous system. It is used for its antispastic (muscle-relaxing) effects.*



bactenecin *any of several highly cationic polypeptides, originally isolated from bovine neutrophil granules but also found in sheep, that exert, *in vitro*, a potent antimicrobial activity, possibly due to inhibition of respiratory chain function. Example, bactenecin 5 (precursor) from *Bos taurus*: database code BCT5_BOVIN, 176 amino acids (20.03 kDa); residues 131–173 are bactenecin 5.*

bacteremia or (*esp. Brit.*) **bacteraemia** *the presence of live bacteria in the blood.*

bacteria *plural of bacterium.*

Bacteria *one of three superkingdoms (domains) of cellular organisms, the others being Archaea and Eukarya. Bacteria are unicellular and anucleate i.e. **prokaryotes**. They embrace a great diversity of forms, major divisions including the Cyanobacteria, Proteobacteria (which includes Gram-negative bacteria), and Gram-positive bacteria.*

bacterial conjugation *the process whereby genetic material is transferred from one bacterial cell (the donor) to another (the recipient). Material is usually transferred via a tubelike pilus, which extends from the donor cell to the recipient cell. The terms 'donor' and 'recipient' are preferred to 'male' and 'female' since the latter refer only to the presence or absence of **F plasmid**.*

bacterial histone *see **H-NS**.*

bacterial photosynthetic reaction centre *see **photosynthetic reaction centre**.*

bacterial porin *see **porin**.*

bacterial recombinant *see **transconjugant**.*

bacterial virus *see **bacteriophage**.*

bactericide or **bacteriocide** *any agent (biological, chemical, or*

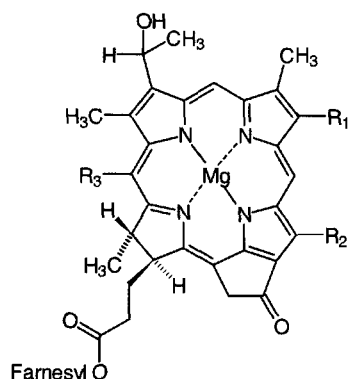


bactericidin

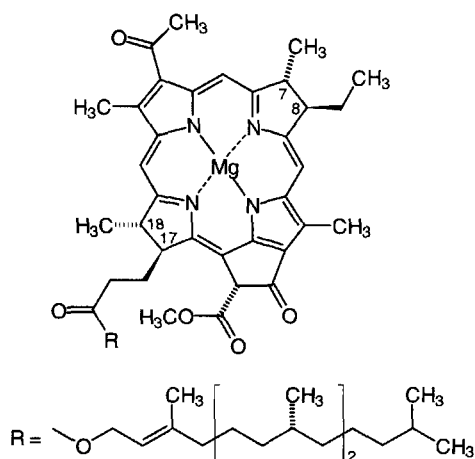
physical) that destroys bacteria. —**bactericidal** or **bacteriocidal** *adj.*

bactericidin or **bacteriocidin** or **bacteriolysin** any antibody that, with **complement**, kills bacteria against which it is active.

bacteriochlorophyll any of the **chlorophylls** of photosynthetic bacteria. They differ structurally from the chlorophylls of higher plants. Bacteriochlorophylls *a* to *g* are known. Bacteriochlorophylls *a* and *b* are the best known, being the photosynthetic pigments of purple bacteria. Their purple colour results from the fact that they are reduced in both rings B and D, and thus may be regarded as tetrahydropyrroles. Tetrapyrrole carbon atoms and carbon rings are numbered according to the Fischer and IUPAC systems as indicated below. The structure of bacteriochlorophyll *a* is shown. Bacteriochlorophyll *b* has $=CH-CH_3$ in place of the ethyl group on ring B of bacteriochlorophyll *a*. In some bacteriochlorophylls *a* the phytyl group may be replaced by a geranylgeranyl group. Bacteriochlorophylls *c*, *d*, and *e* are antenna pigments of Chlorobiaceae, located in chlorobium vesicles (a core structure is shown). These are dihydropyrrole structures, being reduced in ring D only. The esterifying alcohol is farnesol. In these bacteriochlorophylls, the side chain R_1 (see core structure) on ring B varies, being ethyl, *n*-propyl, or isobutyl (in bacteriochlorophyll *d* neopentyl is also possible). On ring C the side chain R_2 is ethyl (in bacteriochlorophyll *d* methyl is also possible). R_3 is methyl in bacteriochlorophylls *c* and *e*, and H in bacteriochlorophyll *d*.



bacteriochlorophyll *c*, *d*, *e* core structure

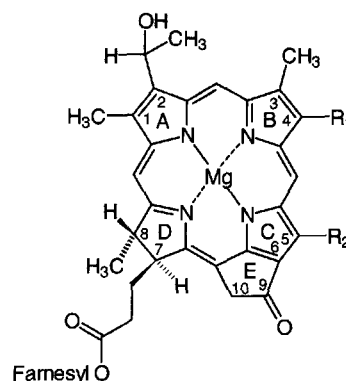


bacteriochlorophyll *a*

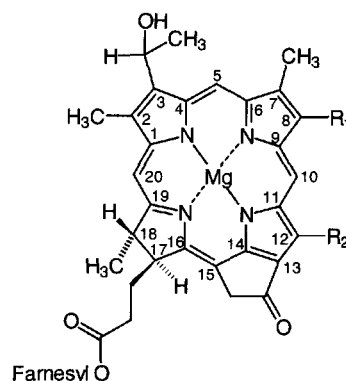
Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.



tetrapyrrole carbon numbering Fischer system



tetrapyrrole carbon numbering IUPAC system

bacteriocide a variant spelling of **bactericide**.

bacteriocidin a variant spelling of **bactericidin**.

bacteriocin any of a heterogeneous group of polypeptide antibiotics that are secreted by certain bacterial strains and are able to kill cells of other susceptible (frequently related) strains after adsorption at specific receptors on the cell surface. They include the colicins, and their mechanisms of action vary. Examples: colicin E1 from *Escherichia coli*; database code CEA1_ECOLI, 522 amino acids (57.21 kDa); this is a channel-forming transmembrane protein that results in membrane depolarization; colicin E3 from *E. coli*; database code CEA3_ECOLI, 541 amino acids (57.90 kDa); this is an rRNA ribonuclease; plantaricin A from *Lactobacillus plantarum*; database code (of precursor) PLNA_LACPL; plantaricin A is a channel-forming peptide dimer of α and β chains; the amino acid sequence of the β chain is:

AYSLQMGATAIKQVKKLFKKWGW;

the α chain lacks the N-terminal A. —**bacteriocinogenic** *adj.*

bacteriocinogenic factor any **plasmid** that carries genetic information for the production of a **bacteriocin**. Generally only a small proportion of cells containing the factor actually produce bacteriocin. *See also* **Col plasmid**.

bacterioferritin a bacterial b_1 -type **cytochrome** of 18 kDa subunit containing 13–20% w/w of iron. It has similar properties to mammalian **ferritin**. It may be concerned with iron or/and electron storage in the bacterium. Similar proteins may be present in other species of bacteria. Example from *Azotobacter*



bacteriology

vinelandii: database code BFR_ECOLI, 158 amino acids (18.49 kDa). These proteins are oligomers of 24 identical sub-units.

bacteriology the branch of science that deals with the study of bacteria.

bacteriolysin an alternative term for **bactericidin**.

bacteriolysis the lysis of bacteria. —**bacteriolytic** *adj.*

bacteriophageophytin a variant spelling of **bacteriophageophytin**.

bacteriophage or **phage** or **bacterial virus** any virus that can infect and multiply in a bacterium. Phages parasitize almost every group of prokaryotes. Commonly they consist of a core of nucleic acid enclosed within a protein coat, the capsid; additional components, e.g. lipid, occur in some phages. Depending on the type of phage, the nucleic acid may be either DNA (single-stranded or double-stranded) or RNA (single-stranded), and may be either linear or circular. Phages may be filamentous, polyhedral, or polyhedral and tailed; the tubular tails, to which one or more tubular tail fibres are attached in certain circumstances, are involved in attachment of the phages to the bacterial surface and for the injection of the DNA into the host cells. The smallest phages, e.g. ϕ X174, are ≈ 25 nm in diameter, whereas tailed polyhedral phages, such as the **T-even phages**, are 200–250 nm long, while filamentous phages measure roughly 800 nm $\times$ 5 nm. In general, a given bacteriophage can infect only one particular species, or strain, or group of closely related strains, though a given strain may be susceptible to infection by a number of different phages. Bacteriophages can bring about **transduction** (def. 1) between bacterial cells. Compare **cyanophage**. See also **(bacterio)phage conversion**, **lysogeny**, **phage induction**, **phage typing**, **prophage**, **temperate phage**, **virulent phage**.

(bacterio)phage conversion or **prophage-mediated conversion** or **lysogenic conversion** the introduction of new properties to a host bacterium by the genome of an infecting lysogenic prophage. Such properties may include resistance to lysis by related phages, changes in antigenic constitution, or the appearance of toxigenicity.

bacteriophage vector see **cloning vector**.

bacteriophageophytin or (*esp. Brit.*) **bacteriophageophytin** a **bacteriochlorophyll** in which Mg^{2+} is replaced by $2 H^+$.

bacteriorhodopsin a retinal-containing protein, resembling animal **rhodopsin**, formed by *Halobacterium halobium* and other halophilic Archaea and inserted into patches of **purple membrane** in the cell surface. The purple membranes serve as light-operated **hydrogen-ion pumps** to translocate hydrons from the inside to the outside of the cells. Example from *H. halobium* (precursor): database code BACR_HALHA, 262 amino acids (28.22 kDa); residues 1–13 form the propeptide and the C-terminal Asp is also missing from the mature protein, which is a seven-helix transmembrane protein of known 3-D structure and a model for **opsin** and related G-protein-linked receptors in eukaryotic cells.

bacteriostatic or **bacteristatic** describing a drug, antibiotic, or other agent that inhibits the growth and multiplication of bacteria. Compare **bactericide**. —**bacteriostasis** *n.*

bacterium (*pl. bacteria*) any of a vast and ubiquitous group of prokaryotic microorganisms that exist as single cells or in clusters or aggregates of single cells. Most authorities now place them in the exclusively prokaryotic kingdom Monera, along with the **cyanobacteria** (blue-green algae). The majority of bacteria possess a rigid cell wall; those lacking this feature are termed **Archaea**. Bacteria, Archaea, and Eukarya constitute the three primary Kingdoms (domains). —**bacterial** *adj.*

Bacteroides a genus of Gram-negative anaerobic rodlike bacteria. Mostly nonmotile, they are commonly present in the alimentary and urogenital tracts of mammals and may behave as opportunistic pathogens.

bactoprenol an alternative name for **undecaprenol**.

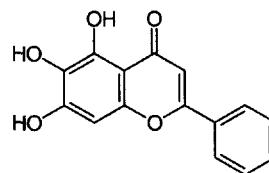
Bactrim see **sulfamethoxazole**.

baculovirus any of a group of DNA viruses that are known to multiply only in invertebrates and are now classified in the

family **Baculoviridae**. Their genome consists of double-stranded circular DNA of 58–100 MDa. Because of their host range they have potential as pest-control agents. Baculovirus vectors are valuable as a means of expressing certain animal proteins (see **expression system**).

BAF *abbr. for* B-cell activating factor (see **interleukin 1**).

baicalein 5,6,7-trihydroxyflavone; a pigment (greenish-brown in dilute alkali) found in the root of *Scutellaria baicalensis*. It is a selective 12-lipoxygenase inhibitor. Baicalein inhibits leukotriene biosynthesis and cellular Ca^{2+} uptake and mobilization.



bait a hybrid protein containing the **GAL4** DNA-binding domain, used in the **yeast two-hybrid system**. See also **bait region**.

bait region a restricted region in the sequence of α_2 -**macroglobulin** (α_2M) that is cleaved by proteinases in the process leading to the conformational changes instrumental in the proteinase inhibitory function of α_2M . The bait region occurs in all α -macroglobulins but shows great variation in length and amino-acid sequence between different α -macroglobulin and animal species. In human α_2M , it covers approximately residues 666–706. Outside of the bait region, α_2M from different species shows very high sequence homology. See also **trap hypothesis**.

bakers' yeast any of various strains of the yeast *Saccharomyces cerevisiae* that are used in baking.

baking soda see **soda**.

BAL *abbr. for* British anti-lewisite (see **dimercaprol**).

balance 1 (*in physiology*) the relation between the intake of a particular nutrient and its excretion (or the excretion of its metabolites). **2** any instrument for determining the equality in mass of two objects (or sets of objects). In cases where one (set) is of known mass the mass of the second (set) is thus obtained; e.g. an analytical balance. **3** any instrument designed to measure the weight of an object, e.g. spring balance, torsion balance.

balanced growth a type of growth such that over a time interval (within the exponential phase) every extensive property of the growing system increases by the same factor. Balanced growth for an individual cell requires that each cell after division is an exact replica of the one of the previous cycle.

balanced salt solution any solution designed to provide a normal environment, in terms of ionic composition, pH, and osmotic pressure, for cells in tissue culture.

balance study any study to determine **balance** (def. 1).

balata see **rubber**.

BALB/c a strain of inbred white mice that readily develop experimental myelomatosis.

Balbani ring a large puff of giant chromosomes present during a greater portion of larval development in members of the Diptera. The structural modification of specific loci is characteristically large and ring-shaped. [After Edouard Gérard Balbani (1823–99), French entomologist who reported it in 1881.]

Baltimore, David (1938–), US microbiologist and molecular biologist distinguished for his studies of viral nucleic-acid biosynthesis, particularly his discovery (at the same time as H. M. Temin) of reverse transcription by the enzyme RNA-directed DNA polymerase in the virions of tumour-producing retroviruses, and for the classification of animal viruses named after him; Nobel Laureate in Physiology or Medicine

Baltimore classification

(1975) jointly with R. Dulbecco and H. M. Temin 'for their discoveries concerning the interaction between tumour viruses and the genetic material of the cell'.

Baltimore classification a method of classification of animal viruses based on the nature of the genome and the mechanism for synthesis of mRNA, which is always regarded as the **plus strand** (def. 2). There are six classes:

Class I Viruses having a double-stranded (ds) DNA genome that make mRNA by asymmetric transcription. In many, different mRNA species come from different DNA strands.

Class II Viruses having a single-stranded (ss) DNA genome of the same polarity as the mRNA. Some have strands of both polarity in different particles.

Class III Viruses having a dsRNA genome that make mRNA by asymmetric transcription. Most have multiple pieces of dsRNA, each of which apparently contains the information for the synthesis of a single protein.

Class IV Viruses having an ssRNA genome that make mRNA of base sequence identical to the genomic RNA.

Class V Viruses having an ssRNA genome that make mRNA with a base sequence complementary to the genomic RNA.

Class VI **Retroviruses** having an ssRNA genome and a DNA intermediate in their growth.

band 1 an **absorption band** in an electromagnetic spectrum. **2** a zone of (macro)molecules such as that obtained in density-gradient centrifugation, zone electrophoresis, isoelectric focusing, chromatography, or other similar method. **3** any of the zones readily distinguishable by microscopy in a **sarcomere** of a muscle. **4** the specific association of a large number of **chromomeres** at the same level in somatically paired **polytenic** giant chromosomes of members of the Diptera. *See also* **cytochrome absorption bands**.

band centrifugation *see* **isopycnic centrifugation**.

band compression inadequate separation of the bands on a gel electrophoretogram, especially in DNA sequencing gels. It results in poor resolution and difficulties in interpretation.

band electrophoresis an alternative name for **zone electrophoresis**.

banding the formation of distinct bands or zones in density-gradient centrifugation, column chromatography, or other separation techniques.

banding density the density of the suspending medium at which a particular type of particle, e.g. a cell organelle, forms a band during centrifugation in a specified density gradient.

band intensity (in **NMR spectrometry**) the area under the band between the signal trace and the baseline. *See* **nuclear magnetic resonance**.

band-pass filter **1** an electric circuit or other device that transmits only frequencies within a selected band. **2** an **interference filter**.

band protein erythrocyte proteins are referred to as bands, numbered according to the position they occupy on electrophoresis gels. Some of the more important bands are: **1** and **2**, α and β **spectrin**; **3**, an anion channel protein; **4.1**, always referred to as such, is involved in spectrin junctions; **5**, **actin**.

band width (in **NMR spectrometry**) the distance between the sides of the band at half the band height of a reasonably smooth and symmetrical band. *See* **nuclear magnetic resonance**.

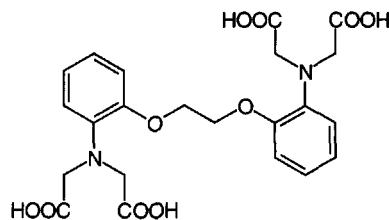
Bangasome or **Bangosome** (colloquially) an alternative name for **liposome**. [After its inventor, Alec Douglas Bangham (1921–), British biophysicist and medical scientist.]

Banting, (Sir) Frederick Grant (1891–1941), Canadian surgeon, physiologist, and endocrinologist distinguished for his discovery (with C. H. Best) of a means of extracting the hormone insulin from pancreatic tissue and for demonstrating its anti-diabetic efficacy; Nobel Laureate in Physiology or Medicine (1923) jointly with J. J. R. Macleod 'for the discovery of insulin'. *See also* **Best, Macleod**.

Bantu siderosis *see* **iron overload**.

BAPTA 1,2-bis(*o*-aminophenoxy)ethane-*N,N,N',N'*-tetraacetate; a Ca^{2+} chelator exhibiting a 10^5 -fold greater affinity (110 nM).

than for Mg^{2+} . Its UV absorption maximum is at 254 nm which, after Ca^{2+} is bound, shifts to 279 nm, a property that can be exploited in its use as an indicator of intracellular Ca^{2+} concentration. Its esterified form can enter cells where it is hydrolysed to BAPTA.



bar a unit of pressure temporarily approved for use with SI units. $1 \text{ bar} = 10^5 \text{ Pa} = 10^5 \text{ N m}^{-2}$.

barbed end *see* **plus end**.

barbital or **barbitone** 5,5-diethylbarbituric acid; 5-diethyl-malonylurea; 5,5-diethyl-2,4,6(1*H*,3*H*,5*H*)-pyrimidinetrione; a long-acting depressant of the central nervous system, widely used as a buffer substance; $\text{pK}_a = 7.98$ (25 °C), 8.06 (20 °C). *One proprietary name:* Veronal.

barbiturate **1** the tautomeric anion derived from barbituric acid (malonylurea; 2,4,6(1*H*,3*H*,5*H*)-pyrimidinetrione). **2** any mixture of free barbituric acid and its anion. **3** any salt of barbituric acid. **4** any of various pharmacologically active derivatives of barbituric acid, including barbitone, pentobarbitone, phenobarbitone, and thiopentone. Barbiturates are potent CNS depressants that potentiate the actions of GABA by binding to the **GABA_A** receptor. Pentobarbitone and thiopentone are used to induce anesthesia.

Barcroft apparatus or **Barcroft respirometer** a differential respirometer for studying gas exchange of cells, tissue slices, or tissue homogenates, now rarely used. It is a closed system comprising two flasks of equal volume connected by a U-shaped manometer (the Barcroft manometer). In operation, both flasks contain the same volumes of liquid and gas; one, the reaction flask, contains the cells or tissue; the other, the compensation flask, is free from cells or tissue and serves to compensate for changes in temperature and barometric pressure during the course of an experiment. The respirometers are arranged so that they may be shaken with the flasks in a constant-temperature bath. The reading of the apparatus is the difference between the levels of the manometer fluid in the two sides of the manometer. [After Joseph Barcroft (1872–1947), British physiologist, who described it in 1908.]

Barfoed's reagent a solution containing cupric acetate in dilute acetic acid that is reduced when heated with monosaccharides but not with disaccharides or oligosaccharides. [After Christen Thomsen Barfoed (1815–99), Swedish physician.]

barn symbol: b; a non-SI unit of area equal to 10^{-28} m^2 . It is used as a unit of atomic nuclear cross section.

barnase or **ribonuclease BA**; a small extracellular endoribonuclease from *Bacillus amyloliquefaciens*. Its action involves formation of a 2':3'-cyclic phosphate (*see* **ribonuclease**). It has been studied as a model for protein folding. The sequence and 3-D structure are known: database code RNBR_BACAM, 157 amino acids (17.47 kDa).

baroreceptor a variant spelling of **baroreceptor**.

barometer an instrument for measuring the pressure of the atmosphere.

barometric pressure the pressure exerted by the atmosphere at a given place and time. It is often expressed in terms of the supported height of a column of mercury, usually measured in millimetres. The SI unit is the **pascal**. Other non-SI units frequently used in addition to the **millimetre of mercury** are the **atmosphere**, the **bar**, and the **torr**.



baroreceptor

baroreceptor or **baroreceptor** any physiological receptor that is sensitive to changes in pressure, especially the sensory nerve endings in the carotid sinus that set off nerve impulses in response to alterations in blood pressure at that point.

Barr body see **sex chromatin**.

barrier an extracellular protein, probably a proteinase, excreted by yeast cells of mating type **a**. Its role may be to act as an antagonist of the α -type mating pheromone and establish optimal pheromone concentration for conjugation. Example (precursor) from *Saccharomyces cerevisiae*: database code BAR1_YEAST, 587 amino acids (63.66 kDa).

Barton, (Sir) Derek Harold Richard (1918–98), British organic chemist; Nobel Laureate in Chemistry (1969), the prize being shared with O. Hassel 'for their contributions to the development of the concept of conformation and its application in chemistry'.

barwin a barley seed protein (possibly a lectin) involved in defence mechanisms. It is closely related to the C-terminal domain of proteins encoded by wound-induced plant genes. It gives its name to a family of such proteins. Example, database code BARW_HORVU, 125 amino acids (13.77 kDa); 3-D structure known.

basal **1** at, of, or being a **base** (def. 4). **2** at, of, or being the minimum level for maintaining the normal or essential functioning of an organism.

basal body **1** (in *prokaryotes*) a structure that apparently attaches the **flagellum** (def. 1) to the cell envelope. The proximal end of the flagellum, the hook, appears curved and thickened and leads into the basal body, which consists of parallel ring-shaped structures, arranged around a rod-shaped core. The rings of the basal body make contact with the layers of the cell envelope. **2** (in *eukaryotes*) a structure at the base of a cilium consisting of spherical granules or short rods arranged in rows below the cell surface. Electron microscopically they are cylindrical bodies, 300–500 nm long and 120–150 nm in diameter, open at one or both ends. They consist of nine sets of triplet microtubules, each triplet containing one complete microtubule fused to two incomplete microtubules.

basal granule an alternative term for **basal body** (esp. in ciliated epithelium).

basal lamina a thin sheet of proteoglycans and glycoproteins, especially **laminin**, secreted by cells as an extracellular matrix, forming a region between the cells and adjacent connective tissue. The basal lamina has important functions in the organization of tissues including an influence on cell polarity, cell differentiation, and cell migration. The basal lamina is subdivided into the **lamina lucida** (an electron-lucent region) immediately adjacent to the cell layer, and the **lamina densa** (an electron-dense region) external to it. Frequently there is a further outermost layer, the **lamina reticularis**, containing collagen fibrils. The three laminae are often collectively referred to as the **basement membrane**.

basal medium any culture medium that will support the growth of nutritionally undemanding chemoorganotrophs.

basal metabolic rate *abbr.*: BMR; the rate of energy metabolism of an animal at rest but not asleep. It is measured with the animal in a 'comfortable' ambient temperature, and in the postabsorptive state, at least 12 hours after taking food. It may be determined from the oxygen consumption and carbon dioxide (and urinary nitrogen) excretion, or from the heat production of the animal.

base **1** (in chemistry) **a** a Brønsted–Lowry base. **b** a Lewis base. See also **acid** (def. 2). **2** (in molecular biology) any purine or pyrimidine that occurs as a component residue in polynucleotides or nucleic acids. Purines and pyrimidines are **nitrogen bases**, hence the terminology. **3** a unit of length of a polynucleotide or nucleic acid, equal to one nucleotide residue. Large molecules are usually measured in kilobases (10^3 bases; *symbol*: kb) or megabases (10^6 bases; *symbol*: Mb). See also **nitrogenous base**. **4** (in mathematics) **a** the number of units in a system of counting that is equivalent to one unit in

the next higher counting place. **b** the number that when raised to a specified power, its **exponent**, has a logarithm equal to the specified power. **5** the bottom or supporting part of anything; the point of attachment of an organ or part to an organism.

base analogue or (esp. US) **base analog** any unnatural purine or pyrimidine base that can be incorporated *in vivo* into DNA but that, because of its different properties, causes altered base-pairing during incorporation or in subsequent DNA replication. Thus 5-bromouracil, an analogue of thymine, pairs with guanine thereby causing transitions of A–T → G–C. Similarly, 2-aminopurine, an analogue of adenine, pairs with cytosine thereby causing transitions of A–T → G–C. Some base analogues have been used as anticancer and antiviral agents.

base calling the automated reading of DNA-sequencing gels using a scanner and appropriate software.

base composition the relative amounts of the various purines and pyrimidines occurring in a specimen of polynucleotide or nucleic acid. It is often expressed as moles percent.

baseline an imaginary line or a standard or reference value, etc. by means of which measurements may be compared, especially the zero line in the tracing of a chromatogram, spectrum, ultracentrifuge pattern, etc.

basement membrane a thin layer of dense material found in various animal tissues interposed between the cells and adjacent connective tissue. It consists of the **basal lamina**, comprising fine fibrillar material in a glycoprotein matrix, plus an associated layer of reticulin fibres. Occurring in, e.g., capillary linings, kidney tubules, lung alveoli, and renal glomeruli, basement membrane has a supportive function in some tissues and may also act as a passive selective filter for substances diffusing in or out of the cells; e.g. in renal glomeruli it retains protein molecules. It gives a strong periodic acid–Schiff reaction.

base pair *symbol*: bp; any of the possible pairings between two bases in opposing strands of double-stranded DNA or RNA molecules. Adenine forms a base pair with thymine (in DNA) or uracil (in RNA) and guanine with cytosine, hence the number of adenine residues equals the number of thymine (and/or uracil) residues while the number of guanine residues equals that of the cytosine residues. See also **Chargaff's rules**. —**base-pair** *vb.*; **base-pairing** *n*.

base-pair ratio see **Chargaff's rules**.

base sequence the sequential order of nucleotide residues in a polynucleotide or nucleic acid molecule.

base stacking the arrangement of base pairs in parallel planes, at an angle to the helix axis, in the interior of a helical double-stranded polynucleotide or nucleic acid molecule.

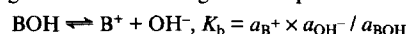
base triplet any sequence of three bases in a nucleic acid that codes for an amino acid, or some other signal, in protein synthesis. Compare **anticodon**; **codon**.

basic **1** of, relating to, containing, or characteristic of a **base** (def. 1). Compare **acidic** (def. 1). **2** having an alkaline reaction; of or relating to an aqueous solution having a pH value >7.0. Compare **acidic** (def. 2).

basic amino acid any amino acid possessing a net positive charge at neutral pH.

basic azo-dye binding protein *abbr.*: B-ABP; a 45 kDa (Sephadex) protein, isolated from the soluble cell supernatant fraction from livers of rats given 4-dimethylaminoazobenzene. It was later named **ligandin**.

basic dissociation constant or **basicity constant** *symbol*: K_b ; the thermodynamic dissociation constant for the dissociation of a **base** (def. 1, 2). For a dilute solution of a weak base, **B**, dissociating in water according to the equilibrium:



where **a** is the **activity** of the species designated by the subscripts. The activity of the water has been omitted from the equation since it may be taken as unity in dilute aqueous solution. K_b is a measure of the strength of a base, i.e. of its ability to accept hydrons from water (compare **acid dissociation con-**

basic dye

stant; *see also* **pK**. The concept is no longer used since the conjugate acid, B^+ , may be considered as an acid and an acid dissociation constant, K_a , for it defined. Also, in aqueous solutions

$$K_a \times K_b = [H^+][OH^-] = K_w,$$

where K_w is the ion product of water, whence K_b may be calculated from K_a or vice versa.

basic dye or **basic stain** any cationic dye that binds to, and hence stains, anionic macromolecules or other materials.

basic salt any salt containing hydroxyl or oxide groups in partial replacement of, or in addition to, ions or groups derived from an acid, e.g. basic lead acetate, $(CH_3COO)_2Pb \cdot Pb(OH)_2$.

basic stain *an alternative term for basic dye*.

basic zipper *see* **bZIP**.

basolateral surface the surface of an epithelial cell that adjoins underlying tissue. *Compare* **apical surface**.

basonuclein a **zinc finger** nuclear protein of epidermis, likely to be a transcription factor specific for squamous epithelium and its keratinocytes prior to terminal differentiation. It is present mainly in the nuclei of the basal cell layer. Example from human: database code BASO_HUMAN, 993 amino acids (110.85 kDa).

basophil or **basophil(ic) leukocyte** a polymorphonuclear phagocytic leucocyte of the myeloid series that is distinguished by the presence of coarse cytoplasmic granules that stain with basic dyes. The granules are believed to contain histamine, heparin, and other vasoactive amines. Basophils are closely related to **mast cells** and constitute <0.5% of blood leukocytes.

basophilia 1 the property of being **basophilic**. 2 an increase in the number of **basophils** circulating in the blood.

basophilic staining readily with basic dyes.

basotropic seeking a basic environment; a term used especially in connection with protein kinases that have a requirement for basic residues near the target residue. *Compare* **acidotropic**.

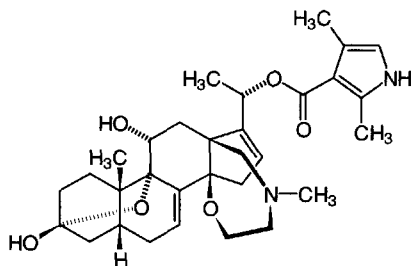
batch a quantity of material produced or processed in one operation on a single occasion, as opposed to a continuous operation; hence batch adsorption, batch culture, etc.

batch elution a method of column chromatography in which conditions are chosen to ensure complete retention of the material under investigation, using the minimal amount of stationary phase, followed by elution with a solvent or buffer that ensures complete and rapid removal in the minimal possible volume.

batch process (*in biotechnology*) a process in which a bioreactor is loaded with substrates inoculated with microorganisms or enzymes and allowed to run to completion, usually without removing biomass and products during the process.

bathochromic (*in absorption spectroscopy*) describing a shift of an absorption band in the direction of longer wavelengths. *Compare* **hypsochromic**.

batrachotoxin a steroidal alkaloid neurotoxin from the skin of Colombian arrow-poison frogs of genus *Phylllobates*. It is the most potent known venom. It binds specifically to the voltage-gated sodium channel and renders the membrane permeable to sodium ions. *Compare* **tetrodotoxin**, which reverses its depolarizing effect.



batroxobin *see* **venombin**.

Batten's disease *see* **H⁺-transporting ATP synthase**.

batyl alcohol 1-*O*-octadecyl-*sn*-glycerol; 3-octadecyloxy-1,2-propanediol; a hydrolysis product of **ether lipids**.

bay region a concave exterior region, bordered by three phenyl rings, of a polycyclic aromatic hydrocarbon when an angularly fused terminal benzo-ring is present. For example, the sterically hindered area between carbon atoms 4 and 5 of phenanthrene, between carbon atoms 1 and 12 of benz[*a*]anthracene, or between carbon atoms 10 and 11 of benzo[*a*]pyrene. It has been suggested that dihydrodiol epoxides in the bay region of such compounds may be responsible for their mutagenicity and carcinogenicity. *See also* **A region**, **B region**, **K region**, **L region**.

B cell 1 or (*formerly*) **beta cell** the principal cell (60–80%) of the islets of Langerhans of the pancreas. It produces, stores (as B granules), and secretes the hormone **insulin**. 2 *an alternative name for B lymphocyte*.

B-cell activation factor *abbr.*: BAF; *an alternative term for interleukin 1*.

B-cell antigen receptor complex a complex consisting of membrane immunoglobulin associated non-covalently with two proteins known as mb-1 (*other names*: Iga, CD79a) and B29 (*other names*: Igβ, CD79b) that binds antigen and initiates B-cell transformation. mb-1 and B29 are often designated α and β respectively, and the complex consists of Ig(αβ)₂. Signalling occurs by activation of tyrosine kinases of the Src family (lyn, blk, fyn, or lck). α and β are themselves phosphorylated on activation. All five immunoglobulin isotypes (IgG, IgM, IgD, IgA, IgE) bind two α/β heterodimers and use a similar signalling pathway, but the glucosylation of the α chain differs according to the isotype with which it associates. Examples (both precursors) from mouse: α chain, database code C79A_MOUSE, 220 amino acids (24.23 kDa); β chain, database code C79B_MOUSE, 228 amino acids (25.70 kDa).

B-cell differentiation factor *abbr.*: BDF; *an alternative term for interleukin 1*.

BCG *abbr. for* bacille Calmette–Guérin; the bacillus from which is made a vaccine composed of live bovine-strain tubercle bacilli, *Mycobacterium bovis*, attenuated by long cultivation on a potato–glycerol–bile medium. [After Léon Charles Albert Calmette (1863–1933), French physician and bacteriologist, and Camille Guérin (1872–1961), French veterinary surgeon and bacteriologist, who described its preparation in 1908.]

B chain 1 the longer of the two polypeptide chains of **insulin**, containing 30 amino-acid residues. 2 the **light chain** of an immunoglobulin.

B chromosome any of a heterogeneous group of accessory, extra, or supernumerary chromosomes found in many species of animals and plants. They differ from normal, or A chromosomes, by being usually smaller and frequently heterochromatic and telocentric, without great effect on the phenotype of their carriers. Both their numbers and their meiotic and mitotic behaviour are highly variable.

BCKDH *abbr. for* branched-chain α-keto acid dehydrogenase.

BCL any of several genes associated with B-cell leukemia and lymphoma. **BCL1** is identical to the gene for human **cyclin D1**; its product activates p34^{cdc2} kinase and is phosphorylated in G₁ of the cell cycle; database code CGD1_HUMAN, 295 amino acids (33.73 kDa). **BCL2** is a protooncogene encoding a growth factor that prolongs survival of hematopoietic cells; its product is a mitochondrial inner membrane protein, is able to block **apoptosis**, and if overexpressed with Myc counteracts the oncogenic action of Myc. Its product exists in two forms, BCL-2 α and β, resulting from differential splicing and expressed in B-cell leukemias and lymphomas. Example α (human); database code BC2A_HUMAN, 239 amino acids (26.24 kDa). **BCL3** encodes a probable transcription factor of the I-κB family; database code BCL3_HUMAN, 446 amino acids (46.80 kDa).

BCR a human gene named after **breakpoint cluster region**, a chro-



mosomal region found in humans and other species. It is a member of a small gene family coding for a protein with protein serine/threonine kinase activity, widely expressed in human hematopoietic and nonhematopoietic cell lines; database code BCR_HUMAN. *See also* **abl**, **Philadelphia chromosome**.

BDF *abbr.* for B-cell differentiation factor interleukin (*see* **interleukin**).

B-DNA *abbr.* for B form of DNA.

BDNF *abbr.* for brain-derived neurotrophic factor.

Be *symbol* for beryllium.

Beadle, George Wells (1903–89), US geneticist notable in particular for formulating the one gene–one reaction hypothesis; Nobel Laureate in Physiology or Medicine (1958) jointly with E. L. Tatum 'for their discovery that genes act by regulating definite chemical events' [prize shared with J. Lederberg].

Beckmann thermometer a sensitive mercury-in-glass thermometer used for measuring small differences or changes of temperature. [After Ernst Otto Beckmann (1853–1923), German chemist.]

Becquerel, Antoine Henri (1852–1908), French physicist; Nobel Laureate in Physics (1903) 'in recognition of the extraordinary services he has rendered by his discovery of spontaneous radioactivity' [prize shared with M. Curie and P. Curie].

becquerel *symbol*: Bq; the SI derived unit of (radioactive) **activity** (def. 2) of a radionuclide; it corresponds to one **nuclear transformation** per second; i.e. 1 Bq = 1 disintegration s⁻¹. *See also* **curie**: 1 Ci = 3.7 × 10¹⁰ Bq or 1 Bq = 2.70 × 10⁻¹¹ Ci.

bed volume in column chromatography, the total volume of material, both solid and liquid, in the column; i.e. the volume of the support particles plus the **void volume**. It is synonymous with **column volume** for a packed column.

Beer–Lambert law or **Beer–Lambert–Bouguer law** or **Lambert–Beer law** a combination of **Beer's law** (def. 1) and **Lambert's law**; put into modern terms, it states that the **absorbance**, *A*, of a beam of collimated monochromatic radiation in a homogeneous isotropic medium is proportional to the absorption pathlength, *l*, and to the amount-of-substance concentration, *c*, (or, in the gas phase, to the pressure) of the absorbing species. The law may be expressed by the relations

$$A = \lg (\Phi_0/\Phi) = \epsilon cl \text{ or } \Phi = \Phi_0 = 10^{-\epsilon cl}$$

where Φ_0 and Φ are respectively the **spectral powers** of the incident and transmitted radiation beams, and the proportionality constant, ϵ , is the molar (decadic) absorption coefficient – *see* **absorption coefficient**.

Beer's law 1 or (sometimes) **Bouguer–Beer law**; a physical law extending **Bouguer's law** to solutions: it states that when light passes through a given thickness of an absorbing solution of concentration *c*, the intensity of the transmitted light, *I*_{tr}, decreases exponentially as *c* increases linearly. It may be expressed as *I*_{tr} = *I*₀ e^{-*k**c*}, where *I*₀ is the intensity of the incident light and *k* is a constant. The law applies in practice only to monochromatic light (or other electromagnetic radiation) and when factors such as reflection or scattering are negligible or can be corrected for. *See also* **Beer–Lambert law**. [After August Beer (1825–63), German physicist who formulated it in 1857.] 2 name commonly but incorrectly applied to **Beer–Lambert law**.

beet sugar an alternative name for **sucrose** when this has been extracted from sugar beet.

Beh *symbol* for the behenoyl group.

behenoyl *symbol*: Beh; *trivial name* for docosanoyl, CH₃–[CH₂]₂₀–CO–, the acyl group derived from the aliphatic acid behenic acid (= docosanoic acid), which occurs naturally in animal tissues and most seed fats.

bek a gene related to that for the **fibroblast growth factor** (FGF) receptor. Named after bacterial expressed kinase, it encodes a plasma membrane protein tyrosine kinase of class IV with an insert in the kinase region that binds FGFA with high affinity; database code Hsfgrbe.

Benacerraf, Baruj (1920–). Venezuelan-born US immunologist. *Bản đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:*

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyenltd.com.

logist notable for his discovery of histocompatibility genes and their role in regulation of the immune response; Nobel Laureate in Physiology or Medicine (1980) jointly with J. Dausset and G. D. Snell 'for their discoveries concerning genetically determined structures on the cell surface that regulate immunological reactions'.

Bence-Jones protein any of a group of **myeloma proteins**, occurring in the urine of patients with multiple myelomatosis and related conditions, that precipitate on heating the urine to 60 °C, but redissolve at higher temperatures. They consist of light chains of **immunoglobulins** synthesized by the myeloma cells. Example, a kappa chain: database code KV1K_HUMAN, 108 amino acids (11.27 kDa). [After Henry Bence-Jones (1813–73), English physician who described them in 1847.]

Benedict's solution 1 **Benedict's qualitative solution**, a reagent used to test for the presence of glucose and other reducing sugars. It contains 0.07 M copper sulfate, 0.67 M sodium citrate, and 0.94 M sodium carbonate. When heated with reducing sugars it is reduced and a red precipitate of cuprous oxide is formed. 2 **Benedict's quantitative solution**, a reagent used in the estimation of glucose and other reducing sugars. It contains 0.072 M copper sulfate, 0.7 M sodium carbonate, 1.29 M potassium thiocyanate, and a trace of potassium ferrocyanide. On boiling, 25 ml of this solution is reduced by 50 mg glucose. Other reducing sugars may be estimated similarly. [After Stanley Rossiter Benedict (1884–1936), US biochemist.]

benign describing a neoplasm that is harmless, localized, and nonmetastasizing, although it may physically impede the function of a nerve or muscle.

bentonite a colloidal, native hydrated aluminium silicate clay consisting principally of montmorillonite, a complex aluminosilicate, Al₂O₃·4SiO₂·H₂O, which has marked adsorptive properties. It is used as an inhibitor of nucleases and also in the **bentonite flocculation test**, a passive agglutination test in which antigen-coated bentonite particles are used to detect specific antibody.

benzidine *trivial name* for 4,4'-diaminobiphenyl, a substance that is carcinogenic for humans. It was formerly used in a highly sensitive test for the presence of blood, giving a green or blue colour in acetic acid solution on addition of hydrogen peroxide. Another former application was in the chromogenic assay of peroxidases. In view of the carcinogenicity, it has been replaced by **2,7-diaminofluorene** and other compounds.

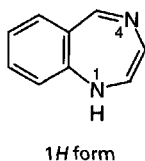
benzo+ or (sometimes before a vowel) **benz+** *comb. form* 1 denoting the fusion of a benzene ring to a cyclic organic molecule. 2 indicating derivation from benzene or benzoic acid; *compare* **phenyl**.

benzoate 1,2-dioxygenase EC 1.14.12.10; *systematic name*: benzoate,NADH: oxygen oxidoreductase (1,2-hydroxylating, decarboxylating); *other name*: benzoate hydroxylase. A bacterial dioxygenase that is involved in the utilization of aromatic compounds. It catalyses a reaction between benzoate, NADH, and dioxygen to form catechol, CO₂, and NAD⁺. Iron is a cofactor. Example from *Acinetobacter calcoaceticus*: this comprises three subunits, two of which, α and β, represent the hydroxylase, while the third, γ, is an electron-transport flavoprotein; α subunit: database code BENA_ACICA, 461 amino acids (52.01 kDa); β subunit (probably involved in substrate specificity): database code BENB_ACICA, 169 amino acids (20.05 kDa); γ subunit (similar to ferredoxin reductase with 2Fe–2S motifs): database code BENC_ACICA, 348 amino acids (38.74 kDa).

benzodiazepine *abbr.*: BZ; 1 2H-1,4-benzodiazepine; a bicyclic compound consisting of a benzene ring fused to a seven-membered partially unsaturated ring containing two nitrogen atoms. 2 any member of a large class of pharmacologically active substances containing the benzodiazepine ring system. They are CNS depressants of low toxicity, having sedative and anxiolytic effects. A number are useful therapeutically; one of the most widely used is **diazepam** (*one proprietary name*: Val-



ium). Specific high-affinity **benzodiazepine binding sites** are present in the CNS. They have been classified as BZ_1 , primarily associated with anxiolytic effects, and BZ_2 . It has been suggested that the effects of benzodiazepines are mediated through enhancement of the inhibitory action of GABA on the brain, and they are known to modulate the effects of GABA at **GABA_A receptors**. Cloning studies of the GABA_A receptor subunits have indicated that the γ subunit confers benzodiazepine sensitivity, and that the combination of different subunit types affects the receptor specificity, the presence of $\alpha 1$ subunits in the complex conferring BZ_1 pharmacology.



benzodiazepine inverse agonist a drug or other compound that binds to benzodiazepine receptors but has effects that are the opposite of those produced by a benzodiazepine agonist, i.e. it causes anxiety and convulsions.

o-benzoquinone 3,5-cyclohexadiene-1,2-dione; 1,2-benzoquinone; *o*-quinone. The oxidized form of **pyrocatechol**.

p-benzoquinone 2,5-cyclohexadiene-1,4-dione; 1,4-benzoquinone; (*p*-)quinone. The oxidized form of **hydroquinone**. See also **quinhydrone**.

benzoyl symbol: Bz or (preferably) PhCO; the C_6H_5 -CO- group; the acyl group derived from benzoic (= benzene-carboxylic) acid.

benzyl symbol: Bzl or (preferably) PhCH₂; the C_6H_5 -CH₂- group; the alkyl group derived from benzyl alcohol (= benzene-methanol).

benzyloxycarbonyl or (formerly) **carbobenzoxy** or **carbobenzyloxy** symbol: Z or Cbz; the C_6H_5 -CH₂-O-CO- group, which is introduced into amino groups of peptides and proteins with reagents such as *N*-(benzyloxycarbonyloxy)succinimide.

Berg, Paul (1926–), US biochemist and molecular geneticist distinguished for his discovery of a means of introducing foreign DNA into bacteria by the use of viruses as vectors; Nobel Laureate in Chemistry (1980) 'for his fundamental studies of the biochemistry of nucleic acids, with particular regard to recombinant-DNA' [prize shared with W. Gilbert and F. Sanger].

Bergström, Sune K. (1916–), Swedish biochemist notable for his isolation and purification of prostaglandins; Nobel Laureate in Physiology or Medicine (1982) jointly with B. I. Samuelsson and J. R. Vane 'for their discoveries concerning prostaglandins and related biologically active substances'.

beriberi a disease due to thiamine (vitamin B₁) deficiency and characterized by degenerative changes in the peripheral nerves, digestive system, and heart accompanied by edema.

Berkfeld filter a (candle) filter, made of diatomaceous earth, used to sterilize solutions. The pore size is relatively large and three porosities are graded V, N, and W (German *viel*, *normal*, and *wenig*) corresponding to coarse, normal, and fine. [After Wilhelm Berkefeld (1836–97), German chemist and manufacturer (the owner of the mine at Kieselguhr, where the earth was mined for production of the filter).]

Bernal chart a means of specifying the position of a spot on an X-ray diffraction photograph by two coordinates. [After John Desmond Bernal (1901–71), British crystallographer who described it in 1926.]

Bernard-Soulier syndrome see **von Willebrand factor**.

Bes or **BES** abbr. for 2-[bis(2-hydroxyethyl)amino]ethanesulfonic acid; a **Good buffer substance**, pK_a (20 °C) = 7.15.

Best, Charles Herbert (1899–1978), Canadian physician and

physiologist notable for discovering the enzyme histaminase and the lipotropic action of choline but particularly for his role (when a research student) in the discovery (with F. G. Banting) of a means of extracting the hormone insulin from pancreatic tissue and demonstrating its antidiabetic efficacy; this achievement was duly recognized by Banting by dividing his share of the Nobel prize with Best.

beta symbol: β (lower case) or B (upper case); the second letter of the Greek alphabet. For uses see **Appendix A**.

beta adrenergic or **β adrenergic** see **adrenergic**.

beta adrenoceptor or **β adrenoceptor** see **adrenoceptor**.

beta barrel or **β barrel** a molecular configuration, more strictly known as an **antiparallel β barrel**, that occurs in the structure of globular proteins. It is characterized by crossing of the beta strands in such a way that the central regions of the strands are buried away from the solvent.

beta bend or **beta turn** or **β bend** or **β turn** a short stretch of polypeptide chain that allows the main direction of the chain to change. It consists of four amino-acid residues in which the CO group of residue *n* is hydrogen bonded to the NH group of residue *n* + 3.

beta blockade the process of inhibiting beta adrenoceptors with an antagonist.

beta blocker an antagonist of beta adrenoceptors.

beta cell or **β cell** (formerly) an alternative name for **B cell** (def. 1).

beta-cell tropin or **β -cell tropin** a peptide fraction obtained from the *pars intermedia* of mouse pituitary that stimulates insulin secretion by perfused rat pancreas or isolated pancreatic islets. It is immunochemically related to, but separable from, **corticotropin-like intermediary lobe peptide**.

betacellulin a potent mitogenic glycoprotein with EGF-like domains, purified from the conditioned medium of mouse pancreatic beta tumour cells, retinal pigment epithelial cells, and vascular smooth muscle cells. It may play a role in the vascular complications associated with diabetes. The C-terminal region shows 50% homology with rat **transforming growth factor α** . The precursor is a type I **membrane protein**, and the mature form is a secretory protein. Example (precursor) from human: database code BTC_HUMAN, 178 amino acids (19.72 kDa).

beta configuration or **β configuration** a stable configuration of a polypeptide chain in which the chain is almost fully extended but slightly puckered and is hydrogen bonded to an adjacent polypeptide chain, thus forming a **beta strand**, the axial distance between adjacent amino-acid residues being $\approx$ 0.35 nm; the adjacent chains may be either **parallel** (running in the same direction) or **antiparallel** (running in opposite directions).

betacyanin see **betalain**.

beta decay or **β decay** decay of a radionuclide by the emission of a **beta particle**.

beta emitter or **β emitter** any nuclide that emits either positive or negative beta particles.

beta globulin or **β globulin** any of a diverse group of **plasma proteins**, including lipoproteins and transferrin, that migrate slower than the **alpha globulins** during electrophoresis at slightly alkaline pH values.

beta granin or **β granin** see **chromogranins**.

beta granule an alternative name for **B granule**.

beta helix or **β helix** a configuration of a polypeptide chain in which each amide group forms hydrogen bonds with the carbonyl group of a residue on average 22 residues removed from it along the chain. The translation along the helix for each residue (the 'pitch') is 0.22 Å and for each turn is 4.8 Å, with 22 residues per turn. The diameter of the helix is thus large (about 27 Å) with a 22 Å hole through the centre of the helix.

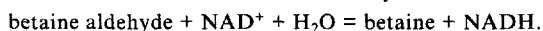
beta hemolysis or **β hemolysis** a type of hemolysis, characteristic of certain strains of streptococci, in which a clear zone is formed around the bacterial colonies when cultured on a blood agar plate. Compare **alpha hemolysis**.



betaine

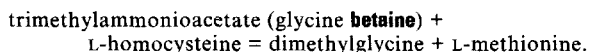
betaine the *N*-trimethyl derivative of an amino acid; the term originally denoted *N*-trimethylglycine, now called **glycine betaine**. Glycine betaine acts as a methyl-group donor in some reactions; *see, e.g., betaine-homocysteine S-methyltransferase*.

betaine-aldehyde dehydrogenase EC 1.2.1.8; an enzyme involved in the formation of **betaine**. It catalyses the reaction:



Example from *Escherichia coli*: database code DHAB_ECOLI, 489 amino acids (52.72 kDa).

betaine-homocysteine S-methyltransferase EC 2.1.1.5; an enzyme that catalyses the reaction:



beta-lactoglobulin or **β lactoglobulin** the principal protein of ruminant milk whey, constituting 50–60% of total whey. It is present as a dimer, $M_r \sim 36\,000$, pI 5.2. Of unknown function, it binds retinol and is a member of the **lipocalin** family. It occurs in milk of other species (e.g. horse), but not human. Analogues occur in marsupial milk. Example (precursor) from *Bos taurus*: database code LACB_BOVIN, 178 amino acids (19.88 kDa). *See also lactalbumin* (def. 1).

betalain any of a group of coloured alkaloids occurring widely in plants of the order Centrospermae. They are divided into two groups, **betacyanins** and **betaxanthins**, and both occur in plant vacuoles. The former impart a red-violet colour and the latter a yellow colour to the plant organs that contain them.

beta lipoprotein or **β lipoprotein** any of the plasma lipoproteins that behave electrophoretically as **beta globulins**, but normally designated **low density lipoprotein**. *See also plasma protein*.

beta oxidation or **β oxidation** the metabolic oxidation of a long-chain fatty acid by successive cycles of reactions during each of which the fatty acid is shortened by a two-carbon-atom fragment removed as acetyl coenzyme A. The name derives from the fact that the β -carbon atom is the point of oxidation. *See also beta-oxidation system, Knoop's hypothesis*.

beta-oxidation system a mitochondrial complex that carries out the reactions of beta oxidation. It contains enzymes that fall into the following four categories:

(1) **Acyl-CoA dehydrogenases**; these include:

(a) **long-chain-acyl-CoA dehydrogenase** (*abbr.*: LCAD) EC 1.3.99.13; this catalyses a reaction between acyl-CoA and electron transferring flavoprotein (ETF) to form 2,3-dehydroacyl-CoA and reduced ETF; FAD is a cofactor. Examples: a mitochondrial matrix protein and a homotetramer, having three isoenzymes with different substrate specificities; deficiency is associated with nonketotic hypoglycemia; precursor (including transit peptide): database code ACDL_HUMAN, 430 amino acids (47.62 kDa);

(b) **acyl-CoA dehydrogenase** is the corresponding medium-chain-specific enzyme, (*abbr.*: MCAD) EC 1.3.99.3; database code ACDM_HUMAN, 421 amino acids (46.54 kDa); and

(c) **butyryl-CoA dehydrogenase** is the short-chain enzyme (*abbr.*: SCAD) EC 1.3.99.2; database code ACDS_HUMAN, 412 amino acids (44.25 kDa).

Both of the latter have corresponding activities and the same subunit composition as LCAD. Deficiencies in MCAD and SCAD are associated with fasting hypoglycemia, hepatic dysfunction, and encephalopathy (often fatal) for the former, and adult lipid storage myopathy and infant acidosis and muscle weakness.

(2) **enoyl-CoA hydratase** EC 4.2.1.17; this catalyses a reaction forming (3*S*)-3-hydroxyacyl-CoA from *trans*-2(or 3)-enoyl-CoA, hydrating the double bond introduced by the enzymes in category 1. Example (rat, precursor): database code ECHM_RAT, 290 amino acids (31.48 kDa); it is a homohexamer and a mitochondrial matrix protein.

(3) **3-hydroxyacyl-CoA dehydrogenase** EC 1.1.1.35; this catalyses the oxidation of (3*S*)-3-hydroxyacyl-CoA by NAD^+ to 3-oxoacyl-CoA with formation of NADH. Example from pig:

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

database code HCDH_PIG, 307 amino acids (33.24 kDa); it is a homodimer and mitochondrial matrix protein.

(4) **3-ketoacyl-CoA thiolase** (*see thiolase*). Example from rat: database code THIM_RAT, 397 amino acids (41.82 kDa); a homotetramer and mitochondrial protein.

beta particle or **β particle** either a (negatively charged) electron or a (positively charged) positron emitted in the **transformation** (def. 5) of a radionuclide.

beta-pleated sheet or **β-pleated sheet** another name for **beta sheet**.

beta radiation or **β radiation** an emission of **beta particles**.

beta receptor or **β receptor** *see adrenoceptor*.

beta sheet or **β sheet** or **β-pleated sheet** an approximately planar array of two or more **beta strands**.

beta strand or **β strand** one of the two basic elements of the **secondary structure** adopted by a polypeptide chain within the hydrophobic core of a globular protein (*see also alpha helix*). It consists of two chains in a **beta configuration**. Two or more strands may interact to form a **beta sheet**.

beta structure or **β structure** any part of the structure of a protein molecule consisting of one or more β strands.

beta turn or **β turn** an alternative name for **beta bend**.

betaxanthin *see betalain*.

B form of DNA *abbr.*: B-DNA or DNA-B; the molecular conformation adopted by fibres of the sodium salt of duplex DNA at high relative humidity and in aqueous solution. It consists of a right-handed double helix containing about ten nucleotide residues per turn with the planes of the base pairs normal to the axis of the helix. It is the basis of the **Watson-Crick model of DNA**, and is thought to resemble the conformation of most of the DNA *in vivo*. *See also A form* (def. 1), **C form**, **Z form**.

B/F ratio *abbr.* for bound-to-free ratio (in ligand binding). A plot of the B/F ratio versus the concentration of free ligand is known as a **Scatchard plot**.

BGG *abbr.* for bovine gamma globulin; the component of bovine serum represented by fraction II of a **Cohn fractionation**.

B granule or **beta granule** any of very numerous membrane-bound vesicles (granules) containing stored insulin that are found in the **B cells** of the pancreatic islets. The granules stain a deep bluish purple with Gomori's aldehyde fuchsin and Ponceau stain.

bhang *see cannabis*.

BHK cell *abbr.* for baby hamster kidney cell; a cultured cell derived from this organ.

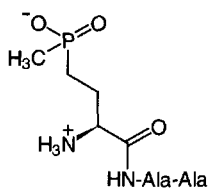
BHT *abbr.* for butylated hydroxytoluene, di-*tert*-butyl-*p*-cresol, an antioxidant.

bi+ 1 *comb. form* meaning two, twice, or double. 2 *prefix* (in organic chemical nomenclature) indicating two identical groups or rings joined by a link; e.g. biacetyl, biphenyl. 3 *prefix* (in inorganic chemical nomenclature) signifying an acid salt of a dibasic acid; e.g. sodium bicarbonate (though now more correctly termed sodium hydrogencarbonate). 4 (*in enzymology*) denoting two kinetically important substrates and/or products of an enzymic reaction (*see reactivity of enzymes*). *See also bis+*, **di+**.

Bi symbol for bismuth.

BIAcore *proprietary name* of an optical biosensor instrument that measures refractive index changes to record in real-time interactions of macromolecules with ligands immobilized on a dextran-coated gold surface. The resulting 'sensorgrams' can be used to derive kinetic parameters of the interactions.

bialaphos a linear tripeptide produced as a secondary metabolite by *Streptomyces hygroscopicus*. The structure is PT-Ala-Ala, where PT is the glutamate analog, phosphinothricin. After hydrolysis of bialaphos by intracellular nonspecific peptidases, PT acts as an inhibitor of **glutamate-ammonia ligase** (glutamine synthetase), and thereby derives herbicidal and antibiotic activities.



bialaphos

Bial's test a colorimetric test for aldopentoses. See **orcinol test**. [After Manfred Bial (1870–1908), German physician.]

bibrototoxin a vasoconstrictor peptide of the **endothelin/sarafotoxin** family, isolated from the venom of the burrowing wasp *Atractaspis bibroni*. The sequence,

CSCADMTDKECLYFCHQDVIW,

is identical to that of sarafotoxin B except that Ala⁴ replaces Lys⁴.

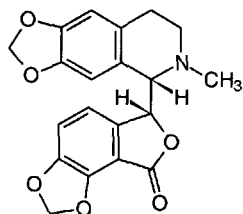
bicarbonate the HCO₃⁻ ion; now more correctly termed hydrogencarbonate.

bichromatic analysis a term sometimes used to describe any analytical method involving observation at two wavelengths.

Bicine or **BICINE** *abbr. for* *N,N*-bis(2-hydroxyethyl)glycine; a **Good buffer substance**; p*K*_a (20 °C) = 8.35.

bicoid protein a segment-polarity protein, discovered in *Drosophila*, that provides positional cues for the development of head and thoracic segments. It regulates the expression of zygotic genes, possibly through its **homeodomain**, and inhibits the activity of other maternal gene products. It forms an anterior–posterior gradient in the embryo and acts as a transcriptional activator. It binds to *caudal* mRNA. There is similarity to other **paired-type** homeobox proteins. Example from *Drosophila melanogaster*: database code HMBC_DROME, 494 amino acids (54.43 kDa).

bicuculline 6-(5,6,7,8-tetrahydro-6-methyl-1,3-dioxolo[4,5-g]isoquinolin-5-yl)furo[3,4-*e*]-1,3-benzodioxol-8(6*H*)-one; a naturally occurring alkaloid with several sources, including *Dicentra cucullaria* and *Adlumia fungosa*. It is a potent convulsant with effects probably mediated by its GABA antagonist activity at GABA_A receptors.



bidentate 1 having two teeth or two toothlike processes. **2** describing a **ligand** molecule that chelates a metal ion by means of two donor atoms.

bifunctional having two functions.

bifunctional antibody an antibody having two combining sites for antigen.

bifunctional catalyst a catalyst that provides both acidic and basic catalytic functions.

bifunctional reagent a compound with two reactive functional groups that can interact with two groups in one molecule or with one group in each of two different molecules.

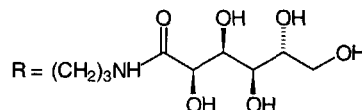
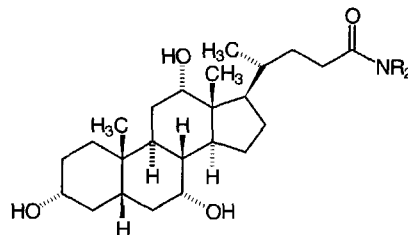
big-big gastrin or **big gastrin** *see* **gastrin**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

BigCHAP *N,N*-bis(3-D-glucanamidopropyl)cholamide; a non-ionic detergent analogue of **CHAPS**, but with reduced electrostatic properties; aggregation number 10, CMC 3.4 mM.



big gastrin *see* **gastrin**.

big glucagon or **big plasma glucagon** material that is separable from normal human plasma and shows glucagon-like immunoreactivity. It has an estimated *M*_r of 160 000.

biglycan bone and cartilage proteoglycan I precursor; a connective tissue glycoprotein of the extracellular matrix related to **decorin** and fibromodulin. Example (precursor) from human: database code PGS1_HUMAN, 368 amino acids (41.61 kDa).

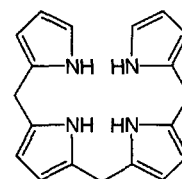
bijou bottle a small glass screw-capped bottle of 5–7 ml capacity, used especially for liquid and solid cultures.

bikunin a plasma glycoprotein found both in the free state and complexed with the heavy chains of the inter- α -inhibitor family, forming the light chain. It is a serine protease inhibitor having a tandem arrangement of Kunitz domains; it may participate in the control of events such as endothelial cell growth and oocyte cumulus expansion and stabilization. It is synthesized from an α_1 -microglobulin/bikunin precursor. Example from human: database code HC_HUMAN, 352 amino acids (38.96 kDa); residues 20–203 form α_1 -microglobulin and 206–352 form inter- α -trypsin inhibitor light chain (the inhibitory subunit).

biladiene the *semisystematic name* for either of two linear tetrapyrroles in which the carbon bridges contain two more double bonds than **bilane** i.e. two of the three carbon bridges in the molecule are methine groups.

bilamellar having two **lamellae**; formed of two plates.

bilane the *systematic name* for a linear tetrapyrrole in which each of the three carbon bridges linking the pyrrole nuclei is saturated, i.e. is a methylene group, and in which, unless otherwise specified, each of the four nitrogen atoms also is saturated. It is the fundamental structure for the naming of linear tetrapyrroles, is defined without oxygen substituents, and is numbered (omitting C₂₀) to agree with the numbering of the unsubstituted porphyrin ring system. *Trivial name*: bilinogen.



bilateral hydrophobic zippers putative α -helical segments of the acidic ribosomal proteins, A1, A2, L44', and L45, of *Saccharomyces cerevisiae*. They have two arrays of 5–7 hydrophobic residues on opposite sides. Compare **leucine zipper**, **bZIP**.

bilatriene

bilatriene an alternative (less-favoured) name for **bilin**.

bilayer a layer that is two molecules thick, as in some membranes. See **lipid bilayer**, **unit membrane**.

bilayer lipid membrane see **black lipid membrane**.

bile a secretion produced by the vertebrate liver and conveyed through the bile duct to the duodenum. Its alkaline nature helps to neutralize acidic gastric secretions passing from the stomach, and the **bile salts** it contains are involved in the digestion of fat. Bile also serves as a medium for the excretion of various substances, notably cholesterol (either as such or in the form of bile salts to which it is converted), pigments (from hemoglobin breakdown), heavy metals, and other waste products (usually of >400 Da).

bile acid any member of a group of steroid carboxylic acids occurring in bile, where they are present as the sodium salts of their amides with glycine or taurine (see **bile salt**). The C_{24} acids are hydroxy or keto derivatives of cholanic or allocholanic acids; some C_{27} bile acids are known in eutherian mammals. In mammals, the C_{24} acids are found conjugated with glycine (see **glycocholate**) as well as with taurine (see **taurocholate**); in the rest of the animal kingdom only taurine conjugates have been found.

bile alcohol any of a group of polyhydroxy derivatives of **cholestane**, present as their sulfuric acid esters in the bile of amphibians and some fishes.

bilene the semisystematic name for either of two tetrapyrroles in which the carbon bridges contain one more double bond than **bilane**; i.e. one of the three bridges in the molecule is a methine group.

bile pigment any bilirubinoid pigment present in bile; i.e. any linear tetrapyrrole derived from a **porphyrin** (which is a cyclic tetrapyrrole). For physiological reasons certain dipyrroles are also sometimes regarded as bile pigments.

bile salt the sodium salt of the conjugate of any **bile acid** with either glycine or taurine. Bile salts are potent surfactants; in the gut they emulsify ingested lipid, promote its hydrolysis by the activation of lipases, and facilitate its absorption, thereafter being excreted in the feces. See also **glycocholate**, **taurocholate**.

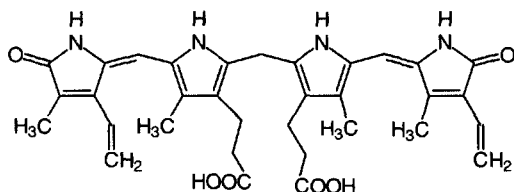
bilial of or relating to **bile**.

bilin 1 the systematic name for the linear tetrapyrrole in which the carbon bridges contain three more double bonds than **bilane**; i.e. all three bridges are methine groups. This name is preferred to **bilatriene**. 2 any coloured **bile pigment** formed by the oxidation of a colourless bile pigment, or **bilinogen** (def. 2). Thus the bilinogens urobilinogen and stercobilinogen form urobilin and stercobilin, respectively, on oxidation.

bilinogen 1 trivial name for **bilane**. 2 any colourless **bile pigment** that may be oxidized to a coloured bile pigment (see **bilin** (def. 2)).

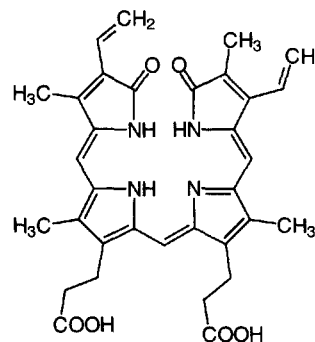
biliprotein an alternative name for **phycobiliprotein**.

bilirubin or (formerly) **biliverdin IX α** the recommended trivial name for the linear tetrapyrrole 8,12-bis(2-carboxyethyl)-2,7,13,17-tetramethyl-3,18-divinylbiladiene-ac-1,19(21*H*,24*H*)-dione. It is produced in the reticuloendothelial system by the reduction of **biliverdin** and transported to the liver as a complex with serum albumin. In the liver, bilirubin is rendered water-soluble by conjugation of the carboxyethyl side-chains to form bilirubin bisglucuronide, most of which is excreted in the bile. Bilirubin accumulates in the blood and tissues in jaundice.



bilirubin oxidase EC 1.3.3.5; an enzyme that catalyses the oxidation by dioxygen of bilirubin to biliverdin and H_2O . Example from the fungus *Myrothecium verrucaria*: database code B48521, 572 amino acids (63.88 kDa).

biliverdin or (formerly) **biliverdin IX α** the recommended trivial name for the linear tetrapyrrole 8,12-bis(2-carboxyethyl)-2,7,13,17-tetramethyl-3,18-divinylbilin-1,19(21*H*,24*H*)-dione. It is formed in the reticuloendothelial system by the first step in heme degradation, catalysed by heme oxygenase (decyclizing), EC 1.14.99.3, a reaction in which the α -methene group of heme is oxidized to carbon monoxide. Biliverdin is then reduced by biliverdin reductase, EC 1.3.1.24, to **bilirubin**.



biliverdin reductase EC 1.3.1.24; an enzyme that catalyses the reaction of bilirubin with $NAD(P)^+$ to form biliverdin and $NAD(P)H$. Example from human: database code PN0159, 204 amino acids (21.83 kDa).

bimodal (in statistics) distribution with two modes.

bimolecular having a **molecularity** of two; relating to, consisting of, or involving two molecular entities.

bimolecular layer a layer that is two molecules thick.

bimolecular reaction any chemical reaction in which there is a transition state composed of the atoms of two separate molecular entities, which may be the same or different.

binial double; twofold.

binary 1 consisting of or forming two things or parts; dual. 2 describing a number system having the numeral 2 as its **base** (def. 4b). See also **bit**.

binary compound a compound consisting of only two chemical elements; e.g. ammonia, sodium chloride, water.

binary digit see **bit**.

binary fission the reproduction of a living cell by division into two equal, or near-equal, parts.

binary operator any arithmetic or logical operator (e.g. +, AND, OR) that operates on two quantities.

binary representation a notation by which alphanumeric symbols are represented by two digits, 0 and 1, in a known sequence. The displacement of one digit position to the left denotes multiplication by a power of 2.

binder protein 1 see **cortisol metabolite binding protein** (large).

bindin a protein of 285 amino acids, M_r 35 000, extracted from the insoluble granular material of sea-urchin sperm, responsible for the attachment of the sperm to the vitelline layer of the egg. Bindins may mediate the species-specific recognition of eggs by sperm. See also **spermadhesin**, **zona pellucida sperm-binding protein**.

binding the act or process by which one molecule attaches to another by noncovalent forces; ligation see **ligate** (def. 3). See **ligand**. See also **binding site**.

binding assay an alternative term for **saturation analysis**.

binding capacity a measure of the quantity of ligand or the number of ligand molecules that can be bound by a given amount of a particular binding agent or system.

binding constant an alternative term for **affinity constant**.



binding curve

binding curve any graphical representation of the extent of binding between two compounds at varying relative concentrations of the reactants.

binding energy the free energy change accompanying the binding of n_i ligand molecules to a macromolecule to produce a particular configuration of the liganded macromolecule.

binding fraction an alternative term for **saturation fraction**.

binding potential *symbol:* Π ; a concept introduced to clarify the definition of linked functions, linkage groups, and linked sites in ligand binding. It is defined by: $\nu_i = \delta \Pi / \delta \mu_i$, where ν_i is the ratio of component i to that of a reference component and μ_i is the chemical potential of i .

binding protein *abbr.:* BiP; a **luminal protein** of the endoplasmic reticulum that is a member of the hsp70 family of **heat-shock proteins** and is a typical **molecular chaperone**. It recognizes incorrectly folded proteins as well as protein subunits that have not yet assembled into their oligomeric complexes and facilitates their oligomerization of folding. BiP activity requires the participation of ATP, which is hydrolysed to ADP and inorganic phosphate. Example (precursor) from human: database code GR78_HUMAN, 653 amino acids (72.03 kDa); this is also known as 78 kDa glucose regulated protein.

binding set the spatially organized collection of all the groups or residues of *one* protomer that are involved in its binding to *one* other protomer.

binding site a specific region in a macromolecule or other molecular entity, or in a membrane, that takes part directly in the combination with a **ligand**. Binding sites may be independent or interacting, depending on whether or not the binding of one ligand to one site affects the binding of other ligands to other sites on the same macromolecule, or membrane. *See also* **co-operative ligand binding**.

binomial **1** a mathematical expression made up of two terms, e.g. $2x + 5y$. **2** a two-part taxonomic name (*see* **binomial nomenclature**).

binomial coefficient the coefficient of any term arising from the expansion of the algebraic binomial $(a + b)^n$.

binomial distribution a statistical function in which the probabilities that an event will or will not occur are given by the successive coefficients of a binomial expansion of the form $(a + b)^n$, where a and b are the probabilities that an event will and will not occur, respectively.

binomial nomenclature a system of naming organisms, devised by the Swedish botanist Carolus Linnaeus (1707–78), that employs two Latin names, the first being the genus and the second the species to which the organism belongs.

bio+ comb. form denoting life or living organisms, or systems derived from them.

bioactivation the activation of a drug or other substance in an organism or by means of a biological or biochemical system.

bioactive an alternative term for **biologically active**.

bioanalytical of or pertaining to biochemical analysis.

bioassay or **biological assay** any quantitative estimation of biologically active substances by measurement of the magnitude of their actions on living organisms or parts of organisms. It is performed under standard conditions and by reference to preparations of defined potency or activity.

bioautograph the record obtained in **bioautography**.

bioautography a method for detecting, in a complex mixture, small amounts of substances that are able to act as essential growth factors for a test organism(s). The components of the mixture are first separated by chromatography. The resulting chromatogram is then placed in contact with a culture of the test organism in a solid medium lacking a specific component. Hence, the organism will only grow where the component is present in the chromatogram, thereby establishing its presence in the original mixture.

bioavailability **1** the relative amount of the dose of a drug or other substance reaching the systemic circulation. **2** the rate and extent to which the therapeutic moiety of a drug is absorbed and becomes available to the site of drug action.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

bioblast any very small mass of protoplasm having formative powers, formerly considered to be the ultimate elementary living particles; it is now thought the particles seen may have been **mitochondria**.

biocatalyst any agent, present in or obtained from living material, that can increase the rate at which a chemical reaction occurs without itself undergoing permanent change; a biological or biochemical **catalyst**. Biocatalysts are commonly **enzymes**. *See also* **ribozyme**. —**biocatalysis** *n*.

biochemical **1** of or pertaining to **biochemistry**. **2** any chemical substance occurring in or isolated from an organism.

biochemical engineering or **bioengineering** the scientific discipline underlying the industrial processing of biological materials, e.g. fermentation-based pharmaceuticals, foods, and effluents.

biochemical evolution the evolution of the biochemical processes and components of living organisms, such as the structure and function of biologically important molecules, metabolic pathways, subcellular structures, and cells.

biochemical genetics the branch of genetics concerned with the chemical nature of hereditary determinants and how they function at the molecular level.

biochemical imprinting the adaptive increase in enzyme activities that can occur along specified metabolic pathways when there is a large and sustained traffic through them.

biochemical lesion any biochemical change, such as the absence or inactivation of an enzyme, that diminishes the fitness of an organism or leads to a pathological condition.

biochemical marker any specific character that may be detected by a biochemical test, e.g. the presence or absence of a particular enzyme.

biochemical oxygen demand or **biological oxygen demand** *abbr.:* BOD; a measure of pollution of water in lakes, rivers, etc. due to the presence of organic matter. Oxidation of the latter by microorganisms removes dissolved oxygen, hence reducing the ability of the water to sustain aquatic life. BOD is the amount of oxygen (mg) taken up by a 1-litre sample of water in 5 days (BOD_5) at 20 °C in the dark; it may be necessary to add additional, uncontaminated, water to augment the amount of oxygen available. Sometimes incubation is for a longer time, e.g. 20 days, in which case it is indicated by BOD_{20} . *Compare* **chemical oxygen demand**.

biochemistry or **biological chemistry** the branch of science dealing with the chemical compounds, reactions, and other processes that occur in living organisms. Lehninger expressed the challenge to the biochemist as follows: 'Living things are composed of lifeless molecules. When these molecules are isolated and examined individually, they conform to all the physical and chemical laws that describe the behaviour of inanimate matter. Yet living organisms possess extraordinary attributes not shown by collections of inanimate molecules.' In this regard Horowitz has proposed a set of criteria for living systems: 'Life possesses the properties of replication, catalysis and mutability.' Biochemists are, therefore, concerned with the manner in which living organisms exhibit these properties. —**biochemist** *n*.

biochip an integrated circuit in which electrical and logical functions are performed by protein molecules appropriately manipulated.

biocide any chemical substance that kills living cells or organisms. —**biocidal** *adj*.

biocompatible compatible with life; having no injurious effect on living organisms. —**biocompatibility** *n*.

bioconversion a technique in which the enzymic apparatus of an organism is used to convert or process a chemical substance, not normally present in the organism, into another, (often commercially) useful one.

biocytin ϵ -N-biotinyl-L-lysine; a naturally occurring complex of **biotin**, found especially in autolysates of rapidly growing yeast. It is liberated from biotin holoenzymes by proteolysis.

biodegradable capable of being broken down by the action of living organisms. —**biodegradability** *n*.



biodegradation

biodegradation 1 the processes by which exogenous, usually waste materials are broken down by living organisms. Sometimes this breakdown involves two or more kinds of organisms acting in cooperation. Such degradative abilities are employed to convert waste materials into more acceptable and manageable forms or to produce useful end products. 2 an alternative term for **catabolism**. Compare **biosynthesis**.

biodeterioration any undesirable change in the properties of a material brought about by the activities of organisms.

bioelectricity electric current or electric potential produced in living organisms. —**bioelectric** *adj.*

bioelectrochemistry the application of electrochemical models and techniques to biological problems.

bioelectronics the application of biomolecular principles to microelectronics, as in **biosensors** and **biochips**.

bioelement any element that is an essential component of living organisms. Bioelements include: as major elements, C, H, O, N, P, and S, found chiefly in combination in organic compounds, and the elements, Na, K, Mg, Ca, and Cl, found mainly as monatomic ions. Many other 'trace elements' occur in relatively minute amounts.

bioenergetics the study of energy transformations in living systems.

bioengineering 1 engineering relating to the operation on an industrial scale of biochemical processes, especially fermentation. This is usually now termed **biochemical engineering**. 2 the application of the physical sciences and engineering to the study of the functioning of the human body and to the treatment and correction of medical conditions.

biofuel cell any fuel cell (a cell in which electricity is produced directly by oxidation of fuel) in which one or both electrode reactions are promoted or catalysed by a biological process.

Bio-Gel *proprietary name* for a group of materials used in gel filtration, consisting of either cross-linked agarose (Bio-Gel A) or porous polyacrylamide (Bio-Gel P).

biogenesis 1 the synthesis of a substance in a living organism; biosynthesis. 2 the principle that a living organism can originate only from a parent or parents that are similar to itself. Compare **abiogenesis**. —**biogenetic**, **biogenetical**, **biogenous** *adj.*

biogenic produced by living organisms.

biogenic amine any of various amines, especially those with neurological activity, isolated from plants or animals.

bioinformatics *see* Appendix E.

biohazard any organism or material derived from a biological source that is hazardous to health or life.

bioinorganic chemistry a branch of inorganic chemistry concerned with the interactions between inorganic substances (especially metallic ions or their complexes) and biological or biochemical systems, and with the properties of related model systems. Compare **inorganic biochemistry**.

biolistics descriptive of the administration of DNA in gold particles, using a device known as a 'gun' to deliver the particles to the cell interior. One use has been in the development of vaccines by the administration of suitable genes to foreign cells.

biolith a rock or other deposit (e.g. peat, humus) formed from organic material.

biological *adj.* 1 of or pertaining to **biology**. 2 of or pertaining to life or living organisms. 3 a biological product that is used prophylactically or therapeutically. —**biologically** *adv.*

biological activity any activity of a substance that is demonstrable in living organisms. Biologically active substances are often of biological origin themselves.

biological clock any biological mechanism that allows the expression of a certain biological structure (e.g. a gene) or a biological function (e.g. sleep) at periodic intervals. The term is also used colloquially to describe ageing, e.g. in respect to a woman's ability to bear a child.

biologically active or **bioactive** active in a living organism.

biological value *abbr.*: B.V.; a measure of the value of a protein in a foodstuff for the maintenance of growth and normal

functioning of any given animal or species. It is based on the amino-acid composition of the protein, its digestibility, and the availability of its digestion products.

biology 1 the scientific study of living organisms, including their structure, functioning, development, distribution, interrelationships, and evolution. 2 the animal and plant life of a particular region considered as a unit. 3 the structure, functioning, etc. of a particular organism or group of organisms. —**biologist** *n.*

bioluminescence the production of light by certain enzyme-catalysed reactions in living organisms. —**bioluminescent** *adj.*

biolysis any decomposition of organic material by biological means. —**biolytic** *adj.*

biomass 1 the total mass of living organisms, of a specified kind, forming a population inhabiting a given region. 2 the material produced by the growth of microorganisms, plants, or animals, especially as a product of or raw material for an industrial process, such as farming or fermentation.

biomedical science or **biomedicine** any branch of science that is relevant to medicine.

biomembrane any membrane surrounding a cell or a subcellular organelle, or derived from these structures.

biometry the statistical study of biology. —**biometric** *adj.*; **biometrically** *adv.*

biomimetic chemistry a branch of organic chemistry that aims to imitate natural reactions and enzymatic processes as a means to improve the power of organic chemistry.

biomolecular engineering a term sometimes used to cover those branches of biotechnology dealing with enzyme engineering and genetic engineering.

biomolecule any molecule, especially a macromolecule, that occurs in or is formed by living organisms. —**biomolecular** *adj.*

bioorganic chemistry a branch of organic chemistry concerned with the study of organic molecules from living systems.

biophotolysis any type of **photolysis** reaction occurring in a biological system; e.g. the lysis of a water molecule in the reactions of **photosystem II** during photosynthesis.

biophysical chemistry the application of the techniques or methods of physical chemistry to biological systems. It is principally concerned with the conformations, shape, structure, conformational changes, dynamics, and interactions of macromolecules and complexes or arrays of macromolecules. Compare **biophysics**, **physical chemistry**.

biophysics the application of physical techniques and physical methods of analysis to biological problems. Traditionally, the discipline has focused on two main areas: first, the transmission of nerve signals and the maintenance of electrical potentials across membranes; and second, large-molecule crystallography and enzyme structure and mechanisms. For the latter X-ray crystallography has been supplemented by a number of physical techniques, including nuclear magnetic resonance spectrometry, mass spectrometry, fluorescence-depolarization measurements, and circular-dichroism studies. —**biophysical** *adj.*; **biophysicist** *n.*

biopolymer any **polymer**, such as a protein, nucleic acid, or polysaccharide, occurring in, or formed by, living systems.

biopotency the extent to which a material (e.g. drug or hormone) influences a biological process.

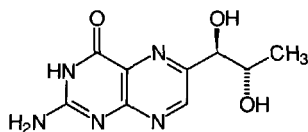
bioproduct any (usually) useful product produced by living organisms or by systems derived from living organisms.

biopsy examination of (usually) a sample of tissue, cells, or fluid from a living human or animal for diagnostic purposes. The term is also now commonly used for the process of removing such a sample. Compare **necropsy**.

biopterin 2-amino-4-hydroxy-6-(1,2-dihydroxypropyl)-pteridine; 1-(2-amino-4-hydroxy-6-pteridyl)-1,2-propanediol; a growth factor for certain protozoans, e.g. *Crithidia fasciculata* and *Trypanosoma platydictyli*. It is widely distributed in tissues and functions in a reduced form, **tetrahydrobiopterin**, as a hydroxylation coenzyme, e.g. in the conversion of phenylalanine to tyrosine. *See* also **phenylalanine hydroxylase**.

bioreactor

nine to tyrosine. It has been isolated from queen-bee jelly and is considered a growth factor for some insects.



bioreactor any reactor containing immobilized enzymes or cells that is used to effect specific chemical reactions in the production of some economically desirable product.

bioregulator omnibus term sometimes used for any substance occurring within a cell or organism that has a specific and reversible controlling action on a particular biochemical event within that cell or organism; it may be, e.g., a hormone, an enzyme, an intermediate metabolite, an inorganic ion, or a cyclic nucleotide.

bioremediation the use of biological systems to remedy environmental damage.

bios the former name for various growth factors required by yeast. Originally believed to be a single factor, it was later fractionated into bios I (shown to be **inositol**), bios IIA (**β -alanine**), and bios IIB (**biotin**).

bioscience(s) collectively all the individual branches of science that are concerned with biology.

biose disaccharide; stem name used in the established trivial names for particular disaccharides whether or not they contain a single type of sugar residue and linkage, as in cellobiose and gentiobiose.

+biose suffix used in nomenclature of glycolipids to indicate a compound containing an oligosaccharide moiety composed of two sugar residues (which may be the same or different), as in galabiose.

biosemiotics the study of signs, of communication, and of information in living organisms.

biosensor a device that uses specific biochemical reactions mediated by isolated enzymes, immunosystems, tissues, organelles, or whole cells to detect chemical compounds, usually by electrical, thermal, or optical signals.

biospecific describing: 1 a reagent, e.g. an antibody or hormone, that can recognize and interact only with cells (e.g. within a mixed population of cells) displaying a particular surface characteristic, e.g. a corresponding antigen or receptor. 2 any reagent that, in a similar way, recognizes and interacts with a particular biospecifically occurring substance. 3 any reagent that interacts specifically with a particular biomolecule (e.g. an enzyme). 4 the reaction between two biospecific reagents.

biospecific-elution chromatography an alternative name for **affinity-elution chromatography** (when the ligand in the eluting solvent is a biospecific macromolecule).

biosphere the parts of the Earth's crust and atmosphere that contain living organisms.

biosynthesis the production of substances by processes occurring in living systems; **anabolism**; **biogenesis**. Compare **biodegradation**.

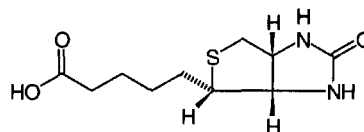
biosynthetic 1 of or relating to **biosynthesis**. 2 any substance produced by biosynthesis.

biosynthetic pathway any metabolic pathway effecting **biosynthesis**; an anabolic pathway.

biotechnology 'the integration of natural sciences and engineering sciences in order to achieve the application of organisms, cells, parts thereof and molecular analogues for products and services' (European Federation of Biotechnology General Assembly, 1989); a field of technological activity in which biochemical, genetic, microbiological, and engineering techniques are combined for the pursuit of technical and applied aspects of research into biological materials and, in particular, into biological processing. It includes traditional technologies such

as fermentation processes, antibiotic production, and sewage treatment, as well as newer ones such as **biomolecular engineering**, and **single-cell protein** production. — **biotechnological** *adj.*

biotin *cis*-tetrahydro-2-oxothieno[3,4-*d*]imidazole-4-valeric acid; hexahydro-2-oxo-1*H*-thieno[3,4-*d*]imidazole-4-pentanoic acid; vitamin H; formerly known as bios IIB and coenzyme RCO₂; a substance the (+) enantiomer of which is very widely distributed in cells and serves as a carrier in a number of enzymic β -carboxylation reactions. In biotin-containing enzymes the biotin is covalently bound at the active site through an amide link to the ϵ -amino group of a lysine residue. It was first found as a growth factor for yeasts. Compare **biocytin**. See also **streptavidin**.



biotin carboxylase see **acetyl-CoA carboxylase**.

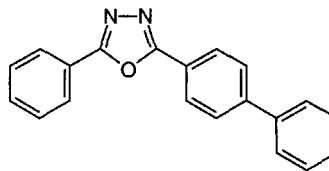
biotinyl the acyl group derived from **biotin**.

biotinylation the act or process of attaching a biotinyl residue by use of an appropriate reactive derivative, e.g. biotin 4-nitrophenyl ester. It is useful for labelling cell-surface functional groups or biologically active molecules. The presence of the label is subsequently detected, or the labelled material is separated, by means of the specific and high-affinity avidin-biotin reaction, using avidin or **streptavidin** conjugated with, e.g., a fluorochrome, an enzyme, an antibody, or an immobilizing agent. — **biotinylate** *vb.*; **biotinylated** *adj.*

biotransformation the conversion or conjugation of a foreign compound introduced into an organism or a culture of cells, especially (in *biotechnology*) with formation of a substance of commercial value. Compare **detoxication**.

BiP *abbr.* for binding protein.

biphenylphenyloxadiazole *abbr.*: PBD; 2-phenyl-5-(4-biphenyl)-1,3,4-oxadiazole; a high-efficiency primary scintillator. The *tert*-butyl derivative, 2-(4-*tert*-butylphenyl)-5-(4-biphenyl)-1,3,4-oxadiazole (*abbr.*: butyl-PBD) has greater solubility and only a slightly lower photon yield. See also **scintillation cocktail**.



biphosphate an obsolete name for the anion H_2PO_4^- or $\text{PO}_2(\text{OH})_2^-$, now correctly termed dihydrogenphosphate.

biradical or **diradical** any molecular entity in an electronic state (described by a formula or by a combination of contributing structures) that contains two unpaired electrons in atomic orbitals on different atoms (compare **carbene**). Both singlet and triplet species are embraced.

birefringence an alternative term for **double refraction**. — **birefringent** *adj.*

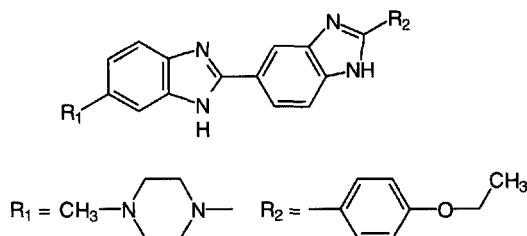
bis+ *comb. form* 1 denoting two, twice, doubled; see also **bi+** (def. 1), **di+** (def. 1). 2 (in *chemical nomenclature*) a indicating the presence in a molecule of two identical organic groups each substituted in the same way, e.g. **bis(2-chloroethyl)sulfide**. b indicating the presence in a molecule of two separate inorganic oxoacid residues, e.g. fructose 1,6-bisphosphate (formerly known as fructose 1,6-diphosphate). Compare **di+** (def. 1).

bisalbuminemia or (*esp. Brit.*) **bisalbuminaemia** a symptom-

bisbenzimidazole

less human condition in which the presence of a variant serum albumin leads to a double albumin band on electrophoresis of serum proteins.

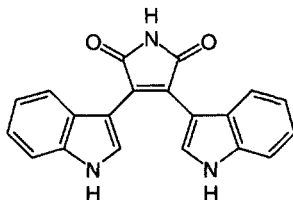
bisbenzimidazole 4-[[5-(4-methyl-1-piperazinyl)-2-benzimidazolyl]-2-benzimidazolyl] phenol (Hoechst 33258) or its ethyl ether (Hoechst 33342); a substance that is useful as a DNA-binding fluorochrome for staining chromosomes within cell nuclei. It is believed to be specific for A-T base pairs.



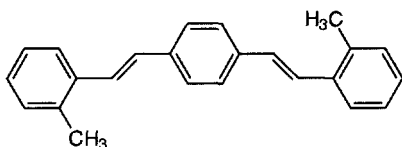
bis(2-chloroethyl)sulfide or (formerly) **mustard gas** ($\text{ClCH}_2\text{-CH}_2\text{)}_2\text{S}$; a bifunctional alkylating agent that causes lethal cross-linking of DNA chains.

Bishop, John Michael (1936–), US microbiologist; Nobel Laureate in Physiology or Medicine (1989) jointly with H. E. Varmus 'for their discovery of the cellular origin of retroviral oncogenes'.

bisindolylmaleimide GF 109203X; a protein kinase inhibitor, structurally similar to **staurosporine**, that is highly selective for **protein kinase C** (PKC), though not between isotypes α , β I, β II, or γ . It acts as a competitive inhibitor at the ATP-binding site on PKC. It exhibits a K_i of 14 nM for the purified enzyme, but its potency on whole cells is much lower (K_i 1–2 μM), possibly owing to high cellular ATP levels.



bismethylstyrylbenzene *abbr.*: bis-MSB; 1,4-bis(2-methylstyryl)-benzene; a secondary scintillator with good resistance to quenching and low reactivity with compounds in samples. See also **scintillation cocktail**.



bis(methylthio)gliotoxin a naturally occurring analogue of **gliotoxin**. It inhibits platelet activating factor (PAF)-induced platelet aggregation ($\text{IC}_{50} = 84 \mu\text{M}$) (but not platelet aggregation induced by arachidonic acid or ADP) and *in vivo* PAF-induced bronchoconstriction; it lacks the antiviral and immunomodulating effect of gliotoxin. (Illustrated top right.)

bis-MSB *abbr.* for bismethylstyrylbenzene.

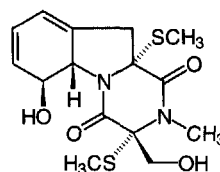
bisphosphate any compound containing two independent phosphoric residues in ester linkages, at positions indicated by locants. Compare **diphosphate** (def. 3, 4).

bisphospho+ prefix to a chemical name indicating the presence of two independent phosphoric residues in ester linkages, at positions indicated by locants. Compare **diphospho+**.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,
Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

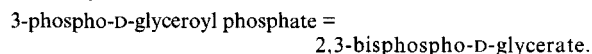
2,5-bis(3,4,5-trimethoxyphenyl)-1,3-dioxolane



bis(methylthio)gliotoxin

2,3-bis(phospho)-D-glycerate a recommended name for (R)-[2,3-dihydroxypropanoate 2,3-bis(dihydrogenphosphate)] (formerly termed 2,3-diphosphoglycerate); *abbr.*: BPG; it decreases the dioxygen affinity of hemoglobin, its concentration in the erythrocyte being regulated by **bisphosphoglycerate mutase** (synthesis) and **bisphosphoglycerate phosphatase** (degradation). It does not pass through the erythrocyte membrane. See also **bisphosphoglycerate pocket**.

bisphosphoglycerate mutase *abbr.*: BPGM; EC 5.4.2.4; *systematic name*: 3-phospho-D-glycerate 1,2-phosphomutase; *other names*: 2,3-bisphosphoglycerate mutase; 2,3-bisphosphoglycerate synthase; diphosphoglycerate mutase; glycerate phosphomutase; bisphosphoglycerate synthase. An enzyme that catalyses the reaction:

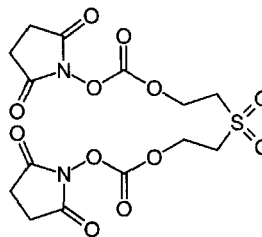


The enzyme plays a major role in regulating the dioxygen affinity of hemoglobin by controlling 2,3-bisphosphoglycerate concentration in the erythrocyte, but with a reduced activity. Example from human: database code PGME_HUMAN, 258 amino acids (29.84 kDa).

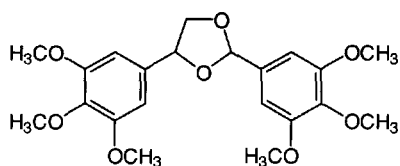
bisphosphoglycerate phosphatase EC 3.1.3.13; *systematic name*: 2,3-bisphospho-D-glycerate 2-phosphohydrolase; an enzyme that catalyses the hydrolysis of 2,3-bisphosphoglycerate to 3-phosphoglycerate and orthophosphate; it thereby assists regulation of the concentration of 2,3-bisphosphoglycerate.

bisphosphoglycerate pocket *abbr.*: BPG pocket; the cavity (pocket) between the two β chains of deoxyhemoglobin in which a single molecule of 2,3-bisphospho-D-glycerate (BPG) binds; the binding of this molecule reduces the affinity of hemoglobin for dioxygen. Residues Val¹ (terminal -NH_3^+), His², Lys⁸², and His¹⁴³ of each β chain participate in this binding. It plays an important part in establishing the difference in oxygen affinity that is found between maternal and fetal hemoglobin; in fetal hemoglobin, the γ chain has a serine in the position occupied by His¹⁴³ of the β chain of the adult hemoglobin, which weakens the affinity of fetal hemoglobin for BPG and thus results in a higher affinity for dioxygen than adult hemoglobin.

bis[2-succinimidocarbonyloxyethyl]sulfone a non cytotoxic, homobifunctional, lysine-specific cross-linking agent that reacts at 0 °C; the cross-links can be reversed at pH 11.6 at 37 °C.

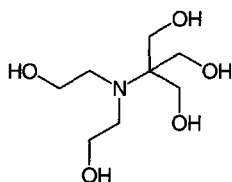


2,5-bis(3,4,5-trimethoxyphenyl)-1,3-dioxolane a substance, the *trans*-isomer of which inhibits platelet activating factor-induced platelet aggregation. The *cis*-isomer is inactive.

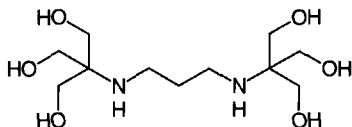


2,5-bis(3,4,5-trimethoxyphenyl)-1,3-dioxolane

Bistris or **bis-tris** *abbr.* for bis(2-hydroxyethyl)imino-tris-(hydroxymethyl)methane; 2-[bis(2-hydroxyethyl)amino]-2-(hydroxymethyl)-propane-1,3-diol. A buffer substance useful for the calibration of glass electrodes and for the preparation of biochemical and biological buffer solutions; pK_a (25 °C) = 6.5. It is inexpensive, stable, and readily crystallizable.



bis-tris propane *abbr.* for 1,3-bis[tris(hydroxymethyl)methylamino]propane; a water-soluble buffer substance used for the preparation of biochemical and biological buffer solutions; pK_a (20 °C) = 6.8.



bit a single binary digit having the value of either 0 or 1; the smallest unit of information in a binary system. A group of (usually) eight bits comprises a **byte**, which can represent numbers between 0 and 11111111 in binary (equal to 0 and 255 in decimal). [*abbr.* for binary digit.]

bithorax complex *see* **BX-C**.

Bittner factor or **Bittner particle** *former name* for murine mammary tumour virus. [After John Joseph Bittner (1904–61), US biologist.]

biuret reaction the formation of a characteristic purple colour when biuret (carbamoylurea) or compounds containing two or more adjacent peptide bonds (e.g. proteins) react with copper sulfate in alkaline solution. It forms the basis of the **biuret test**, a colorimetric method for the qualitative or quantitative determination of proteins.

Bjerrum formation function a function used in studies of the binding of n ligand atoms, ions, or molecules to an acceptor atom, ion, or molecule. It is:

$$\nu = \frac{(K_1[L] + 2K_1K_2[L]^2 + \dots + nK_1K_2\dots K_n[L]^n)}{(1 + K_1[L] + 2K_1K_2[L]^2 + \dots + nK_1K_2\dots K_n[L]^n)},$$

where ν is the average number of ligand molecules bound per molecule of acceptor; $K_1, K_2, \dots, K_n$ are the stepwise formation (stability) constants of the n complexes; and $[L]$ is the free ligand concentration. This function expresses an unambiguous connection between the quantity ν and $[L]$. It is a generalized form of the **Adair equation**. [After Jannik Bjerrum (1909–), Danish chemist who described it in 1941.]

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

Bjerrum plot a plot of ν , the average number of ligand molecules bound per molecule of acceptor, versus $-\log[L]$, where $[L]$ is the free ligand concentration. It is used for evaluating the association constants in a **Bjerrum formation function**.

Bk *symbol* for berkelium.

BK *abbr.* for bradykinin.

BK₁₋₈ *abbr.* for des-Arg⁹-bradykinin (*see* **bradykinin**).

bla *symbol* for β -lactamase gene.

Black, (Sir) James Whyte (1924–), British pharmacologist distinguished for his discoveries of blockade of β -adrenoceptors and of H₂-receptors by novel drugs; Nobel Laureate in Physiology or Medicine (1988) jointly with G. B. Elion and G. H. Hitchings 'for their discoveries of important principles for drug treatment'.

black lipid membrane *abbr.*: BLM; *other names*: black lipid film; bimolecular lipid membrane; bilayer lipid membrane; lipid bilayer. An experimental membrane formed by introducing a small amount of suitable lipid solution on to an opening in the wall of a hydrophobic support immersed in an aqueous solution. The thickness of the black membrane does not differ greatly from the length of two of the lipid molecules used and is often about 4.5 nm.

blank (determination) any analytical mixture from which analyte is deliberately omitted and which undergoes the same procedures as the complete mixture. It is used to ascertain the contribution of extraneous factors, e.g. reagent impurities or photometric absorption by the reagents, to the analytical result.

blast+ a variant form of **blasto+**.

+blast *suffix* denoting a formative cell, e.g. erythroblast, fibroblast, osteoblast.

blast cell or **blastocyte** any undifferentiated embryonic cell. Such cells characteristically have a cytoplasm rich in RNA, and are actively synthesizing DNA.

blast crisis the appearance of heterogeneity in the cell surface characteristics of white blood cells when chronic myeloid leukemia evolves into acute leukemia. This process is virtually inevitable and often rapid.

blasto+ or (*before a vowel*) **blast+** *prefix* denoting 1 an embryo in its early stages; e.g. blastocyst, blastula. 2 a germ cell; e.g. blastocyte.

blastokinin *see* **uteroglobin**.

blast transformation *see* **transformation** (def. 3).

bleb 1 a spherical vesicle, about 10 μ m in diameter, formed when a chloroplast swells on suspension in very hypotonic medium. It is bounded by a single membrane, on the surface of which a part of the **thylakoid** system remains concentrated in a few patches. 2 animal membranes are said to form blebs under special conditions that cause small vesicles to pinch off.

blender or (*sometimes, US*) **blendor** a device for disrupting tissue or any aggregate. It consists of blades rotating at the bottom of a glass or stainless-steel vessel (originally named a **Waring blender**, from a trade name); in an alternative design, known as a top-drive blender, the blades rotate on a spindle from the top of the vessel. The method is indiscriminating and may damage organelle membranes through excessive force or generation of heat. For more precise work, a **Potter-Elvehjem homogenizer** or a **Dounce homogenizer** preserves organelle structure.

bleomycin any of a group of related glycopeptide antibiotics isolated from *Streptomyces verticillus*. Bleomycin acts to stop the cell cycle in the G₂ phase (*see* **cell-division cycle**). It is used to induce synchrony in cell cultures and as an antineoplastic agent, especially in lymphomas.

blepharoplast an alternative term for **basal body** of a flagellate.

Blk a gene encoding a protein tyrosine kinase of the Src family, specifically expressed in the B-cell lineage.

BLM *abbr.* for black lipid membrane.

Bloch I Felix Bloch (1905–83), Swiss-born US physicist notable for his extensive studies in solid-state physics, for his measurement (with L. W. Alvarez) of the magnetic moment of the neutron, and for his discovery (independently of E. M. Purcell) of



block

the phenomenon of **nuclear magnetic resonance** of certain atomic nuclei; Nobel Laureate in Physics (1952) jointly with E. M. Purcell 'for their development of new methods for nuclear magnetic precision measurements and discoveries in connection therewith'. **2 Konrad Emil Bloch** (1912–), German-born US biochemist distinguished for his extensive studies by isotope techniques of the biosynthesis of cholesterol, creatine, and protoporphyrin; Nobel Laureate in Physiology or Medicine (1964) jointly with F. Lynen 'for their discoveries concerning the mechanism and regulation of the cholesterol and fatty acid metabolism'.

block 1 (*in chemistry*) a portion of a polymer molecule comprising many **constitutional base units**; such a portion has a constitutional or configurational feature not present in the adjacent multiple-unit portions. **2** (*in physiology*) an interference in the normal metabolic or physiological functioning of an organ or tissue; e.g. metabolic block, heart block. **3** to impede, retard, or prevent an action or activity, e.g. of an agonist, antigen, chemical reagent, enzyme, or receptor.

blockade 1 (*in pharmacology*) the saturation of a specific type of receptor with an antagonist to its normal agonist. **2** (*in immunology*) the overloading or saturation of the **reticulo-endothelial system** with inert particles, such as carbon particles. **3** to impose any such blockade.

blocker something that blocks an action or activity; an antagonist or inhibitor.

blocking agent 1 a chemical reagent that introduces a **blocking group**. **2** a specific pharmacological antagonist.

blocking antibody 1 an incomplete antibody that does not agglutinate its antigen but attaches to it and so prevents its agglutination by normal, complete antibody. **2** antibody formed during specific **desensitization** in **atopy**.

blocking group 1 or protecting group a chemical group, often easily removable, that is substituted into a reactive part of a molecule to prevent its participation in subsequent chemical or enzymic reactions occurring at another part of the molecule. **2** a chemical group, not easily removable, that is substituted into the active site of an enzyme to prevent its action.

block polymer a polymer whose molecules contain two or more species of **block** (def. 1) attached linearly.

blood a fluid tissue in animals that acts primarily as a transport medium. It is contained in vessels or spaces lined with endothelial cells. Metabolites, nutrients, and excretory substances are carried in solution or suspension from one part of the body to another. Respiratory pigments are often present to assist in the transport of dioxygen. It may also contain suspended cells.

blood-brain barrier the semipermeable membranous barrier that regulates the passage of dissolved materials from the blood into the cerebrospinal fluid that bathes the brain and spinal cord.

blood coagulation or blood clotting a mechanism that prevents blood loss at the site of an injury. It involves the formation of a semisolid mass of material, the blood clot, which plugs the wound. The clot consists of aggregated platelets and a mesh of fibrin molecules. The fibrin components include 13 or 14 plasma proteins, at least one tissue protein, phospholipid membrane surfaces, Ca^{2+} ions, and platelets. The current concept of the mechanism has changed from one involving a linear sequence of activations of proenzymes to one involving protein-protein and protein-lipid- Ca^{2+} interactions between proteinase, protein substrates, and protein cofactors to give discrete complexes and reaction steps. The following factors are involved in coagulation:

Factor I (or fibrinogen) concn. $\approx 2\text{--}4\text{ g L}^{-1}$ human plasma; a soluble fibrous protein of 330 kDa, it has an axial ratio of ≈ 5 and is made up of six disulfide-linked polypeptide chains; two $\text{A}\alpha$ chains (66 kDa); two $\text{B}\beta$ chains (52 kDa); and two γ chains (47 kDa). After proteolysis by thrombin it polymerizes to form an insoluble fibrin clot.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.

Factor II (or prothrombin) concn. 200 mg L^{-1} human plasma, 72.5 kDa; a vitamin K-dependent single-polypeptide-chain glycoprotein containing ten γ -carboxyglutamate residues. It is converted to **thrombin** (or fibrinogenase), EC 3.4.21.5, by the proteinase factor Xa.

Factor III (or thromboplastin or tissue factor) a factor that, when added to blood plasma, markedly shortens the time required for clot formation. Example from human (precursor): database code TF_HUMAN, 295 amino acids (33.07 kDa).

Factor IV (or calcium ion) calcium ions form bridges between vitamin K-dependent proteins and acid phospholipids of cell membranes. They also stabilize factor V and fibrinogen, and are involved in activation of factor XIII.

Factor V (or accelerator globulin or proaccelerin) a glycoprotein of $\approx 330\text{ kDa}$. After activation to **factor Va** (accelerin) it acts with the proteinase factor Xa in activating prothrombin. Example from human (precursor): database code FA5_HUMAN, 2224 amino acids (251.70 kDa).

Factor VI an obsolete term (originally named accelarin).

Factor VII (or proconvertin) concn. 2 mg L^{-1} human plasma; 50 kDa; a vitamin K-dependent single-polypeptide-chain glycoprotein containing γ -carboxyglutamate residues. It is the precursor of **factor VIIa**, convertin, a Ca^{2+} -dependent proteinase, EC 3.4.21.21, that activates factor X and is the first component of the extrinsic pathway of coagulation. Example from rabbit (precursor): database code FA7_RABBIT, 443 amino acids (48.85 kDa).

Factor VIII (or antihemophilic factor); an accessory protein that participates with proteinase factor IXa in the activation of factor X in the intrinsic pathway of coagulation. Example from mouse (precursor): database code FA8_MOUSE, 2319 amino acids (266.15 kDa).

Factor IX (or Christmas factor) concn. $3\text{--}4\text{ mg L}^{-1}$ human plasma; 57 kDa (human), 55.4 kDa (bovine); a vitamin K-dependent single-polypeptide-chain glycoprotein containing 17% carbohydrate (human); 25% in bovine) and γ -carboxyglutamate residues. After proteolytic activation it consists of two chains: **factor IXa**, a serine proteinase, EC 3.4.21.22, of 27 kDa that activates factor X in the intrinsic pathway of coagulation; and a light chain of 16 kDa that contains the γ -carboxyglutamate residues. Example from dog (precursor): database code FA9_CANFA, 452 amino acids (50.83 kDa).

Factor X (or Stuart-Prower factor; other names: Stuart factor; thrombokinas; prothrombase; prothrombinase) concn. $6\text{--}8\text{ mg L}^{-1}$ human plasma; a vitamin K-dependent two-polypeptide-chain glycoprotein, containing 15% carbohydrate (human; 10% in bovine). The 40 kDa chain is a serine proteinase while the 15 kDa chain contains 12 γ -carboxyglutamate residues. Example (precursor) from human: database code FA10_HUMAN, 488 amino-acid residues (54.67 kDa). It is the circulating precursor to the proteinase **factor Xa**, EC 3.4.21.6, that converts prothrombin into thrombin by selective cleavage of Arg-I-Thr then Arg-I-Ile bonds.

Factor XI (or plasma thromboplastin antecedent) 160 kDa (human), 124 kDa (bovine); a protein of two disulfide-linked chains of apparently equal molecular mass. Example from human: database code FA11_HUMAN, 625 amino acids (70.03 kDa). It is the circulating precursor of the serine proteinase **factor XIa**, EC 3.4.21.27, to which it is converted by proteolysis: selective cleavage of Arg-I-Ala and Arg-I-Val bonds in factor IX to form factor IXa triggers the middle phase of the intrinsic pathway of blood coagulation by activating factor IX. Factor XIa converts factor IX into factor IXa.

Factor XII (or Hageman factor) 80 kDa (human), 74 kDa (bovine); a single-polypeptide-chain glycoprotein. It is the circulating precursor of the proteinase **factor XIIa**, which activates factor XI. Example from human (precursor): database code FA12_HUMAN, 615 amino acids (67.82 kDa).



Factor XIII (or plasma transglutaminase) 320 kDa (plasma, tetrameric), 160 kDa (platelets, dimeric). An enzyme precursor whose subunits are proteolytically cleaved by thrombin to form the transglutaminase, **factor XIIIa**, EC 2.3.2.13, which covalently cross-links fibrin monomers into the fibrin polymer.

Prekallikrein a plasma protein, 88 kDa (bovine), that is the precursor of the proteinase **kallikrein**. This consists of a heavy chain (52 kDa) and a light chain (33–38 kDa) and cleaves factor XII to form factor XIIa.

High-molecular-weight kininogen a single-polypeptide-chain glycoprotein of ≈150 kDa (human), 76 kDa (bovine). It is the source of **bradykinin**, which is released from it by **kallikrein**, and acts as an accessory protein in the activation of factor XI by factor XIIa. Example from *Bos taurus* (precursor): database code KNH1_BOVIN, 621 amino acids (68.89 kDa).

Plasminogen 86 kDa (human); a single-polypeptide-chain protein that, on proteolysis, is converted into the protease **plasmin**, which degrades the fibrin clot.

There are two pathways of coagulation: in the **intrinsic pathway**, which occurs during the clotting of platelet-poor plasma, activation of factor X to factor Xa is effected by the combination of factor IXa + factor VIIIa + Ca^{2+} + phospholipid; in the **extrinsic pathway** the earlier stages of the cascade are bypassed and the activation of factor X to factor Xa is effected by the combination of factor VIIa + thromboplastin; this second pathway occurs when tissue extracts are present in optimal amounts and is much more rapid than the intrinsic pathway.

blood group any of various systems of classifying erythrocytes according to their surface isoantigens. In humans 14 different, genetically independent blood-group systems are known. They arise from naturally occurring antibodies and from the **blood-group substances**, antigenic substances present on the surface of the erythrocytes of only certain members of a species. These antigens are also found in a wide variety of other tissues, including endothelial and epithelial cells, and in certain body fluids. The specificities of the blood-group antigens arise in some cases from oligosaccharide chains linked to protein or lipid (ceramide); antigens that fall into this category are those associated with the ABH system, Lewis system, Ii system, and P1/P-related system. In other cases, antigenicity arises directly from the polypeptide chain composition of a surface protein; antigens that fall into this category are associated with the MN system, Ss system, Gerbich system, Rhesus system, Kell system, Duffy system, and Cromer-related system. The blood-group substances were discovered because of the presence in human blood of naturally occurring antibodies (i.e. not resulting from infection or other external cause) that react with these antigens on red blood cells causing them to agglutinate. Only the presence of a blood-group antigen during fetal life ensures that no antibody against it will be produced. Hence if blood from an individual (the donor) who lacks a particular antigen is transfused into an individual (the recipient) who possesses the same antigen, a transfusion reaction will result in the recipient owing to the presence in the donor's blood of antibody against the antigen in question. The blood-group substances have therefore been of interest largely in relation to blood transfusion, and in most cases the true physiological function of these substances is unknown.

blood-group substance any of numerous antigenic substances present on the surface of erythrocytes of certain members of a species but absent from those of other members of the same species. These same antigens are also found in a wide variety of tissues and in certain body fluids, e.g. saliva, milk, gastric juice, seminal fluid, urine, and ovarian-cyst fluid. The specificities of the A, B, and H antigens of the **ABO system** are determined by structures present at the nonreducing ends of oligosaccharides of both glycoproteins and glycolipids. The **Lewis antigens**, Le_a and Le_b , are associated with glycosphingolipids adsorbed from plasma lipoproteins on to adult

cells and are not synthesized in the developing erythrocyte. Antigenic determinants of the **MN system** are associated with glycoproteins rich in sialic acid.

blood plasma 1 the fluid portion of blood in which the red blood cells and other cellular components (the 'formed elements') are suspended. Clotting may be prevented by the addition of disodium hydrogen citrate which removes ionized Ca^{2+} by the formation of an un-ionized complex. Fluoride or oxalate may also be used to precipitate the Ca as an insoluble salt. **Heparin** is often used. When whole blood containing such an anticoagulant is centrifuged, the formed elements sediment in a volume equal to about 45% of the total volume of blood. The supernatant is known as the plasma. In vertebrates it is clear and almost colourless and clots as easily as whole blood. See also **anticoagulant**. Compare **serum**.

blood sugar old term for the glucose concentration in the blood; when estimated by older methods it included small amounts of other reducing substances. See also **glycemia**.

blotting or **blot transfer** a technique useful for identifying similar macromolecules, e.g. sets of DNA or RNA fragments, or mixtures of intact proteins, that are separable by gel electrophoresis. The separated components are then transferred from the gel to a sheet of an appropriate medium while preserving their two-dimensional relationships. This can be achieved either by capillary flow between an underlying wick loaded with buffer solution and an overlay of dry blotting paper, or by applying an electric potential to move separated components on to the medium, to which they bind firmly. Once on the transfer medium the substances may be detected with specific radioactive or fluorescent probes, e.g. complementary polynucleotides or antibodies. The original version of the technique, used for DNA fragments and often referred to as **Southern blotting**, has been extensively developed and adapted for other applications, e.g. **dot hybridization** (or dot blotting), **colony hybridization**, **electrophoretic transfer**, **Northern blotting**, and **Western blotting**.

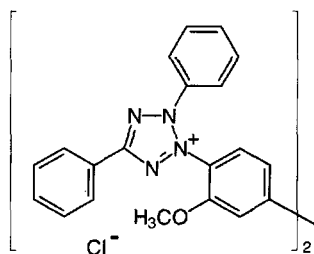
blue agarose any insoluble beaded form of agarose gel that is artificially cross-linked between the individual polysaccharide chains and conjugated with the dye Reactive Blue 2. Carboxymethyl (CM) or diethylaminoethyl (DEAE) groups may also have been introduced. It is useful in **triazine-dye affinity chromatography**.

blue dextran a soluble dextran, of average M_r 10^6 , conjugated with the dye Reactive Blue 2. It is useful in gel filtration for checking column packing and determining the void volume.

blue-green algae an alternative name for **cyanobacteria**.

blue oxidase any of a small number of copper-containing oxidase enzymes in which the product of enzyme action in the reduction of dioxygen is water. Blue oxidases are the most complicated of copper proteins; the blue colour is due to a strong absorption band around 600 nm. Among the blue oxidases are laccase (EC 1.10.3.2), L-ascorbate oxidase (EC 1.10.3.3), and the weak oxidase **ceruloplasmin**.

blue shift any shift of the peaks of absorption or emission of a spectrum of electromagnetic radiation to shorter wavelengths, including from the visible to the ultraviolet region.



Blue Tetrazolium

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.



Blue Tetrazolium or **Tetrazolium Blue** 3,3'-dianisole bis[4,4'-(3,5-diphenyl)tetrazolium chloride]; a dye that yields a dark-blue diformazan pigment on reduction; it has a redox potential of approximately -0.08 V. It is useful as a stain for bacteria and moulds, in histochemical studies to demonstrate oxidoreductase activity, and in the colorimetric assay of some dehydrogenases.

blunt end (of DNA) an alternative term for **flush end** (of DNA).

B lymphocyte or **B cell** or (archaic) **bursocyte** any 'bursa-dependent' lymphocyte concerned in the synthesis of circulating antibody. B lymphocytes arise from primitive lymphoid cells of the bone marrow and develop into the **plasma cell** series. Their maturation is determined by processing within the **bursa of Fabricius** in birds, or equivalent hematopoietic tissue in other vertebrates. B lymphocytes may be differentiated from **T lymphocytes** by their surface markers which contain immunoglobulins. In humans the majority of B lymphocytes are derived from bone marrow stem cells, but a minor population, distinguished by the CD5 marker, appears to form a self-renewing set. Cells in this set respond to a number of common microbial antigens and sometimes generate **autoantibodies**.

Bmi-1 a murine gene encoding a protein that is probably a transcriptional regulator. It is named from B-cell-specific Moloney murine leukemia virus integration site. The protein contains putative **zinc finger** and **helix-loop-helix** domains, as well as C-terminal PEST sequences. See **PEST hypothesis**.

BMP *abbr.* for bone morphogenetic protein.

BMR *abbr.* for basal metabolic rate.

BNP *abbr.* for brain natriuretic peptide (see **natriuretic peptide**).

boat conformation any conformation of a nonplanar six-membered saturated ring compound when ring atoms in relative positions 1, 2, 4, and 5 lie in one plane, and those in relative positions 3 and 6 lie on the same side of that plane. Boat conformations of a monosaccharide or monosaccharide derivative may be designated by the conformational descriptor **-B**. See also cyclic compounds at **conformation**.

Boc or **t-BOC** symbol for the *tert*-butoxycarbonyl group, $(\text{CH}_3)_3\text{C}-\text{O}-\text{CO}-$, a **protecting group** used in polypeptide chemistry. It can be introduced using reagents such as di-*tert*-butyl dicarbonate (Boc anhydride).

BOD *abbr.* for biochemical oxygen demand.

body water the water content of the body of an organism. The **total body water** (*abbr.* TBW) of a man of average build (70 kg) has a volume of approximately 42 L and represents $\approx 60\%$ of the body weight. It may be divided functionally into two parts: **intracellular fluid** and **extracellular fluid**.

Bohr 1 Christian Bohr (1855–1911), father of N. H. D. Bohr (see def. 2), Danish physiologist noted for his researches on the binding of dioxygen to hemoglobin (see **Bohr coefficient**, **Bohr effect**). 2 Niels Henrik David Bohr (1885–1962), son of C. Bohr (see def. 1), Danish physicist distinguished for his theories of atomic and nuclear structure and their applications; Nobel Laureate in Physics (1922) 'for his services in the investigation of the structure of atoms and of the radiation emanating from them'.

Bohr coefficient (for hemoglobin) the ratio of the change in the logarithm of the partial pressure of oxygen when there is half-saturation of the hemoglobin to the change in pH; i.e. $\Delta \log P_{50} / \Delta \text{pH}$. See also **Bohr effect**. [After C. Bohr; see **Bohr** (def. 2).]

Bohr effect the variation of oxygen affinity of hemoglobin with pH (the oxygen affinity increases with increasing pH). It is one of the effects arising from oxygen-linked acid groups in hemoglobin and similar oxygen-carrying proteins, and also encompasses similar effects involving other acid-linked functions, e.g. oxidation loss of protons causing increased dioxygen binding. The effect was discovered by C. Bohr *et al.* following observations that changes in the partial pressure of CO_2 influence the oxygen equilibrium of the blood.

Bohr magneton symbol: μ_B ; the intrinsic magnetic moment of an electron given by $\mu_B = eh/4\pi m_e$, where e is the elementary charge, h is the Planck constant, and m_e is the rest mass of an

electron. It is a fundamental physical constant, of value $9.274\,015\,4(31) \times 10^{-24} \text{ J T}^{-1}$. Compare **nuclear magneton**. [After N. H. D. Bohr; see **Bohr** (def. 2).]

boivinose the D-enantiomer, 2,6-dideoxy-D-xylo-hexose, is a component of some **cardiac glycosides**.

Bollum's enzyme DNA nucleotidyltransferase, EC 2.7.7.31. See **terminal transferase**. [After Frederick James Bollum (1927–), who described it in 1971.]

Bolton and Hunter reagent or **Bolton–Hunter reagent** 3-(4-hydroxyphenyl) propionic acid *N*-hydroxysuccinamide ester, iodinated with ^{125}I . This may be the di-iodo derivative, or the commercially available mono-iodo *N*-succinimidyl 3-(4-hydroxy-5-[^{125}I]iodophenyl)propionate; a reagent for **conjugation labelling** of polypeptides or proteins with iodine-125. It is specific for free amino groups, which under mild conditions become acylated to form (4-hydroxy-5-[^{125}I]iodophenyl)propionylamino groups. [After Anthony E. Bolton and William M. Hunter who described its use in 1972.]

Boltzmann constant symbol: k ; the ratio of the gas constant, R , to the Avogadro constant, N_A . It is given by $k = R/N_A = 1.380\,658\,(12) \times 10^{-23} \text{ J K}^{-1}$. [After Ludwig Boltzmann (1844–1906), Austrian physicist.]

Boltzmann distribution law a law describing the population distribution of a system of particles in different energy states but at thermal equilibrium, given by: $n_i/n_0 = e^{-(E_i - E_0)/kT}$, where n_i is the number of particles in energy state i , n_0 is the number in the lowest energy state, E_i and E_0 are the energies of states of i and 0 respectively, T is the thermodynamic temperature, and k is the **Boltzmann constant**.

bolus 1 a soft, rounded mass, especially of chewed food. 2 a large pill, as used in veterinary or clinical medicine.

bomb calorimeter an apparatus with which the heat of combustion of a substance may be determined. The substance is placed in a metal bomb with dioxygen under pressure, and then ignited electrically. The heat evolved is measured by the rise in temperature of the water in a surrounding jacket.

bombesin a tetradecapeptide amide isolated from frog skin. It has an N-terminal 5-oxoprolyl residue and shows striking sequence homology with other peptides isolated from amphibian skin, all of which have a C-terminal methioninamide residue. The sequence is (toad):

XQRLGNQWAVGHLM-NH₂,

database code (for precursor) BOMB_BOMOR, 119 amino acids (13.86 kDa). In mammals it is a potent releaser of, *inter alia*, gastrin and cholecystokinin, and bombesin-like immunoreactivity has been detected in mammalian gut and brain. It is mitogenic in a number of cell types and stimulates smooth muscle contraction. The peptide is said to occur in gastrointestinal nerves of mammals, while in frogs and birds it occurs in endocrine cells.

bombesin receptor any of several membrane proteins that bind bombesin and mediate its intracellular effects. Binding studies have indicated two receptor subtypes: on gastrin-releasing peptide (GRP)-preferring receptors (e.g. rat pancreas), GRP is more potent than neuromedin B; on other receptors (e.g. rat esophagus) neuromedin B is more potent than GRP. All receptors are of the seven-transmembrane-helix, G-protein-coupled type. The effector pathway involves the phosphoinositide/ Ca^{2+} second messenger system (see **phosphatidylinositol cycle**). Example from human: database code BRS3_HUMAN, 399 amino acids (44.41 kDa).

bombolitin any of five structurally related peptides from the venom of the bumblebee, *Megabombus pennsylvanicus*, that are rich in hydrophobic amino-acid residues. They all lyse erythrocytes and liposomes, release histamine from mast cells, and stimulate phospholipase A₂; their properties are similar to those of **melittin**. Example, bombolitin I: database code BOL1_MEGPE, sequence IKTITTTMLAKLGKVLAVH (1.836 kDa).

bombyxin any of a family of brain secretory peptides of the silkworm that activate prothoracic glands to produce

bond

ecdysone. They contain an A and B chain joined by disulfide bonds, and are homologous to vertebrate insulins. Example, bombyxin A-9 (precursor): database code BXA9_BOMMO, 92 amino acids (10.29 kDa); residues 20–47 are the B chain, 73–92 the A chain.

bond 1 or **covalent bond** a region of high electron density between atoms that arises, at least partly, from sharing of electrons and gives rise to an attractive force and a characteristic internuclear distance between the atoms. *See* **coordination**, **dipolar bond**, **hydrogen bond**. **2** the representation of a covalent bond in an extended chemical formula. **3** to join or link together (esp. chemical entities); to **ligate** (def. 1).

bond angle the angle formed between two bonds at a given polyvalent atom.

bond-dissociation energy *see* **dissociation energy**.

bond energy 1 the average value of the gas-phase **bond dissociation energies** (usually at 298 K) of a given type of bond in the same chemical species: **mean bond energy**. **2** loosely, the free energy change occurring on hydrolysis of a hydrolysable chemical bond.

bonding the act or process of forming a covalent **bond** (def. 1); the state of being so linked. *See also* **domain of bonding**.

bond length the distance between the centres of the nuclei of two bonded atoms.

bone the hard, dense connective tissue that constitutes the skeleton of many animals. It consists of an organic matrix impregnated with bone mineral salts. The matrix is made up of 90–95% type I collagen, a small amount of proteoglycan, and a 49-residue protein containing three γ -carboxyglutamate residues, the formation of which is vitamin K-dependent. This protein binds strongly to the hydroxyapatite crystals of the bone mineral. The inorganic part of the bone consists largely of basic calcium phosphate, which is organized into small crystals of hydroxyapatite, 0.8–1.5 nm thick, 2–4 nm wide, and 20–40 nm long, of approximate composition $\text{Ca}_{10}(\text{PO}_4)_6(\text{OH})_2$. Other anions present are carbonate, fluoride, hydroxide, and citrate. Most of the body's Mg^{2+} , about 25% of its Na^+ , and a smaller proportion of its K^+ is found in bone.

bone GLA-protein (BGP) *see* **osteocalcin**.

bone marrow the soft tissue contained within the internal cavities of bones. **Red marrow** (or myeloid tissue), found in developing bone, ribs, vertebrae, and in parts of the long bones, is actively hemopoietic and contains all the cells of the circulating blood and their precursors together with megakaryocytes, reticulum cells, macrophages, and plasma cells. In adult animals the marrow of many bones, particularly limb bones, becomes filled with fatty tissue known as **yellow marrow**. In adult mammals, B lymphocytes develop and differentiate in the bone marrow.

bone-marrow-derived cell any cell derived from bone marrow tissue, including erythrocytes, lymphocytes, granulocytes, etc. The term is often used misleadingly to refer to the **B lymphocytes**.

bone morphogenetic protein any of several zinc metallo-endopeptidases enzyme belonging to the sub-subclass EC 3.4.24 and having an EGF motif repeat; they are members of the **astacin** subfamily. BMP enzymes are disulfide-linked homodimers of the transforming growth factor- β family and are related to **tolloid**. They are involved in induction of bone and cartilage formation. Example BMP-1 precursor from human: database code BMP1_HUMAN, 730 amino acids (82.81 kDa).

bone sialoprotein 1 *see* **osteopontin**.

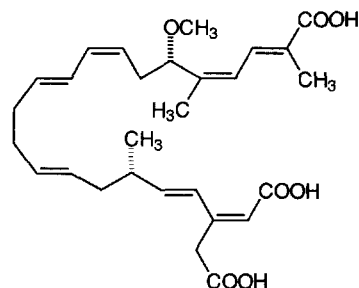
bongkreic acid 3-carboxymethyl-17-methoxy-6,18,21-trimethyldocos-2,4,8,12,14,18,20-heptaenedioic acid; a toxic antibiotic formed by *Pseudomonas cocovenenans* grown on partially defatted coconut. It is named after 'bongkre', a product of mouldy Indonesian coconut that becomes highly toxic when the *Pseudomonas* outgrows the mould. It is an inhibitor of oxidative phosphorylation, an effect ascribable to its inhibitory action on mitochondrial adenine nucleotide.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

translocation, apparently by fixing the adenine nucleotides to the translocation sites thus rendering dissociation of the nucleotides from the translocator rate-limiting in the overall transport process. *Compare* **atractyloside**.



booster dose a second or subsequent dose of antigen, especially in the form of a vaccine, given after the priming dose with the object of stimulating rapid production of large amounts of antibody.

Born, Max (1882–1970), German-born British physicist; Nobel Laureate in Physics (1954) 'for his fundamental research in quantum mechanics, especially for his statistical interpretation of the wavefunction' [prize shared with W. W. G. F. Bothe].

Born equation an equation for the free energy change, ΔG , when a sphere of charge e and radius r is brought into a uniform medium of unvarying dielectric constant D :

$$-\Delta G = (1 - 1/D)(e^2/2r).$$

borohydride any member of a group of compounds with the structure MBH_4 , where M is an alkali metal. These compounds will reduce aldehydes, ketones, and Schiff bases in nonaqueous solvents, and also acids, esters, acid chlorides, and nitriles. ^3H -labelled borohydrides are useful for the facile reductive tritiation of the aforementioned compounds.

boronyl the group $-\text{B}(\text{OH})_2$; it is used attached to a matrix to complex with *cis*-diols (as in some saccharides).

borosilicate (glass) a silicate glass containing at least 5% boric oxide. Such a material has a high melting point and a low coefficient of thermal expansion, and hence is useful for the manufacture of heat-resistant glassware.

boss symbol for bride-of-sevenless gene (*see* **Sevenless protein**).

Bothe, Walther Wilhelm Georg Franz (1891–1957), German experimental physicist noted for his development (with H. Geiger) of the coincidence method of particle counting and for his discovery of the particle later identified as the neutron; Nobel Laureate in Physics (1954) 'for the coincidence method and his discoveries made therewith' [prize shared with M. Born].

bottom yeast the popular name for any of various strains of **brewers' yeast** that effect fermentation at a comparatively low temperature and tend to sediment to the bottom of the fermentation vessel. It is used for the manufacture of light beers, e.g. lager. Bottom yeast is generally distinguishable from **top yeast** by its ability to produce extracellular β -galactosidase, EC 3.2.1.23.

botulinus toxin a mixture of exotoxins formed by *Clostridium botulinum*, the causal agent of botulism, that is a potent inhibitor of acetylcholine release from cholinergic neurons, possibly by inactivation of actin involved in transmitter exocytosis. There are several types, depending on the strain of the producing organism; all comprise a disulfide-linked heterodimer of heavy (H) and light (L) chains. The L chain has the pharmacological activity and the B chain has domains that mediate channel formation (N terminus) and toxin binding (C terminus). Example, type A precursor: database code

Bouguer-Beer law

BXA_CLOBO, 1295 amino acids (149.15 kDa): amino acids 1–447 form the L chain and 448–1295 form the H chain.

Bouguer-Beer law another name for **Beer's law** (def. 1).

Bouguer's law a physical law stating the relation between the amount of light transmitted by an absorbing medium and its thickness, based upon the finding that successive equal thicknesses of sheets of glass absorbed equal fractions of the light incident upon them. It may be expressed in the form $I_{\text{trs}} = I_0 e^{-kd}$, where I_0 is the intensity of the incident light, I_{trs} is the intensity of the transmitted light, d is the thickness of the absorbing medium, and k is a constant. The law was later restated by Lambert — see **Lambert's law (of absorption)** — and extended to solutions by Beer — see **Beer's law** (def. 1). [After Pierre B. Bouguer (1698–1758), French astronomer and mathematician who discovered it in 1728.]

boundary any zone of transition, either between solvent and solution or between two solutions; i.e. a zone in which the composition of a solution changes.

boundary spreading the broadening of a boundary during ultracentrifugation of macromolecules due to the effects of diffusion.

bound-to-free ratio *abbr.*: B/F ratio; the ratio of the amount of a ligand bound by a specific macromolecule or receptor to the amount of ligand free in solution, in a specified system. Plotting [B]/[F] versus [F] yields a **Scatchard plot**.

bouvardin a cyclic hexapeptide isolated from the plant *Bouvardia ternifolia*, used as a drug against dysentery. It has anti-tumour properties and inhibits eukaryotic protein synthesis.

bovine of, belonging to, or characteristic of the ox tribe Bovini, which includes cattle, especially of the genus *Bos*.

bovine albumin or **bovine serum albumin** see **serum albumin**.

bovine pancreatic inhibitor or **bovine pancreatic trypsin inhibitor** or **basic protease inhibitor** a protein from cattle pancreas that inhibits trypsin, kallikrein, chymotrypsin, and plasmin. A popular molecule for studies on protein folding, etc., it contains three disulfide bonds. Database code: NRL_5PTI and others (e.g. NRL_1TGSZ with bound proteinase). The sequence is:

RPDFCLEPPYTGPKARIIRYFYNAGLCQT-
FVYGGCAKRNFKSAEDCMRTCGGA.

(10.903 kDa). The 3-D structure is also known.

bovine spongiform encephalopathy *abbr.*: BSE; a transmissible dementia in cattle, sometimes also known as 'mad cow disease', characterized by a spongy degeneration of neuronal cells in the brain and also enlargement of the astrocytes therein. It appears to be caused by a **prion**, and original outbreaks were thought to be the result of feeding cattle on offal derived from sheep infected with **scrapie**.

Bowman-Birk inhibitor see **soybean trypsin inhibitor**.

Bowman's capsule or **glomerular capsule** the dilated, cup-shaped vesicle at the proximal end of a nephron that encloses a knot of blood capillaries constituting the glomerulus. It is the site of primary filtration of blood into the nephron. [After William Paget Bowman (1816–92), British histologist and ophthalmic surgeon.]

Boyden chamber an apparatus, consisting of two small chambers separated by a micropore filter, used in the study of chemotaxis.

Boyle's law see **gas laws**. [After Robert Boyle (1627–91), Irish chemist and physicist.]

bp symbol for base pair(s).

b.p. *abbr.* for boiling point.

BP *abbr.* for 1 blood pressure. 2 British Pharmacopoeia.

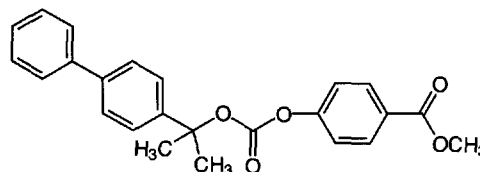
BPC *abbr.* for British Pharmaceutical Codex.

B peptide or **fibrinopeptide B** the 20-residue peptide cleaved from the β chains of a fibrinogen molecule when it is converted into fibrin by the proteolytic action of thrombin.

BPG *abbr.* for 2,3-bis(phospho)-D-glycerate.

BPOC *abbr.* for biphenylisopropoxy carbonyl-; an acid-labelling

group for amino acids, introduced using, e.g., 2-(4-biphenyl)-prop-2-yl 4'-methoxycarbonylphenylcarbonate (reagent structure shown).



Bq symbol for becquerel.

Br symbol for 1 bromine. 2 the butyryl group.

brachyurin EC 3.4.21.32; other name: collagenolytic proteinase; a serine proteinase that catalyses the hydrolysis of proteins with broad specificity for peptide bonds. It degrades native collagen at about 75% of the length of the molecule from the N terminus; it also shows low activity on small-molecule substrates of both trypsin and chymotrypsin. Example from *Uca pugilator* (fiddler crab): database code COGS_UCAPU, 226 amino acids (23.47 kDa).

bradykinin or **kinin-9** *abbr.*: BK; a vasoactive nonapeptide, H-Arg-Pro-Gly-Phe-Ser-Pro-Phe-Arg-OH, formed by the action of **plasma kallikrein**, which hydrolyses the sequence out of the plasma globulin **kininogen**. Glandular kallikrein cleaves kininogen one residue earlier to give the decapeptide Lys-bradykinin (kallidin, *abbr.*: Lys-BK). Met-Lys-bradykinin is also formed, perhaps by the action of leukocyte kallikrein. Pharmacologically important analogues include des-Arg⁹ or BK₁₋₈ and Ile-Ser-bradykinin (or T-kinin), [Hyp³]bradykinin, and [Hyp⁷]bradykinin. Bradykinin is formed in a variety of inflammatory conditions and in experimental anaphylactic shock. It is a powerful blood-vessel dilator, increasing vascular permeability and causing a fall in blood pressure; it is also a constrictor of smooth muscle. See also **lysine(arginine) carboxypeptidase**.

bradykinin receptor any membrane protein that binds bradykinin (BK) and mediates its intracellular effects. Two types of receptor are recognized: B₁ (previously BK₁), on which order of potency is des-Arg⁹-bradykinin (*abbr.*: BK₁₋₈) = kallidin (Lys-BK) > BK; and B₂ (previously BK₂), with order of potency kallidin > BK >> BK₁₋₈. Hence, BK₁₋₈ is a powerful discriminator. B₁ receptors are considerably less common than B₂ receptors, which are present in most tissues. The rat B₂ receptor is a seven-transmembrane-domain protein which has been shown on activation to stimulate phosphoinositide turnover. Examples: B₁, database code BRB1_HUMAN, 353 amino acids (40.44 kDa); B₂, database code BRB2_HUMAN, 364 amino acids (41.44 kDa).

Bragg 1 (Sir) William Henry Bragg (1862–1942), father of W. L. Bragg (see def. 2), British experimental physicist remembered for his invention of the X-ray spectrometer named after him and particularly for his use of it in collaboration with his son in solving many problems of concerning the arrangement of atoms in crystals; Nobel Laureate in Physics (1915) jointly with W. L. Bragg 'for their services in the analysis of crystal structure by means of X-rays'. 2 (Sir) **William Lawrence Bragg** (1890–1971), son of W. H. Bragg (see def. 1) British experimental physicist, born in Australia, remembered for his formulation of Bragg's law of X-ray diffraction and for the X-ray crystallographic studies in collaboration with his father that were thus made possible, for his subsequent concern with the application of X-ray crystallography to biological macromolecules, and for playing an important part in establishing molecular biology as an independent discipline; Nobel Laureate in Physics (1915) jointly with W. H. Bragg (see def. 1).

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYEN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentd.com.



Bragg's law

Bragg's law a law stating that when a crystal is pictured as a set of reflecting planes uniformly spaced at a distance d and a beam of X-rays of wavelength λ strikes the crystal at an angle θ , reinforcement of the reflected waves occurs when $\sin \theta = n\lambda/2d$, where n is an integer known as the **order of reflection**. [After W. L. Bragg; see **Bragg** (see def. 2).]

brain-derived neurotrophic factor *abbr.*: BDNF; a protein that promotes the survival of neuronal populations located either in the central nervous system or directly connected to it. Example (precursor) from chicken: database code BDNF_CHICK, 246 amino acids (27.68 kDa).

brain natriuretic peptide *see* **natriuretic peptide**.

branch 1 a subdivision of the stem or the root of a plant; any smaller structure growing or arising from a larger one. 2 a side chain attached to the main chain of a molecular entity. 3 to form a branch; to bifurcate; to divide into offshoots. — **branched** *adj.*; **branching** *n*.

branched-chain describing any molecular entity with one or more **branches** (def. 2).

branched-chain α -keto-acid dehydrogenase *abbr.*: BCKDH; EC 1.2.4.4; *recommended name*: 3-methyl-2-oxobutanoate dehydrogenase (lipoamide); *other name*: 2-oxoisovalerate dehydrogenase. An enzyme that catalyses the conversion of α -oxo acids to acyl-CoA and CO₂. Thiamine diphosphate is a coenzyme. The enzyme consists of a multi-enzyme complex of branched chain α -keto-acid dehydrogenase (E1), lipoamide acyltransferase (E2), and lipoamide dehydrogenase (E3). Human enzyme details: E1 is a dimer of α and β subunits; precursor database code ODBA_HUMAN, 444 amino acids (50.16 kDa) and ODBB_HUMAN, 392 amino acids (43.07 kDa); E2 database code (precursor) ODB2_HUMAN, 482 amino acids (53.43 kDa).

branched-chain α -keto-acid dehydrogenase kinase *see* [3-methyl-2-oxobutanoate dehydrogenase (lipoamide)] kinase.

branched-chain oxo-acid dehydrogenase phosphatase *see* [3-methyl-2-oxobutanoate dehydrogenase (lipoamide)]-phosphatase.

branched polymer any polymer in which there are side chains attached to a main chain.

brancher-deficiency amylopectinosis *an alternative name for* type IV **glycogen disease**.

branching (*in nuclear physics*) the simultaneous existence of more than one disintegration pathway for a particular radionuclide.

branching enzyme EC 2.4.1.18; *recommended name*: 1,4- α -glucan branching enzyme; *systematic name*: 1,4- α -D-glucan: 1,4- α -D-glucan 6- α -D-(1,4- α -D-glucano)-transferase; *other names*: amylo-(1,4-1,6)-transglycosylase; corn starch branching enzyme II; amylo-transglycosylase; glucanotransferase; starch branching enzyme II; any 1,4- α -glucan branching enzyme that catalyses the transfer of a segment of 1,4- α -glucan chain to a primary C-6 hydroxyl group in a similar glucan chain. Branching enzymes fall into two types: the plant **Q-enzymes** and the mammalian enzymes, which are able to produce glycogen, a more highly branched molecule than the amylopectin produced in plants by Q-enzymes. The name should always be qualified, e.g. glycogen branching enzyme, amylopectin branching enzyme. Example from *Zea mays*: database code MZEGLUCTR, 799 amino acids (90.42 kDa).

branch migration a proposed model to explain the occurrence of apparently branched DNA structures, as seen by electron microscopy, and their conversion into linear duplex DNA. The branched structures may form by association over parts of their lengths of three or more polynucleotide strands. The migration of the branch point is presumed to occur by displacement of a strand from its fellow in one branch by a strand in another. Such a process may occur in transcription or in recombination.

BRCA **BRCA1** and **BRCA2** either of two genes implicated in susceptibility to breast cancer. Up to 80% of women who in-

herit mutated forms of either gene develop breast cancer, and **BRCA1** mutations imply a high risk of ovarian cancer.

brdU or **BrdU** *abbr. for* bromodeoxyuridine (the symbol BrdUrd is recommended).

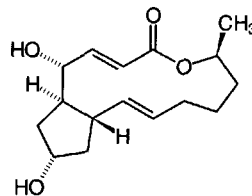
BrdUrd *symbol for* bromodeoxyuridine. *Compare* BrUrd.

breakage and reunion model the classical and generally accepted model of chromosomal crossing-over; it involves breakage of the chromatids and reunion of the alternative fragments.

breakpoint cluster region term originally suggested for a limited region of 5–6 kb in the base sequence of chromosome 22, in which breakpoints involved in formation of the **Philadelphia chromosome** were found to cluster. Molecular cloning of specific areas of DNA (e.g. band 22q11) gave a probe specific for the breakpoint translocation domain. Subsequent work led to this region being identified as the **BCR** gene.

breathing a *colloquial term for* transient dynamic changes in the structure of macromolecules involving expansion and contraction of the 3-D structure.

brefeldin A a fungal metabolite, isolated from *Penicillium brefeldianum*, that inhibits a wide variety of membrane transformations of importance in intracellular transport. It effects a rapid increase in the volume of Golgi cisternae and loss of recognizable Golgi apparatus in treated cells. There is a rapid reversible dissociation of a Golgi-associated peripheral protein identical with a coat protein subunit of non-clathrin-coated vesicles (β -COP). *See* **coatomer**.



B region the site of final epoxidation of carcinogenic polycyclic aromatic hydrocarbons. It is located on the terminal ring of the **bay region** on a bond adjacent to the bay region. *Compare* **A region**, **K region**, **L region**.

brei a finely and uniformly ground tissue suspension containing all the enzymes of the original material but with the normal spatial relationships between them disrupted. *Compare* **homogenate**.

bremstrahlung the **X-rays** produced when an electrically charged particle, usually a β particle or an accelerated electron, is decelerated by the electric field of an atomic nucleus. The yield is directly dependent upon the magnitude of the atomic number of the target element. [German: braking radiation.]

brevetoxin any of the highly toxic lipid-soluble polyether compounds isolated from *Ptychodiscus brevis* (*Gymnodinium breve*), the red-tide organism. They cause the death of fish and are responsible for shellfish poisoning in humans. They are activators of voltage-dependent Na⁺ channels.

brevin *an alternative term for* **gelsolin**.

brevinin a secreted amphibian peptide with antibacterial and hemolytic activities. Example, brevinin-1 from *Rana brevipoda* *porsa*: database code BR1_RANBP, sequence

FLPVLAGIAAKVVPALFCKITKKC.

brewers' yeast the common name for strains of the yeast *Saccharomyces cerevisiae* that are used in brewing. In addition to their ability to produce, and remain viable under, appreciable levels of ethanol, such strains are able to ferment maltose, maltotriose, and other sugars in the wort, and to withstand its relatively high osmotic pressure. *See also* **bottom yeast**.

bridge (*in chemistry*) a valence bond, atom, or unbranched chain of atoms connecting two different parts of a molecule.



bridgehead (*in chemistry*) either of the two atoms, e.g. tertiary carbon atoms, that are connected by a **bridge**.

bridge-receptor any molecular entity with two or more specific binding sites able to interact with and join a corresponding number of complementary ligands, which may or may not be the same.

Briggs-Haldane theory an alternative to the Michaelis-Menten theory of the kinetics of enzyme reactions for cases where breakdown of the enzyme-substrate complex to products is not slow in comparison with its dissociation to enzyme and substrate. *Compare* **Michaelis kinetics**. [After George Edward Briggs (1893–1985), British plant physiologist, and John Burdon Sanderson Haldane (1892–1964), British enzymologist, geneticist, philosopher, and popularizer of science, who put forward their theory in 1925.]

Brij *proprietary name* for any of a series of polyethylene ethers of higher fatty alcohols. They are surfactants useful for the solubilization of membrane fractions. Brij 35 is polyoxyethylene (23) lauryl ether, aggregation number 40, CMC 92 μ M.

British antilewisite *abbr.*: BAL; another name for **dimercaprol**.

British thermal unit *abbr.*: Btu; a non-SI unit given by the energy required to raise the temperature of 1 pound of water through 1 $^{\circ}$ F. Since this depends on the temperature of the water, the temperature range should be specified. The Btu(mean), defined as the 1/180 part of the energy required to raise the temperature of 1 pound of water from 32 $^{\circ}$ F to 212 $^{\circ}$ F, is equal to 1055.06 J.

Brk *abbr.* for Bek-related kinase; an alternative name for **Cek2**. *See* **Cek**.

BRN a family of neurally expressed genes encoding brain-specific transcription factors. They contain a **POU domain**. Brn-3.1 and Brn-3.2 modulate the terminal differentiation of auditory and visual system development respectively. Example, Brn-1 from *Mus musculus*: database code BRN1_MOUSE, 495 amino acids (50.01 kDa).

broad host range relating to the ability of a plasmid to be subjugated as a vector in a broad range of hosts, especially Gram-negative bacteria. Such plasmids are also termed promiscuous.

broad-spectrum antibiotic any antibiotic that is active against a wide range of bacterial species.

Brockmann body a discrete organ, occurring in fish, that contains tissue corresponding to that found in the islets of Langerhans of mammals. One or more Brockmann bodies occur in various teleost species. They are free of pancreatic acinar tissue, which makes them particularly suitable for biochemical studies and a good source of fish insulin. The principal islet weighs in the region of 1–50 mg; smaller secondary islets may also occur.

Brodie's fluid a **manometer fluid** containing 23 g NaCl, 5 g sodium tauroglycerate, 100 mg Evan's Blue, and a few drops of an ethanolic thymol solution, in 1000 mL of water; $D = 1.033$. The standard pressure (760 mmHg) in mm of manometer fluid is 10 000.

brom+ a variant form of **bromo+**.

bromelain or **bromelin** either of two similar **cysteine proteinases**: **stem bromelain**, EC 3.4.22.32, isolated from the stem of the pineapple, *Ananas comosus*; and **fruit bromelain**, EC 3.4.22.33, from the same plant. They are broad-specificity proteinases, similar to other cysteine proteinases but with so far unique geometry and reactivity at the active site. The two types of enzyme are distinguished by the substrate Z-Arg-Arg-NHMe, which is hydrolysed by stem bromelain but not by fruit bromelain. Stem enzyme: database code BROM_ANACO, 212 amino acids (22.81 kDa).

bromination the introduction of one or more bromine atoms (bromo residues) into an organic molecule, whether by addition or substitution. —**brominate** *vb*.

bromine *symbol*: Br; a halogen element of group 17 of the IUPAC **periodic table**; atomic number 35; relative atomic mass 79.909. Natural bromine consists of the two isotopes ^{79}Br and ^{81}Br at almost equal relative abundances.

Bạn đang đọc một quyển sách được tặng bởi HUY NGUYỄN. Nếu bạn có nhu cầu về sách sinh học, xin liên hệ:

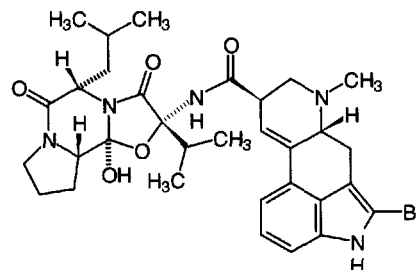
Huynhnguyen Co. LTD. Biotechnology, Trades and Services, 201 Luong Nhu Hoc, Dist 5, HCMC,

Phone/Fax: (84-8) 9509938, 8551478, 8574867 Handphone: 0918190535, email: huynhnguyen@huynhnguyentld.com.

bromine-82 the radioactive nuclide ^{82}Br ; half-life 35.3 h; it emits beta radiation (0.444 MeV) and gamma radiation of various energies.

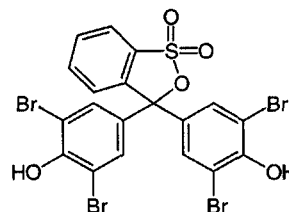
bromo+ or (*sometimes before a vowel*) **brom+** *prefix* denoting a bromine atom in organic linkage.

bromocriptine or **bromoergocriptine** 2-bromoergocryptine; an ergotoxin **ergot** alkaloid derivative that acts as a dopamine (D_2) receptor agonist. It inhibits the secretion of prolactin and somatotropin by the pituitary gland, and has potential applications in the treatment of Parkinson's disease.

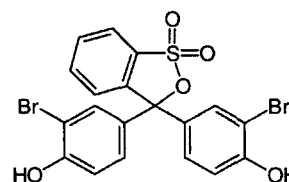


bromodeoxyuridine *symbol*: BrdUrd; 5-bromo-2'-deoxyuridine; 5-bromouracil 2'-deoxyriboside; a synthetic analogue of thymidine. It acts as an inhibitor of cell differentiation, and also has antiviral properties. *See* **bromouracil**.

Bromophenol Blue a dye used as a pH indicator, changing from yellow to blue over the pH range 3.0–4.6



Bromophenol Red a dye used as a pH indicator, changing from yellow to red over the pH range 5.2–6.8



Bromothymol Blue a dye used as a pH indicator, changing from yellow to blue over the pH range 6.0–7.6

